

# libbfd

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The Binary File Descriptor Library

First Edition—BFD version < 3.0 % Since no product is stable before version 3.0 :-)

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Cygnus Support

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# 1 Introduction

BFD is a package which allows applications to use the same routines to operate on object files whatever the object file format. A new object file format can be supported simply by creating a new BFD back end and adding it to the library.

BFD is split into two parts: the front end, and the back ends (one for each object file format).

- The front end of BFD provides the interface to the user. It manages memory and various canonical data structures. The front end also decides which back end to use and when to call back end routines.
- The back ends provide BFD its view of the real world. Each back end provides a set of calls which the BFD front end can use to maintain its canonical form. The back ends also may keep around information for their own use, for greater efficiency.

## 1.1 History

One spur behind BFD was the desire, on the part of the GNU 960 team at Intel Oregon, for interoperability of applications on their COFF and b.out file formats. Cygnus was providing GNU support for the team, and was contracted to provide the required functionality.

The name came from a conversation David Wallace was having with Richard Stallman about the library: RMS said that it would be quite hard—David said “BFD”. Stallman was right, but the name stuck.

At the same time, Ready Systems wanted much the same thing, but for different object file formats: IEEE-695, Oasys, Srecords, a.out and 68k coff.

BFD was first implemented by members of Cygnus Support; Steve Chamberlain ([sac@cygnus.com](mailto:sac@cygnus.com)), John Gilmore ([gnu@cygnus.com](mailto:gnu@cygnus.com)), K. Richard Pixley ([rich@cygnus.com](mailto:rich@cygnus.com)) and David Henkel-Wallace ([gumby@cygnus.com](mailto:gumby@cygnus.com)).

## 1.2 How To Use BFD

To use the library, include `bfd.h` and link with `libbfd.a`.

BFD provides a common interface to the parts of an object file for a calling application.

When an application successfully opens a target file (object, archive, or whatever), a pointer to an internal structure is returned. This pointer points to a structure called `bfd`, described in `bfd.h`. Our convention is to call this pointer a BFD, and instances of it within code `abfd`. All operations on the target object file are applied as methods to the BFD. The mapping is defined within `bfd.h` in a set of macros, all beginning with ‘`bfd_`’ to reduce namespace pollution.

For example, this sequence does what you would probably expect: return the number of sections in an object file attached to a BFD `abfd`.

```
#include "bfd.h"

unsigned int number_of_sections (abfd)
bfd *abfd;
{
```

```
    return bfd_count_sections (abfd);  
}
```

The abstraction used within BFD is that an object file has:

- a header,
- a number of sections containing raw data (see Section 2.6 [Sections], page 25),
- a set of relocations (see Section 2.10 [Relocations], page 50), and
- some symbol information (see Section 2.7 [Symbols], page 40).

Also, BFDs opened for archives have the additional attribute of an index and contain subordinate BFDs. This approach is fine for `a.out` and `coff`, but loses efficiency when applied to formats such as S-records and IEEE-695.

## 1.3 What BFD Version 2 Can Do

When an object file is opened, BFD subroutines automatically determine the format of the input object file. They then build a descriptor in memory with pointers to routines that will be used to access elements of the object file's data structures.

As different information from the object files is required, BFD reads from different sections of the file and processes them. For example, a very common operation for the linker is processing symbol tables. Each BFD back end provides a routine for converting between the object file's representation of symbols and an internal canonical format. When the linker asks for the symbol table of an object file, it calls through a memory pointer to the routine from the relevant BFD back end which reads and converts the table into a canonical form. The linker then operates upon the canonical form. When the link is finished and the linker writes the output file's symbol table, another BFD back end routine is called to take the newly created symbol table and convert it into the chosen output format.

### 1.3.1 Information Loss

*Information can be lost during output.* The output formats supported by BFD do not provide identical facilities, and information which can be described in one form has nowhere to go in another format. One example of this is alignment information in `b.out`. There is nowhere in an `a.out` format file to store alignment information on the contained data, so when a file is linked from `b.out` and an `a.out` image is produced, alignment information will not propagate to the output file. (The linker will still use the alignment information internally, so the link is performed correctly).

Another example is COFF section names. COFF files may contain an unlimited number of sections, each one with a textual section name. If the target of the link is a format which does not have many sections (e.g., `a.out`) or has sections without names (e.g., the Oasys format), the link cannot be done simply. You can circumvent this problem by describing the desired input-to-output section mapping with the linker command language.

*Information can be lost during canonicalization.* The BFD internal canonical form of the external formats is not exhaustive; there are structures in input formats for which there is no direct representation internally. This means that the BFD back ends cannot maintain all possible data richness through the transformation between external to internal and back to external formats.

This limitation is only a problem when an application reads one format and writes another. Each BFD back end is responsible for maintaining as much data as possible, and the internal BFD canonical form has structures which are opaque to the BFD core, and exported only to the back ends. When a file is read in one format, the canonical form is generated for BFD and the application. At the same time, the back end saves away any information which may otherwise be lost. If the data is then written back in the same format, the back end routine will be able to use the canonical form provided by the BFD core as well as the information it prepared earlier. Since there is a great deal of commonality between back ends, there is no information lost when linking or copying big endian COFF to little endian COFF, or `a.out` to `b.out`. When a mixture of formats is linked, the information is only lost from the files whose format differs from the destination.

### 1.3.2 The BFD canonical object-file format

The greatest potential for loss of information occurs when there is the least overlap between the information provided by the source format, that stored by the canonical format, and that needed by the destination format. A brief description of the canonical form may help you understand which kinds of data you can count on preserving across conversions.

*files* Information stored on a per-file basis includes target machine architecture, particular implementation format type, a demand pageable bit, and a write protected bit. Information like Unix magic numbers is not stored here—only the magic numbers’ meaning, so a `ZMAGIC` file would have both the demand pageable bit and the write protected text bit set. The byte order of the target is stored on a per-file basis, so that big- and little-endian object files may be used with one another.

*sections* Each section in the input file contains the name of the section, the section’s original address in the object file, size and alignment information, various flags, and pointers into other BFD data structures.

*symbols* Each symbol contains a pointer to the information for the object file which originally defined it, its name, its value, and various flag bits. When a BFD back end reads in a symbol table, it relocates all symbols to make them relative to the base of the section where they were defined. Doing this ensures that each symbol points to its containing section. Each symbol also has a varying amount of hidden private data for the BFD back end. Since the symbol points to the original file, the private data format for that symbol is accessible. `ld` can operate on a collection of symbols of wildly different formats without problems. Normal global and simple local symbols are maintained on output, so an output file (no matter its format) will retain symbols pointing to functions and to global, static, and common variables. Some symbol information is not worth retaining; in `a.out`, type information is stored in the symbol table as long symbol names. This information would be useless to most COFF debuggers; the linker has command-line switches to allow users to throw it away.

There is one word of type information within the symbol, so if the format supports symbol type information within symbols (for example, COFF, Oasys) and the type is simple enough to fit within one word (nearly everything but aggregates), the information will be preserved.

*relocation level*

Each canonical BFD relocation record contains a pointer to the symbol to relocate to, the offset of the data to relocate, the section the data is in, and a pointer to a relocation type descriptor. Relocation is performed by passing messages through the relocation type descriptor and the symbol pointer. Therefore, relocations can be performed on output data using a relocation method that is only available in one of the input formats. For instance, Oasys provides a byte relocation format. A relocation record requesting this relocation type would point indirectly to a routine to perform this, so the relocation may be performed on a byte being written to a 68k COFF file, even though 68k COFF has no such relocation type.

*line numbers*

Object formats can contain, for debugging purposes, some form of mapping between symbols, source line numbers, and addresses in the output file. These addresses have to be relocated along with the symbol information. Each symbol with an associated list of line number records points to the first record of the list. The head of a line number list consists of a pointer to the symbol, which allows finding out the address of the function whose line number is being described. The rest of the list is made up of pairs: offsets into the section and line numbers. Any format which can simply derive this information can pass it successfully between formats.

## 2 BFD Front End

### 2.1 typedef bfd

A BFD has type `bfd`; objects of this type are the cornerstone of any application using BFD. Using BFD consists of making references though the BFD and to data in the BFD.

Here is the structure that defines the type `bfd`. It contains the major data about the file and pointers to the rest of the data.

```
struct bfd
{
    /* The filename the application opened the BFD with.  */
    const char *filename;

    /* A pointer to the target jump table.  */
    const struct bfd_target *xvec;

    /* The IOSTREAM, and corresponding IO vector that provide access
       to the file backing the BFD.  */
    void *iostream;
    const struct bfd_iovec *iovec;

    /* The caching routines use these to maintain a
       least-recently-used list of BFDs.  */
    struct bfd *lru_prev, *lru_next;

    /* Track current file position (or current buffer offset for
       in-memory BFDs).  When a file is closed by the caching routines,
       BFD retains state information on the file here.  */
    ufile_ptr where;

    /* File modified time, if mtime_set is TRUE.  */
    long mtime;

    /* A unique identifier of the BFD  */
    unsigned int id;

    /* Format_specific flags.  */
    flagword flags;

    /* Values that may appear in the flags field of a BFD.  These also
       appear in the object_flags field of the bfd_target structure, where
       they indicate the set of flags used by that backend (not all flags
       are meaningful for all object file formats) (FIXME: at the moment,
       the object_flags values have mostly just been copied from backend
       to another, and are not necessarily correct).  */

```

```
#define BFD_NO_FLAGS                0x0

/* BFD contains relocation entries. */
#define HAS_RELOC                    0x1

/* BFD is directly executable. */
#define EXEC_P                        0x2

/* BFD has line number information (basically used for F_LNNO in a
   COFF header). */
#define HAS_LINENO                    0x4

/* BFD has debugging information. */
#define HAS_DEBUG                      0x08

/* BFD has symbols. */
#define HAS_SYMS                      0x10

/* BFD has local symbols (basically used for F_LSYMS in a COFF
   header). */
#define HAS_LOCALS                    0x20

/* BFD is a dynamic object. */
#define DYNAMIC                      0x40

/* Text section is write protected (if D_PAGED is not set, this is
   like an a.out NMAGIC file) (the linker sets this by default, but
   clears it for -r or -N). */
#define WP_TEXT                      0x80

/* BFD is dynamically paged (this is like an a.out ZMAGIC file) (the
   linker sets this by default, but clears it for -r or -n or -N). */
#define D_PAGED                      0x100

/* BFD is relaxable (this means that bfd_relax_section may be able to
   do something) (sometimes bfd_relax_section can do something even if
   this is not set). */
#define BFD_IS_RELAXABLE              0x200

/* This may be set before writing out a BFD to request using a
   traditional format. For example, this is used to request that when
   writing out an a.out object the symbols not be hashed to eliminate
   duplicates. */
#define BFD_TRADITIONAL_FORMAT        0x400

/* This flag indicates that the BFD contents are actually cached
   in memory. If this is set, iostream points to a malloc'd
```

```

        bfd_in_memory struct. */
#define BFD_IN_MEMORY                0x800

        /* This BFD has been created by the linker and doesn't correspond
           to any input file. */
#define BFD_LINKER_CREATED            0x1000

        /* This may be set before writing out a BFD to request that it
           be written using values for UIDs, GIDs, timestamps, etc. that
           will be consistent from run to run. */
#define BFD_DETERMINISTIC_OUTPUT     0x2000

        /* Compress sections in this BFD. */
#define BFD_COMPRESS                  0x4000

        /* Decompress sections in this BFD. */
#define BFD_DECOMPRESS                0x8000

        /* BFD is a dummy, for plugins. */
#define BFD_PLUGIN                     0x10000

        /* Compress sections in this BFD with SHF_COMPRESSED from gABI. */
#define BFD_COMPRESS_GABI              0x20000

        /* Convert ELF common symbol type to STT_COMMON or STT_OBJECT in this
           BFD. */
#define BFD_CONVERT_ELF_COMMON        0x40000

        /* Use the ELF STT_COMMON type in this BFD. */
#define BFD_USE_ELF_STT_COMMON        0x80000

        /* Put pathnames into archives (non-POSIX). */
#define BFD_ARCHIVE_FULL_PATH         0x100000

#define BFD_CLOSED_BY_CACHE           0x200000
        /* Compress sections in this BFD with SHF_COMPRESSED zstd. */
#define BFD_COMPRESS_ZSTD             0x400000

        /* Don't generate ELF section header. */
#define BFD_NO_SECTION_HEADER         0x800000

        /* Flags bits which are for BFD use only. */
#define BFD_FLAGS_FOR_BFD_USE_MASK \
        (BFD_IN_MEMORY | BFD_COMPRESS | BFD_DECOMPRESS | BFD_LINKER_CREATED \
         | BFD_PLUGIN | BFD_TRADITIONAL_FORMAT | BFD_DETERMINISTIC_OUTPUT \
         | BFD_COMPRESS_GABI | BFD_CONVERT_ELF_COMMON | BFD_USE_ELF_STT_COMMON \
         | BFD_NO_SECTION_HEADER)

```



```
/* The format which belongs to the BFD. (object, core, etc.) */
ENUM_BITFIELD (bfd_format) format : 3;

/* The direction with which the BFD was opened. */
ENUM_BITFIELD (bfd_direction) direction : 2;

/* POSIX.1-2017 (IEEE Std 1003.1) says of fopen : "When a file is
   opened with update mode ('+' as the second or third character in
   the mode argument), both input and output may be performed on
   the associated stream. However, the application shall ensure
   that output is not directly followed by input without an
   intervening call to fflush() or to a file positioning function
   (fseek(), fsetpos(), or rewind()), and input is not directly
   followed by output without an intervening call to a file
   positioning function, unless the input operation encounters
   end-of-file."
   This field tracks the last IO operation, so that bfd can insert
   a seek when IO direction changes. */
ENUM_BITFIELD (bfd_last_io) last_io : 2;

/* Is the file descriptor being cached? That is, can it be closed as
   needed, and re-opened when accessed later? */
unsigned int cacheable : 1;

/* Marks whether there was a default target specified when the
   BFD was opened. This is used to select which matching algorithm
   to use to choose the back end. */
unsigned int target_defaulted : 1;

/* ... and here: ('once' means at least once). */
unsigned int opened_once : 1;

/* Set if we have a locally maintained mtime value, rather than
   getting it from the file each time. */
unsigned int mtime_set : 1;

/* Flag set if symbols from this BFD should not be exported. */
unsigned int no_export : 1;

/* Remember when output has begun, to stop strange things
   from happening. */
unsigned int output_has_begun : 1;

/* Have archive map. */
unsigned int has_armap : 1;
```

```
/* Accept a mapless archive for link. */
unsigned int link_mapless : 1;

/* Set if this is a thin archive. */
unsigned int is_thin_archive : 1;

/* Set if this is a collection of files pretending to be an archive. */
unsigned int is_fake_archive : 1;

/* Set if this archive should not cache element positions. */
unsigned int no_element_cache : 1;

/* Set if only required symbols should be added in the link hash table for
   this object. Used by VMS linkers. */
unsigned int selective_search : 1;

/* Set if this is the linker output BFD. */
unsigned int is_linker_output : 1;

/* Set if this is the linker input BFD. */
unsigned int is_linker_input : 1;

/* If this is an input for a compiler plug-in library. */
ENUM_BITFIELD (bfd_plugin_format) plugin_format : 2;

/* Set if this is a plugin output file. */
unsigned int lto_output : 1;

/* Do not attempt to modify this file. Set when detecting errors
   that BFD is not prepared to handle for objcopy/strip. */
unsigned int read_only : 1;

/* LTO object type. */
ENUM_BITFIELD (bfd_lto_object_type) lto_type : 3;

/* Set if this BFD is currently being processed by
   bfd_check_format_matches. This is checked by the cache to
   avoid closing the BFD in this case. This should only be
   examined or modified while the BFD lock is held. */
unsigned int in_format_matches : 1;

/* Set to dummy BFD created when claimed by a compiler plug-in
   library. */
bfd *plugin_dummy_bfd;

/* The offset of this bfd in the file, typically 0 if it is not
   contained in an archive. */
```

```
ufile_ptr origin;

/* A reference in the archive for the proxy entry as follows:

    1. For regular archives this will be the same as origin.

    2. For thin archives it will contain the current offset
       of the proxy in the thin archive rather than the offset
       of the bfd in its actual container.

    3. For fake archives it will contain the next archive member's
       BFD reference or a NULL pointer if this is the last member. */
ufile_ptr_or_bfd proxy_handle;

/* A hash table for section names. */
struct bfd_hash_table section_htab;

/* Pointer to linked list of sections. */
struct bfd_section *sections;

/* The last section on the section list. */
struct bfd_section *section_last;

/* The object-only section on the section list. */
struct bfd_section *object_only_section;

/* The number of sections. */
unsigned int section_count;

/* The archive plugin file descriptor. */
int archive_plugin_fd;

/* The number of opens on the archive plugin file descriptor. */
unsigned int archive_plugin_fd_open_count;

/* A field used by _bfd_generic_link_add_archive_symbols. This will
   be used only for archive elements. */
int archive_pass;

/* The total size of memory from bfd_alloc. */
bfd_size_type alloc_size;

/* Stuff only useful for object files:
   The start address. */
bfd_vma start_address;

/* Symbol table for output BFD (with symcount entries).
```

```

    Also used by the linker to cache input BFD symbols.  */
    struct bfd_symbol **outsymbols;

    /* Used for input and output.  */
    unsigned int symcount;

    /* Used for slurped dynamic symbol tables.  */
    unsigned int dynsymcount;

    /* Pointer to structure which contains architecture information.  */
    const struct bfd_arch_info *arch_info;

    /* Cached length of file for bfd_get_size.  0 until bfd_get_size is
       called, 1 if stat returns an error or the file size is too large to
       return in ufile_ptr.  Both 0 and 1 should be treated as "unknown".  */
    ufile_ptr size;

    /* Stuff only useful for archives.  */
    void *arelt_data;
    struct bfd *my_archive;      /* The containing archive BFD.  */
    struct bfd *archive_next;    /* The next BFD in the archive.  */
    struct bfd *archive_head;    /* The first BFD in the archive.  */
    struct bfd *nested_archives; /* List of nested archive in a flattened
                                   thin archive.  */

    union {
        /* For input BFDs, a chain of BFDs involved in a link.  */
        struct bfd *next;
        /* For output BFD, the linker hash table.  */
        struct bfd_link_hash_table *hash;
    } link;

    /* Used by the back end to hold private data.  */
    union
    {
        struct aout_data_struct *aout_data;
        struct artdat *aout_ar_data;
        struct coff_tdata *coff_obj_data;
        struct pe_tdata *pe_obj_data;
        struct xcoff_tdata *xcoff_obj_data;
        struct ecoff_tdata *ecoff_obj_data;
        struct srec_data_struct *srec_data;
        struct verilog_data_struct *verilog_data;
        struct ihex_data_struct *ihex_data;
        struct tekhex_data_struct *tekhex_data;
        struct elf_obj_tdata *elf_obj_data;
        struct mmo_data_struct *mmo_data;
    }

```

```

    struct trad_core_struct *trad_core_data;
    struct som_data_struct *som_data;
    struct hpux_core_struct *hpux_core_data;
    struct hppabsd_core_struct *hppabsd_core_data;
    struct sgi_core_struct *sgi_core_data;
    struct lynx_core_struct *lynx_core_data;
    struct osf_core_struct *osf_core_data;
    struct cisco_core_struct *cisco_core_data;
    struct netbsd_core_struct *netbsd_core_data;
    struct mach_o_data_struct *mach_o_data;
    struct mach_o_fat_data_struct *mach_o_fat_data;
    struct plugin_data_struct *plugin_data;
    struct bfd_pef_data_struct *pef_data;
    struct bfd_pef_xlib_data_struct *pef_xlib_data;
    struct bfd_sym_data_struct *sym_data;
    void *any;
}
tdata;

/* Used by the application to hold private data. */
void *usrdata;

/* Where all the allocated stuff under this BFD goes. This is a
   struct objalloc *, but we use void * to avoid requiring the inclusion
   of objalloc.h. */
void *memory;

/* For input BFDs, the build ID, if the object has one. */
const struct bfd_build_id *build_id;

/* For input BFDs, mmapped entries. */
struct bfd_mmapped *mmapped;
};

```

## 2.2 Error reporting

Most BFD functions return nonzero on success (check their individual documentation for precise semantics). On an error, they call `bfd_set_error` to set an error condition that callers can check by calling `bfd_get_error`. If that returns `bfd_error_system_call`, then check `errno`.

The easiest way to report a BFD error to the user is to use `bfd_perror`.

The BFD error is thread-local.

### 2.2.1 Type `bfd_error_type`

The values returned by `bfd_get_error` are defined by the enumerated type `bfd_error_type`.

```

typedef enum bfd_error
{
    bfd_error_no_error = 0,
    bfd_error_system_call,
    bfd_error_invalid_target,
    bfd_error_wrong_format,
    bfd_error_wrong_object_format,
    bfd_error_invalid_operation,
    bfd_error_no_memory,
    bfd_error_no_symbols,
    bfd_error_no_armap,
    bfd_error_no_more_archived_files,
    bfd_error_malformed_archive,
    bfd_error_missing_dso,
    bfd_error_file_not_recognized,
    bfd_error_file_ambiguously_recognized,
    bfd_error_no_contents,
    bfd_error_nonrepresentable_section,
    bfd_error_no_debug_section,
    bfd_error_bad_value,
    bfd_error_file_truncated,
    bfd_error_file_too_big,
    bfd_error_sorry,
    bfd_error_on_input,
    bfd_error_invalid_error_code
}
bfd_error_type;

```

### 2.2.1.1 bfd\_get\_error

`bfd_error_type bfd_get_error (void);` [Function]  
 Return the current BFD error condition.

### 2.2.1.2 bfd\_set\_error

`void bfd_set_error (bfd_error_type error_tag);` [Function]  
 Set the BFD error condition to be *error\_tag*.  
*error\_tag* must not be `bfd_error_on_input`. Use `bfd_set_input_error` for input errors instead.

### 2.2.1.3 bfd\_set\_input\_error

`void bfd_set_input_error (bfd *input, bfd_error_type error_tag);` [Function]  
 Set the BFD error condition to be `bfd_error_on_input`. *input* is the input bfd where the error occurred, and *error\_tag* the `bfd_error_type` error.

### 2.2.1.4 bfd\_errmsg

**const char \*bfd\_errmsg (bfd\_error\_type error\_tag);** [Function]  
 Return a string describing the error *error\_tag*, or the system error if *error\_tag* is *bfd\_error\_system\_call*.

### 2.2.1.5 bfd\_perror

**void bfd\_perror (const char \*message);** [Function]  
 Print to the standard error stream a string describing the last BFD error that occurred, or the last system error if the last BFD error was a system call failure. If *message* is non-NULL and non-empty, the error string printed is preceded by *message*, a colon, and a space. It is followed by a newline.

### 2.2.1.6 bfd\_asprintf

**char \*bfd\_asprintf (const char \*fmt, ...);** [Function]  
 Primarily for error reporting, this function is like *liberty's xasprintf* except that it can return NULL on no memory and the returned string should not be freed. Uses a thread-local *malloc'd* buffer managed by *libbfd*, *\_bfd\_error\_buf*. Be aware that a call to this function frees the result of any previous call. *bfd\_errmsg (bfd\_error\_on\_input)* also calls this function.

## 2.2.2 BFD error handler

Some BFD functions want to print messages describing the problem. They call a BFD error handler function. This function may be overridden by the program.

The BFD error handler acts like *vprintf*.

```
typedef void (*bfd_error_handler_type) (const char *, va_list);
```

```
typedef int (*bfd_print_callback) (void *, const char *, ...);
```

### 2.2.2.1 bfd\_print\_error

**void bfd\_print\_error (bfd\_print\_callback print\_func, void \*stream, const char \*fmt, va\_list ap);** [Function]  
 This formats *FMT* and *AP* according to BFD "printf" rules, sending the output to *STREAM* by repeated calls to *PRINT\_FUNC*. *PRINT\_FUNC* is a printf-like function; it does not need to implement the BFD printf format extensions. This can be used in a callback that is set via *bfd\_set\_error\_handler* to turn the error into ordinary output.

### 2.2.2.2 \_bfd\_error\_handler

**void \_bfd\_error\_handler (const char \*fmt, ...)** [Function]  
*ATTRIBUTE\_PRINTF\_1;*  
 This is the default routine to handle BFD error messages. Like *fprintf (stderr, ...)*, but also handles some extra format specifiers.  
*%pA* section name from section. For group components, prints group name too. *%pB* file name from *bfd*. For archive components, prints archive too.  
 Beware: Only supports a maximum of 9 format arguments.

### 2.2.2.3 bfd\_set\_error\_handler

`bfd_error_handler_type` `bfd_set_error_handler` [Function]  
     (*bfd\_error\_handler\_type*);

Set the BFD error handler function. Returns the previous function.

### 2.2.2.4 \_bfd\_set\_error\_handler\_caching

`struct per_xvec_messages` `*_bfd_set_error_handler_caching` (*struct* [Function]  
     *per\_xvec\_messages* \*);

Set the BFD error handler function to one that stores messages to the `per_xvec_messages` object. Returns the previous object to which messages are stored. Note that two sequential calls to this with a non-NULL argument will cause output to be dropped, rather than gathered.

### 2.2.2.5 \_bfd\_restore\_error\_handler\_caching

`void` `_bfd_restore_error_handler_caching` (*struct* [Function]  
     *per\_xvec\_messages* \*);

Reset the BFD error handler object to an earlier value.

### 2.2.2.6 bfd\_set\_error\_program\_name

`void` `bfd_set_error_program_name` (*const char* \*); [Function]

Set the program name to use when printing a BFD error. This is printed before the error message followed by a colon and space. The string must not be changed after it is passed to this function.

### 2.2.2.7 \_bfd\_get\_error\_program\_name

`const char` `*_bfd_get_error_program_name` (*void*); [Function]

Get the program name used when printing a BFD error.

## 2.2.3 BFD assert handler

If BFD finds an internal inconsistency, the bfd assert handler is called with information on the BFD version, BFD source file and line. If this happens, most programs linked against BFD are expected to want to exit with an error, or mark the current BFD operation as failed, so it is recommended to override the default handler, which just calls `_bfd_error_handler` and continues.

```
typedef void (*bfd_assert_handler_type) (const char *bfd_formatmsg,
                                         const char *bfd_version,
                                         const char *bfd_file,
                                         int bfd_line);
```

### 2.2.3.1 bfd\_set\_assert\_handler

`bfd_assert_handler_type` `bfd_set_assert_handler` [Function]  
     (*bfd\_assert\_handler\_type*);

Set the BFD assert handler function. Returns the previous function.



### 2.2.3.2 bfd\_init

`unsigned int bfd_init (void);` [Function]

This routine must be called before any other BFD function to initialize magical internal data structures. Returns a magic number, which may be used to check that the bfd library is configured as expected by users.

```
/* Value returned by bfd_init. */
#define BFD_INIT_MAGIC (sizeof (struct bfd_section))
```

## 2.3 Threading

BFD has limited support for thread-safety. Most BFD globals are protected by locks, while the error-related globals are thread-local. A given BFD cannot safely be used from two threads at the same time; it is up to the application to do any needed locking. However, it is ok for different threads to work on different BFD objects at the same time.

### 2.3.1 Thread functions.

```
typedef bool (*bfd_lock_unlock_fn_type) (void *);
```

#### 2.3.1.1 \_bfd\_threading\_enabled

`bool _bfd_threading_enabled (void);` [Function]

Return true if threading is enabled, false if not.

#### 2.3.1.2 bfd\_thread\_init

`bool bfd_thread_init (bfd_lock_unlock_fn_type lock, [Function]  
bfd_lock_unlock_fn_type unlock, void *data);`

Initialize BFD threading. The functions passed in will be used to lock and unlock global data structures. This may only be called a single time in a given process. Returns true on success and false on error. On error, the caller should assume that BFD cannot be used by multiple threads. DATA is passed verbatim to the lock and unlock functions. The lock and unlock functions should return true on success, or set the BFD error and return false on failure. Note also that the lock must be a recursive lock: BFD may attempt to acquire the lock when it is already held by the current thread.

#### 2.3.1.3 bfd\_thread\_cleanup

`void bfd_thread_cleanup (void);` [Function]

Clean up any thread-local state. This should be called by a thread that uses any BFD functions, before the thread exits. It is fine to call this multiple times, or to call it and then later call BFD functions on the same thread again.

#### 2.3.1.4 bfd\_lock

`bool bfd_lock (void);` [Function]

Acquire the global BFD lock, if needed. Returns true on success, false on error.

### 2.3.1.5 bfd\_unlock

`bool bfd_unlock (void);` [Function]  
 Release the global BFD lock, if needed. Returns true on success, false on error.

## 2.4 Miscellaneous

### 2.4.1 Miscellaneous functions

#### 2.4.1.1 bfd\_get\_reloc\_upper\_bound

`long bfd_get_reloc_upper_bound (bfd *abfd, asection *sect);` [Function]  
 Return the number of bytes required to store the relocation information associated with section *sect* attached to bfd *abfd*. If an error occurs, return -1.

#### 2.4.1.2 bfd\_canonicalize\_reloc

`long bfd_canonicalize_reloc (bfd *abfd, asection *sec, arelent **loc, asymbol **syms);` [Function]  
 Call the back end associated with the open BFD *abfd* and translate the external form of the relocation information attached to *sec* into the internal canonical form. Place the table into memory at *loc*, which has been preallocated, usually by a call to `bfd_get_reloc_upper_bound`. Returns the number of relocs, or -1 on error.  
 The *syms* table is also needed for horrible internal magic reasons.

#### 2.4.1.3 bfd\_finalize\_section\_relocs

`bool bfd_finalize_section_relocs (bfd *abfd, asection *sec, arelent **rel, unsigned int count);` [Function]  
 Set the relocation pointer and count within section *sec* to the values *rel* and *count*, and take any other actions required at the conclusion of section relocation processing.

```
#define bfd_finalize_section_relocs(abfd, asect, location, count) \
    BFD_SEND (abfd, _bfd_finalize_section_relocs, \
              (abfd, asect, location, count))
```

#### 2.4.1.4 bfd\_set\_file\_flags

`bool bfd_set_file_flags (bfd *abfd, flagword flags);` [Function]  
 Set the flag word in the BFD *abfd* to the value *flags*.

Possible errors are:

- `bfd_error_wrong_format` - The target bfd was not of object format.
- `bfd_error_invalid_operation` - The target bfd was open for reading.
- `bfd_error_invalid_operation` - The flag word contained a bit which was not applicable to the type of file. E.g., an attempt was made to set the `D_PAGED` bit on a BFD format which does not support demand paging.

### 2.4.1.5 bfd\_get\_arch\_size

**int bfd\_get\_arch\_size (bfd \*abfd);** [Function]  
 Returns the normalized architecture address size, in bits, as determined by the object file's format. By normalized, we mean either 32 or 64. For ELF, this information is included in the header. Use `bfd_arch_bits_per_address` for number of bits in the architecture address.  
 Returns the arch size in bits if known, -1 otherwise.

### 2.4.1.6 bfd\_get\_sign\_extend\_vma

**int bfd\_get\_sign\_extend\_vma (bfd \*abfd);** [Function]  
 Indicates if the target architecture "naturally" sign extends an address. Some architectures implicitly sign extend address values when they are converted to types larger than the size of an address. For instance, `bfd_get_start_address()` will return an address sign extended to fill a `bfd_vma` when this is the case.  
 Returns 1 if the target architecture is known to sign extend addresses, 0 if the target architecture is known to not sign extend addresses, and -1 otherwise.

### 2.4.1.7 bfd\_set\_start\_address

**bool bfd\_set\_start\_address (bfd \*abfd, bfd\_vma vma);** [Function]  
 Make *vma* the entry point of output BFD *abfd*.  
 Returns TRUE on success, FALSE otherwise.

### 2.4.1.8 bfd\_get\_gp\_size

**unsigned int bfd\_get\_gp\_size (bfd \*abfd);** [Function]  
 Return the maximum size of objects to be optimized using the GP register under MIPS ECOFF. This is typically set by the `-G` argument to the compiler, assembler or linker.

### 2.4.1.9 bfd\_set\_gp\_size

**void bfd\_set\_gp\_size (bfd \*abfd, unsigned int i);** [Function]  
 Set the maximum size of objects to be optimized using the GP register under ECOFF or MIPS ELF. This is typically set by the `-G` argument to the compiler, assembler or linker.

### 2.4.1.10 bfd\_set\_gp\_value

**void bfd\_set\_gp\_value (bfd \*abfd, bfd\_vma v);** [Function]  
 Allow external access to the function to set the GP value. This is specifically added for gdb-compile support.

### 2.4.1.11 bfd\_scan\_vma

**bfd\_vma bfd\_scan\_vma (const char \*string, const char \*\*end, int base);** [Function]  
 Convert, like `strtoul` or `strtoull` depending on the size of a `bfd_vma`, a numerical expression *string* into a `bfd_vma` integer, and return that integer.

#### 2.4.1.12 bfd\_copy\_private\_header\_data

`bool bfd_copy_private_header_data (bfd *ibfd, bfd *obfd);` [Function]

Copy private BFD header information from the BFD *ibfd* to the the BFD *obfd*. This copies information that may require sections to exist, but does not require symbol tables. Return `true` on success, `false` on error. Possible error returns are:

- `bfd_error_no_memory` - Not enough memory exists to create private data for *obfd*.

```
#define bfd_copy_private_header_data(ibfd, obfd) \
    BFD_SEND (obfd, _bfd_copy_private_header_data, \
              (ibfd, obfd))
```

#### 2.4.1.13 bfd\_copy\_private\_bfd\_data

`bool bfd_copy_private_bfd_data (bfd *ibfd, bfd *obfd);` [Function]

Copy private BFD information from the BFD *ibfd* to the the BFD *obfd*. Return `TRUE` on success, `FALSE` on error. Possible error returns are:

- `bfd_error_no_memory` - Not enough memory exists to create private data for *obfd*.

```
#define bfd_copy_private_bfd_data(ibfd, obfd) \
    BFD_SEND (obfd, _bfd_copy_private_bfd_data, \
              (ibfd, obfd))
```

#### 2.4.1.14 bfd\_set\_private\_flags

`bool bfd_set_private_flags (bfd *abfd, flagword flags);` [Function]

Set private BFD flag information in the BFD *abfd*. Return `TRUE` on success, `FALSE` on error. Possible error returns are:

- `bfd_error_no_memory` - Not enough memory exists to create private data for *obfd*.

```
#define bfd_set_private_flags(abfd, flags) \
    BFD_SEND (abfd, _bfd_set_private_flags, (abfd, flags))
```

#### 2.4.1.15 Other functions

The following functions exist but have not yet been documented.

```
#define bfd_sizeof_headers(abfd, info) \
    BFD_SEND (abfd, _bfd_sizeof_headers, (abfd, info))

#define bfd_find_nearest_line(abfd, sec, syms, off, file, func, line) \
    BFD_SEND (abfd, _bfd_find_nearest_line, \
              (abfd, syms, sec, off, file, func, line, NULL))

#define bfd_find_nearest_line_with_alt(abfd, alt_filename, sec, syms, off, \
                                         file, func, line, disc) \
    BFD_SEND (abfd, _bfd_find_nearest_line_with_alt, \
              (abfd, alt_filename, syms, sec, off, file, func, line, disc))
```

```

#define bfd_find_nearest_line_discriminator(abfd, sec, syms, off, file, func, \
                                           line, disc) \
    BFD_SEND (abfd, _bfd_find_nearest_line, \
              (abfd, syms, sec, off, file, func, line, disc))

#define bfd_find_line(abfd, syms, sym, file, line) \
    BFD_SEND (abfd, _bfd_find_line, \
              (abfd, syms, sym, file, line))

#define bfd_find_inliner_info(abfd, file, func, line) \
    BFD_SEND (abfd, _bfd_find_inliner_info, \
              (abfd, file, func, line))

#define bfd_debug_info_start(abfd) \
    BFD_SEND (abfd, _bfd_debug_info_start, (abfd))

#define bfd_debug_info_end(abfd) \
    BFD_SEND (abfd, _bfd_debug_info_end, (abfd))

#define bfd_debug_info_accumulate(abfd, section) \
    BFD_SEND (abfd, _bfd_debug_info_accumulate, (abfd, section))

#define bfd_stat_arch_elt(abfd, stat) \
    BFD_SEND (abfd->my_archive ? abfd->my_archive : abfd, \
              _bfd_stat_arch_elt, (abfd, stat))

#define bfd_update_armap_timestamp(abfd) \
    BFD_SEND (abfd, _bfd_update_armap_timestamp, (abfd))

#define bfd_set_arch_mach(abfd, arch, mach)\
    BFD_SEND ( abfd, _bfd_set_arch_mach, (abfd, arch, mach))

#define bfd_relax_section(abfd, section, link_info, again) \
    BFD_SEND (abfd, _bfd_relax_section, (abfd, section, link_info, again))

#define bfd_gc_sections(abfd, link_info) \
    BFD_SEND (abfd, _bfd_gc_sections, (abfd, link_info))

#define bfd_lookup_section_flags(link_info, flag_info, section) \
    BFD_SEND (abfd, _bfd_lookup_section_flags, (link_info, flag_info, section))

#define bfd_is_group_section(abfd, sec) \
    BFD_SEND (abfd, _bfd_is_group_section, (abfd, sec))

#define bfd_group_name(abfd, sec) \
    BFD_SEND (abfd, _bfd_group_name, (abfd, sec))

```

```

#define bfd_discard_group(abfd, sec) \
    BFD_SEND (abfd, _bfd_discard_group, (abfd, sec))

#define bfd_link_hash_table_create(abfd) \
    BFD_SEND (abfd, _bfd_link_hash_table_create, (abfd))

#define bfd_link_add_symbols(abfd, info) \
    BFD_SEND (abfd, _bfd_link_add_symbols, (abfd, info))

#define bfd_link_just_syms(abfd, sec, info) \
    BFD_SEND (abfd, _bfd_link_just_syms, (sec, info))

#define bfd_final_link(abfd, info) \
    BFD_SEND (abfd, _bfd_final_link, (abfd, info))

#define bfd_free_cached_info(abfd) \
    BFD_SEND (abfd, _bfd_free_cached_info, (abfd))

#define bfd_get_dynamic_symtab_upper_bound(abfd) \
    BFD_SEND (abfd, _bfd_get_dynamic_symtab_upper_bound, (abfd))

#define bfd_print_private_bfd_data(abfd, file)\
    BFD_SEND (abfd, _bfd_print_private_bfd_data, (abfd, file))

#define bfd_canonicalize_dynamic_symtab(abfd, asymbols) \
    BFD_SEND (abfd, _bfd_canonicalize_dynamic_symtab, (abfd, asymbols))■

#define bfd_get_synthetic_symtab(abfd, count, syms, dyncount, dynsyms, ret) \
    BFD_SEND (abfd, _bfd_get_synthetic_symtab, (abfd, count, syms, \
        dyncount, dynsyms, ret))■

#define bfd_get_dynamic_reloc_upper_bound(abfd) \
    BFD_SEND (abfd, _bfd_get_dynamic_reloc_upper_bound, (abfd))

#define bfd_canonicalize_dynamic_reloc(abfd, arels, asyms) \
    BFD_SEND (abfd, _bfd_canonicalize_dynamic_reloc, (abfd, arels, asyms))■

```

#### 2.4.1.16 bfd\_get\_relocated\_section\_contents

`bfd_byte *bfd_get_relocated_section_contents` (*bfd \**, *struct* [Function]  
*bfd\_link\_info \**, *struct bfd\_link\_order \**, *bfd\_byte \**, *bool*, *asymbol \*\**);

Read and relocate the indirect link\_order section, into DATA (if non-NULL) or to a malloc'd buffer. Return the buffer, or NULL on errors.

**2.4.1.17 bfd\_record\_phdr**

`bool bfd_record_phdr (bfd *, unsigned long, bool, flagword, bool, bfd_vma, bool, bool, unsigned int, struct bfd_section **);` [Function]

Record information about an ELF program header.

**2.4.1.18 bfd\_sprintf\_vma**

`void bfd_sprintf_vma (bfd *, char *, bfd_vma); void bfd_fprintf_vma (bfd *, void *, bfd_vma);` [Function]

`bfd_sprintf_vma` and `bfd_fprintf_vma` display an address in the target's address size.

**2.4.1.19 bfd\_alt\_mach\_code**

`bool bfd_alt_mach_code (bfd *abfd, int alternative);` [Function]

When more than one machine code number is available for the same machine type, this function can be used to switch between the preferred one (`alternative == 0`) and any others. Currently, only ELF supports this feature, with up to two alternate machine codes.

**2.4.1.20 bfd\_emul\_get\_maxpagesize**

`bfd_vma bfd_emul_get_maxpagesize (const char *);` [Function]

Returns the maximum page size, in bytes, as determined by emulation.

**2.4.1.21 bfd\_emul\_get\_commonpagesize**

`bfd_vma bfd_emul_get_commonpagesize (const char *);` [Function]

Returns the common page size, in bytes, as determined by emulation.

**2.4.1.22 bfd\_demangle**

`char *bfd_demangle (bfd *, const char *, int);` [Function]

Wrapper around `cplus_demangle`. Strips leading underscores and other such chars that would otherwise confuse the demangler. If passed a g++ v3 ABI mangled name, returns a buffer allocated with `malloc` holding the demangled name. Returns `NULL` otherwise and on memory alloc failure.

**2.4.1.23 bfd\_group\_signature**

`asymbol *bfd_group_signature (asection *group, asymbol **isympp);` [Function]

Return a pointer to the symbol used as a signature for GROUP.

**2.4.1.24 struct bfd\_iovec**

The `struct bfd_iovec` contains the internal file I/O class. Each BFD has an instance of this class and all file I/O is routed through it (it is assumed that the instance implements all methods listed below).

`struct bfd_iovec`

```

{
    /* To avoid problems with macros, a "b" rather than "f"
       prefix is prepended to each method name.  */
    /* Attempt to read/write NBYTES on ABFD's IOSTREAM storing/fetching
       bytes starting at PTR.  Return the number of bytes actually
       transfered (a read past end-of-file returns less than NBYTES),
       or -1 (setting bfd_error) if an error occurs.  */
    file_ptr (*bread) (struct bfd *abfd, void *ptr, file_ptr nbytes);
    file_ptr (*bwrite) (struct bfd *abfd, const void *ptr,
                       file_ptr nbytes);
    /* Return the current IOSTREAM file offset, or -1 (setting bfd_error
       if an error occurs.  */
    file_ptr (*btell) (struct bfd *abfd);
    /* For the following, on successful completion a value of 0 is returned.
       Otherwise, a value of -1 is returned (and bfd_error is set).  */
    int (*bseek) (struct bfd *abfd, file_ptr offset, int whence);
    int (*bclose) (struct bfd *abfd);
    int (*bflush) (struct bfd *abfd);
    int (*bstat) (struct bfd *abfd, struct stat *sb);
    /* Mmap a part of the files. ADDR, LEN, PROT, FLAGS and OFFSET are the usual
       mmap parameter, except that LEN and OFFSET do not need to be page
       aligned.  Returns MAP_FAILED on failure, mmapped address on success.
       Also write in MAP_ADDR the address of the page aligned buffer and in
       MAP_LEN the size mapped (a page multiple).  Use unmap with MAP_ADDR and
       MAP_LEN to unmap.  */
    void *(*bmmap) (struct bfd *abfd, void *addr, size_t len,
                   int prot, int flags, file_ptr offset,
                   void **map_addr, size_t *map_len);
};
extern const struct bfd_iovec _bfd_memory_iovec;

```

#### 2.4.1.25 bfd\_read

`bfd_size_type bfd_read (void *, bfd_size_type, bfd *)` [Function]  
*ATTRIBUTE\_WARN\_UNUSED\_RESULT;*

Attempt to read SIZE bytes from ABFD's iostream to PTR. Return the amount read.

#### 2.4.1.26 bfd\_write

`bfd_size_type bfd_write (const void *, bfd_size_type, bfd *)` [Function]  
*ATTRIBUTE\_WARN\_UNUSED\_RESULT;*

Attempt to write SIZE bytes to ABFD's iostream from PTR. Return the amount written.



**2.4.1.27 bfd\_tell**

`file_ptr bfd_tell (bfd *)` [Function]  
*ATTRIBUTE\_WARN\_UNUSED\_RESULT;*  
 Return ABFD's iostream file position.

**2.4.1.28 bfd\_flush**

`int bfd_flush (bfd *);` [Function]  
 Flush ABFD's iostream pending IO.

**2.4.1.29 bfd\_stat**

`int bfd_stat (bfd *, struct stat *)` [Function]  
*ATTRIBUTE\_WARN\_UNUSED\_RESULT;*  
 Call fstat on ABFD's iostream. Return 0 on success, and a negative value on failure.

**2.4.1.30 bfd\_seek**

`int bfd_seek (bfd *, file_ptr, int)` [Function]  
*ATTRIBUTE\_WARN\_UNUSED\_RESULT;*  
 Call fseek on ABFD's iostream. Return 0 on success, and a negative value on failure.

**2.4.1.31 bfd\_get\_mtime**

`long bfd_get_mtime (bfd *abfd);` [Function]  
 Return the file modification time (as read from the file system, or from the archive header for archive members).

**2.4.1.32 bfd\_get\_size**

`ufile_ptr bfd_get_size (bfd *abfd);` [Function]  
 Return the file size (as read from file system) for the file associated with BFD *abfd*.

The initial motivation for, and use of, this routine is not so we can get the exact size of the object the BFD applies to, since that might not be generally possible (archive members for example). It would be ideal if someone could eventually modify it so that such results were guaranteed.

Instead, we want to ask questions like "is this NNN byte sized object I'm about to try read from file offset YYY reasonable?" As an example of where we might do this, some object formats use string tables for which the first `sizeof (long)` bytes of the table contain the size of the table itself, including the size bytes. If an application tries to read what it thinks is one of these string tables, without some way to validate the size, and for some reason the size is wrong (byte swapping error, wrong location for the string table, etc.), the only clue is likely to be a read error when it tries to read the table, or a "virtual memory exhausted" error when it tries to allocate 15 bazillion bytes of space for the 15 bazillion byte table it is about to read. This function at least allows us to answer the question, "is the size reasonable?".

A return value of zero indicates the file size is unknown.

### 2.4.1.33 `bfd_get_file_size`

`ufile_ptr bfd_get_file_size (bfd *abfd);` [Function]  
 Return the file size (as read from file system) for the file associated with BFD *abfd*.  
 It supports both normal files and archive elements.

### 2.4.1.34 `bfd_mmap`

`void *bfd_mmap (bfd *abfd, void *addr, size_t len, int prot, int flags, file_ptr offset, void **map_addr, size_t *map_len)` [Function]  
`ATTRIBUTE_WARN_UNUSED_RESULT;`  
 Return mmap(ed) region of the file, if possible and implemented. LEN and OFFSET do not need to be page aligned. The page aligned address and length are written to MAP\_ADDR and MAP\_LEN. Returns MAP\_FAILED on failure.

### 2.4.1.35 `bfd_get_current_time`

`time_t bfd_get_current_time (time_t now);` [Function]  
 Returns the current time.  
 If the environment variable SOURCE\_DATE\_EPOCH is defined then this is parsed and its value is returned. Otherwise if the paramter NOW is non-zero, then that is returned. Otherwise the result of the system call "time(NULL)" is returned.

## 2.5 Memory Usage

BFD keeps all of its internal structures in obstacks. There is one obstack per open BFD file, into which the current state is stored. When a BFD is closed, the obstack is deleted, and so everything which has been allocated by BFD for the closing file is thrown away.

BFD does not free anything created by an application, but pointers into `bfd` structures become invalid on a `bfd_close`; for example, after a `bfd_close` the vector passed to `bfd_canonicalize_symtab` is still around, since it has been allocated by the application, but the data that it pointed to are lost.

The general rule is to not close a BFD until all operations dependent upon data from the BFD have been completed, or all the data from within the file has been copied. To help with the management of memory, there is a function (`bfd_alloc_size`) which returns the number of bytes in obstacks associated with the supplied BFD. This could be used to select the greediest open BFD, close it to reclaim the memory, perform some operation and reopen the BFD again, to get a fresh copy of the data structures.

## 2.6 Sections

The raw data contained within a BFD is maintained through the section abstraction. A single BFD may have any number of sections. It keeps hold of them by pointing to the first; each one points to the next in the list.

Sections are supported in BFD in `section.c`.

### 2.6.1 Section input

When a BFD is opened for reading, the section structures are created and attached to the BFD.

Each section has a name which describes the section in the outside world—for example, `a.out` would contain at least three sections, called `.text`, `.data` and `.bss`.

Names need not be unique; for example a COFF file may have several sections named `.data`. Sometimes a BFD will contain more than the “natural” number of sections. A back end may attach other sections containing constructor data, or an application may add a section (using `bfd_make_section`) to the sections attached to an already open BFD. For example, the linker creates an extra section `COMMON` for each input file’s BFD to hold information about common storage.

The raw data is not necessarily read in when the section descriptor is created. Some targets may leave the data in place until a `bfd_get_section_contents` call is made. Other back ends may read in all the data at once. For example, an S-record file has to be read once to determine the size of the data.

### 2.6.2 Section output

To write a new object style BFD, the various sections to be written have to be created. They are attached to the BFD in the same way as input sections; data is written to the sections using `bfd_set_section_contents`.

Any program that creates or combines sections (e.g., the assembler and linker) must use the `asection` fields `output_section` and `output_offset` to indicate the file sections to which each section must be written. (If the section is being created from scratch, `output_section` should probably point to the section itself and `output_offset` should probably be zero.)

The data to be written comes from input sections attached (via `output_section` pointers) to the output sections. The output section structure can be considered a filter for the input section: the output section determines the vma of the output data and the name, but the input section determines the offset into the output section of the data to be written.

E.g., to create a section “O”, starting at 0x100, 0x123 long, containing two subsections, “A” at offset 0x0 (i.e., at vma 0x100) and “B” at offset 0x20 (i.e., at vma 0x120) the `asection` structures would look like:

|                |        |              |       |
|----------------|--------|--------------|-------|
| section name   | "A"    |              |       |
| output_offset  | 0x00   |              |       |
| size           | 0x20   |              |       |
| output_section | -----> | section name | "O"   |
|                |        | vma          | 0x100 |
| section name   | "B"    | size         | 0x123 |
| output_offset  | 0x20   |              |       |
| size           | 0x103  |              |       |
| output_section | -----  |              |       |

### 2.6.3 Link orders

The data within a section is stored in a *link\_order*. These are much like the fixups in `gas`. The `link_order` abstraction allows a section to grow and shrink within itself.

A `link_order` knows how big it is, and which is the next `link_order` and where the raw data for it is; it also points to a list of relocations which apply to it.

The `link_order` is used by the linker to perform relaxing on final code. The compiler creates code which is as big as necessary to make it work without relaxing, and the user can select whether to relax. Sometimes relaxing takes a lot of time. The linker runs around the relocations to see if any are attached to data which can be shrunk, if so it does it on a `link_order` by `link_order` basis.

## 2.6.4 typedef asection

Here is the section structure:

```
typedef struct bfd_section
{
    /* The name of the section; the name isn't a copy, the pointer is
       the same as that passed to bfd_make_section.  */
    const char *name;

    /* The next section in the list belonging to the BFD, or NULL.  */
    struct bfd_section *next;

    /* The previous section in the list belonging to the BFD, or NULL.  */
    struct bfd_section *prev;

    /* A unique sequence number.  */
    unsigned int id;

    /* A unique section number which can be used by assembler to
       distinguish different sections with the same section name.  */
    unsigned int section_id;

    /* Which section in the bfd; 0..n-1 as sections are created in a bfd.  */
    unsigned int index;

    /* The field flags contains attributes of the section. Some
       flags are read in from the object file, and some are
       synthesized from other information.  */
    flagword flags;

#define SEC_NO_FLAGS                0x0

    /* Tells the OS to allocate space for this section when loading.
       This is clear for a section containing debug information only.  */
#define SEC_ALLOC                   0x1

    /* Tells the OS to load the section from the file when loading.
       This is clear for a .bss section.  */
#define SEC_LOAD                    0x2
```

```

    /* The section contains data still to be relocated, so there is
       some relocation information too. */
#define SEC_RELOC                                0x4

    /* A signal to the OS that the section contains read only data. */
#define SEC_READONLY                            0x8

    /* The section contains code only. */
#define SEC_CODE                                0x10

    /* The section contains data only. */
#define SEC_DATA                                0x20

    /* The section will reside in ROM. */
#define SEC_ROM                                0x40

    /* The section contains constructor information. This section
       type is used by the linker to create lists of constructors and
       destructors used by g++. When a back end sees a symbol
       which should be used in a constructor list, it creates a new
       section for the type of name (e.g., __CTOR_LIST__), attaches
       the symbol to it, and builds a relocation. To build the lists
       of constructors, all the linker has to do is catenate all the
       sections called __CTOR_LIST__ and relocate the data
       contained within - exactly the operations it would perform on
       standard data. */
#define SEC_CONSTRUCTOR                        0x80

    /* The section has contents - a data section could be
       SEC_ALLOC | SEC_HAS_CONTENTS; a debug section could be
       SEC_HAS_CONTENTS */
#define SEC_HAS_CONTENTS                      0x100

    /* An instruction to the linker to not output the section
       even if it has information which would normally be written. */
#define SEC_NEVER_LOAD                        0x200

    /* The section contains thread local data. */
#define SEC_THREAD_LOCAL                      0x400

    /* The section's size is fixed. Generic linker code will not
       recalculate it and it is up to whoever has set this flag to
       get the size right. */
#define SEC_FIXED_SIZE                        0x800

    /* The section contains common symbols (symbols may be defined

```

```

        multiple times, the value of a symbol is the amount of
        space it requires, and the largest symbol value is the one
        used). Most targets have exactly one of these (which we
        translate to bfd_com_section_ptr), but ECOFF has two. */
#define SEC_IS_COMMON                0x1000

    /* The section contains only debugging information. For
       example, this is set for ELF .debug and .stab sections.
       strip tests this flag to see if a section can be
       discarded. */
#define SEC_DEBUGGING                0x2000

    /* The contents of this section are held in memory pointed to
       by the contents field. This is checked by bfd_get_section_contents,
       and the data is retrieved from memory if appropriate. */
#define SEC_IN_MEMORY                0x4000

    /* The contents of this section are to be excluded by the
       linker for executable and shared objects unless those
       objects are to be further relocated. */
#define SEC_EXCLUDE                  0x8000

    /* The contents of this section are to be sorted based on the sum of
       the symbol and addend values specified by the associated relocation
       entries. Entries without associated relocation entries will be
       appended to the end of the section in an unspecified order. */
#define SEC_SORT_ENTRIES             0x10000

    /* When linking, duplicate sections of the same name should be
       discarded, rather than being combined into a single section as
       is usually done. This is similar to how common symbols are
       handled. See SEC_LINK_DUPLICATES below. */
#define SEC_LINK_ONCE                 0x20000

    /* If SEC_LINK_ONCE is set, this bitfield describes how the linker
       should handle duplicate sections. */
#define SEC_LINK_DUPLICATES          0xc0000

    /* This value for SEC_LINK_DUPLICATES means that duplicate
       sections with the same name should simply be discarded. */
#define SEC_LINK_DUPLICATES_DISCARD  0x0

    /* This value for SEC_LINK_DUPLICATES means that the linker
       should warn if there are any duplicate sections, although
       it should still only link one copy. */
#define SEC_LINK_DUPLICATES_ONE_ONLY 0x40000

```

```

/* This value for SEC_LINK_DUPLICATES means that the linker
   should warn if any duplicate sections are a different size. */
#define SEC_LINK_DUPLICATES_SAME_SIZE 0x80000

/* This value for SEC_LINK_DUPLICATES means that the linker
   should warn if any duplicate sections contain different
   contents. */
#define SEC_LINK_DUPLICATES_SAME_CONTENTS \
  (SEC_LINK_DUPLICATES_ONE_ONLY | SEC_LINK_DUPLICATES_SAME_SIZE)

/* This section was created by the linker as part of dynamic
   relocation or other arcane processing. It is skipped when
   going through the first-pass output, trusting that someone
   else up the line will take care of it later. */
#define SEC_LINKER_CREATED 0x100000

/* This section contains a section ID to distinguish different
   sections with the same section name. */
#define SEC_ASSEMBLER_SECTION_ID 0x100000

/* This section should not be subject to garbage collection.
   Also set to inform the linker that this section should not be
   listed in the link map as discarded. */
#define SEC_KEEP 0x200000

/* This section contains "short" data, and should be placed
   "near" the GP. */
#define SEC_SMALL_DATA 0x400000

/* Attempt to merge identical entities in the section.
   Entity size is given in the entsize field. */
#define SEC_MERGE 0x800000

/* If given with SEC_MERGE, entities to merge are zero terminated
   strings where entsize specifies character size instead of fixed
   size entries. */
#define SEC_STRINGS 0x1000000

/* This section contains data about section groups. */
#define SEC_GROUP 0x2000000

/* The section is a COFF shared library section. This flag is
   only for the linker. If this type of section appears in
   the input file, the linker must copy it to the output file
   without changing the vma or size. FIXME: Although this
   was originally intended to be general, it really is COFF
   specific (and the flag was renamed to indicate this). It

```

```

        might be cleaner to have some more general mechanism to
        allow the back end to control what the linker does with
        sections. */
#define SEC_COFF_SHARED_LIBRARY      0x4000000

    /* This input section should be copied to output in reverse order
       as an array of pointers. This is for ELF linker internal use
       only. */
#define SEC_ELF_REVERSE_COPY         0x4000000

    /* This section contains data which may be shared with other
       executables or shared objects. This is for COFF only. */
#define SEC_COFF_SHARED              0x8000000

    /* Indicate that section has the purecode flag set. */
#define SEC_ELF_PURECODE             0x8000000

    /* When a section with this flag is being linked, then if the size of
       the input section is less than a page, it should not cross a page
       boundary. If the size of the input section is one page or more,
       it should be aligned on a page boundary. This is for TI
       TMS320C54X only. */
#define SEC_TIC54X_BLOCK             0x10000000

    /* This section has the SHF_X86_64_LARGE flag. This is ELF x86-64 only. */
#define SEC_ELF_LARGE                0x10000000

    /* Conditionally link this section; do not link if there are no
       references found to any symbol in the section. This is for TI
       TMS320C54X only. */
#define SEC_TIC54X_CLINK             0x20000000

    /* This section contains vliw code. This is for Toshiba MeP only. */
#define SEC_MEP_VLIW                 0x20000000

    /* All symbols, sizes and relocations in this section are octets
       instead of bytes. Required for DWARF debug sections as DWARF
       information is organized in octets, not bytes. */
#define SEC_ELF_OCTETS               0x40000000

    /* Indicate that section has the no read flag set. This happens
       when memory read flag isn't set. */
#define SEC_COFF_NOREAD              0x40000000

    /* End of section flags. */

    /* Some internal packed boolean fields. */

```



```
/* See the vma field. */
unsigned int user_set_vma : 1;

/* A mark flag used by some of the linker backends. */
unsigned int linker_mark : 1;

/* Another mark flag used by some of the linker backends. Set for
   output sections that have an input section. */
unsigned int linker_has_input : 1;

/* Mark flag used by some linker backends for garbage collection. */
unsigned int gc_mark : 1;

/* Section compression status. */
unsigned int compress_status : 2;
#define COMPRESS_SECTION_NONE 0
#define COMPRESS_SECTION_DONE 1
#define DECOMPRESS_SECTION_ZLIB 2
#define DECOMPRESS_SECTION_ZSTD 3

/* The following flags are used by the ELF linker. */

/* Mark sections which have been allocated to segments. */
unsigned int segment_mark : 1;

/* Type of sec_info information. */
unsigned int sec_info_type:3;
#define SEC_INFO_TYPE_NONE 0
#define SEC_INFO_TYPE_STABS 1
#define SEC_INFO_TYPE_MERGE 2
#define SEC_INFO_TYPE_EH_FRAME 3
#define SEC_INFO_TYPE_JUST_SYMS 4
#define SEC_INFO_TYPE_TARGET 5
#define SEC_INFO_TYPE_EH_FRAME_ENTRY 6
#define SEC_INFO_TYPE_SFRAME 7

/* Nonzero if this section uses RELA relocations, rather than REL. */
unsigned int use_rela_p:1;

/* Nonzero if section contents are mmaped. */
unsigned int mmaped_p:1;

/* Nonzero if section contents should not be freed. */
unsigned int allocated:1;

/* Bits used by various backends. The generic code doesn't touch
```

```

    these fields. */

unsigned int sec_flg0:1;
unsigned int sec_flg1:1;
unsigned int sec_flg2:1;
unsigned int sec_flg3:1;
unsigned int sec_flg4:1;
unsigned int sec_flg5:1;

/* End of internal packed boolean fields. */

/* The virtual memory address of the section - where it will be
   at run time. The symbols are relocated against this. The
   user_set_vma flag is maintained by bfd; if it's not set, the
   backend can assign addresses (for example, in a.out, where
   the default address for .data is dependent on the specific
   target and various flags). */
bfd_vma vma;

/* The load address of the section - where it would be in a
   rom image; really only used for writing section header
   information. */
bfd_vma lma;

/* The size of the section in *octets*, as it will be output.
   Contains a value even if the section has no contents (e.g., the
   size of .bss). */
bfd_size_type size;

/* For input sections, the original size on disk of the section, in
   octets. This field should be set for any section whose size is
   changed by linker relaxation. It is required for sections where
   the linker relaxation scheme doesn't cache altered section and
   reloc contents (stabs, eh_frame, SEC_MERGE, some coff relaxing
   targets), and thus the original size needs to be kept to read the
   section multiple times. For output sections, rawsize holds the
   section size calculated on a previous linker relaxation pass. */
bfd_size_type rawsize;

/* The compressed size of the section in octets. */
bfd_size_type compressed_size;

/* If this section is going to be output, then this value is the
   offset in *bytes* into the output section of the first byte in the
   input section (byte ==> smallest addressable unit on the
   target). In most cases, if this was going to start at the
   100th octet (8-bit quantity) in the output section, this value

```

```

        would be 100.  However, if the target byte size is 16 bits
        (bfd_octets_per_byte is "2"), this value would be 50.  */
bfd_vma output_offset;

/* The output section through which to map on output.  */
struct bfd_section *output_section;

/* If an input section, a pointer to a vector of relocation
   records for the data in this section.  */
struct reloc_cache_entry *relocation;

/* If an output section, a pointer to a vector of pointers to
   relocation records for the data in this section.  */
struct reloc_cache_entry **orelocation;

/* The number of relocation records in one of the above.  */
unsigned reloc_count;

/* The alignment requirement of the section, as an exponent of 2 -
   e.g., 3 aligns to 2^3 (or 8).  */
unsigned int alignment_power;

/* Information below is back end specific - and not always used
   or updated.  */

/* File position of section data.  */
file_ptr filepos;

/* File position of relocation info.  */
file_ptr rel_filepos;

/* File position of line data.  */
file_ptr line_filepos;

/* Pointer to data for applications.  */
void *userdata;

/* If the SEC_IN_MEMORY flag is set, this points to the actual
   contents.  */
bfd_byte *contents;

/* Attached line number information.  */
alint *lineno;

/* Number of line number records.  */
unsigned int lineno_count;

```

```

/* Entity size for merging purposes.  */
unsigned int entsize;

/* Points to the kept section if this section is a link-once section,
   and is discarded.  */
struct bfd_section *kept_section;

/* When a section is being output, this value changes as more
   linenumbers are written out.  */
file_ptr moving_line_filepos;

/* What the section number is in the target world.  */
int target_index;

void *used_by_bfd;

/* If this is a constructor section then here is a list of the
   relocations created to relocate items within it.  */
struct relent_chain *constructor_chain;

/* The BFD which owns the section.  */
bfd *owner;

/* A symbol which points at this section only.  */
struct bfd_symbol *symbol;

/* Early in the link process, map_head and map_tail are used to build
   a list of input sections attached to an output section.  Later,
   output sections use these fields for a list of bfd_link_order
   structs.  The linked_to_symbol_name field is for ELF assembler
   internal use.  */
union {
    struct bfd_link_order *link_order;
    struct bfd_section *s;
    const char *linked_to_symbol_name;
} map_head, map_tail;

/* Points to the output section this section is already assigned to,
   if any.  This is used when support for non-contiguous memory
   regions is enabled.  */
struct bfd_section *already_assigned;

/* A pointer used for various section optimizations.  sec_info_type
   qualifies which one it is.  */
void *sec_info;

/* Explicitly specified section type, if non-zero.  */

```

```

    unsigned int type;

} asection;

```

## 2.6.5 Section prototypes

These are the functions exported by the section handling part of BFD.

### 2.6.5.1 bfd\_section\_list\_clear

```
void bfd_section_list_clear (bfd *);
```

 [Function]  
 Clears the section list, and also resets the section count and hash table entries.

### 2.6.5.2 bfd\_get\_section\_by\_name

```
asection *bfd_get_section_by_name (bfd *abfd, const char
```

 [Function]  
                                   \*name);  
 Return the most recently created section attached to *abfd* named *name*. Return NULL if no such section exists.

### 2.6.5.3 bfd\_get\_next\_section\_by\_name

```
asection *bfd_get_next_section_by_name (bfd *ibfd, asection
```

 [Function]  
                                   \*sec);  
 Given *sec* is a section returned by `bfd_get_section_by_name`, return the next most recently created section attached to the same BFD with the same name, or if no such section exists in the same BFD and IBFD is non-NULL, the next section with the same name in any input BFD following IBFD. Return NULL on finding no section.

### 2.6.5.4 bfd\_get\_linker\_section

```
asection *bfd_get_linker_section (bfd *abfd, const char *name);
```

 [Function]  
 Return the linker created section attached to *abfd* named *name*. Return NULL if no such section exists.

### 2.6.5.5 bfd\_get\_section\_by\_name\_if

```
asection *bfd_get_section_by_name_if (bfd *abfd, const char
```

 [Function]  
                                   \*name, bool (\*func) (bfd \*abfd, asection \*sect, void \*obj), void \*obj);  
 Call the provided function *func* for each section attached to the BFD *abfd* whose name matches *name*, passing *obj* as an argument. The function will be called as if by  
                                   func (abfd, the\_section, obj);  
 It returns the first section for which *func* returns true, otherwise NULL.

### 2.6.5.6 bfd\_get\_unique\_section\_name

```
char *bfd_get_unique_section_name (bfd *abfd, const char
```

 [Function]  
                                   \*templat, int \*count);  
 Invent a section name that is unique in *abfd* by tacking a dot and a digit suffix onto the original *templat*. If *count* is non-NULL, then it specifies the first number tried as

a suffix to generate a unique name. The value pointed to by *count* will be incremented in this case.

### 2.6.5.7 bfd\_make\_section\_old\_way

`asection *bfd_make_section_old_way (bfd *abfd, const char *name);` [Function]

Create a new empty section called *name* and attach it to the end of the chain of sections for the BFD *abfd*. An attempt to create a section with a name which is already in use returns its pointer without changing the section chain.

It has the funny name since this is the way it used to be before it was rewritten....

Possible errors are:

- `bfd_error_invalid_operation` - If output has already started for this BFD.
- `bfd_error_no_memory` - If memory allocation fails.

### 2.6.5.8 bfd\_make\_section\_anyway\_with\_flags

`asection *bfd_make_section_anyway_with_flags (bfd *abfd, const char *name, flagword flags);` [Function]

Create a new empty section called *name* and attach it to the end of the chain of sections for *abfd*. Create a new section even if there is already a section with that name. Also set the attributes of the new section to the value *flags*.

Return NULL and set `bfd_error` on error; possible errors are:

- `bfd_error_invalid_operation` - If output has already started for *abfd*.
- `bfd_error_no_memory` - If memory allocation fails.

### 2.6.5.9 bfd\_make\_section\_anyway

`asection *bfd_make_section_anyway (bfd *abfd, const char *name);` [Function]

Create a new empty section called *name* and attach it to the end of the chain of sections for *abfd*. Create a new section even if there is already a section with that name.

Return NULL and set `bfd_error` on error; possible errors are:

- `bfd_error_invalid_operation` - If output has already started for *abfd*.
- `bfd_error_no_memory` - If memory allocation fails.

### 2.6.5.10 bfd\_make\_section\_with\_flags

`asection *bfd_make_section_with_flags (bfd *, const char *name, flagword flags);` [Function]

Like `bfd_make_section_anyway`, but return NULL (without calling `bfd_set_error()`) without changing the section chain if there is already a section named *name*. Also set the attributes of the new section to the value *flags*. If there is an error, return NULL and set `bfd_error`.

### 2.6.5.11 bfd\_make\_section

**asection \*bfd\_make\_section** (*bfd \**, *const char \*name*); [Function]

Like `bfd_make_section_anyway`, but return `NULL` (without calling `bfd_set_error ()`) without changing the section chain if there is already a section named *name*. If there is an error, return `NULL` and set `bfd_error`.

### 2.6.5.12 bfd\_set\_section\_flags

**bool bfd\_set\_section\_flags** (*asection \*sec*, *flagword flags*); [Function]

Set the attributes of the section *sec* to the value *flags*. Return `TRUE` on success, `FALSE` on error. Possible error returns are:

- `bfd_error_invalid_operation` - The section cannot have one or more of the attributes requested. For example, a `.bss` section in `a.out` may not have the `SEC_HAS_CONTENTS` field set.

### 2.6.5.13 bfd\_rename\_section

**void bfd\_rename\_section** (*asection \*sec*, *const char \*newname*); [Function]

Rename section *sec* to *newname*.

### 2.6.5.14 bfd\_map\_over\_sections

**void bfd\_map\_over\_sections** (*bfd \*abfd*, *void (\*func) (bfd \*abfd, asection \*sect, void \*obj), void \*obj*); [Function]

Call the provided function *func* for each section attached to the BFD *abfd*, passing *obj* as an argument. The function will be called as if by

```
func (abfd, the_section, obj);
```

This is the preferred method for iterating over sections; an alternative would be to use a loop:

```
asection *p;
for (p = abfd->sections; p != NULL; p = p->next)
    func (abfd, p, ...)
```

### 2.6.5.15 bfd\_sections\_find\_if

**asection \*bfd\_sections\_find\_if** (*bfd \*abfd*, *bool (\*operation) (bfd \*abfd, asection \*sect, void \*obj), void \*obj*); [Function]

Call the provided function *operation* for each section attached to the BFD *abfd*, passing *obj* as an argument. The function will be called as if by

```
operation (abfd, the_section, obj);
```

It returns the first section for which *operation* returns true.

### 2.6.5.16 bfd\_set\_section\_size

**bool bfd\_set\_section\_size** (*asection \*sec*, *bfd\_size\_type val*); [Function]

Set *sec* to the size *val*. If the operation is ok, then `TRUE` is returned, else `FALSE`.

Possible error returns:

- `bfd_error_invalid_operation` - Writing has started to the BFD, so setting the size is invalid.

### 2.6.5.17 `bfd_set_section_contents`

`bool bfd_set_section_contents (bfd *abfd, asection *section, [Function]  
const void *data, file_ptr offset, bfd_size_type count);`

Sets the contents of the section *section* in BFD *abfd* to the data starting in memory at *location*. The data is written to the output section starting at offset *offset* for *count* octets.

Normally TRUE is returned, but FALSE is returned if there was an error. Possible error returns are:

- `bfd_error_no_contents` - The output section does not have the SEC\_HAS\_CONTENTS attribute, so nothing can be written to it.
- `bfd_error_bad_value` - The section is unable to contain all of the data.
- `bfd_error_invalid_operation` - The BFD is not writeable.
- and some more too.

This routine is front end to the back end function `_bfd_set_section_contents`.

### 2.6.5.18 `bfd_get_section_contents`

`bool bfd_get_section_contents (bfd *abfd, asection *section, void [Function]  
*location, file_ptr offset, bfd_size_type count);`

Read data from *section* in BFD *abfd* into memory starting at *location*. The data is read at an offset of *offset* from the start of the input section, and is read for *count* bytes.

If the contents of a constructor with the SEC\_CONSTRUCTOR flag set are requested or if the section does not have the SEC\_HAS\_CONTENTS flag set, then the *location* is filled with zeroes. If no errors occur, TRUE is returned, else FALSE.

### 2.6.5.19 `bfd_malloc_and_get_section`

`bool bfd_malloc_and_get_section (bfd *abfd, asection *section, [Function]  
bfd_byte **buf);`

Read all data from *section* in BFD *abfd* into a buffer, *\*buf*, malloc'd by this function. Return true on success, false on failure in which case *\*buf* will be NULL.

### 2.6.5.20 `bfd_copy_private_section_data`

Copy private section information from *isec* in the BFD *ibfd* to the section *osec* in the BFD *obfd*. Return TRUE on success, FALSE on error. Possible error returns are:

- `bfd_error_no_memory` - Not enough memory exists to create private data for *osec*.

```
#define bfd_copy_private_section_data(ibfd, isec, obfd, osec, link_info) \
    BFD_SEND (obfd, _bfd_copy_private_section_data, \
              (ibfd, isec, obfd, osec, link_info))
```



### 2.6.5.21 `bfd_generic_is_group_section`

`bool bfd_generic_is_group_section (bfd *, const asection *sec);` [Function]  
Returns TRUE if *sec* is a member of a group.

### 2.6.5.22 `bfd_generic_group_name`

`const char *bfd_generic_group_name (bfd *, const asection *sec);` [Function]  
Returns group name if *sec* is a member of a group.

### 2.6.5.23 `bfd_generic_discard_group`

`bool bfd_generic_discard_group (bfd *abfd, asection *group);` [Function]  
Remove all members of *group* from the output.

### 2.6.5.24 `bfd_section_size_insane`

`bool bfd_section_size_insane (bfd *abfd, asection *sec);` [Function]  
Returns true if the given section has a size that indicates it cannot be read from file.  
Return false if the size is OK or\* this function can't say one way or the other.

## 2.7 Symbols

BFD tries to maintain as much symbol information as it can when it moves information from file to file. BFD passes information to applications through the `asymbol` structure. When the application requests the symbol table, BFD reads the table in the native form and translates parts of it into the internal format. To maintain more than the information passed to applications, some targets keep some information “behind the scenes” in a structure only the particular back end knows about. For example, the coff back end keeps the original symbol table structure as well as the canonical structure when a BFD is read in. On output, the coff back end can reconstruct the output symbol table so that no information is lost, even information unique to coff which BFD doesn't know or understand. If a coff symbol table were read, but were written through an a.out back end, all the coff specific information would be lost. The symbol table of a BFD is not necessarily read in until a canonicalize request is made. Then the BFD back end fills in a table provided by the application with pointers to the canonical information. To output symbols, the application provides BFD with a table of pointers to pointers to `asymbols`. This allows applications like the linker to output a symbol as it was read, since the “behind the scenes” information will be still available.

### 2.7.1 Reading symbols

There are two stages to reading a symbol table from a BFD: allocating storage, and the actual reading process. This is an excerpt from an application which reads the symbol table:

```
long storage_needed;
asymbol **symbol_table;
long number_of_symbols;
long i;
```

```

storage_needed = bfd_get_symtab_upper_bound (abfd);

if (storage_needed < 0)
    FAIL

if (storage_needed == 0)
    return;

symbol_table = xmalloc (storage_needed);
...
number_of_symbols =
    bfd_canonicalize_symtab (abfd, symbol_table);

if (number_of_symbols < 0)
    FAIL

for (i = 0; i < number_of_symbols; i++)
    process_symbol (symbol_table[i]);

```

All storage for the symbols themselves is in an objalloc connected to the BFD; it is freed when the BFD is closed.

## 2.7.2 Writing symbols

Writing of a symbol table is automatic when a BFD open for writing is closed. The application attaches a vector of pointers to pointers to symbols to the BFD being written, and fills in the symbol count. The close and cleanup code reads through the table provided and performs all the necessary operations. The BFD output code must always be provided with an “owned” symbol: one which has come from another BFD, or one which has been created using `bfd_make_empty_symbol`. Here is an example showing the creation of a symbol table with only one element:

```

#include "sysdep.h"
#include "bfd.h"
int main (void)
{
    bfd *abfd;
    asymbol *ptrs[2];
    asymbol *new;

    abfd = bfd_openw ("foo", "a.out-sunos-big");
    bfd_set_format (abfd, bfd_object);
    new = bfd_make_empty_symbol (abfd);
    new->name = "dummy_symbol";
    new->section = bfd_make_section_old_way (abfd, ".text");
    new->flags = BSF_GLOBAL;
    new->value = 0x12345;

    ptrs[0] = new;

```

```

    ptrs[1] = 0;

    bfd_set_symtab (abfd, ptrs, 1);
    bfd_close (abfd);
    return 0;
}

./makesym
nm foo
00012345 A dummy_symbol

```

Many formats cannot represent arbitrary symbol information; for instance, the `a.out` object format does not allow an arbitrary number of sections. A symbol pointing to a section which is not one of `.text`, `.data` or `.bss` cannot be described.

### 2.7.3 Mini Symbols

Mini symbols provide read-only access to the symbol table. They use less memory space, but require more time to access. They can be useful for tools like `nm` or `objdump`, which may have to handle symbol tables of extremely large executables.

The `bfd_read_minisymbols` function will read the symbols into memory in an internal form. It will return a `void *` pointer to a block of memory, a symbol count, and the size of each symbol. The pointer is allocated using `malloc`, and should be freed by the caller when it is no longer needed.

The function `bfd_minisymbol_to_symbol` will take a pointer to a minisymbol, and a pointer to a structure returned by `bfd_make_empty_symbol`, and return a `asymbol` structure. The return value may or may not be the same as the value from `bfd_make_empty_symbol` which was passed in.

### 2.7.4 typedef asymbol

An `asymbol` has the form:

```

typedef struct bfd_symbol
{
    /* A pointer to the BFD which owns the symbol. This information
       is necessary so that a back end can work out what additional
       information (invisible to the application writer) is carried
       with the symbol.

       This field is *almost* redundant, since you can use section->owner
       instead, except that some symbols point to the global sections
       bfd_{abs,com,und}_section. This could be fixed by making
       these globals be per-bfd (or per-target-flavor).  FIXME.  */
    struct bfd *the_bfd; /* Use bfd_asymbol_bfd(sym) to access this field. */

    /* The text of the symbol. The name is left alone, and not copied; the
       application may not alter it.  */
    const char *name;

```

```

/* The value of the symbol.  This really should be a union of a
   numeric value with a pointer, since some flags indicate that
   a pointer to another symbol is stored here.  */
symvalue value;

/* Attributes of a symbol.  */
#define BSF_NO_FLAGS                0

/* The symbol has local scope; static in C. The value
   is the offset into the section of the data.  */
#define BSF_LOCAL                    (1 << 0)

/* The symbol has global scope; initialized data in C. The
   value is the offset into the section of the data.  */
#define BSF_GLOBAL                    (1 << 1)

/* The symbol has global scope and is exported. The value is
   the offset into the section of the data.  */
#define BSF_EXPORT                    BSF_GLOBAL /* No real difference.  */

/* A normal C symbol would be one of:
   BSF_LOCAL, BSF_UNDEFINED or BSF_GLOBAL.  */

/* The symbol is a debugging record. The value has an arbitrary
   meaning, unless BSF_DEBUGGING_RELOC is also set.  */
#define BSF_DEBUGGING                (1 << 2)

/* The symbol denotes a function entry point.  Used in ELF,
   perhaps others someday.  */
#define BSF_FUNCTION                  (1 << 3)

/* Used by the linker.  */
#define BSF_KEEP                      (1 << 5)

/* An ELF common symbol.  */
#define BSF_ELF_COMMON                (1 << 6)

/* A weak global symbol, overridable without warnings by
   a regular global symbol of the same name.  */
#define BSF_WEAK                      (1 << 7)

/* This symbol was created to point to a section, e.g. ELF's
   STT_SECTION symbols.  */
#define BSF_SECTION_SYM              (1 << 8)

/* The symbol used to be a common symbol, but now it is
   allocated.  */

```

```
#define BSF_OLD_COMMON          (1 << 9)

/* In some files the type of a symbol sometimes alters its
   location in an output file - ie in coff a ISFCN symbol
   which is also C_EXT symbol appears where it was
   declared and not at the end of a section. This bit is set
   by the target BFD part to convey this information. */
#define BSF_NOT_AT_END          (1 << 10)

/* Signal that the symbol is the label of constructor section. */
#define BSF_CONSTRUCTOR         (1 << 11)

/* Signal that the symbol is a warning symbol. The name is a
   warning. The name of the next symbol is the one to warn about;
   if a reference is made to a symbol with the same name as the next
   symbol, a warning is issued by the linker. */
#define BSF_WARNING             (1 << 12)

/* Signal that the symbol is indirect. This symbol is an indirect
   pointer to the symbol with the same name as the next symbol. */
#define BSF_INDIRECT            (1 << 13)

/* BSF_FILE marks symbols that contain a file name. This is used
   for ELF STT_FILE symbols. */
#define BSF_FILE                (1 << 14)

/* Symbol is from dynamic linking information. */
#define BSF_DYNAMIC             (1 << 15)

/* The symbol denotes a data object. Used in ELF, and perhaps
   others someday. */
#define BSF_OBJECT              (1 << 16)

/* This symbol is a debugging symbol. The value is the offset
   into the section of the data. BSF_DEBUGGING should be set
   as well. */
#define BSF_DEBUGGING_RELOC     (1 << 17)

/* This symbol is thread local. Used in ELF. */
#define BSF_THREAD_LOCAL        (1 << 18)

/* This symbol represents a complex relocation expression,
   with the expression tree serialized in the symbol name. */
#define BSF_RELC                (1 << 19)

/* This symbol represents a signed complex relocation expression,
   with the expression tree serialized in the symbol name. */
```

```

#define BSF_SRELC                (1 << 20)

/* This symbol was created by bfd_get_synthetic_symtab. */
#define BSF_SYNTHETIC            (1 << 21)

/* This symbol is an indirect code object.  Unrelated to BSF_INDIRECT.
   The dynamic linker will compute the value of this symbol by
   calling the function that it points to.  BSF_FUNCTION must
   also be also set. */
#define BSF_GNU_INDIRECT_FUNCTION (1 << 22)
/* This symbol is a globally unique data object.  The dynamic linker
   will make sure that in the entire process there is just one symbol
   with this name and type in use.  BSF_OBJECT must also be set. */
#define BSF_GNU_UNIQUE           (1 << 23)

/* This section symbol should be included in the symbol table. */
#define BSF_SECTION_SYM_USED     (1 << 24)

/* This symbol underwent section merge resolution. */
#define BSF_MERGE_RESOLVED       (1 << 25)

flagword flags;

/* A pointer to the section to which this symbol is
   relative.  This will always be non NULL, there are special
   sections for undefined and absolute symbols. */
struct bfd_section *section;

/* Back end special data. */
union
{
    void *p;
    bfd_vma i;
}
udata;
}
asymbol;

```

## 2.7.5 Symbol handling functions

### 2.7.5.1 bfd\_get\_symtab\_upper\_bound

Return the number of bytes required to store a vector of pointers to `asymbols` for all the symbols in the BFD *abfd*, including a terminal NULL pointer. If there are no symbols in the BFD, then return 0. If an error occurs, return -1.

```

#define bfd_get_symtab_upper_bound(abfd) \

```

```
BFD_SEND (abfd, _bfd_get_symtab_upper_bound, (abfd))
```

### 2.7.5.2 bfd\_is\_local\_label

```
bool bfd_is_local_label (bfd *abfd, asymbol *sym);           [Function]
```

Return TRUE if the given symbol *sym* in the BFD *abfd* is a compiler generated local label, else return FALSE.

### 2.7.5.3 bfd\_is\_local\_label\_name

```
bool bfd_is_local_label_name (bfd *abfd, const char *name); [Function]
```

Return TRUE if a symbol with the name *name* in the BFD *abfd* is a compiler generated local label, else return FALSE. This just checks whether the name has the form of a local label.

```
#define bfd_is_local_label_name(abfd, name) \
    BFD_SEND (abfd, _bfd_is_local_label_name, (abfd, name))
```

### 2.7.5.4 bfd\_is\_target\_special\_symbol

```
bool bfd_is_target_special_symbol (bfd *abfd, asymbol *sym); [Function]
```

Return TRUE iff a symbol *sym* in the BFD *abfd* is something special to the particular target represented by the BFD. Such symbols should normally not be mentioned to the user.

```
#define bfd_is_target_special_symbol(abfd, sym) \
    BFD_SEND (abfd, _bfd_is_target_special_symbol, (abfd, sym))
```

### 2.7.5.5 bfd\_canonicalize\_symtab

Read the symbols from the BFD *abfd*, and fills in the vector *location* with pointers to the symbols and a trailing NULL. Return the actual number of symbol pointers, not including the NULL.

```
#define bfd_canonicalize_symtab(abfd, location) \
    BFD_SEND (abfd, _bfd_canonicalize_symtab, (abfd, location))
```

### 2.7.5.6 bfd\_set\_symtab

```
bool bfd_set_symtab (bfd *abfd, asymbol **location, unsigned int [Function]
                    count);
```

Arrange that when the output BFD *abfd* is closed, the table *location* of *count* pointers to symbols will be written.

### 2.7.5.7 bfd\_print\_symbol\_vandf

```
void bfd_print_symbol_vandf (bfd *abfd, void *file, asymbol [Function]
                             *symbol);
```

Print the value and flags of the *symbol* supplied to the stream *file*.

### 2.7.5.8 bfd\_make\_empty\_symbol

Create a new `asymbol` structure for the BFD *abfd* and return a pointer to it.

This routine is necessary because each back end has private information surrounding the `asymbol`. Building your own `asymbol` and pointing to it will not create the private information, and will cause problems later on.

```
#define bfd_make_empty_symbol(abfd) \
    BFD_SEND (abfd, _bfd_make_empty_symbol, (abfd))
```

### 2.7.5.9 \_bfd\_generic\_make\_empty\_symbol

`asymbol *_bfd_generic_make_empty_symbol (bfd *);` [Function]

Create a new `asymbol` structure for the BFD *abfd* and return a pointer to it. Used by core file routines, binary back-end and anywhere else where no private info is needed.

### 2.7.5.10 bfd\_make\_debug\_symbol

Create a new `asymbol` structure for the BFD *abfd*, to be used as a debugging symbol.

```
#define bfd_make_debug_symbol(abfd) \
    BFD_SEND (abfd, _bfd_make_debug_symbol, (abfd))
```

### 2.7.5.11 bfd\_decode\_symclass

`int bfd_decode_symclass (asymbol *symbol);` [Function]

Return a character corresponding to the symbol class of *symbol*, or '?' for an unknown class.

### 2.7.5.12 bfd\_is\_undefined\_symclass

`bool bfd_is_undefined_symclass (int symclass);` [Function]

Returns non-zero if the class symbol returned by `bfd_decode_symclass` represents an undefined symbol. Returns zero otherwise.

### 2.7.5.13 bfd\_symbol\_info

`void bfd_symbol_info (asymbol *symbol, symbol_info *ret);` [Function]

Fill in the basic info about symbol that nm needs. Additional info may be added by the back-ends after calling this function.

### 2.7.5.14 bfd\_copy\_private\_symbol\_data

Copy private symbol information from *isym* in the BFD *ibfd* to the symbol *osym* in the BFD *obfd*. Return TRUE on success, FALSE on error. Possible error returns are:

- `bfd_error_no_memory` - Not enough memory exists to create private data for *osym*.

```
#define bfd_copy_private_symbol_data(ibfd, isymbol, obfd, osymbol) \
    BFD_SEND (obfd, _bfd_copy_private_symbol_data, \
              (ibfd, isymbol, obfd, osymbol))
```



## 2.8 Archives

An archive (or library) is just another BFD. It has a symbol table, although there's not much a user program will do with it.

The big difference between an archive BFD and an ordinary BFD is that the archive doesn't have sections. Instead it has a chain of BFDs that are considered its contents. These BFDs can be manipulated like any other. The BFDs contained in an archive opened for reading will all be opened for reading. You may put either input or output BFDs into an archive opened for output; they will be handled correctly when the archive is closed.

Use `bfd_openr_next_archived_file` to step through the contents of an archive opened for input. You don't have to read the entire archive if you don't want to! Read it until you find what you want.

A BFD returned by `bfd_openr_next_archived_file` can be closed manually with `bfd_close`. If you do not close it, then a second iteration through the members of an archive may return the same BFD. If you close the archive BFD, then all the member BFDs will automatically be closed as well.

Archive contents of output BFDs are chained through the `archive_next` pointer in a BFD. The first one is findable through the `archive_head` slot of the archive. Set it with `bfd_set_archive_head` (q.v.). A given BFD may be in only one open output archive at a time.

As expected, the BFD archive code is more general than the archive code of any given environment. BFD archives may contain files of different formats (e.g., a.out and coff) and even different architectures. You may even place archives recursively into archives!

This can cause unexpected confusion, since some archive formats are more expressive than others. For instance, Intel COFF archives can preserve long filenames; SunOS a.out archives cannot. If you move a file from the first to the second format and back again, the filename may be truncated. Likewise, different a.out environments have different conventions as to how they truncate filenames, whether they preserve directory names in filenames, etc. When interoperating with native tools, be sure your files are homogeneous.

Beware: most of these formats do not react well to the presence of spaces in filenames. We do the best we can, but can't always handle this case due to restrictions in the format of archives. Many Unix utilities are braindead in regards to spaces and such in filenames anyway, so this shouldn't be much of a restriction.

Archives are supported in BFD in `archive.c`.

### 2.8.1 Archive functions

#### 2.8.1.1 `bfd_get_next_mapent`

`symindex bfd_get_next_mapent (bfd *abfd, symindex previous, [Function] carsym **sym);`

Step through archive *abfd*'s symbol table (if it has one). Successively update *sym* with the next symbol's information, returning that symbol's (internal) index into the symbol table.

Supply `BFD_NO_MORE_SYMBOLS` as the *previous* entry to get the first one; returns `BFD_NO_MORE_SYMBOLS` when you've already got the last one.

A `carsym` is a canonical archive symbol. The only user-visible element is its name, a null-terminated string.

### 2.8.1.2 `bfd_set_archive_head`

`bool bfd_set_archive_head (bfd *output, bfd *new_head);` [Function]  
Set the head of the chain of BFDs contained in the archive *output* to *new\_head*.

### 2.8.1.3 `bfd_openr_next_archived_file`

`bfd *bfd_openr_next_archived_file (bfd *archive, bfd *previous);` [Function]  
Provided a BFD, *archive*, containing an archive and NULL, open an input BFD on the first contained element and returns that. Subsequent calls should pass the archive and the previous return value to return a created BFD to the next contained element. NULL is returned when there are no more. Note - if you want to process the bfd returned by this call be sure to call `bfd_check_format()` on it first.

## 2.9 File formats

A format is a BFD concept of high level file contents type. The formats supported by BFD are:

- `bfd_object`

The BFD may contain data, symbols, relocations and debug info.

- `bfd_archive`

The BFD contains other BFDs and an optional index.

- `bfd_core`

The BFD contains the result of an executable core dump.

### 2.9.1 File format functions

#### 2.9.1.1 `bfd_check_format`

`bool bfd_check_format (bfd *abfd, bfd_format format);` [Function]  
Verify if the file attached to the BFD *abfd* is compatible with the format *format* (i.e., one of `bfd_object`, `bfd_archive` or `bfd_core`).

If the BFD has been set to a specific target before the call, only the named target and format combination is checked. If the target has not been set, or has been set to `default`, then all the known target backends is interrogated to determine a match. If the default target matches, it is used. If not, exactly one target must recognize the file, or an error results.

The function returns `TRUE` on success, otherwise `FALSE` with one of the following error codes:

- `bfd_error_invalid_operation` - if format is not one of `bfd_object`, `bfd_archive` or `bfd_core`.
- `bfd_error_system_call` - if an error occurred during a read - even some file mismatches can cause `bfd_error_system_calls`.

- `file_not_recognised` - none of the backends recognised the file format.
- `bfd_error_file_ambiguously_recognized` - more than one backend recognised the file format.

When calling `bfd_check_format` (or `bfd_check_format_matches`), any underlying file descriptor will be kept open for the duration of the call. This is done to avoid races when another thread calls `bfd_cache_close_all`. In this scenario, the thread calling `bfd_check_format` must call `bfd_cache_close` itself.

### 2.9.1.2 `bfd_check_format_matches`

```
bool bfd_check_format_matches (bfd *abfd, bfd_format format,      [Function]
                             char ***matching);
```

Like `bfd_check_format`, except when it returns `FALSE` with `bfd_errno` set to `bfd_error_file_ambiguously_recognized`. In that case, if *matching* is not `NULL`, it will be filled in with a `NULL`-terminated list of the names of the formats that matched, allocated with `malloc`. Then the user may choose a format and try again.

When done with the list that *matching* points to, the caller should free it.

### 2.9.1.3 `bfd_set_format`

```
bool bfd_set_format (bfd *abfd, bfd_format format);                [Function]
```

This function sets the file format of the BFD *abfd* to the format *format*. If the target set in the BFD does not support the format requested, the format is invalid, or the BFD is not open for writing, then an error occurs.

### 2.9.1.4 `bfd_format_string`

```
const char *bfd_format_string (bfd_format format);                 [Function]
```

Return a pointer to a const string `invalid`, `object`, `archive`, `core`, or `unknown`, depending upon the value of *format*.

## 2.10 Relocations

BFD maintains relocations in much the same way it maintains symbols: they are left alone until required, then read in en-masse and translated into an internal form. A common routine `bfd_perform_relocation` acts upon the canonical form to do the fixup.

Relocations are maintained on a per section basis, while symbols are maintained on a per BFD basis.

All that a back end has to do to fit the BFD interface is to create a `struct reloc_cache_entry` for each relocation in a particular section, and fill in the right bits of the structures.

### 2.10.1 `typedef arelent`

This is the structure of a relocation entry:

```
struct reloc_cache_entry
{
    /* A pointer into the canonical table of pointers. */
    struct bfd_symbol **sym_ptr_ptr;
```

```

/* offset in section. */
bfd_size_type address;

/* addend for relocation value. */
bfd_vma addend;

/* Pointer to how to perform the required relocation. */
reloc_howto_type *howto;

};

```

Here is a description of each of the fields within an `arelent`:

- `sym_ptr_ptr`

The symbol table pointer points to a pointer to the symbol associated with the relocation request. It is the pointer into the table returned by the back end's `canonicalize_symtab` action. See Section 2.7 [Symbols], page 40. The symbol is referenced through a pointer to a pointer so that tools like the linker can fix up all the symbols of the same name by modifying only one pointer. The relocation routine looks in the symbol and uses the base of the section the symbol is attached to and the value of the symbol as the initial relocation offset. If the symbol pointer is zero, then the section provided is looked up.

- `address`

The `address` field gives the offset in bytes from the base of the section data which owns the relocation record to the first byte of relocatable information. The actual data relocated will be relative to this point; for example, a relocation type which modifies the bottom two bytes of a four byte word would not touch the first byte pointed to in a big endian world.

- `addend`

The `addend` is a value provided by the back end to be added (!) to the relocation offset. Its interpretation is dependent upon the howto. For example, on the 68k the code:

```

char foo[];
main()
{
    return foo[0x12345678];
}

```

Could be compiled into:

```

linkw fp,#-4
moveb @#12345678,d0
extbl d0
unlk fp
rts

```

This could create a reloc pointing to `foo`, but leave the offset in the data, something like:

```

RELOCATION RECORDS FOR [.text]:
offset    type        value

```

```

00000006 32      _foo

00000000 4e56 fffc      ; linkw fp,#-4
00000004 1039 1234 5678  ; moveb @#12345678,d0
0000000a 49c0      ; extbl d0
0000000c 4e5e      ; unlk fp
0000000e 4e75      ; rts

```

Using coff and an 88k, some instructions don't have enough space in them to represent the full address range, and pointers have to be loaded in two parts. So you'd get something like:

```

or.u      r13,r0,hi16(_foo+0x12345678)
ld.b      r2,r13,lo16(_foo+0x12345678)
jmp       r1

```

This should create two relocs, both pointing to `_foo`, and with 0x12340000 in their addend field. The data would consist of:

```

RELOCATION RECORDS FOR [.text]:
offset  type      value
00000002 HVRT16    _foo+0x12340000
00000006 LVRT16    _foo+0x12340000

00000000 5da05678      ; or.u r13,r0,0x5678
00000004 1c4d5678      ; ld.b r2,r13,0x5678
00000008 f400c001      ; jmp r1

```

The relocation routine digs out the value from the data, adds it to the addend to get the original offset, and then adds the value of `_foo`. Note that all 32 bits have to be kept around somewhere, to cope with carry from bit 15 to bit 16.

One further example is the sparc and the a.out format. The sparc has a similar problem to the 88k, in that some instructions don't have room for an entire offset, but on the sparc the parts are created in odd sized lumps. The designers of the a.out format chose to not use the data within the section for storing part of the offset; all the offset is kept within the reloc. Anything in the data should be ignored.

```

save %sp,-112,%sp
sethi %hi(_foo+0x12345678),%g2
ldsb [%g2+%lo(_foo+0x12345678)],%i0
ret
restore

```

Both relocs contain a pointer to `foo`, and the offsets contain junk.

```

RELOCATION RECORDS FOR [.text]:
offset  type      value
00000004 HI22      _foo+0x12345678
00000008 L010      _foo+0x12345678

00000000 9de3bf90      ; save %sp,-112,%sp
00000004 05000000      ; sethi %hi(_foo+0),%g2
00000008 f048a000      ; ldsb [%g2+%lo(_foo+0)],%i0

```

```

0000000c 81c7e008    ; ret
00000010 81e80000    ; restore

```

- **howto**

The **howto** field can be imagined as a relocation instruction. It is a pointer to a structure which contains information on what to do with all of the other information in the reloc record and data section. A back end would normally have a relocation instruction set and turn relocations into pointers to the correct structure on input - but it would be possible to create each **howto** field on demand.

### 2.10.1.1 enum complain\_overflow

Indicates what sort of overflow checking should be done when performing a relocation.

```

enum complain_overflow
{
    /* Do not complain on overflow.  */
    complain_overflow_dont,

    /* Complain if the value overflows when considered as a signed
       number one bit larger than the field.  ie. A bitfield of N bits
       is allowed to represent -2**n to 2**n-1.  */
    complain_overflow_bitfield,

    /* Complain if the value overflows when considered as a signed
       number.  */
    complain_overflow_signed,

    /* Complain if the value overflows when considered as an
       unsigned number.  */
    complain_overflow_unsigned
};

```

### 2.10.1.2 reloc\_howto\_type

The **reloc\_howto\_type** is a structure which contains all the information that libbfd needs to know to tie up a back end's data.

```

struct reloc_howto_struct
{
    /* The type field has mainly a documentary use - the back end can
       do what it wants with it, though normally the back end's idea of
       an external reloc number is stored in this field.  */
    unsigned int type;

    /* The size of the item to be relocated in bytes.  */
    unsigned int size:4;

    /* The number of bits in the field to be relocated.  This is used

```

```

    when doing overflow checking.  */
    unsigned int bitsize:7;

    /* The value the final relocation is shifted right by.  This drops
       unwanted data from the relocation.  */
    unsigned int rightshift:6;

    /* The bit position of the reloc value in the destination.
       The relocated value is left shifted by this amount.  */
    unsigned int bitpos:6;

    /* What type of overflow error should be checked for when
       relocating.  */
    ENUM_BITFIELD (complain_overflow) complain_on_overflow:2;

    /* The relocation value should be negated before applying.  */
    unsigned int negate:1;

    /* The relocation is relative to the item being relocated.  */
    unsigned int pc_relative:1;

    /* Some formats record a relocation addend in the section contents
       rather than with the relocation.  For ELF formats this is the
       distinction between USE_REL and USE_RELA (though the code checks
       for USE_REL == 1/0).  The value of this field is TRUE if the
       addend is recorded with the section contents; when performing a
       partial link (ld -r) the section contents (the data) will be
       modified.  The value of this field is FALSE if addends are
       recorded with the relocation (in arelent.addend); when performing
       a partial link the relocation will be modified.
       All relocations for all ELF USE_RELA targets should set this field
       to FALSE (values of TRUE should be looked on with suspicion).
       However, the converse is not true: not all relocations of all ELF
       USE_REL targets set this field to TRUE.  Why this is so is peculiar
       to each particular target.  For relocs that aren't used in partial
       links (e.g. GOT stuff) it doesn't matter what this is set to.  */
    unsigned int partial_inplace:1;

    /* When some formats create PC relative instructions, they leave
       the value of the pc of the place being relocated in the offset
       slot of the instruction, so that a PC relative relocation can
       be made just by adding in an ordinary offset (e.g., sun3 a.out).
       Some formats leave the displacement part of an instruction
       empty (e.g., ELF); this flag signals the fact.  */
    unsigned int pcrel_offset:1;

    /* Whether bfd_install_relocation should just install the addend,

```

```

        or should follow the practice of some older object formats and
        install a value including the symbol.  */
    unsigned int install_addend:1;

    /* src_mask selects the part of the instruction (or data) to be used
       in the relocation sum.  If the target relocations don't have an
       addend in the reloc, eg. ELF USE_REL, src_mask will normally equal
       dst_mask to extract the addend from the section contents.  If
       relocations do have an addend in the reloc, eg. ELF USE_RELA, this
       field should normally be zero.  Non-zero values for ELF USE_RELA
       targets should be viewed with suspicion as normally the value in
       the dst_mask part of the section contents should be ignored.  */
    bfd_vma src_mask;

    /* dst_mask selects which parts of the instruction (or data) are
       replaced with a relocated value.  */
    bfd_vma dst_mask;

    /* If this field is non null, then the supplied function is
       called rather than the normal function.  This allows really
       strange relocation methods to be accommodated.  */
    bfd_reloc_status_type (*special_function)
        (bfd *, arelent *, struct bfd_symbol *, void *, asection *,
         bfd *, char **);

    /* The textual name of the relocation type.  */
    const char *name;
};

```

### 2.10.1.3 The HOWTO Macro

The HOWTO macro fills in a `reloc_howto_type` (a typedef for `const struct reloc_howto_struct`).

```

#define HOWTO_INSTALL_ADDEND 0
#define HOWTO_RSIZE(sz) ((sz) < 0 ? -(sz) : (sz))
#define HOWTO(type, right, size, bits, pcrel, left, ovf, func, name, \
               inplace, src_mask, dst_mask, pcrel_off) \
{ (unsigned) type, HOWTO_RSIZE (size), bits, right, left, ovf, \
  size < 0, pcrel, inplace, pcrel_off, HOWTO_INSTALL_ADDEND, \
  src_mask, dst_mask, func, name }

```

This is used to fill in an empty howto entry in an array.

```

#define EMPTY_HOWTO(C) \
    HOWTO ((C), 0, 1, 0, false, 0, complain_overflow_dont, NULL, \
          NULL, false, 0, 0, false)

```

```

static inline unsigned int

```



```

bfd_get_reloc_size (reloc_howto_type *howto)
{
    return howto->size;
}

```

#### 2.10.1.4 arelent\_chain

How relocs are tied together in an asection:

```

typedef struct relent_chain
{
    arelent relent;
    struct relent_chain *next;
}
arelent_chain;

```

#### 2.10.1.5 bfd\_check\_overflow

`bfd_reloc_status_type bfd_check_overflow` (enum [Function]  
*complain\_overflow how, unsigned int bitsize, unsigned int rightshift,*  
*unsigned int addrsz, bfd\_vma relocation);*

Perform overflow checking on *relocation* which has *bitsize* significant bits and will be shifted right by *rightshift* bits, on a machine with addresses containing *addrsz* significant bits. The result is either of `bfd_reloc_ok` or `bfd_reloc_overflow`.

#### 2.10.1.6 bfd\_reloc\_offset\_in\_range

`bool bfd_reloc_offset_in_range` (*reloc\_howto\_type \*howto, bfd* [Function]  
*\*abfd, asection \*section, bfd\_size\_type offset);*

Returns TRUE if the reloc described by *HOWTO* can be applied at *OFFSET* octets in *SECTION*.

#### 2.10.1.7 bfd\_perform\_relocation

`bfd_reloc_status_type bfd_perform_relocation` (*bfd \*abfd,* [Function]  
*arelent \*reloc\_entry, void \*data, asection \*input\_section, bfd*  
*\*output\_bfd, char \*\*error\_message);*

If *output\_bfd* is supplied to this function, the generated image will be relocatable; the relocations are copied to the output file after they have been changed to reflect the new state of the world. There are two ways of reflecting the results of partial linkage in an output file: by modifying the output data in place, and by modifying the relocation record. Some native formats (e.g., basic a.out and basic coff) have no way of specifying an addend in the relocation type, so the addend has to go in the output data. This is no big deal since in these formats the output data slot will always be big enough for the addend. Complex reloc types with addends were invented to solve just this problem. The *error\_message* argument is set to an error message if this return `bfd_reloc_dangerous`.

### 2.10.1.8 bfd\_install\_relocation

```
bfd_reloc_status_type bfd_install_relocation (bfd *abfd,          [Function]
      arelent *reloc_entry, void *data, bfd_vma data_start, asection
      *input_section, char **error_message);
```

This looks remarkably like `bfd_perform_relocation`, except it does not expect that the section contents have been filled in. I.e., it's suitable for use when creating, rather than applying a relocation.

For now, this function should be considered reserved for the assembler.

## 2.10.2 The howto manager

When an application wants to create a relocation, but doesn't know what the target machine might call it, it can find out by using this bit of code.

### 2.10.2.1 bfd\_reloc\_code\_real\_type

The insides of a reloc code. The idea is that, eventually, there will be one enumerator for every type of relocation we ever do. Pass one of these values to `bfd_reloc_type_lookup`, and it'll return a howto pointer.

This does mean that the application must determine the correct enumerator value; you can't get a howto pointer from a random set of attributes.

Here are the possible values for `enum bfd_reloc_code_real`:

```
BFD_RELOC_64
BFD_RELOC_32
BFD_RELOC_26
BFD_RELOC_24
BFD_RELOC_16
BFD_RELOC_14
BFD_RELOC_8
```

Basic absolute relocations of N bits.

```
BFD_RELOC_64_PCREL
BFD_RELOC_32_PCREL
BFD_RELOC_24_PCREL
BFD_RELOC_16_PCREL
BFD_RELOC_12_PCREL
BFD_RELOC_8_PCREL
```

PC-relative relocations. Sometimes these are relative to the address of the relocation itself; sometimes they are relative to the start of the section containing the relocation. It depends on the specific target.

```
BFD_RELOC_32_SECREL
BFD_RELOC_16_SECIDX
```

Section relative relocations. Some targets need this for DWARF2.

```
BFD_RELOC_32_GOT_PCREL
BFD_RELOC_16_GOT_PCREL
BFD_RELOC_8_GOT_PCREL
```

BFD\_RELOC\_32\_GOTOFF  
BFD\_RELOC\_16\_GOTOFF  
BFD\_RELOC\_L016\_GOTOFF  
BFD\_RELOC\_HI16\_GOTOFF  
BFD\_RELOC\_HI16\_S\_GOTOFF  
BFD\_RELOC\_8\_GOTOFF  
BFD\_RELOC\_64\_PLT\_PCREL  
BFD\_RELOC\_32\_PLT\_PCREL  
BFD\_RELOC\_24\_PLT\_PCREL  
BFD\_RELOC\_16\_PLT\_PCREL  
BFD\_RELOC\_8\_PLT\_PCREL  
BFD\_RELOC\_64\_PLTOFF  
BFD\_RELOC\_32\_PLTOFF  
BFD\_RELOC\_16\_PLTOFF  
BFD\_RELOC\_L016\_PLTOFF  
BFD\_RELOC\_HI16\_PLTOFF  
BFD\_RELOC\_HI16\_S\_PLTOFF  
BFD\_RELOC\_8\_PLTOFF  
BFD\_RELOC\_COPY  
BFD\_RELOC\_GLOB\_DAT  
BFD\_RELOC\_JMP\_SLOT  
BFD\_RELOC\_RELATIVE  
BFD\_RELOC\_IRELATIVE

For ELF.

BFD\_RELOC\_SIZE32  
BFD\_RELOC\_SIZE64

Size relocations.

BFD\_RELOC\_68K\_TLS\_GD32  
BFD\_RELOC\_68K\_TLS\_GD16  
BFD\_RELOC\_68K\_TLS\_GD8  
BFD\_RELOC\_68K\_TLS\_LDM32  
BFD\_RELOC\_68K\_TLS\_LDM16  
BFD\_RELOC\_68K\_TLS\_LDM8  
BFD\_RELOC\_68K\_TLS\_LD032  
BFD\_RELOC\_68K\_TLS\_LD016  
BFD\_RELOC\_68K\_TLS\_LD08  
BFD\_RELOC\_68K\_TLS\_IE32  
BFD\_RELOC\_68K\_TLS\_IE16  
BFD\_RELOC\_68K\_TLS\_IE8  
BFD\_RELOC\_68K\_TLS\_LE32  
BFD\_RELOC\_68K\_TLS\_LE16  
BFD\_RELOC\_68K\_TLS\_LE8

Relocations used by 68K ELF.

BFD\_RELOC\_32\_BASEREL  
BFD\_RELOC\_16\_BASEREL

BFD\_RELOC\_L016\_BASEREL  
 BFD\_RELOC\_HI16\_BASEREL  
 BFD\_RELOC\_HI16\_S\_BASEREL  
 BFD\_RELOC\_8\_BASEREL  
 BFD\_RELOC\_RVA

Linkage-table relative.

BFD\_RELOC\_8\_FFnn

Absolute 8-bit relocation, but used to form an address like 0xFFnn.

BFD\_RELOC\_32\_PCREL\_S2  
 BFD\_RELOC\_16\_PCREL\_S2  
 BFD\_RELOC\_23\_PCREL\_S2

These PC-relative relocations are stored as word displacements – i.e., byte displacements shifted right two bits. The 30-bit word displacement (<<32\_PCREL\_S2>> – 32 bits, shifted 2) is used on the SPARC. (SPARC tools generally refer to this as <<WDISP30>>.) The signed 16-bit displacement is used on the MIPS, and the 23-bit displacement is used on the Alpha.

BFD\_RELOC\_HI22  
 BFD\_RELOC\_LO10

High 22 bits and low 10 bits of 32-bit value, placed into lower bits of the target word. These are used on the SPARC.

BFD\_RELOC\_GPREL16  
 BFD\_RELOC\_GPREL32

For systems that allocate a Global Pointer register, these are displacements off that register. These relocation types are handled specially, because the value the register will have is decided relatively late.

BFD\_RELOC\_NONE  
 BFD\_RELOC\_SPARC\_WDISP22  
 BFD\_RELOC\_SPARC22  
 BFD\_RELOC\_SPARC13  
 BFD\_RELOC\_SPARC\_GOT10  
 BFD\_RELOC\_SPARC\_GOT13  
 BFD\_RELOC\_SPARC\_GOT22  
 BFD\_RELOC\_SPARC\_PC10  
 BFD\_RELOC\_SPARC\_PC22  
 BFD\_RELOC\_SPARC\_WPLT30  
 BFD\_RELOC\_SPARC\_UA16  
 BFD\_RELOC\_SPARC\_UA32  
 BFD\_RELOC\_SPARC\_UA64  
 BFD\_RELOC\_SPARC\_GOTDATA\_HIX22  
 BFD\_RELOC\_SPARC\_GOTDATA\_LOX10  
 BFD\_RELOC\_SPARC\_GOTDATA\_OP\_HIX22  
 BFD\_RELOC\_SPARC\_GOTDATA\_OP\_LOX10  
 BFD\_RELOC\_SPARC\_GOTDATA\_OP

BFD\_RELOC\_SPARC\_JMP\_IREL

SPARC ELF relocations. There is probably some overlap with other relocation types already defined.

BFD\_RELOC\_SPARC\_BASE13

BFD\_RELOC\_SPARC\_BASE22

I think these are specific to SPARC a.out (e.g., Sun 4).

BFD\_RELOC\_SPARC\_64

BFD\_RELOC\_SPARC\_10

BFD\_RELOC\_SPARC\_11

BFD\_RELOC\_SPARC\_OL010

BFD\_RELOC\_SPARC\_HH22

BFD\_RELOC\_SPARC\_HM10

BFD\_RELOC\_SPARC\_LM22

BFD\_RELOC\_SPARC\_PC\_HH22

BFD\_RELOC\_SPARC\_PC\_HM10

BFD\_RELOC\_SPARC\_PC\_LM22

BFD\_RELOC\_SPARC\_WDISP16

BFD\_RELOC\_SPARC\_WDISP19

BFD\_RELOC\_SPARC\_7

BFD\_RELOC\_SPARC\_6

BFD\_RELOC\_SPARC\_5

BFD\_RELOC\_SPARC\_DISP64

BFD\_RELOC\_SPARC\_HIX22

BFD\_RELOC\_SPARC\_LOX10

BFD\_RELOC\_SPARC\_H44

BFD\_RELOC\_SPARC\_M44

BFD\_RELOC\_SPARC\_L44

BFD\_RELOC\_SPARC\_REGISTER

BFD\_RELOC\_SPARC\_H34

BFD\_RELOC\_SPARC\_SIZE32

BFD\_RELOC\_SPARC\_SIZE64

BFD\_RELOC\_SPARC\_WDISP10

SPARC64 relocations.

BFD\_RELOC\_SPARC\_REV32

SPARC little endian relocation.

BFD\_RELOC\_SPARC\_TLS\_GD\_HI22

BFD\_RELOC\_SPARC\_TLS\_GD\_L010

BFD\_RELOC\_SPARC\_TLS\_GD\_ADD

BFD\_RELOC\_SPARC\_TLS\_GD\_CALL

BFD\_RELOC\_SPARC\_TLS\_LDM\_HI22

BFD\_RELOC\_SPARC\_TLS\_LDM\_L010

BFD\_RELOC\_SPARC\_TLS\_LDM\_ADD

BFD\_RELOC\_SPARC\_TLS\_LDM\_CALL

BFD\_RELOC\_SPARC\_TLS\_LDO\_HIX22

BFD\_RELOC\_SPARC\_TLS\_LDO\_LOX10  
 BFD\_RELOC\_SPARC\_TLS\_LDO\_ADD  
 BFD\_RELOC\_SPARC\_TLS\_IE\_HI22  
 BFD\_RELOC\_SPARC\_TLS\_IE\_LO10  
 BFD\_RELOC\_SPARC\_TLS\_IE\_LD  
 BFD\_RELOC\_SPARC\_TLS\_IE\_LDX  
 BFD\_RELOC\_SPARC\_TLS\_IE\_ADD  
 BFD\_RELOC\_SPARC\_TLS\_LE\_HIX22  
 BFD\_RELOC\_SPARC\_TLS\_LE\_LOX10  
 BFD\_RELOC\_SPARC\_TLS\_DTPMOD32  
 BFD\_RELOC\_SPARC\_TLS\_DTPMOD64  
 BFD\_RELOC\_SPARC\_TLS\_DTPOFF32  
 BFD\_RELOC\_SPARC\_TLS\_DTPOFF64  
 BFD\_RELOC\_SPARC\_TLS\_TPOFF32  
 BFD\_RELOC\_SPARC\_TLS\_TPOFF64

SPARC TLS relocations.

BFD\_RELOC\_SPU\_IMM7  
 BFD\_RELOC\_SPU\_IMM8  
 BFD\_RELOC\_SPU\_IMM10  
 BFD\_RELOC\_SPU\_IMM10W  
 BFD\_RELOC\_SPU\_IMM16  
 BFD\_RELOC\_SPU\_IMM16W  
 BFD\_RELOC\_SPU\_IMM18  
 BFD\_RELOC\_SPU\_PCREL9a  
 BFD\_RELOC\_SPU\_PCREL9b  
 BFD\_RELOC\_SPU\_PCREL16  
 BFD\_RELOC\_SPU\_LO16  
 BFD\_RELOC\_SPU\_HI16  
 BFD\_RELOC\_SPU\_PPU32  
 BFD\_RELOC\_SPU\_PPU64  
 BFD\_RELOC\_SPU\_ADD\_PIC

SPU Relocations.

BFD\_RELOC\_ALPHA\_GPDISP\_HI16

Alpha ECOFF and ELF relocations. Some of these treat the symbol or "addend" in some special way. For GPDISP\_HI16 ("gpdisp") relocations, the symbol is ignored when writing; when reading, it will be the absolute section symbol. The addend is the displacement in bytes of the "lda" instruction from the "ldah" instruction (which is at the address of this reloc).

BFD\_RELOC\_ALPHA\_GPDISP\_LO16

For GPDISP\_LO16 ("ignore") relocations, the symbol is handled as with GPDISP\_HI16 relocations. The addend is ignored when writing the relocations out, and is filled in with the file's GP value on reading, for convenience.

BFD\_RELOC\_ALPHA\_GPDISP

The ELF GPDISP relocation is exactly the same as the GPDISP\_HI16 relocation except that there is no accompanying GPDISP\_LO16 relocation.

**BFD\_RELOC\_ALPHA\_LITERAL****BFD\_RELOC\_ALPHA\_ELF\_LITERAL****BFD\_RELOC\_ALPHA\_LITUSE**

The Alpha LITERAL/LITUSE relocs are produced by a symbol reference; the assembler turns it into a LDQ instruction to load the address of the symbol, and then fills in a register in the real instruction.

The LITERAL reloc, at the LDQ instruction, refers to the .lita section symbol. The addend is ignored when writing, but is filled in with the file's GP value on reading, for convenience, as with the GPDISP\_LO16 reloc.

The ELF\_LITERAL reloc is somewhere between 16\_GOTOFF and GPDISP\_LO16. It should refer to the symbol to be referenced, as with 16\_GOTOFF, but it generates output not based on the position within the .got section, but relative to the GP value chosen for the file during the final link stage.

The LITUSE reloc, on the instruction using the loaded address, gives information to the linker that it might be able to use to optimize away some literal section references. The symbol is ignored (read as the absolute section symbol), and the "addend" indicates the type of instruction using the register: 1 - "memory" fmt insn 2 - byte-manipulation (byte offset reg) 3 - jsr (target of branch)

**BFD\_RELOC\_ALPHA\_HINT**

The HINT relocation indicates a value that should be filled into the "hint" field of a jmp/jsr/ret instruction, for possible branch- prediction logic which may be provided on some processors.

**BFD\_RELOC\_ALPHA\_LINKAGE**

The LINKAGE relocation outputs a linkage pair in the object file, which is filled by the linker.

**BFD\_RELOC\_ALPHA\_CODEADDR**

The CODEADDR relocation outputs a STO\_CA in the object file, which is filled by the linker.

**BFD\_RELOC\_ALPHA\_GPREL\_HI16****BFD\_RELOC\_ALPHA\_GPREL\_LO16**

The GPREL\_HI/LO relocations together form a 32-bit offset from the GP register.

**BFD\_RELOC\_ALPHA\_BRSGP**

Like BFD\_RELOC\_23\_PCREL\_S2, except that the source and target must share a common GP, and the target address is adjusted for STO\_ALPHA\_STD\_GPLOAD.

**BFD\_RELOC\_ALPHA\_NOP**

The NOP relocation outputs a NOP if the longword displacement between two procedure entry points is  $< 2^{21}$ .

**BFD\_RELOC\_ALPHA\_BSR**

The BSR relocation outputs a BSR if the longword displacement between two procedure entry points is  $< 2^{21}$ .

**BFD\_RELOC\_ALPHA\_LDA**

The LDA relocation outputs a LDA if the longword displacement between two procedure entry points is  $< 2^{16}$ .

**BFD\_RELOC\_ALPHA\_BOH**

The BOH relocation outputs a BSR if the longword displacement between two procedure entry points is  $< 2^{21}$ , or else a hint.

**BFD\_RELOC\_ALPHA\_TLSGD****BFD\_RELOC\_ALPHA\_TLSDM****BFD\_RELOC\_ALPHA\_DTPMOD64****BFD\_RELOC\_ALPHA\_GOTDTPREL16****BFD\_RELOC\_ALPHA\_DTPREL64****BFD\_RELOC\_ALPHA\_DTPREL\_HI16****BFD\_RELOC\_ALPHA\_DTPREL\_L016****BFD\_RELOC\_ALPHA\_DTPREL16****BFD\_RELOC\_ALPHA\_GOTTREL16****BFD\_RELOC\_ALPHA\_TPREL64****BFD\_RELOC\_ALPHA\_TPREL\_HI16****BFD\_RELOC\_ALPHA\_TPREL\_L016****BFD\_RELOC\_ALPHA\_TPREL16**

Alpha thread-local storage relocations.

**BFD\_RELOC\_MIPS\_JMP****BFD\_RELOC\_MICROMIPS\_JMP**

The MIPS jump instruction.

**BFD\_RELOC\_MIPS16\_JMP**

The MIPS16 jump instruction.

**BFD\_RELOC\_MIPS16\_GPREL**

MIPS16 GP relative reloc.

**BFD\_RELOC\_HI16**

High 16 bits of 32-bit value; simple reloc.

**BFD\_RELOC\_HI16\_S**

High 16 bits of 32-bit value but the low 16 bits will be sign extended and added to form the final result. If the low 16 bits form a negative number, we need to add one to the high value to compensate for the borrow when the low bits are added.

**BFD\_RELOC\_L016**

Low 16 bits.

**BFD\_RELOC\_HI16\_PCREL**

High 16 bits of 32-bit pc-relative value.

**BFD\_RELOC\_HI16\_S\_PCREL**

High 16 bits of 32-bit pc-relative value, adjusted.



**BFD\_RELOC\_LO16\_PCREL**

Low 16 bits of pc-relative value.

**BFD\_RELOC\_MIPS16\_GOT16**

**BFD\_RELOC\_MIPS16\_CALL16**

Equivalent of BFD\_RELOC\_MIPS\_\*, but with the MIPS16 layout of 16-bit immediate fields.

**BFD\_RELOC\_MIPS16\_HI16**

MIPS16 high 16 bits of 32-bit value.

**BFD\_RELOC\_MIPS16\_HI16\_S**

MIPS16 high 16 bits of 32-bit value but the low 16 bits will be sign extended and added to form the final result. If the low 16 bits form a negative number, we need to add one to the high value to compensate for the borrow when the low bits are added.

**BFD\_RELOC\_MIPS16\_LO16**

MIPS16 low 16 bits.

**BFD\_RELOC\_MIPS16\_TLS\_GD**

**BFD\_RELOC\_MIPS16\_TLS\_LDM**

**BFD\_RELOC\_MIPS16\_TLS\_DTPREL\_HI16**

**BFD\_RELOC\_MIPS16\_TLS\_DTPREL\_LO16**

**BFD\_RELOC\_MIPS16\_TLS\_GOTTPREL**

**BFD\_RELOC\_MIPS16\_TLS\_TPREL\_HI16**

**BFD\_RELOC\_MIPS16\_TLS\_TPREL\_LO16**

MIPS16 TLS relocations.

**BFD\_RELOC\_MIPS\_LITERAL**

**BFD\_RELOC\_MICROMIPS\_LITERAL**

Relocation against a MIPS literal section.

**BFD\_RELOC\_MICROMIPS\_7\_PCREL\_S1**

**BFD\_RELOC\_MICROMIPS\_10\_PCREL\_S1**

**BFD\_RELOC\_MICROMIPS\_16\_PCREL\_S1**

microMIPS PC-relative relocations.

**BFD\_RELOC\_MIPS16\_16\_PCREL\_S1**

MIPS16 PC-relative relocation.

**BFD\_RELOC\_MIPS\_21\_PCREL\_S2**

**BFD\_RELOC\_MIPS\_26\_PCREL\_S2**

**BFD\_RELOC\_MIPS\_18\_PCREL\_S3**

**BFD\_RELOC\_MIPS\_19\_PCREL\_S2**

MIPS PC-relative relocations.

**BFD\_RELOC\_MICROMIPS\_GPREL16**

**BFD\_RELOC\_MICROMIPS\_HI16**

**BFD\_RELOC\_MICROMIPS\_HI16\_S**

**BFD\_RELOC\_MICROMIPS\_LO16**

microMIPS versions of generic BFD relocs.

BFD\_RELOC\_MIPS\_GOT16  
BFD\_RELOC\_MICROMIPS\_GOT16  
BFD\_RELOC\_MIPS\_CALL16  
BFD\_RELOC\_MICROMIPS\_CALL16  
BFD\_RELOC\_MIPS\_GOT\_HI16  
BFD\_RELOC\_MICROMIPS\_GOT\_HI16  
BFD\_RELOC\_MIPS\_GOT\_LO16  
BFD\_RELOC\_MICROMIPS\_GOT\_LO16  
BFD\_RELOC\_MIPS\_CALL\_HI16  
BFD\_RELOC\_MICROMIPS\_CALL\_HI16  
BFD\_RELOC\_MIPS\_CALL\_LO16  
BFD\_RELOC\_MICROMIPS\_CALL\_LO16  
BFD\_RELOC\_MIPS\_SUB  
BFD\_RELOC\_MICROMIPS\_SUB  
BFD\_RELOC\_MIPS\_GOT\_PAGE  
BFD\_RELOC\_MICROMIPS\_GOT\_PAGE  
BFD\_RELOC\_MIPS\_GOT\_OFST  
BFD\_RELOC\_MICROMIPS\_GOT\_OFST  
BFD\_RELOC\_MIPS\_GOT\_DISP  
BFD\_RELOC\_MICROMIPS\_GOT\_DISP  
BFD\_RELOC\_MIPS\_SHIFT5  
BFD\_RELOC\_MIPS\_SHIFT6  
BFD\_RELOC\_MIPS\_INSERT\_A  
BFD\_RELOC\_MIPS\_INSERT\_B  
BFD\_RELOC\_MIPS\_DELETE  
BFD\_RELOC\_MIPS\_HIGHEST  
BFD\_RELOC\_MICROMIPS\_HIGHEST  
BFD\_RELOC\_MIPS\_HIGHER  
BFD\_RELOC\_MICROMIPS\_HIGHER  
BFD\_RELOC\_MIPS\_SCN\_DISP  
BFD\_RELOC\_MICROMIPS\_SCN\_DISP  
BFD\_RELOC\_MIPS\_16  
BFD\_RELOC\_MIPS\_RELGOT  
BFD\_RELOC\_MIPS\_JALR  
BFD\_RELOC\_MICROMIPS\_JALR  
BFD\_RELOC\_MIPS\_TLS\_DTPMOD32  
BFD\_RELOC\_MIPS\_TLS\_DTPREL32  
BFD\_RELOC\_MIPS\_TLS\_DTPMOD64  
BFD\_RELOC\_MIPS\_TLS\_DTPREL64  
BFD\_RELOC\_MIPS\_TLS\_GD  
BFD\_RELOC\_MICROMIPS\_TLS\_GD  
BFD\_RELOC\_MIPS\_TLS\_LDM  
BFD\_RELOC\_MICROMIPS\_TLS\_LDM  
BFD\_RELOC\_MIPS\_TLS\_DTPREL\_HI16  
BFD\_RELOC\_MICROMIPS\_TLS\_DTPREL\_HI16  
BFD\_RELOC\_MIPS\_TLS\_DTPREL\_LO16  
BFD\_RELOC\_MICROMIPS\_TLS\_DTPREL\_LO16

BFD\_RELOC\_MIPS\_TLS\_GOTTPREL  
BFD\_RELOC\_MICROMIPS\_TLS\_GOTTPREL  
BFD\_RELOC\_MIPS\_TLS\_TPREL32  
BFD\_RELOC\_MIPS\_TLS\_TPREL64  
BFD\_RELOC\_MIPS\_TLS\_TPREL\_HI16  
BFD\_RELOC\_MICROMIPS\_TLS\_TPREL\_HI16  
BFD\_RELOC\_MIPS\_TLS\_TPREL\_L016  
BFD\_RELOC\_MICROMIPS\_TLS\_TPREL\_L016  
BFD\_RELOC\_MIPS\_EH

MIPS ELF relocations.

BFD\_RELOC\_MOXIE\_10\_PCREL

Moxie ELF relocations.

BFD\_RELOC\_FT32\_10  
BFD\_RELOC\_FT32\_20  
BFD\_RELOC\_FT32\_17  
BFD\_RELOC\_FT32\_18  
BFD\_RELOC\_FT32\_RELAX  
BFD\_RELOC\_FT32\_SC0  
BFD\_RELOC\_FT32\_SC1  
BFD\_RELOC\_FT32\_15  
BFD\_RELOC\_FT32\_DIFF32

FT32 ELF relocations.

BFD\_RELOC\_FRV\_LABEL16  
BFD\_RELOC\_FRV\_LABEL24  
BFD\_RELOC\_FRV\_L016  
BFD\_RELOC\_FRV\_HI16  
BFD\_RELOC\_FRV\_GPREL12  
BFD\_RELOC\_FRV\_GPRELU12  
BFD\_RELOC\_FRV\_GPREL32  
BFD\_RELOC\_FRV\_GPRELHI  
BFD\_RELOC\_FRV\_GPRELLO  
BFD\_RELOC\_FRV\_GOT12  
BFD\_RELOC\_FRV\_GOTHI  
BFD\_RELOC\_FRV\_GOTLO  
BFD\_RELOC\_FRV\_FUNCDESC  
BFD\_RELOC\_FRV\_FUNCDESC\_GOT12  
BFD\_RELOC\_FRV\_FUNCDESC\_GOTHI  
BFD\_RELOC\_FRV\_FUNCDESC\_GOTLO  
BFD\_RELOC\_FRV\_FUNCDESC\_VALUE  
BFD\_RELOC\_FRV\_FUNCDESC\_GTOFF12  
BFD\_RELOC\_FRV\_FUNCDESC\_GTOFFHI  
BFD\_RELOC\_FRV\_FUNCDESC\_GTOFFLO  
BFD\_RELOC\_FRV\_GTOFF12  
BFD\_RELOC\_FRV\_GTOFFHI  
BFD\_RELOC\_FRV\_GTOFFLO

BFD\_RELOC\_FRV\_GETTLSOFF  
 BFD\_RELOC\_FRV\_TLSDESC\_VALUE  
 BFD\_RELOC\_FRV\_GOTTLSDESC12  
 BFD\_RELOC\_FRV\_GOTTLSDESCHI  
 BFD\_RELOC\_FRV\_GOTTLSDESCLO  
 BFD\_RELOC\_FRV\_TLSMOFF12  
 BFD\_RELOC\_FRV\_TLSMOFFHI  
 BFD\_RELOC\_FRV\_TLSMOFFLO  
 BFD\_RELOC\_FRV\_GOTTLSOFF12  
 BFD\_RELOC\_FRV\_GOTTLSOFFHI  
 BFD\_RELOC\_FRV\_GOTTLSOFFLO  
 BFD\_RELOC\_FRV\_TLSOFF  
 BFD\_RELOC\_FRV\_TLSDESC\_RELAX  
 BFD\_RELOC\_FRV\_GETTLSOFF\_RELAX  
 BFD\_RELOC\_FRV\_TLSOFF\_RELAX  
 BFD\_RELOC\_FRV\_TLSMOFF

Fujitsu Frv Relocations.

BFD\_RELOC\_MN10300\_GOTOFF24

This is a 24bit GOT-relative reloc for the mn10300.

BFD\_RELOC\_MN10300\_GOT32

This is a 32bit GOT-relative reloc for the mn10300, offset by two bytes in the instruction.

BFD\_RELOC\_MN10300\_GOT24

This is a 24bit GOT-relative reloc for the mn10300, offset by two bytes in the instruction.

BFD\_RELOC\_MN10300\_GOT16

This is a 16bit GOT-relative reloc for the mn10300, offset by two bytes in the instruction.

BFD\_RELOC\_MN10300\_SYM\_DIFF

Together with another reloc targeted at the same location, allows for a value that is the difference of two symbols in the same section.

BFD\_RELOC\_MN10300\_ALIGN

The addend of this reloc is an alignment power that must be honoured at the offset's location, regardless of linker relaxation.

BFD\_RELOC\_MN10300\_TLS\_GD  
 BFD\_RELOC\_MN10300\_TLS\_LD  
 BFD\_RELOC\_MN10300\_TLS\_LDO  
 BFD\_RELOC\_MN10300\_TLS\_GOTIE  
 BFD\_RELOC\_MN10300\_TLS\_IE  
 BFD\_RELOC\_MN10300\_TLS\_LE  
 BFD\_RELOC\_MN10300\_TLS\_DTPMOD  
 BFD\_RELOC\_MN10300\_TLS\_DTPOFF

BFD\_RELOC\_MN10300\_TLS\_TPOFF

Various TLS-related relocations.

BFD\_RELOC\_MN10300\_32\_PCREL

This is a 32bit pcrel reloc for the mn10300, offset by two bytes in the instruction.

BFD\_RELOC\_MN10300\_16\_PCREL

This is a 16bit pcrel reloc for the mn10300, offset by two bytes in the instruction.

BFD\_RELOC\_386\_GOT32

BFD\_RELOC\_386\_PLT32

BFD\_RELOC\_386\_GOTOFF

BFD\_RELOC\_386\_GOTPC

BFD\_RELOC\_386\_TLS\_TPOFF

BFD\_RELOC\_386\_TLS\_IE

BFD\_RELOC\_386\_TLS\_GOTIE

BFD\_RELOC\_386\_TLS\_LE

BFD\_RELOC\_386\_TLS\_GD

BFD\_RELOC\_386\_TLS\_LDM

BFD\_RELOC\_386\_TLS\_LDO\_32

BFD\_RELOC\_386\_TLS\_IE\_32

BFD\_RELOC\_386\_TLS\_LE\_32

BFD\_RELOC\_386\_TLS\_DTPMOD32

BFD\_RELOC\_386\_TLS\_DTPOFF32

BFD\_RELOC\_386\_TLS\_TPOFF32

BFD\_RELOC\_386\_TLS\_GOTDESC

BFD\_RELOC\_386\_TLS\_DESC\_CALL

BFD\_RELOC\_386\_TLS\_DESC

BFD\_RELOC\_386\_GOT32X

i386/elf relocations.

BFD\_RELOC\_X86\_64\_GOT32

BFD\_RELOC\_X86\_64\_GOTPCREL

BFD\_RELOC\_X86\_64\_32S

BFD\_RELOC\_X86\_64\_DTPMOD64

BFD\_RELOC\_X86\_64\_DTPOFF64

BFD\_RELOC\_X86\_64\_TPOFF64

BFD\_RELOC\_X86\_64\_TLSD

BFD\_RELOC\_X86\_64\_TLSLD

BFD\_RELOC\_X86\_64\_DTPOFF32

BFD\_RELOC\_X86\_64\_GOTTPOFF

BFD\_RELOC\_X86\_64\_TPOFF32

BFD\_RELOC\_X86\_64\_GOTOFF64

BFD\_RELOC\_X86\_64\_GOTPC32

BFD\_RELOC\_X86\_64\_GOT64

BFD\_RELOC\_X86\_64\_GOTPCREL64

BFD\_RELOC\_X86\_64\_GOTPC64

BFD\_RELOC\_X86\_64\_GOTPLT64

```

BFD_RELOC_X86_64_GOTPC32_TLSDESC
BFD_RELOC_X86_64_TLSDESC_CALL
BFD_RELOC_X86_64_TLSDESC
BFD_RELOC_X86_64_PC32_BND
BFD_RELOC_X86_64_PLT32_BND
BFD_RELOC_X86_64_GOTPCRELX
BFD_RELOC_X86_64_REX_GOTPCRELX
BFD_RELOC_X86_64_CODE_4_GOTPCRELX
BFD_RELOC_X86_64_CODE_4_GOTTPOFF
BFD_RELOC_X86_64_CODE_4_GOTPC32_TLSDESC
BFD_RELOC_X86_64_CODE_5_GOTPCRELX
BFD_RELOC_X86_64_CODE_5_GOTTPOFF
BFD_RELOC_X86_64_CODE_5_GOTPC32_TLSDESC
BFD_RELOC_X86_64_CODE_6_GOTPCRELX
BFD_RELOC_X86_64_CODE_6_GOTTPOFF
BFD_RELOC_X86_64_CODE_6_GOTPC32_TLSDESC

```

x86-64/elf relocations.

```

BFD_RELOC_NS32K_IMM_8
BFD_RELOC_NS32K_IMM_16
BFD_RELOC_NS32K_IMM_32
BFD_RELOC_NS32K_IMM_8_PCREL
BFD_RELOC_NS32K_IMM_16_PCREL
BFD_RELOC_NS32K_IMM_32_PCREL
BFD_RELOC_NS32K_DISP_8
BFD_RELOC_NS32K_DISP_16
BFD_RELOC_NS32K_DISP_32
BFD_RELOC_NS32K_DISP_8_PCREL
BFD_RELOC_NS32K_DISP_16_PCREL
BFD_RELOC_NS32K_DISP_32_PCREL

```

ns32k relocations.

```

BFD_RELOC_PDP11_DISP_8_PCREL
BFD_RELOC_PDP11_DISP_6_PCREL

```

PDP11 relocations.

```

BFD_RELOC_PJ_CODE_HI16
BFD_RELOC_PJ_CODE_LO16
BFD_RELOC_PJ_CODE_DIR16
BFD_RELOC_PJ_CODE_DIR32
BFD_RELOC_PJ_CODE_REL16
BFD_RELOC_PJ_CODE_REL32

```

Picojava relocs. Not all of these appear in object files.

```

BFD_RELOC_PPC_B26
BFD_RELOC_PPC_BA26
BFD_RELOC_PPC_TOC16
BFD_RELOC_PPC_TOC16_LO

```

BFD\_RELOC\_PPC\_TOC16\_HI  
BFD\_RELOC\_PPC\_B16  
BFD\_RELOC\_PPC\_B16\_BRTAKEN  
BFD\_RELOC\_PPC\_B16\_BRNTAKEN  
BFD\_RELOC\_PPC\_BA16  
BFD\_RELOC\_PPC\_BA16\_BRTAKEN  
BFD\_RELOC\_PPC\_BA16\_BRNTAKEN  
BFD\_RELOC\_PPC\_LOCAL24PC  
BFD\_RELOC\_PPC\_EMB\_NADDR32  
BFD\_RELOC\_PPC\_EMB\_NADDR16  
BFD\_RELOC\_PPC\_EMB\_NADDR16\_LO  
BFD\_RELOC\_PPC\_EMB\_NADDR16\_HI  
BFD\_RELOC\_PPC\_EMB\_NADDR16\_HA  
BFD\_RELOC\_PPC\_EMB\_SDAI16  
BFD\_RELOC\_PPC\_EMB\_SDA2I16  
BFD\_RELOC\_PPC\_EMB\_SDA2REL  
BFD\_RELOC\_PPC\_EMB\_SDA21  
BFD\_RELOC\_PPC\_EMB\_MRKREF  
BFD\_RELOC\_PPC\_EMB\_RELSEC16  
BFD\_RELOC\_PPC\_EMB\_RELST\_LO  
BFD\_RELOC\_PPC\_EMB\_RELST\_HI  
BFD\_RELOC\_PPC\_EMB\_RELST\_HA  
BFD\_RELOC\_PPC\_EMB\_BIT\_FLD  
BFD\_RELOC\_PPC\_EMB\_RELSDA  
BFD\_RELOC\_PPC\_VLE\_REL8  
BFD\_RELOC\_PPC\_VLE\_REL15  
BFD\_RELOC\_PPC\_VLE\_REL24  
BFD\_RELOC\_PPC\_VLE\_LO16A  
BFD\_RELOC\_PPC\_VLE\_LO16D  
BFD\_RELOC\_PPC\_VLE\_HI16A  
BFD\_RELOC\_PPC\_VLE\_HI16D  
BFD\_RELOC\_PPC\_VLE\_HA16A  
BFD\_RELOC\_PPC\_VLE\_HA16D  
BFD\_RELOC\_PPC\_VLE\_SDA21  
BFD\_RELOC\_PPC\_VLE\_SDA21\_LO  
BFD\_RELOC\_PPC\_VLE\_SDAREL\_LO16A  
BFD\_RELOC\_PPC\_VLE\_SDAREL\_LO16D  
BFD\_RELOC\_PPC\_VLE\_SDAREL\_HI16A  
BFD\_RELOC\_PPC\_VLE\_SDAREL\_HI16D  
BFD\_RELOC\_PPC\_VLE\_SDAREL\_HA16A  
BFD\_RELOC\_PPC\_VLE\_SDAREL\_HA16D  
BFD\_RELOC\_PPC\_16DX\_HA  
BFD\_RELOC\_PPC\_REL16DX\_HA  
BFD\_RELOC\_PPC\_NEG  
BFD\_RELOC\_PPC64\_HIGHER  
BFD\_RELOC\_PPC64\_HIGHER\_S  
BFD\_RELOC\_PPC64\_HIGHEST

BFD\_RELOC\_PPC64\_HIGHEST\_S  
BFD\_RELOC\_PPC64\_TOC16\_LO  
BFD\_RELOC\_PPC64\_TOC16\_HI  
BFD\_RELOC\_PPC64\_TOC16\_HA  
BFD\_RELOC\_PPC64\_TOC  
BFD\_RELOC\_PPC64\_PLTGOT16  
BFD\_RELOC\_PPC64\_PLTGOT16\_LO  
BFD\_RELOC\_PPC64\_PLTGOT16\_HI  
BFD\_RELOC\_PPC64\_PLTGOT16\_HA  
BFD\_RELOC\_PPC64\_ADDR16\_DS  
BFD\_RELOC\_PPC64\_ADDR16\_LO\_DS  
BFD\_RELOC\_PPC64\_GOT16\_DS  
BFD\_RELOC\_PPC64\_GOT16\_LO\_DS  
BFD\_RELOC\_PPC64\_PLT16\_LO\_DS  
BFD\_RELOC\_PPC64\_SECTOFF\_DS  
BFD\_RELOC\_PPC64\_SECTOFF\_LO\_DS  
BFD\_RELOC\_PPC64\_TOC16\_DS  
BFD\_RELOC\_PPC64\_TOC16\_LO\_DS  
BFD\_RELOC\_PPC64\_PLTGOT16\_DS  
BFD\_RELOC\_PPC64\_PLTGOT16\_LO\_DS  
BFD\_RELOC\_PPC64\_ADDR16\_HIGH  
BFD\_RELOC\_PPC64\_ADDR16\_HIGHA  
BFD\_RELOC\_PPC64\_REL16\_HIGH  
BFD\_RELOC\_PPC64\_REL16\_HIGHA  
BFD\_RELOC\_PPC64\_REL16\_HIGHER  
BFD\_RELOC\_PPC64\_REL16\_HIGHERA  
BFD\_RELOC\_PPC64\_REL16\_HIGHEST  
BFD\_RELOC\_PPC64\_REL16\_HIGHESTA  
BFD\_RELOC\_PPC64\_ADDR64\_LOCAL  
BFD\_RELOC\_PPC64\_ENTRY  
BFD\_RELOC\_PPC64\_REL24\_NOTOC  
BFD\_RELOC\_PPC64\_REL24\_P9NOTOC  
BFD\_RELOC\_PPC64\_D34  
BFD\_RELOC\_PPC64\_D34\_LO  
BFD\_RELOC\_PPC64\_D34\_HI30  
BFD\_RELOC\_PPC64\_D34\_HA30  
BFD\_RELOC\_PPC64\_PCREL34  
BFD\_RELOC\_PPC64\_GOT\_PCREL34  
BFD\_RELOC\_PPC64\_PLT\_PCREL34  
BFD\_RELOC\_PPC64\_ADDR16\_HIGHER34  
BFD\_RELOC\_PPC64\_ADDR16\_HIGHERA34  
BFD\_RELOC\_PPC64\_ADDR16\_HIGHEST34  
BFD\_RELOC\_PPC64\_ADDR16\_HIGHESTA34  
BFD\_RELOC\_PPC64\_REL16\_HIGHER34  
BFD\_RELOC\_PPC64\_REL16\_HIGHERA34  
BFD\_RELOC\_PPC64\_REL16\_HIGHEST34  
BFD\_RELOC\_PPC64\_REL16\_HIGHESTA34



BFD\_RELOC\_PPC64\_D28

BFD\_RELOC\_PPC64\_PCREL28

Power(rs6000) and PowerPC relocations.

BFD\_RELOC\_PPC\_TLS

BFD\_RELOC\_PPC\_TLSGD

BFD\_RELOC\_PPC\_TLSLD

BFD\_RELOC\_PPC\_TLSLE

BFD\_RELOC\_PPC\_TLSIE

BFD\_RELOC\_PPC\_TLSM

BFD\_RELOC\_PPC\_TLSML

BFD\_RELOC\_PPC\_DTPMOD

BFD\_RELOC\_PPC\_TPREL16

BFD\_RELOC\_PPC\_TPREL16\_LO

BFD\_RELOC\_PPC\_TPREL16\_HI

BFD\_RELOC\_PPC\_TPREL16\_HA

BFD\_RELOC\_PPC\_TPREL

BFD\_RELOC\_PPC\_DTPREL16

BFD\_RELOC\_PPC\_DTPREL16\_LO

BFD\_RELOC\_PPC\_DTPREL16\_HI

BFD\_RELOC\_PPC\_DTPREL16\_HA

BFD\_RELOC\_PPC\_DTPREL

BFD\_RELOC\_PPC\_GOT\_TLSGD16

BFD\_RELOC\_PPC\_GOT\_TLSGD16\_LO

BFD\_RELOC\_PPC\_GOT\_TLSGD16\_HI

BFD\_RELOC\_PPC\_GOT\_TLSGD16\_HA

BFD\_RELOC\_PPC\_GOT\_TLSLD16

BFD\_RELOC\_PPC\_GOT\_TLSLD16\_LO

BFD\_RELOC\_PPC\_GOT\_TLSLD16\_HI

BFD\_RELOC\_PPC\_GOT\_TLSLD16\_HA

BFD\_RELOC\_PPC\_GOT\_TPREL16

BFD\_RELOC\_PPC\_GOT\_TPREL16\_LO

BFD\_RELOC\_PPC\_GOT\_TPREL16\_HI

BFD\_RELOC\_PPC\_GOT\_TPREL16\_HA

BFD\_RELOC\_PPC\_GOT\_DTPREL16

BFD\_RELOC\_PPC\_GOT\_DTPREL16\_LO

BFD\_RELOC\_PPC\_GOT\_DTPREL16\_HI

BFD\_RELOC\_PPC\_GOT\_DTPREL16\_HA

BFD\_RELOC\_PPC64\_TLSGD

BFD\_RELOC\_PPC64\_TLSLD

BFD\_RELOC\_PPC64\_TLSLE

BFD\_RELOC\_PPC64\_TLSIE

BFD\_RELOC\_PPC64\_TLSM

BFD\_RELOC\_PPC64\_TLSML

BFD\_RELOC\_PPC64\_TPREL16\_DS

BFD\_RELOC\_PPC64\_TPREL16\_LO\_DS

BFD\_RELOC\_PPC64\_TPREL16\_HIGH

BFD\_RELOC\_PPC64\_TPREL16\_HIGHA  
 BFD\_RELOC\_PPC64\_TPREL16\_HIGHER  
 BFD\_RELOC\_PPC64\_TPREL16\_HIGHERA  
 BFD\_RELOC\_PPC64\_TPREL16\_HIGHEST  
 BFD\_RELOC\_PPC64\_TPREL16\_HIGHESTA  
 BFD\_RELOC\_PPC64\_DTPREL16\_DS  
 BFD\_RELOC\_PPC64\_DTPREL16\_LO\_DS  
 BFD\_RELOC\_PPC64\_DTPREL16\_HIGH  
 BFD\_RELOC\_PPC64\_DTPREL16\_HIGHA  
 BFD\_RELOC\_PPC64\_DTPREL16\_HIGHER  
 BFD\_RELOC\_PPC64\_DTPREL16\_HIGHERA  
 BFD\_RELOC\_PPC64\_DTPREL16\_HIGHEST  
 BFD\_RELOC\_PPC64\_DTPREL16\_HIGHESTA  
 BFD\_RELOC\_PPC64\_TPREL34  
 BFD\_RELOC\_PPC64\_DTPREL34  
 BFD\_RELOC\_PPC64\_GOT\_TLSGD\_PCREL34  
 BFD\_RELOC\_PPC64\_GOT\_TLSD\_PCREL34  
 BFD\_RELOC\_PPC64\_GOT\_TPREL\_PCREL34  
 BFD\_RELOC\_PPC64\_GOT\_DTPREL\_PCREL34  
 BFD\_RELOC\_PPC64\_TLS\_PCREL

PowerPC and PowerPC64 thread-local storage relocations.

**BFD\_RELOC\_CTOR**

The type of reloc used to build a constructor table - at the moment probably a 32 bit wide absolute relocation, but the target can choose. It generally does map to one of the other relocation types.

**BFD\_RELOC\_ARM\_PCREL\_BRANCH**

ARM 26 bit pc-relative branch. The lowest two bits must be zero and are not stored in the instruction.

**BFD\_RELOC\_ARM\_PCREL\_BLX**

ARM 26 bit pc-relative branch. The lowest bit must be zero and is not stored in the instruction. The 2nd lowest bit comes from a 1 bit field in the instruction.

**BFD\_RELOC\_THUMB\_PCREL\_BLX**

Thumb 22 bit pc-relative branch. The lowest bit must be zero and is not stored in the instruction. The 2nd lowest bit comes from a 1 bit field in the instruction.

**BFD\_RELOC\_ARM\_PCREL\_CALL**

ARM 26-bit pc-relative branch for an unconditional BL or BLX instruction.

**BFD\_RELOC\_ARM\_PCREL\_JUMP**

ARM 26-bit pc-relative branch for B or conditional BL instruction.

**BFD\_RELOC\_THUMB\_PCREL\_BRANCH5**

ARM 5-bit pc-relative branch for Branch Future instructions.

**BFD\_RELOC\_THUMB\_PCREL\_BFCSEL**

ARM 6-bit pc-relative branch for BFCSEL instruction.

**BFD\_RELOC\_ARM\_THUMB\_BF17**

ARM 17-bit pc-relative branch for Branch Future instructions.

**BFD\_RELOC\_ARM\_THUMB\_BF13**

ARM 13-bit pc-relative branch for BFCSEL instruction.

**BFD\_RELOC\_ARM\_THUMB\_BF19**

ARM 19-bit pc-relative branch for Branch Future Link instruction.

**BFD\_RELOC\_ARM\_THUMB\_LOOP12**

ARM 12-bit pc-relative branch for Low Overhead Loop instructions.

**BFD\_RELOC\_THUMB\_PCREL\_BRANCH7**

**BFD\_RELOC\_THUMB\_PCREL\_BRANCH9**

**BFD\_RELOC\_THUMB\_PCREL\_BRANCH12**

**BFD\_RELOC\_THUMB\_PCREL\_BRANCH20**

**BFD\_RELOC\_THUMB\_PCREL\_BRANCH23**

**BFD\_RELOC\_THUMB\_PCREL\_BRANCH25**

Thumb 7-, 9-, 12-, 20-, 23-, and 25-bit pc-relative branches. The lowest bit must be zero and is not stored in the instruction. Note that the corresponding ELF R\_ARM\_THM\_JUMPnn constant has an "nn" one smaller in all cases. Note further that BRANCH23 corresponds to R\_ARM\_THM\_CALL.

**BFD\_RELOC\_ARM\_OFFSET\_IMM**

12-bit immediate offset, used in ARM-format ldr and str instructions.

**BFD\_RELOC\_ARM\_THUMB\_OFFSET**

5-bit immediate offset, used in Thumb-format ldr and str instructions.

**BFD\_RELOC\_ARM\_TARGET1**

Pc-relative or absolute relocation depending on target. Used for entries in .init\_array sections.

**BFD\_RELOC\_ARM\_ROSEGREL32**

Read-only segment base relative address.

**BFD\_RELOC\_ARM\_SBREL32**

Data segment base relative address.

**BFD\_RELOC\_ARM\_TARGET2**

This reloc is used for references to RTTI data from exception handling tables. The actual definition depends on the target. It may be a pc-relative or some form of GOT-indirect relocation.

**BFD\_RELOC\_ARM\_PREL31**

31-bit PC relative address.

**BFD\_RELOC\_ARM\_MOVW**

**BFD\_RELOC\_ARM\_MOVT**

**BFD\_RELOC\_ARM\_MOVW\_PCREL**

**BFD\_RELOC\_ARM\_MOVT\_PCREL**

```

BFD_RELOC_ARM_THUMB_MOVW
BFD_RELOC_ARM_THUMB_MOVT
BFD_RELOC_ARM_THUMB_MOVW_PCREL
BFD_RELOC_ARM_THUMB_MOVT_PCREL

```

Low and High halfword relocations for MOVW and MOVT instructions.

```

BFD_RELOC_ARM_GOTFUNCDESC
BFD_RELOC_ARM_GOTOFFFUNCDESC
BFD_RELOC_ARM_FUNCDESC
BFD_RELOC_ARM_FUNCDESC_VALUE
BFD_RELOC_ARM_TLS_GD32_FDPIC
BFD_RELOC_ARM_TLS_LDM32_FDPIC
BFD_RELOC_ARM_TLS_IE32_FDPIC

```

ARM FDPIC specific relocations.

```

BFD_RELOC_ARM_GOT32
BFD_RELOC_ARM_GOTOFF
BFD_RELOC_ARM_GOTPC
BFD_RELOC_ARM_GOT_PREL

```

Relocations for setting up GOTs and PLTs for shared libraries.

```

BFD_RELOC_ARM_TLS_GD32
BFD_RELOC_ARM_TLS_LD032
BFD_RELOC_ARM_TLS_LDM32
BFD_RELOC_ARM_TLS_DTPOFF32
BFD_RELOC_ARM_TLS_DTPMOD32
BFD_RELOC_ARM_TLS_TPOFF32
BFD_RELOC_ARM_TLS_IE32
BFD_RELOC_ARM_TLS_LE32
BFD_RELOC_ARM_TLS_GOTDESC
BFD_RELOC_ARM_TLS_CALL
BFD_RELOC_ARM_THM_TLS_CALL
BFD_RELOC_ARM_TLS_DESCSEQ
BFD_RELOC_ARM_THM_TLS_DESCSEQ
BFD_RELOC_ARM_TLS_DESC

```

ARM thread-local storage relocations.

```

BFD_RELOC_ARM_ALU_PC_GO_NC
BFD_RELOC_ARM_ALU_PC_GO
BFD_RELOC_ARM_ALU_PC_G1_NC
BFD_RELOC_ARM_ALU_PC_G1
BFD_RELOC_ARM_ALU_PC_G2
BFD_RELOC_ARM_LDR_PC_GO
BFD_RELOC_ARM_LDR_PC_G1
BFD_RELOC_ARM_LDR_PC_G2
BFD_RELOC_ARM_LDRS_PC_GO
BFD_RELOC_ARM_LDRS_PC_G1
BFD_RELOC_ARM_LDRS_PC_G2

```

BFD\_RELOC\_ARM\_LDC\_PC\_G0  
 BFD\_RELOC\_ARM\_LDC\_PC\_G1  
 BFD\_RELOC\_ARM\_LDC\_PC\_G2  
 BFD\_RELOC\_ARM\_ALU\_SB\_G0\_NC  
 BFD\_RELOC\_ARM\_ALU\_SB\_G0  
 BFD\_RELOC\_ARM\_ALU\_SB\_G1\_NC  
 BFD\_RELOC\_ARM\_ALU\_SB\_G1  
 BFD\_RELOC\_ARM\_ALU\_SB\_G2  
 BFD\_RELOC\_ARM\_LDR\_SB\_G0  
 BFD\_RELOC\_ARM\_LDR\_SB\_G1  
 BFD\_RELOC\_ARM\_LDR\_SB\_G2  
 BFD\_RELOC\_ARM\_LDRS\_SB\_G0  
 BFD\_RELOC\_ARM\_LDRS\_SB\_G1  
 BFD\_RELOC\_ARM\_LDRS\_SB\_G2  
 BFD\_RELOC\_ARM\_LDC\_SB\_G0  
 BFD\_RELOC\_ARM\_LDC\_SB\_G1  
 BFD\_RELOC\_ARM\_LDC\_SB\_G2

ARM group relocations.

BFD\_RELOC\_ARM\_V4BX

Annotation of BX instructions.

BFD\_RELOC\_ARM\_THUMB\_ALU\_ABS\_G0\_NC  
 BFD\_RELOC\_ARM\_THUMB\_ALU\_ABS\_G1\_NC  
 BFD\_RELOC\_ARM\_THUMB\_ALU\_ABS\_G2\_NC  
 BFD\_RELOC\_ARM\_THUMB\_ALU\_ABS\_G3\_NC

Thumb1 relocations to support execute-only code.

BFD\_RELOC\_ARM\_IMMEDIATE  
 BFD\_RELOC\_ARM\_ADRL\_IMMEDIATE  
 BFD\_RELOC\_ARM\_T32\_IMMEDIATE  
 BFD\_RELOC\_ARM\_T32\_ADD\_IMM  
 BFD\_RELOC\_ARM\_T32\_IMM12  
 BFD\_RELOC\_ARM\_T32\_ADD\_PC12  
 BFD\_RELOC\_ARM\_SHIFT\_IMM  
 BFD\_RELOC\_ARM\_SMC  
 BFD\_RELOC\_ARM\_HVC  
 BFD\_RELOC\_ARM\_SWI  
 BFD\_RELOC\_ARM\_MULTII  
 BFD\_RELOC\_ARM\_CP\_OFF\_IMM  
 BFD\_RELOC\_ARM\_CP\_OFF\_IMM\_S2  
 BFD\_RELOC\_ARM\_T32\_CP\_OFF\_IMM  
 BFD\_RELOC\_ARM\_T32\_CP\_OFF\_IMM\_S2  
 BFD\_RELOC\_ARM\_T32\_VLDR\_VSTR\_OFF\_IMM  
 BFD\_RELOC\_ARM\_ADR\_IMM  
 BFD\_RELOC\_ARM\_LDR\_IMM  
 BFD\_RELOC\_ARM\_LITERAL  
 BFD\_RELOC\_ARM\_IN\_POOL

BFD\_RELOC\_ARM\_OFFSET\_IMM8  
BFD\_RELOC\_ARM\_T32\_OFFSET\_U8  
BFD\_RELOC\_ARM\_T32\_OFFSET\_IMM  
BFD\_RELOC\_ARM\_HWLITERAL  
BFD\_RELOC\_ARM\_THUMB\_ADD  
BFD\_RELOC\_ARM\_THUMB\_IMM  
BFD\_RELOC\_ARM\_THUMB\_SHIFT

These relocs are only used within the ARM assembler. They are not (at present) written to any object files.

BFD\_RELOC\_SH\_PCDISP8BY2  
BFD\_RELOC\_SH\_PCDISP12BY2  
BFD\_RELOC\_SH\_IMM3  
BFD\_RELOC\_SH\_IMM3U  
BFD\_RELOC\_SH\_DISP12  
BFD\_RELOC\_SH\_DISP12BY2  
BFD\_RELOC\_SH\_DISP12BY4  
BFD\_RELOC\_SH\_DISP12BY8  
BFD\_RELOC\_SH\_DISP20  
BFD\_RELOC\_SH\_DISP20BY8  
BFD\_RELOC\_SH\_IMM4  
BFD\_RELOC\_SH\_IMM4BY2  
BFD\_RELOC\_SH\_IMM4BY4  
BFD\_RELOC\_SH\_IMM8  
BFD\_RELOC\_SH\_IMM8BY2  
BFD\_RELOC\_SH\_IMM8BY4  
BFD\_RELOC\_SH\_PCRELIMM8BY2  
BFD\_RELOC\_SH\_PCRELIMM8BY4  
BFD\_RELOC\_SH\_SWITCH16  
BFD\_RELOC\_SH\_SWITCH32  
BFD\_RELOC\_SH\_USES  
BFD\_RELOC\_SH\_COUNT  
BFD\_RELOC\_SH\_ALIGN  
BFD\_RELOC\_SH\_CODE  
BFD\_RELOC\_SH\_DATA  
BFD\_RELOC\_SH\_LABEL  
BFD\_RELOC\_SH\_LOOP\_START  
BFD\_RELOC\_SH\_LOOP\_END  
BFD\_RELOC\_SH\_GOTPC  
BFD\_RELOC\_SH\_GOT\_LOW16  
BFD\_RELOC\_SH\_GOT\_MEDLOW16  
BFD\_RELOC\_SH\_GOT\_MEDHI16  
BFD\_RELOC\_SH\_GOT\_HI16  
BFD\_RELOC\_SH\_GOTPLT\_LOW16  
BFD\_RELOC\_SH\_GOTPLT\_MEDLOW16  
BFD\_RELOC\_SH\_GOTPLT\_MEDHI16  
BFD\_RELOC\_SH\_GOTPLT\_HI16

BFD\_RELOC\_SH\_PLT\_LOW16  
BFD\_RELOC\_SH\_PLT\_MEDLOW16  
BFD\_RELOC\_SH\_PLT\_MEDHI16  
BFD\_RELOC\_SH\_PLT\_HI16  
BFD\_RELOC\_SH\_GOTOFF\_LOW16  
BFD\_RELOC\_SH\_GOTOFF\_MEDLOW16  
BFD\_RELOC\_SH\_GOTOFF\_MEDHI16  
BFD\_RELOC\_SH\_GOTOFF\_HI16  
BFD\_RELOC\_SH\_GOTPC\_LOW16  
BFD\_RELOC\_SH\_GOTPC\_MEDLOW16  
BFD\_RELOC\_SH\_GOTPC\_MEDHI16  
BFD\_RELOC\_SH\_GOTPC\_HI16  
BFD\_RELOC\_SH\_COPY64  
BFD\_RELOC\_SH\_GLOB\_DAT64  
BFD\_RELOC\_SH\_JMP\_SLOT64  
BFD\_RELOC\_SH\_RELATIVE64  
BFD\_RELOC\_SH\_GOT10BY4  
BFD\_RELOC\_SH\_GOT10BY8  
BFD\_RELOC\_SH\_GOTPLT10BY4  
BFD\_RELOC\_SH\_GOTPLT10BY8  
BFD\_RELOC\_SH\_GOTPLT32  
BFD\_RELOC\_SH\_SHMEDIA\_CODE  
BFD\_RELOC\_SH\_IMMU5  
BFD\_RELOC\_SH\_IMMS6  
BFD\_RELOC\_SH\_IMMS6BY32  
BFD\_RELOC\_SH\_IMMU6  
BFD\_RELOC\_SH\_IMMS10  
BFD\_RELOC\_SH\_IMMS10BY2  
BFD\_RELOC\_SH\_IMMS10BY4  
BFD\_RELOC\_SH\_IMMS10BY8  
BFD\_RELOC\_SH\_IMMS16  
BFD\_RELOC\_SH\_IMMU16  
BFD\_RELOC\_SH\_IMM\_LOW16  
BFD\_RELOC\_SH\_IMM\_LOW16\_PCREL  
BFD\_RELOC\_SH\_IMM\_MEDLOW16  
BFD\_RELOC\_SH\_IMM\_MEDLOW16\_PCREL  
BFD\_RELOC\_SH\_IMM\_MEDHI16  
BFD\_RELOC\_SH\_IMM\_MEDHI16\_PCREL  
BFD\_RELOC\_SH\_IMM\_HI16  
BFD\_RELOC\_SH\_IMM\_HI16\_PCREL  
BFD\_RELOC\_SH\_PT\_16  
BFD\_RELOC\_SH\_TLS\_GD\_32  
BFD\_RELOC\_SH\_TLS\_LD\_32  
BFD\_RELOC\_SH\_TLS\_LDO\_32  
BFD\_RELOC\_SH\_TLS\_IE\_32  
BFD\_RELOC\_SH\_TLS\_LE\_32  
BFD\_RELOC\_SH\_TLS\_DTPMOD32

BFD\_RELOC\_SH\_TLS\_DTPOFF32  
BFD\_RELOC\_SH\_TLS\_TPOFF32  
BFD\_RELOC\_SH\_GOT20  
BFD\_RELOC\_SH\_GOTOFF20  
BFD\_RELOC\_SH\_GOTFUNCDESC  
BFD\_RELOC\_SH\_GOTFUNCDESC20  
BFD\_RELOC\_SH\_GOTOFFFUNCDESC  
BFD\_RELOC\_SH\_GOTOFFFUNCDESC20  
BFD\_RELOC\_SH\_FUNCDESC

Renesas / SuperH SH relocs. Not all of these appear in object files.

BFD\_RELOC\_ARC\_N8  
BFD\_RELOC\_ARC\_N16  
BFD\_RELOC\_ARC\_N24  
BFD\_RELOC\_ARC\_N32  
BFD\_RELOC\_ARC\_SDA  
BFD\_RELOC\_ARC\_SECTOFF  
BFD\_RELOC\_ARC\_S21H\_PCREL  
BFD\_RELOC\_ARC\_S21W\_PCREL  
BFD\_RELOC\_ARC\_S25H\_PCREL  
BFD\_RELOC\_ARC\_S25W\_PCREL  
BFD\_RELOC\_ARC\_SDA32  
BFD\_RELOC\_ARC\_SDA\_LDST  
BFD\_RELOC\_ARC\_SDA\_LDST1  
BFD\_RELOC\_ARC\_SDA\_LDST2  
BFD\_RELOC\_ARC\_SDA16\_LD  
BFD\_RELOC\_ARC\_SDA16\_LD1  
BFD\_RELOC\_ARC\_SDA16\_LD2  
BFD\_RELOC\_ARC\_S13\_PCREL  
BFD\_RELOC\_ARC\_W  
BFD\_RELOC\_ARC\_32\_ME  
BFD\_RELOC\_ARC\_32\_ME\_S  
BFD\_RELOC\_ARC\_N32\_ME  
BFD\_RELOC\_ARC\_SECTOFF\_ME  
BFD\_RELOC\_ARC\_SDA32\_ME  
BFD\_RELOC\_ARC\_W\_ME  
BFD\_RELOC\_AC\_SECTOFF\_U8  
BFD\_RELOC\_AC\_SECTOFF\_U8\_1  
BFD\_RELOC\_AC\_SECTOFF\_U8\_2  
BFD\_RELOC\_AC\_SECTOFF\_S9  
BFD\_RELOC\_AC\_SECTOFF\_S9\_1  
BFD\_RELOC\_AC\_SECTOFF\_S9\_2  
BFD\_RELOC\_ARC\_SECTOFF\_ME\_1  
BFD\_RELOC\_ARC\_SECTOFF\_ME\_2  
BFD\_RELOC\_ARC\_SECTOFF\_1  
BFD\_RELOC\_ARC\_SECTOFF\_2  
BFD\_RELOC\_ARC\_SDA\_12



BFD\_RELOC\_ARC\_SDA16\_ST2  
BFD\_RELOC\_ARC\_32\_PCREL  
BFD\_RELOC\_ARC\_GOT32  
BFD\_RELOC\_ARC\_GOTPC32  
BFD\_RELOC\_ARC\_GOTOFF  
BFD\_RELOC\_ARC\_GOTPC  
BFD\_RELOC\_ARC\_S21W\_PCREL\_PLT  
BFD\_RELOC\_ARC\_S25H\_PCREL\_PLT  
BFD\_RELOC\_ARC\_TLS\_DTPMOD  
BFD\_RELOC\_ARC\_TLS\_TPOFF  
BFD\_RELOC\_ARC\_TLS\_GD\_GOT  
BFD\_RELOC\_ARC\_TLS\_GD\_LD  
BFD\_RELOC\_ARC\_TLS\_GD\_CALL  
BFD\_RELOC\_ARC\_TLS\_IE\_GOT  
BFD\_RELOC\_ARC\_TLS\_DTPOFF  
BFD\_RELOC\_ARC\_TLS\_DTPOFF\_S9  
BFD\_RELOC\_ARC\_TLS\_LE\_S9  
BFD\_RELOC\_ARC\_TLS\_LE\_32  
BFD\_RELOC\_ARC\_S25W\_PCREL\_PLT  
BFD\_RELOC\_ARC\_S21H\_PCREL\_PLT  
BFD\_RELOC\_ARC\_NPS\_CMEM16  
BFD\_RELOC\_ARC\_JLI\_SECTOFF

ARC relocs.

BFD\_RELOC\_BFIN\_16\_IMM

ADI Blackfin 16 bit immediate absolute reloc.

BFD\_RELOC\_BFIN\_16\_HIGH

ADI Blackfin 16 bit immediate absolute reloc higher 16 bits.

BFD\_RELOC\_BFIN\_4\_PCREL

ADI Blackfin 'a' part of LSETUP.

BFD\_RELOC\_BFIN\_5\_PCREL

ADI Blackfin.

BFD\_RELOC\_BFIN\_16\_LOW

ADI Blackfin 16 bit immediate absolute reloc lower 16 bits.

BFD\_RELOC\_BFIN\_10\_PCREL

ADI Blackfin.

BFD\_RELOC\_BFIN\_11\_PCREL

ADI Blackfin 'b' part of LSETUP.

BFD\_RELOC\_BFIN\_12\_PCREL\_JUMP

ADI Blackfin.

BFD\_RELOC\_BFIN\_12\_PCREL\_JUMP\_S

ADI Blackfin Short jump, pcrel.

BFD\_RELOC\_BFIN\_24\_PCREL\_CALL\_X  
ADI Blackfin Call.x not implemented.

BFD\_RELOC\_BFIN\_24\_PCREL\_JUMP\_L  
ADI Blackfin Long Jump pcrel.

BFD\_RELOC\_BFIN\_GOT17M4  
BFD\_RELOC\_BFIN\_GOTHI  
BFD\_RELOC\_BFIN\_GOTLO  
BFD\_RELOC\_BFIN\_FUNCDESC  
BFD\_RELOC\_BFIN\_FUNCDESC\_GOT17M4  
BFD\_RELOC\_BFIN\_FUNCDESC\_GOTHI  
BFD\_RELOC\_BFIN\_FUNCDESC\_GOTLO  
BFD\_RELOC\_BFIN\_FUNCDESC\_VALUE  
BFD\_RELOC\_BFIN\_FUNCDESC\_GOTOFF17M4  
BFD\_RELOC\_BFIN\_FUNCDESC\_GOTOFFHI  
BFD\_RELOC\_BFIN\_FUNCDESC\_GOTOFFLO  
BFD\_RELOC\_BFIN\_GOTOFF17M4  
BFD\_RELOC\_BFIN\_GOTOFFHI  
BFD\_RELOC\_BFIN\_GOTOFFLO  
ADI Blackfin FD-PIC relocations.

BFD\_RELOC\_BFIN\_GOT  
ADI Blackfin GOT relocation.

BFD\_RELOC\_BFIN\_PLTPC  
ADI Blackfin PLTPC relocation.

BFD\_ARELOC\_BFIN\_PUSH  
ADI Blackfin arithmetic relocation.

BFD\_ARELOC\_BFIN\_CONST  
ADI Blackfin arithmetic relocation.

BFD\_ARELOC\_BFIN\_ADD  
ADI Blackfin arithmetic relocation.

BFD\_ARELOC\_BFIN\_SUB  
ADI Blackfin arithmetic relocation.

BFD\_ARELOC\_BFIN\_MULT  
ADI Blackfin arithmetic relocation.

BFD\_ARELOC\_BFIN\_DIV  
ADI Blackfin arithmetic relocation.

BFD\_ARELOC\_BFIN\_MOD  
ADI Blackfin arithmetic relocation.

BFD\_ARELOC\_BFIN\_LSHIFT  
ADI Blackfin arithmetic relocation.

**BFD\_ARELOC\_BFIN\_RSHIFT**

ADI Blackfin arithmetic relocation.

**BFD\_ARELOC\_BFIN\_AND**

ADI Blackfin arithmetic relocation.

**BFD\_ARELOC\_BFIN\_OR**

ADI Blackfin arithmetic relocation.

**BFD\_ARELOC\_BFIN\_XOR**

ADI Blackfin arithmetic relocation.

**BFD\_ARELOC\_BFIN\_LAND**

ADI Blackfin arithmetic relocation.

**BFD\_ARELOC\_BFIN\_LOR**

ADI Blackfin arithmetic relocation.

**BFD\_ARELOC\_BFIN\_LEN**

ADI Blackfin arithmetic relocation.

**BFD\_ARELOC\_BFIN\_NEG**

ADI Blackfin arithmetic relocation.

**BFD\_ARELOC\_BFIN\_COMP**

ADI Blackfin arithmetic relocation.

**BFD\_ARELOC\_BFIN\_PAGE**

ADI Blackfin arithmetic relocation.

**BFD\_ARELOC\_BFIN\_HWPAGE**

ADI Blackfin arithmetic relocation.

**BFD\_ARELOC\_BFIN\_ADDR**

ADI Blackfin arithmetic relocation.

**BFD\_RELOC\_D10V\_10\_PCREL\_R**

Mitsubishi D10V relocs. This is a 10-bit reloc with the right 2 bits assumed to be 0.

**BFD\_RELOC\_D10V\_10\_PCREL\_L**

Mitsubishi D10V relocs. This is a 10-bit reloc with the right 2 bits assumed to be 0. This is the same as the previous reloc except it is in the left container, i.e., shifted left 15 bits.

**BFD\_RELOC\_D10V\_18**

This is an 18-bit reloc with the right 2 bits assumed to be 0.

**BFD\_RELOC\_D10V\_18\_PCREL**

This is an 18-bit reloc with the right 2 bits assumed to be 0.

**BFD\_RELOC\_D30V\_6**

Mitsubishi D30V relocs. This is a 6-bit absolute reloc.

**BFD\_RELOC\_D30V\_9\_PCREL**

This is a 6-bit pc-relative reloc with the right 3 bits assumed to be 0.

**BFD\_RELOC\_D30V\_9\_PCREL\_R**

This is a 6-bit pc-relative reloc with the right 3 bits assumed to be 0. Same as the previous reloc but on the right side of the container.

**BFD\_RELOC\_D30V\_15**

This is a 12-bit absolute reloc with the right 3 bits assumed to be 0.

**BFD\_RELOC\_D30V\_15\_PCREL**

This is a 12-bit pc-relative reloc with the right 3 bits assumed to be 0.

**BFD\_RELOC\_D30V\_15\_PCREL\_R**

This is a 12-bit pc-relative reloc with the right 3 bits assumed to be 0. Same as the previous reloc but on the right side of the container.

**BFD\_RELOC\_D30V\_21**

This is an 18-bit absolute reloc with the right 3 bits assumed to be 0.

**BFD\_RELOC\_D30V\_21\_PCREL**

This is an 18-bit pc-relative reloc with the right 3 bits assumed to be 0.

**BFD\_RELOC\_D30V\_21\_PCREL\_R**

This is an 18-bit pc-relative reloc with the right 3 bits assumed to be 0. Same as the previous reloc but on the right side of the container.

**BFD\_RELOC\_D30V\_32**

This is a 32-bit absolute reloc.

**BFD\_RELOC\_D30V\_32\_PCREL**

This is a 32-bit pc-relative reloc.

**BFD\_RELOC\_DLX\_HI16\_S****BFD\_RELOC\_DLX\_LO16****BFD\_RELOC\_DLX\_JMP26**

DLX relocs.

**BFD\_RELOC\_M32C\_HI8****BFD\_RELOC\_M32C\_RL\_JUMP****BFD\_RELOC\_M32C\_RL\_1ADDR****BFD\_RELOC\_M32C\_RL\_2ADDR**

Renesas M16C/M32C Relocations.

**BFD\_RELOC\_M32R\_24**

Renesas M32R (formerly Mitsubishi M32R) relocs. This is a 24 bit absolute address.

**BFD\_RELOC\_M32R\_10\_PCREL**

This is a 10-bit pc-relative reloc with the right 2 bits assumed to be 0.

**BFD\_RELOC\_M32R\_18\_PCREL**

This is an 18-bit reloc with the right 2 bits assumed to be 0.

**BFD\_RELOC\_M32R\_26\_PCREL**

This is a 26-bit reloc with the right 2 bits assumed to be 0.

**BFD\_RELOC\_M32R\_HI16\_ULO**

This is a 16-bit reloc containing the high 16 bits of an address used when the lower 16 bits are treated as unsigned.

**BFD\_RELOC\_M32R\_HI16\_SLO**

This is a 16-bit reloc containing the high 16 bits of an address used when the lower 16 bits are treated as signed.

**BFD\_RELOC\_M32R\_LO16**

This is a 16-bit reloc containing the lower 16 bits of an address.

**BFD\_RELOC\_M32R\_SDA16**

This is a 16-bit reloc containing the small data area offset for use in add3, load, and store instructions.

**BFD\_RELOC\_M32R\_GOT24****BFD\_RELOC\_M32R\_26\_PLTREL****BFD\_RELOC\_M32R\_GOTOFF****BFD\_RELOC\_M32R\_GOTOFF\_HI\_ULO****BFD\_RELOC\_M32R\_GOTOFF\_HI\_SLO****BFD\_RELOC\_M32R\_GOTOFF\_LO****BFD\_RELOC\_M32R\_GOTPC24****BFD\_RELOC\_M32R\_GOT16\_HI\_ULO****BFD\_RELOC\_M32R\_GOT16\_HI\_SLO****BFD\_RELOC\_M32R\_GOT16\_LO****BFD\_RELOC\_M32R\_GOTPC\_HI\_ULO****BFD\_RELOC\_M32R\_GOTPC\_HI\_SLO****BFD\_RELOC\_M32R\_GOTPC\_LO**

For PIC.

**BFD\_RELOC\_NDS32\_20**

NDS32 relocs. This is a 20 bit absolute address.

**BFD\_RELOC\_NDS32\_9\_PCREL**

This is a 9-bit pc-relative reloc with the right 1 bit assumed to be 0.

**BFD\_RELOC\_NDS32\_WORD\_9\_PCREL**

This is a 9-bit pc-relative reloc with the right 1 bit assumed to be 0.

**BFD\_RELOC\_NDS32\_15\_PCREL**

This is an 15-bit reloc with the right 1 bit assumed to be 0.

**BFD\_RELOC\_NDS32\_17\_PCREL**

This is an 17-bit reloc with the right 1 bit assumed to be 0.

**BFD\_RELOC\_NDS32\_25\_PCREL**

This is a 25-bit reloc with the right 1 bit assumed to be 0.

**BFD\_RELOC\_NDS32\_HI20**

This is a 20-bit reloc containing the high 20 bits of an address used with the lower 12 bits.

**BFD\_RELOC\_NDS32\_L012S3**

This is a 12-bit reloc containing the lower 12 bits of an address then shift right by 3. This is used with ldi,sdi.

**BFD\_RELOC\_NDS32\_L012S2**

This is a 12-bit reloc containing the lower 12 bits of an address then shift left by 2. This is used with lwi,swi.

**BFD\_RELOC\_NDS32\_L012S1**

This is a 12-bit reloc containing the lower 12 bits of an address then shift left by 1. This is used with lhi,shi.

**BFD\_RELOC\_NDS32\_L012S0**

This is a 12-bit reloc containing the lower 12 bits of an address then shift left by 0. This is used with lbi,sbi.

**BFD\_RELOC\_NDS32\_L012S0\_ORI**

This is a 12-bit reloc containing the lower 12 bits of an address then shift left by 0. This is only used with branch relaxations.

**BFD\_RELOC\_NDS32\_SDA15S3**

This is a 15-bit reloc containing the small data area 18-bit signed offset and shift left by 3 for use in ldi, sdi.

**BFD\_RELOC\_NDS32\_SDA15S2**

This is a 15-bit reloc containing the small data area 17-bit signed offset and shift left by 2 for use in lwi, swi.

**BFD\_RELOC\_NDS32\_SDA15S1**

This is a 15-bit reloc containing the small data area 16-bit signed offset and shift left by 1 for use in lhi, shi.

**BFD\_RELOC\_NDS32\_SDA15S0**

This is a 15-bit reloc containing the small data area 15-bit signed offset and shift left by 0 for use in lbi, sbi.

**BFD\_RELOC\_NDS32\_SDA16S3**

This is a 16-bit reloc containing the small data area 16-bit signed offset and shift left by 3.

**BFD\_RELOC\_NDS32\_SDA17S2**

This is a 17-bit reloc containing the small data area 17-bit signed offset and shift left by 2 for use in lwi.gp, swi.gp.

**BFD\_RELOC\_NDS32\_SDA18S1**

This is a 18-bit reloc containing the small data area 18-bit signed offset and shift left by 1 for use in lhi.gp, shi.gp.

**BFD\_RELOC\_NDS32\_SDA19S0**

This is a 19-bit reloc containing the small data area 19-bit signed offset and shift left by 0 for use in lbi.gp, sbi.gp.

BFD\_RELOC\_NDS32\_GOT20  
BFD\_RELOC\_NDS32\_9\_PLTREL  
BFD\_RELOC\_NDS32\_25\_PLTREL  
BFD\_RELOC\_NDS32\_GOTOFF  
BFD\_RELOC\_NDS32\_GOTOFF\_HI20  
BFD\_RELOC\_NDS32\_GOTOFF\_L012  
BFD\_RELOC\_NDS32\_GOTPC20  
BFD\_RELOC\_NDS32\_GOT\_HI20  
BFD\_RELOC\_NDS32\_GOT\_L012  
BFD\_RELOC\_NDS32\_GOTPC\_HI20  
BFD\_RELOC\_NDS32\_GOTPC\_L012

For PIC.

BFD\_RELOC\_NDS32\_INSN16  
BFD\_RELOC\_NDS32\_LABEL  
BFD\_RELOC\_NDS32\_LONGCALL1  
BFD\_RELOC\_NDS32\_LONGCALL2  
BFD\_RELOC\_NDS32\_LONGCALL3  
BFD\_RELOC\_NDS32\_LONGJUMP1  
BFD\_RELOC\_NDS32\_LONGJUMP2  
BFD\_RELOC\_NDS32\_LONGJUMP3  
BFD\_RELOC\_NDS32\_LOADSTORE  
BFD\_RELOC\_NDS32\_9\_FIXED  
BFD\_RELOC\_NDS32\_15\_FIXED  
BFD\_RELOC\_NDS32\_17\_FIXED  
BFD\_RELOC\_NDS32\_25\_FIXED  
BFD\_RELOC\_NDS32\_LONGCALL4  
BFD\_RELOC\_NDS32\_LONGCALL5  
BFD\_RELOC\_NDS32\_LONGCALL6  
BFD\_RELOC\_NDS32\_LONGJUMP4  
BFD\_RELOC\_NDS32\_LONGJUMP5  
BFD\_RELOC\_NDS32\_LONGJUMP6  
BFD\_RELOC\_NDS32\_LONGJUMP7

For relax.

BFD\_RELOC\_NDS32\_PLTREL\_HI20  
BFD\_RELOC\_NDS32\_PLTREL\_L012  
BFD\_RELOC\_NDS32\_PLT\_GOTREL\_HI20  
BFD\_RELOC\_NDS32\_PLT\_GOTREL\_L012

For PIC.

BFD\_RELOC\_NDS32\_SDA12S2\_DP  
BFD\_RELOC\_NDS32\_SDA12S2\_SP  
BFD\_RELOC\_NDS32\_L012S2\_DP

BFD\_RELOC\_NDS32\_L012S2\_SP

For floating point.

BFD\_RELOC\_NDS32\_DWARF2\_OP1

BFD\_RELOC\_NDS32\_DWARF2\_OP2

BFD\_RELOC\_NDS32\_DWARF2\_LEB

For dwarf2 debug\_line.

BFD\_RELOC\_NDS32\_UPDATE\_TA

For eliminating 16-bit instructions.

BFD\_RELOC\_NDS32\_PLT\_GOTREL\_L020

BFD\_RELOC\_NDS32\_PLT\_GOTREL\_L015

BFD\_RELOC\_NDS32\_PLT\_GOTREL\_L019

BFD\_RELOC\_NDS32\_GOT\_L015

BFD\_RELOC\_NDS32\_GOT\_L019

BFD\_RELOC\_NDS32\_GOTOFF\_L015

BFD\_RELOC\_NDS32\_GOTOFF\_L019

BFD\_RELOC\_NDS32\_GOT15S2

BFD\_RELOC\_NDS32\_GOT17S2

For PIC object relaxation.

BFD\_RELOC\_NDS32\_5

NDS32 relocs. This is a 5 bit absolute address.

BFD\_RELOC\_NDS32\_10\_UPCREL

This is a 10-bit unsigned pc-relative reloc with the right 1 bit assumed to be 0.

BFD\_RELOC\_NDS32\_SDA\_FP7U2\_RELA

If fp were omitted, fp can be used as another gp.

BFD\_RELOC\_NDS32\_RELAX\_ENTRY

BFD\_RELOC\_NDS32\_GOT\_SUFF

BFD\_RELOC\_NDS32\_GOTOFF\_SUFF

BFD\_RELOC\_NDS32\_PLT\_GOT\_SUFF

BFD\_RELOC\_NDS32\_MULCALL\_SUFF

BFD\_RELOC\_NDS32\_PTR

BFD\_RELOC\_NDS32\_PTR\_COUNT

BFD\_RELOC\_NDS32\_PTR\_RESOLVED

BFD\_RELOC\_NDS32\_PLTBLOCK

BFD\_RELOC\_NDS32\_RELAX\_REGION\_BEGIN

BFD\_RELOC\_NDS32\_RELAX\_REGION\_END

BFD\_RELOC\_NDS32\_MINUEND

BFD\_RELOC\_NDS32\_SUBTRAHEND

BFD\_RELOC\_NDS32\_DIFF8

BFD\_RELOC\_NDS32\_DIFF16

BFD\_RELOC\_NDS32\_DIFF32

BFD\_RELOC\_NDS32\_DIFF\_ULEB128

BFD\_RELOC\_NDS32\_EMPTY

Relaxation relative relocation types.



BFD\_RELOC\_NDS32\_25\_ABS

This is a 25 bit absolute address.

BFD\_RELOC\_NDS32\_DATA

BFD\_RELOC\_NDS32\_TRAN

BFD\_RELOC\_NDS32\_17IFC\_PCREL

BFD\_RELOC\_NDS32\_10IFCU\_PCREL

For ex9 and ifc using.

BFD\_RELOC\_NDS32\_TPOFF

BFD\_RELOC\_NDS32\_GOTTPOFF

BFD\_RELOC\_NDS32\_TLS\_LE\_HI20

BFD\_RELOC\_NDS32\_TLS\_LE\_L012

BFD\_RELOC\_NDS32\_TLS\_LE\_20

BFD\_RELOC\_NDS32\_TLS\_LE\_15S0

BFD\_RELOC\_NDS32\_TLS\_LE\_15S1

BFD\_RELOC\_NDS32\_TLS\_LE\_15S2

BFD\_RELOC\_NDS32\_TLS\_LE\_ADD

BFD\_RELOC\_NDS32\_TLS\_LE\_LS

BFD\_RELOC\_NDS32\_TLS\_IE\_HI20

BFD\_RELOC\_NDS32\_TLS\_IE\_L012

BFD\_RELOC\_NDS32\_TLS\_IE\_L012S2

BFD\_RELOC\_NDS32\_TLS\_IEGP\_HI20

BFD\_RELOC\_NDS32\_TLS\_IEGP\_L012

BFD\_RELOC\_NDS32\_TLS\_IEGP\_L012S2

BFD\_RELOC\_NDS32\_TLS\_IEGP\_LW

BFD\_RELOC\_NDS32\_TLS\_DESC

BFD\_RELOC\_NDS32\_TLS\_DESC\_HI20

BFD\_RELOC\_NDS32\_TLS\_DESC\_L012

BFD\_RELOC\_NDS32\_TLS\_DESC\_20

BFD\_RELOC\_NDS32\_TLS\_DESC\_SDA17S2

BFD\_RELOC\_NDS32\_TLS\_DESC\_ADD

BFD\_RELOC\_NDS32\_TLS\_DESC\_FUNC

BFD\_RELOC\_NDS32\_TLS\_DESC\_CALL

BFD\_RELOC\_NDS32\_TLS\_DESC\_MEM

BFD\_RELOC\_NDS32\_REMOVE

BFD\_RELOC\_NDS32\_GROUP

For TLS.

BFD\_RELOC\_NDS32\_LSI

For floating load store relaxation.

BFD\_RELOC\_V850\_9\_PCREL

This is a 9-bit reloc.

BFD\_RELOC\_V850\_22\_PCREL

This is a 22-bit reloc.

BFD\_RELOC\_V850\_SDA\_16\_16\_OFFSET

This is a 16 bit offset from the short data area pointer.

**BFD\_RELOC\_V850\_SDA\_15\_16\_OFFSET**

This is a 16 bit offset (of which only 15 bits are used) from the short data area pointer.

**BFD\_RELOC\_V850\_ZDA\_16\_16\_OFFSET**

This is a 16 bit offset from the zero data area pointer.

**BFD\_RELOC\_V850\_ZDA\_15\_16\_OFFSET**

This is a 16 bit offset (of which only 15 bits are used) from the zero data area pointer.

**BFD\_RELOC\_V850\_TDA\_6\_8\_OFFSET**

This is an 8 bit offset (of which only 6 bits are used) from the tiny data area pointer.

**BFD\_RELOC\_V850\_TDA\_7\_8\_OFFSET**

This is an 8bit offset (of which only 7 bits are used) from the tiny data area pointer.

**BFD\_RELOC\_V850\_TDA\_7\_7\_OFFSET**

This is a 7 bit offset from the tiny data area pointer.

**BFD\_RELOC\_V850\_TDA\_16\_16\_OFFSET**

This is a 16 bit offset from the tiny data area pointer.

**BFD\_RELOC\_V850\_TDA\_4\_5\_OFFSET**

This is a 5 bit offset (of which only 4 bits are used) from the tiny data area pointer.

**BFD\_RELOC\_V850\_TDA\_4\_4\_OFFSET**

This is a 4 bit offset from the tiny data area pointer.

**BFD\_RELOC\_V850\_SDA\_16\_16\_SPLIT\_OFFSET**

This is a 16 bit offset from the short data area pointer, with the bits placed non-contiguously in the instruction.

**BFD\_RELOC\_V850\_ZDA\_16\_16\_SPLIT\_OFFSET**

This is a 16 bit offset from the zero data area pointer, with the bits placed non-contiguously in the instruction.

**BFD\_RELOC\_V850\_CALLT\_6\_7\_OFFSET**

This is a 6 bit offset from the call table base pointer.

**BFD\_RELOC\_V850\_CALLT\_16\_16\_OFFSET**

This is a 16 bit offset from the call table base pointer.

**BFD\_RELOC\_V850\_LONGCALL**

Used for relaxing indirect function calls.

**BFD\_RELOC\_V850\_LONGJUMP**

Used for relaxing indirect jumps.

**BFD\_RELOC\_V850\_ALIGN**

Used to maintain alignment whilst relaxing.

**BFD\_RELOC\_V850\_LO16\_SPLIT\_OFFSET**

This is a variation of BFD\_RELOC\_LO16 that can be used in v850e ld.bu instructions.

**BFD\_RELOC\_V850\_16\_PCREL**

This is a 16-bit reloc.

**BFD\_RELOC\_V850\_17\_PCREL**

This is a 17-bit reloc.

**BFD\_RELOC\_V850\_23**

This is a 23-bit reloc.

**BFD\_RELOC\_V850\_32\_PCREL**

This is a 32-bit reloc.

**BFD\_RELOC\_V850\_32\_ABS**

This is a 32-bit reloc.

**BFD\_RELOC\_V850\_16\_SPLIT\_OFFSET**

This is a 16-bit reloc.

**BFD\_RELOC\_V850\_16\_S1**

This is a 16-bit reloc.

**BFD\_RELOC\_V850\_L016\_S1**

Low 16 bits. 16 bit shifted by 1.

**BFD\_RELOC\_V850\_CALLT\_15\_16\_OFFSET**

This is a 16 bit offset from the call table base pointer.

**BFD\_RELOC\_V850\_32\_GOTPCREL**

**BFD\_RELOC\_V850\_16\_GOT**

**BFD\_RELOC\_V850\_32\_GOT**

**BFD\_RELOC\_V850\_22\_PLT\_PCREL**

**BFD\_RELOC\_V850\_32\_PLT\_PCREL**

**BFD\_RELOC\_V850\_16\_GOTOFF**

**BFD\_RELOC\_V850\_32\_GOTOFF**

DSO relocations.

**BFD\_RELOC\_V850\_CODE**

Start code.

**BFD\_RELOC\_V850\_DATA**

Start data in text.

**BFD\_RELOC\_TIC30\_LDP**

This is a 8bit DP reloc for the tms320c30, where the most significant 8 bits of a 24 bit word are placed into the least significant 8 bits of the opcode.

**BFD\_RELOC\_TIC54X\_PARTLS7**

This is a 7bit reloc for the tms320c54x, where the least significant 7 bits of a 16 bit word are placed into the least significant 7 bits of the opcode.

**BFD\_RELOC\_TIC54X\_PARTMS9**

This is a 9bit DP reloc for the tms320c54x, where the most significant 9 bits of a 16 bit word are placed into the least significant 9 bits of the opcode.

**BFD\_RELOC\_TIC54X\_23**

This is an extended address 23-bit reloc for the tms320c54x.

**BFD\_RELOC\_TIC54X\_16\_OF\_23**

This is a 16-bit reloc for the tms320c54x, where the least significant 16 bits of a 23-bit extended address are placed into the opcode.

**BFD\_RELOC\_TIC54X\_MS7\_OF\_23**

This is a reloc for the tms320c54x, where the most significant 7 bits of a 23-bit extended address are placed into the opcode.

**BFD\_RELOC\_C6000\_PCR\_S21****BFD\_RELOC\_C6000\_PCR\_S12****BFD\_RELOC\_C6000\_PCR\_S10****BFD\_RELOC\_C6000\_PCR\_S7****BFD\_RELOC\_C6000\_ABS\_S16****BFD\_RELOC\_C6000\_ABS\_L16****BFD\_RELOC\_C6000\_ABS\_H16****BFD\_RELOC\_C6000\_SBR\_U15\_B****BFD\_RELOC\_C6000\_SBR\_U15\_H****BFD\_RELOC\_C6000\_SBR\_U15\_W****BFD\_RELOC\_C6000\_SBR\_S16****BFD\_RELOC\_C6000\_SBR\_L16\_B****BFD\_RELOC\_C6000\_SBR\_L16\_H****BFD\_RELOC\_C6000\_SBR\_L16\_W****BFD\_RELOC\_C6000\_SBR\_H16\_B****BFD\_RELOC\_C6000\_SBR\_H16\_H****BFD\_RELOC\_C6000\_SBR\_H16\_W****BFD\_RELOC\_C6000\_SBR\_GOT\_U15\_W****BFD\_RELOC\_C6000\_SBR\_GOT\_L16\_W****BFD\_RELOC\_C6000\_SBR\_GOT\_H16\_W****BFD\_RELOC\_C6000\_DSBT\_INDEX****BFD\_RELOC\_C6000\_PREL31****BFD\_RELOC\_C6000\_EHTYPE****BFD\_RELOC\_C6000\_PCR\_H16****BFD\_RELOC\_C6000\_PCR\_L16****BFD\_RELOC\_C6000\_ALIGN****BFD\_RELOC\_C6000\_FPHEAD****BFD\_RELOC\_C6000\_NOCMP**

TMS320C6000 relocations.

**BFD\_RELOC\_FR30\_48**

This is a 48 bit reloc for the FR30 that stores 32 bits.

**BFD\_RELOC\_FR30\_20**

This is a 32 bit reloc for the FR30 that stores 20 bits split up into two sections.

**BFD\_RELOC\_FR30\_6\_IN\_4**

This is a 16 bit reloc for the FR30 that stores a 6 bit word offset in 4 bits.

BFD\_RELOC\_FR30\_8\_IN\_8

This is a 16 bit reloc for the FR30 that stores an 8 bit byte offset into 8 bits.

BFD\_RELOC\_FR30\_9\_IN\_8

This is a 16 bit reloc for the FR30 that stores a 9 bit short offset into 8 bits.

BFD\_RELOC\_FR30\_10\_IN\_8

This is a 16 bit reloc for the FR30 that stores a 10 bit word offset into 8 bits.

BFD\_RELOC\_FR30\_9\_PCREL

This is a 16 bit reloc for the FR30 that stores a 9 bit pc relative short offset into 8 bits.

BFD\_RELOC\_FR30\_12\_PCREL

This is a 16 bit reloc for the FR30 that stores a 12 bit pc relative short offset into 11 bits.

BFD\_RELOC\_MCORE\_PCREL\_IMM8BY4

BFD\_RELOC\_MCORE\_PCREL\_IMM11BY2

BFD\_RELOC\_MCORE\_PCREL\_IMM4BY2

BFD\_RELOC\_MCORE\_PCREL\_32

BFD\_RELOC\_MCORE\_PCREL\_JSR\_IMM11BY2

BFD\_RELOC\_MCORE\_RVA

Motorola Mcore relocations.

BFD\_RELOC\_MEP\_8

BFD\_RELOC\_MEP\_16

BFD\_RELOC\_MEP\_32

BFD\_RELOC\_MEP\_PCREL8A2

BFD\_RELOC\_MEP\_PCREL12A2

BFD\_RELOC\_MEP\_PCREL17A2

BFD\_RELOC\_MEP\_PCREL24A2

BFD\_RELOC\_MEP\_PCABS24A2

BFD\_RELOC\_MEP\_LOW16

BFD\_RELOC\_MEP\_HI16U

BFD\_RELOC\_MEP\_HI16S

BFD\_RELOC\_MEP\_GPREL

BFD\_RELOC\_MEP\_TPREL

BFD\_RELOC\_MEP\_TPREL7

BFD\_RELOC\_MEP\_TPREL7A2

BFD\_RELOC\_MEP\_TPREL7A4

BFD\_RELOC\_MEP\_UIMM24

BFD\_RELOC\_MEP\_ADDR24A4

BFD\_RELOC\_MEP\_GNU\_VTINHERIT

BFD\_RELOC\_MEP\_GNU\_VTENTRY

Toshiba Media Processor Relocations.

BFD\_RELOC\_METAG\_HIADDR16

BFD\_RELOC\_METAG\_LOADADDR16

```

BFD_RELOC_METAG_RELBRANCH
BFD_RELOC_METAG_GETSETOFF
BFD_RELOC_METAG_HIOG
BFD_RELOC_METAG_LOOG
BFD_RELOC_METAG_REL8
BFD_RELOC_METAG_REL16
BFD_RELOC_METAG_HI16_GOTOFF
BFD_RELOC_METAG_LO16_GOTOFF
BFD_RELOC_METAG_GETSET_GOTOFF
BFD_RELOC_METAG_GETSET_GOT
BFD_RELOC_METAG_HI16_GOTPC
BFD_RELOC_METAG_LO16_GOTPC
BFD_RELOC_METAG_HI16_PLT
BFD_RELOC_METAG_LO16_PLT
BFD_RELOC_METAG_RELBRANCH_PLT
BFD_RELOC_METAG_GOTOFF
BFD_RELOC_METAG_PLT
BFD_RELOC_METAG_TLS_GD
BFD_RELOC_METAG_TLS_LDM
BFD_RELOC_METAG_TLS_LDO_HI16
BFD_RELOC_METAG_TLS_LDO_LO16
BFD_RELOC_METAG_TLS_LDO
BFD_RELOC_METAG_TLS_IE
BFD_RELOC_METAG_TLS_IENONPIC
BFD_RELOC_METAG_TLS_IENONPIC_HI16
BFD_RELOC_METAG_TLS_IENONPIC_LO16
BFD_RELOC_METAG_TLS_TPOFF
BFD_RELOC_METAG_TLS_DTPMOD
BFD_RELOC_METAG_TLS_DTPOFF
BFD_RELOC_METAG_TLS_LE
BFD_RELOC_METAG_TLS_LE_HI16
BFD_RELOC_METAG_TLS_LE_LO16

```

Imagination Technologies Meta relocations.

```

BFD_RELOC_MMIX_GETA
BFD_RELOC_MMIX_GETA_1
BFD_RELOC_MMIX_GETA_2
BFD_RELOC_MMIX_GETA_3

```

These are relocations for the GETA instruction.

```

BFD_RELOC_MMIX_CBRANCH
BFD_RELOC_MMIX_CBRANCH_J
BFD_RELOC_MMIX_CBRANCH_1
BFD_RELOC_MMIX_CBRANCH_2
BFD_RELOC_MMIX_CBRANCH_3

```

These are relocations for a conditional branch instruction.

BFD\_RELOC\_MMIX\_PUSHJ  
BFD\_RELOC\_MMIX\_PUSHJ\_1  
BFD\_RELOC\_MMIX\_PUSHJ\_2  
BFD\_RELOC\_MMIX\_PUSHJ\_3  
BFD\_RELOC\_MMIX\_PUSHJ\_STUBBABLE

These are relocations for the PUSHJ instruction.

BFD\_RELOC\_MMIX\_JMP  
BFD\_RELOC\_MMIX\_JMP\_1  
BFD\_RELOC\_MMIX\_JMP\_2  
BFD\_RELOC\_MMIX\_JMP\_3

These are relocations for the JMP instruction.

BFD\_RELOC\_MMIX\_ADDR19

This is a relocation for a relative address as in a GETA instruction or a branch.

BFD\_RELOC\_MMIX\_ADDR27

This is a relocation for a relative address as in a JMP instruction.

BFD\_RELOC\_MMIX\_REG\_OR\_BYTE

This is a relocation for an instruction field that may be a general register or a value 0..255.

BFD\_RELOC\_MMIX\_REG

This is a relocation for an instruction field that may be a general register.

BFD\_RELOC\_MMIX\_BASE\_PLUS\_OFFSET

This is a relocation for two instruction fields holding a register and an offset, the equivalent of the relocation.

BFD\_RELOC\_MMIX\_LOCAL

This relocation is an assertion that the expression is not allocated as a global register. It does not modify contents.

BFD\_RELOC\_AVR\_7\_PCREL

This is a 16 bit reloc for the AVR that stores 8 bit pc relative short offset into 7 bits.

BFD\_RELOC\_AVR\_13\_PCREL

This is a 16 bit reloc for the AVR that stores 13 bit pc relative short offset into 12 bits.

BFD\_RELOC\_AVR\_16\_PM

This is a 16 bit reloc for the AVR that stores 17 bit value (usually program memory address) into 16 bits.

BFD\_RELOC\_AVR\_LO8\_LDI

This is a 16 bit reloc for the AVR that stores 8 bit value (usually data memory address) into 8 bit immediate value of LDI insn.

BFD\_RELOC\_AVR\_HI8\_LDI

This is a 16 bit reloc for the AVR that stores 8 bit value (high 8 bit of data memory address) into 8 bit immediate value of LDI insn.

**BFD\_RELOC\_AVR\_HH8\_LDI**

This is a 16 bit reloc for the AVR that stores 8 bit value (most high 8 bit of program memory address) into 8 bit immediate value of LDI insn.

**BFD\_RELOC\_AVR\_MS8\_LDI**

This is a 16 bit reloc for the AVR that stores 8 bit value (most high 8 bit of 32 bit value) into 8 bit immediate value of LDI insn.

**BFD\_RELOC\_AVR\_LO8\_LDI\_NEG**

This is a 16 bit reloc for the AVR that stores negated 8 bit value (usually data memory address) into 8 bit immediate value of SUBI insn.

**BFD\_RELOC\_AVR\_HI8\_LDI\_NEG**

This is a 16 bit reloc for the AVR that stores negated 8 bit value (high 8 bit of data memory address) into 8 bit immediate value of SUBI insn.

**BFD\_RELOC\_AVR\_HH8\_LDI\_NEG**

This is a 16 bit reloc for the AVR that stores negated 8 bit value (most high 8 bit of program memory address) into 8 bit immediate value of LDI or SUBI insn.

**BFD\_RELOC\_AVR\_MS8\_LDI\_NEG**

This is a 16 bit reloc for the AVR that stores negated 8 bit value (msb of 32 bit value) into 8 bit immediate value of LDI insn.

**BFD\_RELOC\_AVR\_LO8\_LDI\_PM**

This is a 16 bit reloc for the AVR that stores 8 bit value (usually command address) into 8 bit immediate value of LDI insn.

**BFD\_RELOC\_AVR\_LO8\_LDI\_GS**

This is a 16 bit reloc for the AVR that stores 8 bit value (command address) into 8 bit immediate value of LDI insn. If the address is beyond the 128k boundary, the linker inserts a jump stub for this reloc in the lower 128k.

**BFD\_RELOC\_AVR\_HI8\_LDI\_PM**

This is a 16 bit reloc for the AVR that stores 8 bit value (high 8 bit of command address) into 8 bit immediate value of LDI insn.

**BFD\_RELOC\_AVR\_HI8\_LDI\_GS**

This is a 16 bit reloc for the AVR that stores 8 bit value (high 8 bit of command address) into 8 bit immediate value of LDI insn. If the address is beyond the 128k boundary, the linker inserts a jump stub for this reloc below 128k.

**BFD\_RELOC\_AVR\_HH8\_LDI\_PM**

This is a 16 bit reloc for the AVR that stores 8 bit value (most high 8 bit of command address) into 8 bit immediate value of LDI insn.

**BFD\_RELOC\_AVR\_LO8\_LDI\_PM\_NEG**

This is a 16 bit reloc for the AVR that stores negated 8 bit value (usually command address) into 8 bit immediate value of SUBI insn.



**BFD\_RELOC\_AVR\_HI8\_LDI\_PM\_NEG**

This is a 16 bit reloc for the AVR that stores negated 8 bit value (high 8 bit of 16 bit command address) into 8 bit immediate value of SUBI insn.

**BFD\_RELOC\_AVR\_HH8\_LDI\_PM\_NEG**

This is a 16 bit reloc for the AVR that stores negated 8 bit value (high 6 bit of 22 bit command address) into 8 bit immediate value of SUBI insn.

**BFD\_RELOC\_AVR\_CALL**

This is a 32 bit reloc for the AVR that stores 23 bit value into 22 bits.

**BFD\_RELOC\_AVR\_LDI**

This is a 16 bit reloc for the AVR that stores all needed bits for absolute addressing with ldi with overflow check to linktime.

**BFD\_RELOC\_AVR\_6**

This is a 6 bit reloc for the AVR that stores offset for ldd/std instructions.

**BFD\_RELOC\_AVR\_6\_ADIW**

This is a 6 bit reloc for the AVR that stores offset for adiw/sbiw instructions.

**BFD\_RELOC\_AVR\_8\_LO**

This is a 8 bit reloc for the AVR that stores bits 0..7 of a symbol in .byte lo8(symbol).

**BFD\_RELOC\_AVR\_8\_HI**

This is a 8 bit reloc for the AVR that stores bits 8..15 of a symbol in .byte hi8(symbol).

**BFD\_RELOC\_AVR\_8\_HLO**

This is a 8 bit reloc for the AVR that stores bits 16..23 of a symbol in .byte hlo8(symbol).

**BFD\_RELOC\_AVR\_DIFF8****BFD\_RELOC\_AVR\_DIFF16****BFD\_RELOC\_AVR\_DIFF32**

AVR relocations to mark the difference of two local symbols. These are only needed to support linker relaxation and can be ignored when not relaxing. The field is set to the value of the difference assuming no relaxation. The relocation encodes the position of the second symbol so the linker can determine whether to adjust the field value.

**BFD\_RELOC\_AVR\_LDS\_STS\_16**

This is a 7 bit reloc for the AVR that stores SRAM address for 16bit lds and sts instructions supported only tiny core.

**BFD\_RELOC\_AVR\_PORT6**

This is a 6 bit reloc for the AVR that stores an I/O register number for the IN and OUT instructions.

**BFD\_RELOC\_AVR\_PORT5**

This is a 5 bit reloc for the AVR that stores an I/O register number for the SBIC, SBIS, SBI and CBI instructions.

BFD\_RELOC\_RISCV\_HI20  
BFD\_RELOC\_RISCV\_PCREL\_HI20  
BFD\_RELOC\_RISCV\_PCREL\_L012\_I  
BFD\_RELOC\_RISCV\_PCREL\_L012\_S  
BFD\_RELOC\_RISCV\_L012\_I  
BFD\_RELOC\_RISCV\_L012\_S  
BFD\_RELOC\_RISCV\_GPREL12\_I  
BFD\_RELOC\_RISCV\_GPREL12\_S  
BFD\_RELOC\_RISCV\_TPREL\_HI20  
BFD\_RELOC\_RISCV\_TPREL\_L012\_I  
BFD\_RELOC\_RISCV\_TPREL\_L012\_S  
BFD\_RELOC\_RISCV\_TPREL\_ADD  
BFD\_RELOC\_RISCV\_CALL  
BFD\_RELOC\_RISCV\_CALL\_PLT  
BFD\_RELOC\_RISCV\_ADD8  
BFD\_RELOC\_RISCV\_ADD16  
BFD\_RELOC\_RISCV\_ADD32  
BFD\_RELOC\_RISCV\_ADD64  
BFD\_RELOC\_RISCV\_SUB8  
BFD\_RELOC\_RISCV\_SUB16  
BFD\_RELOC\_RISCV\_SUB32  
BFD\_RELOC\_RISCV\_SUB64  
BFD\_RELOC\_RISCV\_GOT\_HI20  
BFD\_RELOC\_RISCV\_TLS\_GOT\_HI20  
BFD\_RELOC\_RISCV\_TLS\_GD\_HI20  
BFD\_RELOC\_RISCV\_JMP  
BFD\_RELOC\_RISCV\_TLS\_DTPMOD32  
BFD\_RELOC\_RISCV\_TLS\_DTPREL32  
BFD\_RELOC\_RISCV\_TLS\_DTPMOD64  
BFD\_RELOC\_RISCV\_TLS\_DTPREL64  
BFD\_RELOC\_RISCV\_TLS\_TPREL32  
BFD\_RELOC\_RISCV\_TLS\_TPREL64  
BFD\_RELOC\_RISCV\_TLSDESC\_HI20  
BFD\_RELOC\_RISCV\_TLSDESC\_LOAD\_L012  
BFD\_RELOC\_RISCV\_TLSDESC\_ADD\_L012  
BFD\_RELOC\_RISCV\_TLSDESC\_CALL  
BFD\_RELOC\_RISCV\_ALIGN  
BFD\_RELOC\_RISCV\_RVC\_BRANCH  
BFD\_RELOC\_RISCV\_RVC\_JUMP  
BFD\_RELOC\_RISCV\_RELAX  
BFD\_RELOC\_RISCV\_CFA  
BFD\_RELOC\_RISCV\_SUB6  
BFD\_RELOC\_RISCV\_SET6  
BFD\_RELOC\_RISCV\_SET8  
BFD\_RELOC\_RISCV\_SET16  
BFD\_RELOC\_RISCV\_SET32  
BFD\_RELOC\_RISCV\_32\_PCREL

BFD\_RELOC\_RISCV\_SET\_ULEB128  
BFD\_RELOC\_RISCV\_SUB\_ULEB128

RISC-V relocations.

BFD\_RELOC\_RL78\_NEG8  
BFD\_RELOC\_RL78\_NEG16  
BFD\_RELOC\_RL78\_NEG24  
BFD\_RELOC\_RL78\_NEG32  
BFD\_RELOC\_RL78\_16\_OP  
BFD\_RELOC\_RL78\_24\_OP  
BFD\_RELOC\_RL78\_32\_OP  
BFD\_RELOC\_RL78\_8U  
BFD\_RELOC\_RL78\_16U  
BFD\_RELOC\_RL78\_24U  
BFD\_RELOC\_RL78\_DIR3U\_PCREL  
BFD\_RELOC\_RL78\_DIFF  
BFD\_RELOC\_RL78\_GPRELB  
BFD\_RELOC\_RL78\_GPRELW  
BFD\_RELOC\_RL78\_GPRELL  
BFD\_RELOC\_RL78\_SYM  
BFD\_RELOC\_RL78\_OP\_SUBTRACT  
BFD\_RELOC\_RL78\_OP\_NEG  
BFD\_RELOC\_RL78\_OP\_AND  
BFD\_RELOC\_RL78\_OP\_SHRA  
BFD\_RELOC\_RL78\_ABS8  
BFD\_RELOC\_RL78\_ABS16  
BFD\_RELOC\_RL78\_ABS16\_REV  
BFD\_RELOC\_RL78\_ABS32  
BFD\_RELOC\_RL78\_ABS32\_REV  
BFD\_RELOC\_RL78\_ABS16U  
BFD\_RELOC\_RL78\_ABS16UW  
BFD\_RELOC\_RL78\_ABS16UL  
BFD\_RELOC\_RL78\_RELAX  
BFD\_RELOC\_RL78\_HI16  
BFD\_RELOC\_RL78\_HI8  
BFD\_RELOC\_RL78\_LO16  
BFD\_RELOC\_RL78\_CODE  
BFD\_RELOC\_RL78\_SADDR

Renesas RL78 Relocations.

BFD\_RELOC\_RX\_NEG8  
BFD\_RELOC\_RX\_NEG16  
BFD\_RELOC\_RX\_NEG24  
BFD\_RELOC\_RX\_NEG32  
BFD\_RELOC\_RX\_16\_OP  
BFD\_RELOC\_RX\_24\_OP  
BFD\_RELOC\_RX\_32\_OP  
BFD\_RELOC\_RX\_8U

BFD\_RELOC\_RX\_16U  
BFD\_RELOC\_RX\_24U  
BFD\_RELOC\_RX\_DIR3U\_PCREL  
BFD\_RELOC\_RX\_DIFF  
BFD\_RELOC\_RX\_GPRELB  
BFD\_RELOC\_RX\_GPRELW  
BFD\_RELOC\_RX\_GPRELL  
BFD\_RELOC\_RX\_SYM  
BFD\_RELOC\_RX\_OP\_SUBTRACT  
BFD\_RELOC\_RX\_OP\_NEG  
BFD\_RELOC\_RX\_ABS8  
BFD\_RELOC\_RX\_ABS16  
BFD\_RELOC\_RX\_ABS16\_REV  
BFD\_RELOC\_RX\_ABS32  
BFD\_RELOC\_RX\_ABS32\_REV  
BFD\_RELOC\_RX\_ABS16U  
BFD\_RELOC\_RX\_ABS16UW  
BFD\_RELOC\_RX\_ABS16UL  
BFD\_RELOC\_RX\_RELAX

Renesas RX Relocations.

BFD\_RELOC\_390\_12

Direct 12 bit.

BFD\_RELOC\_390\_GOT12

12 bit GOT offset.

BFD\_RELOC\_390\_GOTPC

32 bit PC relative offset to GOT.

BFD\_RELOC\_390\_GOT16

16 bit GOT offset.

BFD\_RELOC\_390\_PC12DBL

PC relative 12 bit shifted by 1.

BFD\_RELOC\_390\_PLT12DBL

12 bit PC rel. PLT shifted by 1.

BFD\_RELOC\_390\_PC16DBL

PC relative 16 bit shifted by 1.

BFD\_RELOC\_390\_PLT16DBL

16 bit PC rel. PLT shifted by 1.

BFD\_RELOC\_390\_PC24DBL

PC relative 24 bit shifted by 1.

BFD\_RELOC\_390\_PLT24DBL

24 bit PC rel. PLT shifted by 1.

BFD\_RELOC\_390\_PC32DBL  
PC relative 32 bit shifted by 1.

BFD\_RELOC\_390\_PLT32DBL  
32 bit PC rel. PLT shifted by 1.

BFD\_RELOC\_390\_GOTPCDBL  
32 bit PC rel. GOT shifted by 1.

BFD\_RELOC\_390\_GOT64  
64 bit GOT offset.

BFD\_RELOC\_390\_GOTENT  
32 bit rel. offset to GOT entry.

BFD\_RELOC\_390\_GOTOFF64  
64 bit offset to GOT.

BFD\_RELOC\_390\_GOTPLT12  
12-bit offset to symbol-entry within GOT, with PLT handling.

BFD\_RELOC\_390\_GOTPLT16  
16-bit offset to symbol-entry within GOT, with PLT handling.

BFD\_RELOC\_390\_GOTPLT32  
32-bit offset to symbol-entry within GOT, with PLT handling.

BFD\_RELOC\_390\_GOTPLT64  
64-bit offset to symbol-entry within GOT, with PLT handling.

BFD\_RELOC\_390\_GOTPLTENT  
32-bit rel. offset to symbol-entry within GOT, with PLT handling.

BFD\_RELOC\_390\_PLTOFF16  
16-bit rel. offset from the GOT to a PLT entry.

BFD\_RELOC\_390\_PLTOFF32  
32-bit rel. offset from the GOT to a PLT entry.

BFD\_RELOC\_390\_PLTOFF64  
64-bit rel. offset from the GOT to a PLT entry.

BFD\_RELOC\_390\_TLS\_LOAD  
BFD\_RELOC\_390\_TLS\_GDCALL  
BFD\_RELOC\_390\_TLS\_LDCALL  
BFD\_RELOC\_390\_TLS\_GD32  
BFD\_RELOC\_390\_TLS\_GD64  
BFD\_RELOC\_390\_TLS\_GOTIE12  
BFD\_RELOC\_390\_TLS\_GOTIE32  
BFD\_RELOC\_390\_TLS\_GOTIE64  
BFD\_RELOC\_390\_TLS\_LDM32  
BFD\_RELOC\_390\_TLS\_LDM64

BFD\_RELOC\_390\_TLS\_IE32  
BFD\_RELOC\_390\_TLS\_IE64  
BFD\_RELOC\_390\_TLS\_IEENT  
BFD\_RELOC\_390\_TLS\_LE32  
BFD\_RELOC\_390\_TLS\_LE64  
BFD\_RELOC\_390\_TLS\_LD032  
BFD\_RELOC\_390\_TLS\_LD064  
BFD\_RELOC\_390\_TLS\_DTPMOD  
BFD\_RELOC\_390\_TLS\_DTPOFF  
BFD\_RELOC\_390\_TLS\_TPOFF

s390 tls relocations.

BFD\_RELOC\_390\_20  
BFD\_RELOC\_390\_GOT20  
BFD\_RELOC\_390\_GOTPLT20  
BFD\_RELOC\_390\_TLS\_GOTIE20

Long displacement extension.

BFD\_RELOC\_SCORE\_GPREL15

Score relocations. Low 16 bit for load/store.

BFD\_RELOC\_SCORE\_DUMMY2

BFD\_RELOC\_SCORE\_JMP

This is a 24-bit reloc with the right 1 bit assumed to be 0.

BFD\_RELOC\_SCORE\_BRANCH

This is a 19-bit reloc with the right 1 bit assumed to be 0.

BFD\_RELOC\_SCORE\_IMM30

This is a 32-bit reloc for 48-bit instructions.

BFD\_RELOC\_SCORE\_IMM32

This is a 32-bit reloc for 48-bit instructions.

BFD\_RELOC\_SCORE16\_JMP

This is a 11-bit reloc with the right 1 bit assumed to be 0.

BFD\_RELOC\_SCORE16\_BRANCH

This is a 8-bit reloc with the right 1 bit assumed to be 0.

BFD\_RELOC\_SCORE\_BCMP

This is a 9-bit reloc with the right 1 bit assumed to be 0.

BFD\_RELOC\_SCORE\_GOT15

BFD\_RELOC\_SCORE\_GOT\_L016

BFD\_RELOC\_SCORE\_CALL15

BFD\_RELOC\_SCORE\_DUMMY\_HI16

Undocumented Score relocs.

BFD\_RELOC\_IP2K\_FR9

Scenix IP2K - 9-bit register number / data address.

**BFD\_RELOC\_IP2K\_BANK**

Scenix IP2K - 4-bit register/data bank number.

**BFD\_RELOC\_IP2K\_ADDR16CJP**

Scenix IP2K - low 13 bits of instruction word address.

**BFD\_RELOC\_IP2K\_PAGE3**

Scenix IP2K - high 3 bits of instruction word address.

**BFD\_RELOC\_IP2K\_LO8DATA**

**BFD\_RELOC\_IP2K\_HI8DATA**

**BFD\_RELOC\_IP2K\_EX8DATA**

Scenix IP2K - ext/low/high 8 bits of data address.

**BFD\_RELOC\_IP2K\_LO8INSN**

**BFD\_RELOC\_IP2K\_HI8INSN**

Scenix IP2K - low/high 8 bits of instruction word address.

**BFD\_RELOC\_IP2K\_PC\_SKIP**

Scenix IP2K - even/odd PC modifier to modify snb pcl.0.

**BFD\_RELOC\_IP2K\_TEXT**

Scenix IP2K - 16 bit word address in text section.

**BFD\_RELOC\_IP2K\_FR\_OFFSET**

Scenix IP2K - 7-bit sp or dp offset.

**BFD\_RELOC\_VTABLE\_INHERIT**

**BFD\_RELOC\_VTABLE\_ENTRY**

These two relocations are used by the linker to determine which of the entries in a C++ virtual function table are actually used. When the `-gc-sections` option is given, the linker will zero out the entries that are not used, so that the code for those functions need not be included in the output.

**VTABLE\_INHERIT** is a zero-space relocation used to describe to the linker the inheritance tree of a C++ virtual function table. The relocation's symbol should be the parent class' vtable, and the relocation should be located at the child vtable.

**VTABLE\_ENTRY** is a zero-space relocation that describes the use of a virtual function table entry. The reloc's symbol should refer to the table of the class mentioned in the code. Off of that base, an offset describes the entry that is being used. For Rela hosts, this offset is stored in the reloc's addend. For Rel hosts, we are forced to put this offset in the reloc's section offset.

**BFD\_RELOC\_IA64\_IMM14**

**BFD\_RELOC\_IA64\_IMM22**

**BFD\_RELOC\_IA64\_IMM64**

**BFD\_RELOC\_IA64\_DIR32MSB**

**BFD\_RELOC\_IA64\_DIR32LSB**

**BFD\_RELOC\_IA64\_DIR64MSB**

**BFD\_RELOC\_IA64\_DIR64LSB**

**BFD\_RELOC\_IA64\_GPREL22**

BFD\_RELOC\_IA64\_GPREL64I  
BFD\_RELOC\_IA64\_GPREL32MSB  
BFD\_RELOC\_IA64\_GPREL32LSB  
BFD\_RELOC\_IA64\_GPREL64MSB  
BFD\_RELOC\_IA64\_GPREL64LSB  
BFD\_RELOC\_IA64\_LTOFF22  
BFD\_RELOC\_IA64\_LTOFF64I  
BFD\_RELOC\_IA64\_PLTOFF22  
BFD\_RELOC\_IA64\_PLTOFF64I  
BFD\_RELOC\_IA64\_PLTOFF64MSB  
BFD\_RELOC\_IA64\_PLTOFF64LSB  
BFD\_RELOC\_IA64\_FPTR64I  
BFD\_RELOC\_IA64\_FPTR32MSB  
BFD\_RELOC\_IA64\_FPTR32LSB  
BFD\_RELOC\_IA64\_FPTR64MSB  
BFD\_RELOC\_IA64\_FPTR64LSB  
BFD\_RELOC\_IA64\_PCREL21B  
BFD\_RELOC\_IA64\_PCREL21BI  
BFD\_RELOC\_IA64\_PCREL21M  
BFD\_RELOC\_IA64\_PCREL21F  
BFD\_RELOC\_IA64\_PCREL22  
BFD\_RELOC\_IA64\_PCREL60B  
BFD\_RELOC\_IA64\_PCREL64I  
BFD\_RELOC\_IA64\_PCREL32MSB  
BFD\_RELOC\_IA64\_PCREL32LSB  
BFD\_RELOC\_IA64\_PCREL64MSB  
BFD\_RELOC\_IA64\_PCREL64LSB  
BFD\_RELOC\_IA64\_LTOFF\_FPTR22  
BFD\_RELOC\_IA64\_LTOFF\_FPTR64I  
BFD\_RELOC\_IA64\_LTOFF\_FPTR32MSB  
BFD\_RELOC\_IA64\_LTOFF\_FPTR32LSB  
BFD\_RELOC\_IA64\_LTOFF\_FPTR64MSB  
BFD\_RELOC\_IA64\_LTOFF\_FPTR64LSB  
BFD\_RELOC\_IA64\_SEGREL32MSB  
BFD\_RELOC\_IA64\_SEGREL32LSB  
BFD\_RELOC\_IA64\_SEGREL64MSB  
BFD\_RELOC\_IA64\_SEGREL64LSB  
BFD\_RELOC\_IA64\_SECREL32MSB  
BFD\_RELOC\_IA64\_SECREL32LSB  
BFD\_RELOC\_IA64\_SECREL64MSB  
BFD\_RELOC\_IA64\_SECREL64LSB  
BFD\_RELOC\_IA64\_REL32MSB  
BFD\_RELOC\_IA64\_REL32LSB  
BFD\_RELOC\_IA64\_REL64MSB  
BFD\_RELOC\_IA64\_REL64LSB  
BFD\_RELOC\_IA64\_LTV32MSB  
BFD\_RELOC\_IA64\_LTV32LSB



BFD\_RELOC\_IA64\_LTV64MSB  
 BFD\_RELOC\_IA64\_LTV64LSB  
 BFD\_RELOC\_IA64\_IPLTMSB  
 BFD\_RELOC\_IA64\_IPLTLSB  
 BFD\_RELOC\_IA64\_LTOFF22X  
 BFD\_RELOC\_IA64\_LDXMOV  
 BFD\_RELOC\_IA64\_TPREL14  
 BFD\_RELOC\_IA64\_TPREL22  
 BFD\_RELOC\_IA64\_TPREL64I  
 BFD\_RELOC\_IA64\_TPREL64MSB  
 BFD\_RELOC\_IA64\_TPREL64LSB  
 BFD\_RELOC\_IA64\_LTOFF\_TPREL22  
 BFD\_RELOC\_IA64\_DTPMOD64MSB  
 BFD\_RELOC\_IA64\_DTPMOD64LSB  
 BFD\_RELOC\_IA64\_LTOFF\_DTPMOD22  
 BFD\_RELOC\_IA64\_DTPREL14  
 BFD\_RELOC\_IA64\_DTPREL22  
 BFD\_RELOC\_IA64\_DTPREL64I  
 BFD\_RELOC\_IA64\_DTPREL32MSB  
 BFD\_RELOC\_IA64\_DTPREL32LSB  
 BFD\_RELOC\_IA64\_DTPREL64MSB  
 BFD\_RELOC\_IA64\_DTPREL64LSB  
 BFD\_RELOC\_IA64\_LTOFF\_DTPREL22

Intel IA64 Relocations.

**BFD\_RELOC\_M68HC11\_HI8**

Motorola 68HC11 reloc. This is the 8 bit high part of an absolute address.

**BFD\_RELOC\_M68HC11\_LO8**

Motorola 68HC11 reloc. This is the 8 bit low part of an absolute address.

**BFD\_RELOC\_M68HC11\_3B**

Motorola 68HC11 reloc. This is the 3 bit of a value.

**BFD\_RELOC\_M68HC11\_RL\_JUMP**

Motorola 68HC11 reloc. This reloc marks the beginning of a jump/call instruction. It is used for linker relaxation to correctly identify beginning of instruction and change some branches to use PC-relative addressing mode.

**BFD\_RELOC\_M68HC11\_RL\_GROUP**

Motorola 68HC11 reloc. This reloc marks a group of several instructions that gcc generates and for which the linker relaxation pass can modify and/or remove some of them.

**BFD\_RELOC\_M68HC11\_LO16**

Motorola 68HC11 reloc. This is the 16-bit lower part of an address. It is used for 'call' instruction to specify the symbol address without any special transformation (due to memory bank window).

**BFD\_RELOC\_M68HC11\_PAGE**

Motorola 68HC11 reloc. This is a 8-bit reloc that specifies the page number of an address. It is used by 'call' instruction to specify the page number of the symbol.

**BFD\_RELOC\_M68HC11\_24**

Motorola 68HC11 reloc. This is a 24-bit reloc that represents the address with a 16-bit value and a 8-bit page number. The symbol address is transformed to follow the 16K memory bank of 68HC12 (seen as mapped in the window).

**BFD\_RELOC\_M68HC12\_5B**

Motorola 68HC12 reloc. This is the 5 bits of a value.

**BFD\_RELOC\_XGATE\_RL\_JUMP**

Freescale XGATE reloc. This reloc marks the beginning of a bra/jal instruction.

**BFD\_RELOC\_XGATE\_RL\_GROUP**

Freescale XGATE reloc. This reloc marks a group of several instructions that gcc generates and for which the linker relaxation pass can modify and/or remove some of them.

**BFD\_RELOC\_XGATE\_L016**

Freescale XGATE reloc. This is the 16-bit lower part of an address. It is used for the '16-bit' instructions.

**BFD\_RELOC\_XGATE\_GPAGE**

Freescale XGATE reloc.

**BFD\_RELOC\_XGATE\_24**

Freescale XGATE reloc.

**BFD\_RELOC\_XGATE\_PCREL\_9**

Freescale XGATE reloc. This is a 9-bit pc-relative reloc.

**BFD\_RELOC\_XGATE\_PCREL\_10**

Freescale XGATE reloc. This is a 10-bit pc-relative reloc.

**BFD\_RELOC\_XGATE\_IMM8\_LO**

Freescale XGATE reloc. This is the 16-bit lower part of an address. It is used for the '16-bit' instructions.

**BFD\_RELOC\_XGATE\_IMM8\_HI**

Freescale XGATE reloc. This is the 16-bit higher part of an address. It is used for the '16-bit' instructions.

**BFD\_RELOC\_XGATE\_IMM3**

Freescale XGATE reloc. This is a 3-bit pc-relative reloc.

**BFD\_RELOC\_XGATE\_IMM4**

Freescale XGATE reloc. This is a 4-bit pc-relative reloc.

**BFD\_RELOC\_XGATE\_IMM5**

Freescale XGATE reloc. This is a 5-bit pc-relative reloc.

BFD\_RELOC\_M68HC12\_9B

Motorola 68HC12 reloc. This is the 9 bits of a value.

BFD\_RELOC\_M68HC12\_16B

Motorola 68HC12 reloc. This is the 16 bits of a value.

BFD\_RELOC\_M68HC12\_9\_PCREL

Motorola 68HC12/XGATE reloc. This is a PCREL9 branch.

BFD\_RELOC\_M68HC12\_10\_PCREL

Motorola 68HC12/XGATE reloc. This is a PCREL10 branch.

BFD\_RELOC\_M68HC12\_LO8XG

Motorola 68HC12/XGATE reloc. This is the 8 bit low part of an absolute address and immediately precedes a matching HI8XG part.

BFD\_RELOC\_M68HC12\_HI8XG

Motorola 68HC12/XGATE reloc. This is the 8 bit high part of an absolute address and immediately follows a matching LO8XG part.

BFD\_RELOC\_CR16\_NUM8

BFD\_RELOC\_CR16\_NUM16

BFD\_RELOC\_CR16\_NUM32

BFD\_RELOC\_CR16\_NUM32a

BFD\_RELOC\_CR16\_REGREL0

BFD\_RELOC\_CR16\_REGREL4

BFD\_RELOC\_CR16\_REGREL4a

BFD\_RELOC\_CR16\_REGREL14

BFD\_RELOC\_CR16\_REGREL14a

BFD\_RELOC\_CR16\_REGREL16

BFD\_RELOC\_CR16\_REGREL20

BFD\_RELOC\_CR16\_REGREL20a

BFD\_RELOC\_CR16\_ABS20

BFD\_RELOC\_CR16\_ABS24

BFD\_RELOC\_CR16\_IMM4

BFD\_RELOC\_CR16\_IMM8

BFD\_RELOC\_CR16\_IMM16

BFD\_RELOC\_CR16\_IMM20

BFD\_RELOC\_CR16\_IMM24

BFD\_RELOC\_CR16\_IMM32

BFD\_RELOC\_CR16\_IMM32a

BFD\_RELOC\_CR16\_DISP4

BFD\_RELOC\_CR16\_DISP8

BFD\_RELOC\_CR16\_DISP16

BFD\_RELOC\_CR16\_DISP20

BFD\_RELOC\_CR16\_DISP24

BFD\_RELOC\_CR16\_DISP24a

BFD\_RELOC\_CR16\_SWITCH8

BFD\_RELOC\_CR16\_SWITCH16

BFD\_RELOC\_CR16\_SWITCH32  
BFD\_RELOC\_CR16\_GOT\_REGREL20  
BFD\_RELOC\_CR16\_GOTC\_REGREL20  
NS CR16 Relocations.

BFD\_RELOC\_CRX\_REL4  
BFD\_RELOC\_CRX\_REL8  
BFD\_RELOC\_CRX\_REL8\_CMP  
BFD\_RELOC\_CRX\_REL16  
BFD\_RELOC\_CRX\_REL24  
BFD\_RELOC\_CRX\_REL32  
BFD\_RELOC\_CRX\_REGREL12  
BFD\_RELOC\_CRX\_REGREL22  
BFD\_RELOC\_CRX\_REGREL28  
BFD\_RELOC\_CRX\_REGREL32  
BFD\_RELOC\_CRX\_ABS16  
BFD\_RELOC\_CRX\_ABS32  
BFD\_RELOC\_CRX\_NUM8  
BFD\_RELOC\_CRX\_NUM16  
BFD\_RELOC\_CRX\_NUM32  
BFD\_RELOC\_CRX\_IMM16  
BFD\_RELOC\_CRX\_IMM32  
BFD\_RELOC\_CRX\_SWITCH8  
BFD\_RELOC\_CRX\_SWITCH16  
BFD\_RELOC\_CRX\_SWITCH32  
NS CRX Relocations.

BFD\_RELOC\_CRIS\_BDISP8  
BFD\_RELOC\_CRIS\_UNSIGNED\_5  
BFD\_RELOC\_CRIS\_SIGNED\_6  
BFD\_RELOC\_CRIS\_UNSIGNED\_6  
BFD\_RELOC\_CRIS\_SIGNED\_8  
BFD\_RELOC\_CRIS\_UNSIGNED\_8  
BFD\_RELOC\_CRIS\_SIGNED\_16  
BFD\_RELOC\_CRIS\_UNSIGNED\_16  
BFD\_RELOC\_CRIS\_LAPCQ\_OFFSET  
BFD\_RELOC\_CRIS\_UNSIGNED\_4

These relocs are only used within the CRIS assembler. They are not (at present) written to any object files.

BFD\_RELOC\_CRIS\_32\_GOT  
32-bit offset to symbol-entry within GOT.

BFD\_RELOC\_CRIS\_16\_GOT  
16-bit offset to symbol-entry within GOT.

BFD\_RELOC\_CRIS\_32\_GOTPLT  
32-bit offset to symbol-entry within GOT, with PLT handling.

BFD\_RELOC\_CRIS\_16\_GOTPLT

16-bit offset to symbol-entry within GOT, with PLT handling.

BFD\_RELOC\_CRIS\_32\_GOTREL

32-bit offset to symbol, relative to GOT.

BFD\_RELOC\_CRIS\_32\_PLT\_GOTREL

32-bit offset to symbol with PLT entry, relative to GOT.

BFD\_RELOC\_CRIS\_32\_PLT\_PCREL

32-bit offset to symbol with PLT entry, relative to this relocation.

BFD\_RELOC\_CRIS\_32\_GOT\_GD

BFD\_RELOC\_CRIS\_16\_GOT\_GD

BFD\_RELOC\_CRIS\_32\_GD

BFD\_RELOC\_CRIS\_DTP

BFD\_RELOC\_CRIS\_32\_DTPREL

BFD\_RELOC\_CRIS\_16\_DTPREL

BFD\_RELOC\_CRIS\_32\_GOT\_TPREL

BFD\_RELOC\_CRIS\_16\_GOT\_TPREL

BFD\_RELOC\_CRIS\_32\_TPREL

BFD\_RELOC\_CRIS\_16\_TPREL

BFD\_RELOC\_CRIS\_DTPMOD

BFD\_RELOC\_CRIS\_32\_IE

Relocs used in TLS code for CRIS.

BFD\_RELOC\_OR1K\_REL\_26

BFD\_RELOC\_OR1K\_SL016

BFD\_RELOC\_OR1K\_PCREL\_PG21

BFD\_RELOC\_OR1K\_L013

BFD\_RELOC\_OR1K\_SL013

BFD\_RELOC\_OR1K\_GOTPC\_HI16

BFD\_RELOC\_OR1K\_GOTPC\_L016

BFD\_RELOC\_OR1K\_GOT\_AHI16

BFD\_RELOC\_OR1K\_GOT16

BFD\_RELOC\_OR1K\_GOT\_PG21

BFD\_RELOC\_OR1K\_GOT\_L013

BFD\_RELOC\_OR1K\_PLT26

BFD\_RELOC\_OR1K\_PLTA26

BFD\_RELOC\_OR1K\_GOTOFF\_SL016

BFD\_RELOC\_OR1K\_TLS\_GD\_HI16

BFD\_RELOC\_OR1K\_TLS\_GD\_L016

BFD\_RELOC\_OR1K\_TLS\_GD\_PG21

BFD\_RELOC\_OR1K\_TLS\_GD\_L013

BFD\_RELOC\_OR1K\_TLS\_LDM\_HI16

BFD\_RELOC\_OR1K\_TLS\_LDM\_L016

BFD\_RELOC\_OR1K\_TLS\_LDM\_PG21

BFD\_RELOC\_OR1K\_TLS\_LDM\_L013

BFD\_RELOC\_OR1K\_TLS\_LDO\_HI16

BFD\_RELOC\_OR1K\_TLS\_LDO\_L016  
BFD\_RELOC\_OR1K\_TLS\_IE\_HI16  
BFD\_RELOC\_OR1K\_TLS\_IE\_AHI16  
BFD\_RELOC\_OR1K\_TLS\_IE\_L016  
BFD\_RELOC\_OR1K\_TLS\_IE\_PG21  
BFD\_RELOC\_OR1K\_TLS\_IE\_L013  
BFD\_RELOC\_OR1K\_TLS\_LE\_HI16  
BFD\_RELOC\_OR1K\_TLS\_LE\_AHI16  
BFD\_RELOC\_OR1K\_TLS\_LE\_L016  
BFD\_RELOC\_OR1K\_TLS\_LE\_SL016  
BFD\_RELOC\_OR1K\_TLS\_TPOFF  
BFD\_RELOC\_OR1K\_TLS\_DTPOFF  
BFD\_RELOC\_OR1K\_TLS\_DTPMOD

OpenRISC 1000 Relocations.

BFD\_RELOC\_H8\_DIR16A8  
BFD\_RELOC\_H8\_DIR16R8  
BFD\_RELOC\_H8\_DIR24A8  
BFD\_RELOC\_H8\_DIR24R8  
BFD\_RELOC\_H8\_DIR32A16  
BFD\_RELOC\_H8\_DISP32A16

H8 elf Relocations.

BFD\_RELOC\_XSTORMY16\_REL\_12  
BFD\_RELOC\_XSTORMY16\_12  
BFD\_RELOC\_XSTORMY16\_24  
BFD\_RELOC\_XSTORMY16\_FPTR16

Sony Xstormy16 Relocations.

BFD\_RELOC\_RELC

Self-describing complex relocations.

BFD\_RELOC\_MT\_PC16

Morpho MT - 16 bit immediate relocation.

BFD\_RELOC\_MT\_HI16

Morpho MT - Hi 16 bits of an address.

BFD\_RELOC\_MT\_LO16

Morpho MT - Low 16 bits of an address.

BFD\_RELOC\_MT\_GNU\_VTINHERIT

Morpho MT - Used to tell the linker which vtable entries are used.

BFD\_RELOC\_MT\_GNU\_VTENTRY

Morpho MT - Used to tell the linker which vtable entries are used.

BFD\_RELOC\_MT\_PCINSN8

Morpho MT - 8 bit immediate relocation.

BFD\_RELOC\_MSP430\_10\_PCREL  
 BFD\_RELOC\_MSP430\_16\_PCREL  
 BFD\_RELOC\_MSP430\_16  
 BFD\_RELOC\_MSP430\_16\_PCREL\_BYTE  
 BFD\_RELOC\_MSP430\_16\_BYTE  
 BFD\_RELOC\_MSP430\_2X\_PCREL  
 BFD\_RELOC\_MSP430\_RL\_PCREL  
 BFD\_RELOC\_MSP430\_ABS8  
 BFD\_RELOC\_MSP430X\_PCR20\_EXT\_SRC  
 BFD\_RELOC\_MSP430X\_PCR20\_EXT\_DST  
 BFD\_RELOC\_MSP430X\_PCR20\_EXT\_ODST  
 BFD\_RELOC\_MSP430X\_ABS20\_EXT\_SRC  
 BFD\_RELOC\_MSP430X\_ABS20\_EXT\_DST  
 BFD\_RELOC\_MSP430X\_ABS20\_EXT\_ODST  
 BFD\_RELOC\_MSP430X\_ABS20\_ADR\_SRC  
 BFD\_RELOC\_MSP430X\_ABS20\_ADR\_DST  
 BFD\_RELOC\_MSP430X\_PCR16  
 BFD\_RELOC\_MSP430X\_PCR20\_CALL  
 BFD\_RELOC\_MSP430X\_ABS16  
 BFD\_RELOC\_MSP430\_ABS\_HI16  
 BFD\_RELOC\_MSP430\_PREL31  
 BFD\_RELOC\_MSP430\_SYM\_DIFF  
 BFD\_RELOC\_MSP430\_SET\_ULEB128  
 BFD\_RELOC\_MSP430\_SUB\_ULEB128

msp430 specific relocation codes.

BFD\_RELOC\_PRU\_U16

PRU LDI 16-bit unsigned data-memory relocation.

BFD\_RELOC\_PRU\_U16\_PMEMIMM

PRU LDI 16-bit unsigned instruction-memory relocation.

BFD\_RELOC\_PRU\_LDI32

PRU relocation for two consecutive LDI load instructions that load a 32 bit value into a register. If the higher bits are all zero, then the second instruction may be relaxed.

BFD\_RELOC\_PRU\_S10\_PCREL

PRU QBBx 10-bit signed PC-relative relocation.

BFD\_RELOC\_PRU\_U8\_PCREL

PRU 8-bit unsigned relocation used for the LOOP instruction.

BFD\_RELOC\_PRU\_32\_PMEM

BFD\_RELOC\_PRU\_16\_PMEM

PRU Program Memory relocations. Used to convert from byte addressing to 32-bit word addressing.

BFD\_RELOC\_PRU\_GNU\_DIFF8

BFD\_RELOC\_PRU\_GNU\_DIFF16

BFD\_RELOC\_PRU\_GNU\_DIFF32  
 BFD\_RELOC\_PRU\_GNU\_DIFF16\_PMEM  
 BFD\_RELOC\_PRU\_GNU\_DIFF32\_PMEM

PRU relocations to mark the difference of two local symbols. These are only needed to support linker relaxation and can be ignored when not relaxing. The field is set to the value of the difference assuming no relaxation. The relocation encodes the position of the second symbol so the linker can determine whether to adjust the field value. The PMEM variants encode the word difference, instead of byte difference between symbols.

BFD\_RELOC\_IQ2000\_OFFSET\_16  
 BFD\_RELOC\_IQ2000\_OFFSET\_21  
 BFD\_RELOC\_IQ2000\_UHI16  
 IQ2000 Relocations.

BFD\_RELOC\_XTENSA\_RTLD  
 Special Xtensa relocation used only by PLT entries in ELF shared objects to indicate that the runtime linker should set the value to one of its own internal functions or data structures.

BFD\_RELOC\_XTENSA\_PLT  
 Xtensa relocation used in ELF object files for symbols that may require PLT entries. Otherwise, this is just a generic 32-bit relocation.

BFD\_RELOC\_XTENSA\_DIFF8  
 BFD\_RELOC\_XTENSA\_DIFF16  
 BFD\_RELOC\_XTENSA\_DIFF32  
 Xtensa relocations for backward compatibility. These have been replaced by BFD\_RELOC\_XTENSA\_PDIFF and BFD\_RELOC\_XTENSA\_NDIFF. Xtensa relocations to mark the difference of two local symbols. These are only needed to support linker relaxation and can be ignored when not relaxing. The field is set to the value of the difference assuming no relaxation. The relocation encodes the position of the first symbol so the linker can determine whether to adjust the field value.

BFD\_RELOC\_XTENSA\_SLOT0\_OP  
 BFD\_RELOC\_XTENSA\_SLOT1\_OP  
 BFD\_RELOC\_XTENSA\_SLOT2\_OP  
 BFD\_RELOC\_XTENSA\_SLOT3\_OP  
 BFD\_RELOC\_XTENSA\_SLOT4\_OP  
 BFD\_RELOC\_XTENSA\_SLOT5\_OP  
 BFD\_RELOC\_XTENSA\_SLOT6\_OP  
 BFD\_RELOC\_XTENSA\_SLOT7\_OP  
 BFD\_RELOC\_XTENSA\_SLOT8\_OP  
 BFD\_RELOC\_XTENSA\_SLOT9\_OP  
 BFD\_RELOC\_XTENSA\_SLOT10\_OP  
 BFD\_RELOC\_XTENSA\_SLOT11\_OP  
 BFD\_RELOC\_XTENSA\_SLOT12\_OP  
 BFD\_RELOC\_XTENSA\_SLOT13\_OP



**BFD\_RELOC\_XTENSA\_SLOT14\_OP**

Generic Xtensa relocations for instruction operands. Only the slot number is encoded in the relocation. The relocation applies to the last PC-relative immediate operand, or if there are no PC-relative immediates, to the last immediate operand.

**BFD\_RELOC\_XTENSA\_SLOT0\_ALT**  
**BFD\_RELOC\_XTENSA\_SLOT1\_ALT**  
**BFD\_RELOC\_XTENSA\_SLOT2\_ALT**  
**BFD\_RELOC\_XTENSA\_SLOT3\_ALT**  
**BFD\_RELOC\_XTENSA\_SLOT4\_ALT**  
**BFD\_RELOC\_XTENSA\_SLOT5\_ALT**  
**BFD\_RELOC\_XTENSA\_SLOT6\_ALT**  
**BFD\_RELOC\_XTENSA\_SLOT7\_ALT**  
**BFD\_RELOC\_XTENSA\_SLOT8\_ALT**  
**BFD\_RELOC\_XTENSA\_SLOT9\_ALT**  
**BFD\_RELOC\_XTENSA\_SLOT10\_ALT**  
**BFD\_RELOC\_XTENSA\_SLOT11\_ALT**  
**BFD\_RELOC\_XTENSA\_SLOT12\_ALT**  
**BFD\_RELOC\_XTENSA\_SLOT13\_ALT**  
**BFD\_RELOC\_XTENSA\_SLOT14\_ALT**

Alternate Xtensa relocations. Only the slot is encoded in the relocation. The meaning of these relocations is opcode-specific.

**BFD\_RELOC\_XTENSA\_OP0**  
**BFD\_RELOC\_XTENSA\_OP1**  
**BFD\_RELOC\_XTENSA\_OP2**

Xtensa relocations for backward compatibility. These have all been replaced by **BFD\_RELOC\_XTENSA\_SLOT0\_OP**.

**BFD\_RELOC\_XTENSA\_ASM\_EXPAND**

Xtensa relocation to mark that the assembler expanded the instructions from an original target. The expansion size is encoded in the reloc size.

**BFD\_RELOC\_XTENSA\_ASM\_SIMPLIFY**

Xtensa relocation to mark that the linker should simplify assembler-expanded instructions. This is commonly used internally by the linker after analysis of a **BFD\_RELOC\_XTENSA\_ASM\_EXPAND**.

**BFD\_RELOC\_XTENSA\_TLSDESC\_FN**  
**BFD\_RELOC\_XTENSA\_TLSDESC\_ARG**  
**BFD\_RELOC\_XTENSA\_TLS\_DTPOFF**  
**BFD\_RELOC\_XTENSA\_TLS\_TPOFF**  
**BFD\_RELOC\_XTENSA\_TLS\_FUNC**  
**BFD\_RELOC\_XTENSA\_TLS\_ARG**  
**BFD\_RELOC\_XTENSA\_TLS\_CALL**

Xtensa TLS relocations.

**BFD\_RELOC\_XTENSA\_PDIFF8**  
**BFD\_RELOC\_XTENSA\_PDIFF16**

BFD\_RELOC\_XTENSA\_PDIFF32

BFD\_RELOC\_XTENSA\_NDIFF8

BFD\_RELOC\_XTENSA\_NDIFF16

BFD\_RELOC\_XTENSA\_NDIFF32

Xtensa relocations to mark the difference of two local symbols. These are only needed to support linker relaxation and can be ignored when not relaxing. The field is set to the value of the difference assuming no relaxation. The relocation encodes the position of the subtracted symbol so the linker can determine whether to adjust the field value. PDIFF relocations are used for positive differences, NDIFF relocations are used for negative differences. The difference value is treated as unsigned with these relocation types, giving full 8/16 value ranges.

BFD\_RELOC\_Z80\_DISP8

8 bit signed offset in (ix+d) or (iy+d).

BFD\_RELOC\_Z80\_BYTE0

First 8 bits of multibyte (32, 24 or 16 bit) value.

BFD\_RELOC\_Z80\_BYTE1

Second 8 bits of multibyte (32, 24 or 16 bit) value.

BFD\_RELOC\_Z80\_BYTE2

Third 8 bits of multibyte (32 or 24 bit) value.

BFD\_RELOC\_Z80\_BYTE3

Fourth 8 bits of multibyte (32 bit) value.

BFD\_RELOC\_Z80\_WORD0

Lowest 16 bits of multibyte (32 or 24 bit) value.

BFD\_RELOC\_Z80\_WORD1

Highest 16 bits of multibyte (32 or 24 bit) value.

BFD\_RELOC\_Z80\_16\_BE

Like BFD\_RELOC\_16 but big-endian.

BFD\_RELOC\_Z8K\_DISP7

DJNZ offset.

BFD\_RELOC\_Z8K\_CALLR

CALR offset.

BFD\_RELOC\_Z8K\_IMM4L

4 bit value.

BFD\_RELOC\_LM32\_CALL

BFD\_RELOC\_LM32\_BRANCH

BFD\_RELOC\_LM32\_16\_GOT

BFD\_RELOC\_LM32\_GOTOFF\_HI16

BFD\_RELOC\_LM32\_GOTOFF\_LO16

Lattice Mico32 relocations.

**BFD\_RELOC\_MACH\_O\_SECTDIFF**

Difference between two section addresses. Must be followed by a BFD\_RELOC\_MACH\_O\_PAIR.

**BFD\_RELOC\_MACH\_O\_LOCAL\_SECTDIFF**

Like BFD\_RELOC\_MACH\_O\_SECTDIFF but with a local symbol.

**BFD\_RELOC\_MACH\_O\_PAIR**

Pair of relocation. Contains the first symbol.

**BFD\_RELOC\_MACH\_O\_SUBTRACTOR32**

Symbol will be subtracted. Must be followed by a BFD\_RELOC\_32.

**BFD\_RELOC\_MACH\_O\_SUBTRACTOR64**

Symbol will be subtracted. Must be followed by a BFD\_RELOC\_64.

**BFD\_RELOC\_MACH\_O\_X86\_64\_BRANCH32****BFD\_RELOC\_MACH\_O\_X86\_64\_BRANCH8**

PCREL relocations. They are marked as branch to create PLT entry if required.

**BFD\_RELOC\_MACH\_O\_X86\_64\_GOT**

Used when referencing a GOT entry.

**BFD\_RELOC\_MACH\_O\_X86\_64\_GOT\_LOAD**

Used when loading a GOT entry with movq. It is specially marked so that the linker could optimize the movq to a leaq if possible.

**BFD\_RELOC\_MACH\_O\_X86\_64\_PCREL32\_1**

Same as BFD\_RELOC\_32\_PCREL but with an implicit -1 addend.

**BFD\_RELOC\_MACH\_O\_X86\_64\_PCREL32\_2**

Same as BFD\_RELOC\_32\_PCREL but with an implicit -2 addend.

**BFD\_RELOC\_MACH\_O\_X86\_64\_PCREL32\_4**

Same as BFD\_RELOC\_32\_PCREL but with an implicit -4 addend.

**BFD\_RELOC\_MACH\_O\_X86\_64\_TLV**

Used when referencing a TLV entry.

**BFD\_RELOC\_MACH\_O\_ARM64\_ADDEND**

Addend for PAGE or PAGEOFF.

**BFD\_RELOC\_MACH\_O\_ARM64\_GOT\_LOAD\_PAGE21**

Relative offset to page of GOT slot.

**BFD\_RELOC\_MACH\_O\_ARM64\_GOT\_LOAD\_PAGEOFF12**

Relative offset within page of GOT slot.

**BFD\_RELOC\_MACH\_O\_ARM64\_POINTER\_TO\_GOT**

Address of a GOT entry.

**BFD\_RELOC\_MICROBLAZE\_32\_LO**

This is a 32 bit reloc for the microblaze that stores the low 16 bits of a value.

**BFD\_RELOC\_MICROBLAZE\_32\_LO\_PCREL**

This is a 32 bit pc-relative reloc for the microblaze that stores the low 16 bits of a value.

**BFD\_RELOC\_MICROBLAZE\_32\_ROSDA**

This is a 32 bit reloc for the microblaze that stores a value relative to the read-only small data area anchor.

**BFD\_RELOC\_MICROBLAZE\_32\_RWSDA**

This is a 32 bit reloc for the microblaze that stores a value relative to the read-write small data area anchor.

**BFD\_RELOC\_MICROBLAZE\_32\_SYM\_OP\_SYM**

This is a 32 bit reloc for the microblaze to handle expressions of the form "Symbol Op Symbol".

**BFD\_RELOC\_MICROBLAZE\_32\_NONE**

This is a 32 bit reloc that stores the 32 bit pc relative value in two words (with an imm instruction). No relocation is done here - only used for relaxing.

**BFD\_RELOC\_MICROBLAZE\_64\_NONE**

This is a 64 bit reloc that stores the 32 bit pc relative value in two words (with an imm instruction). No relocation is done here - only used for relaxing.

**BFD\_RELOC\_MICROBLAZE\_64\_GOTPC**

This is a 64 bit reloc that stores the 32 bit pc relative value in two words (with an imm instruction). The relocation is PC-relative GOT offset.

**BFD\_RELOC\_MICROBLAZE\_64\_GOT**

This is a 64 bit reloc that stores the 32 bit pc relative value in two words (with an imm instruction). The relocation is GOT offset.

**BFD\_RELOC\_MICROBLAZE\_64\_PLT**

This is a 64 bit reloc that stores the 32 bit pc relative value in two words (with an imm instruction). The relocation is PC-relative offset into PLT.

**BFD\_RELOC\_MICROBLAZE\_64\_GOTOFF**

This is a 64 bit reloc that stores the 32 bit GOT relative value in two words (with an imm instruction). The relocation is relative offset from `_GLOBAL_OFFSET_TABLE_`.

**BFD\_RELOC\_MICROBLAZE\_32\_GOTOFF**

This is a 32 bit reloc that stores the 32 bit GOT relative value in a word. The relocation is relative offset from `_GLOBAL_OFFSET_TABLE_`.

**BFD\_RELOC\_MICROBLAZE\_64\_TLS**

Unused Reloc.

**BFD\_RELOC\_MICROBLAZE\_64\_TLSD**

This is a 64 bit reloc that stores the 32 bit GOT relative value of the GOT TLS GD info entry in two words (with an imm instruction). The relocation is GOT offset.

**BFD\_RELOC\_MICROBLAZE\_64\_TLSLD**

This is a 64 bit reloc that stores the 32 bit GOT relative value of the GOT TLS LD info entry in two words (with an imm instruction). The relocation is GOT offset.

**BFD\_RELOC\_MICROBLAZE\_32\_TLSDTPMOD**

This is a 32 bit reloc that stores the Module ID to GOT(n).

**BFD\_RELOC\_MICROBLAZE\_32\_TLSDTPREL**

This is a 32 bit reloc that stores TLS offset to GOT(n+1).

**BFD\_RELOC\_MICROBLAZE\_64\_TLSDTPREL**

This is a 32 bit reloc for storing TLS offset to two words (uses imm instruction).

**BFD\_RELOC\_MICROBLAZE\_64\_TLSGOTTPREL**

This is a 64 bit reloc that stores 32-bit thread pointer relative offset to two words (uses imm instruction).

**BFD\_RELOC\_MICROBLAZE\_64\_TLSTPREL**

This is a 64 bit reloc that stores 32-bit thread pointer relative offset to two words (uses imm instruction).

**BFD\_RELOC\_MICROBLAZE\_64\_TEXTPCREL**

This is a 64 bit reloc that stores the 32 bit pc relative value in two words (with an imm instruction). The relocation is PC-relative offset from start of TEXT.

**BFD\_RELOC\_MICROBLAZE\_64\_TEXTREL**

This is a 64 bit reloc that stores the 32 bit offset value in two words (with an imm instruction). The relocation is relative offset from start of TEXT.

**BFD\_RELOC\_KVX\_RELOC\_START**

KVX pseudo relocation code to mark the start of the KVX relocation enumerators. N.B. the order of the enumerators is important as several tables in the KVX bfd backend are indexed by these enumerators; make sure they are all synced.

**BFD\_RELOC\_KVX\_NONE**

KVX null relocation code.

**BFD\_RELOC\_KVX\_16****BFD\_RELOC\_KVX\_32****BFD\_RELOC\_KVX\_64****BFD\_RELOC\_KVX\_S16\_PCREL****BFD\_RELOC\_KVX\_PCREL17****BFD\_RELOC\_KVX\_PCREL27****BFD\_RELOC\_KVX\_32\_PCREL****BFD\_RELOC\_KVX\_S37\_PCREL\_L010****BFD\_RELOC\_KVX\_S37\_PCREL\_UP27****BFD\_RELOC\_KVX\_S43\_PCREL\_L010****BFD\_RELOC\_KVX\_S43\_PCREL\_UP27****BFD\_RELOC\_KVX\_S43\_PCREL\_EX6****BFD\_RELOC\_KVX\_S64\_PCREL\_L010**

BFD\_RELOC\_KVX\_S64\_PCREL\_UP27  
BFD\_RELOC\_KVX\_S64\_PCREL\_EX27  
BFD\_RELOC\_KVX\_64\_PCREL  
BFD\_RELOC\_KVX\_S16  
BFD\_RELOC\_KVX\_S32\_L05  
BFD\_RELOC\_KVX\_S32\_UP27  
BFD\_RELOC\_KVX\_S37\_L010  
BFD\_RELOC\_KVX\_S37\_UP27  
BFD\_RELOC\_KVX\_S37\_GOTOFF\_L010  
BFD\_RELOC\_KVX\_S37\_GOTOFF\_UP27  
BFD\_RELOC\_KVX\_S43\_GOTOFF\_L010  
BFD\_RELOC\_KVX\_S43\_GOTOFF\_UP27  
BFD\_RELOC\_KVX\_S43\_GOTOFF\_EX6  
BFD\_RELOC\_KVX\_32\_GOTOFF  
BFD\_RELOC\_KVX\_64\_GOTOFF  
BFD\_RELOC\_KVX\_32\_GOT  
BFD\_RELOC\_KVX\_S37\_GOT\_L010  
BFD\_RELOC\_KVX\_S37\_GOT\_UP27  
BFD\_RELOC\_KVX\_S43\_GOT\_L010  
BFD\_RELOC\_KVX\_S43\_GOT\_UP27  
BFD\_RELOC\_KVX\_S43\_GOT\_EX6  
BFD\_RELOC\_KVX\_64\_GOT  
BFD\_RELOC\_KVX\_GLOB\_DAT  
BFD\_RELOC\_KVX\_COPY  
BFD\_RELOC\_KVX\_JMP\_SLOT  
BFD\_RELOC\_KVX\_RELATIVE  
BFD\_RELOC\_KVX\_S43\_L010  
BFD\_RELOC\_KVX\_S43\_UP27  
BFD\_RELOC\_KVX\_S43\_EX6  
BFD\_RELOC\_KVX\_S64\_L010  
BFD\_RELOC\_KVX\_S64\_UP27  
BFD\_RELOC\_KVX\_S64\_EX27  
BFD\_RELOC\_KVX\_S37\_GOTADDR\_L010  
BFD\_RELOC\_KVX\_S37\_GOTADDR\_UP27  
BFD\_RELOC\_KVX\_S43\_GOTADDR\_L010  
BFD\_RELOC\_KVX\_S43\_GOTADDR\_UP27  
BFD\_RELOC\_KVX\_S43\_GOTADDR\_EX6  
BFD\_RELOC\_KVX\_S64\_GOTADDR\_L010  
BFD\_RELOC\_KVX\_S64\_GOTADDR\_UP27  
BFD\_RELOC\_KVX\_S64\_GOTADDR\_EX27  
BFD\_RELOC\_KVX\_64\_DTPMOD  
BFD\_RELOC\_KVX\_64\_DTPOFF  
BFD\_RELOC\_KVX\_S37\_TLS\_DTPOFF\_L010  
BFD\_RELOC\_KVX\_S37\_TLS\_DTPOFF\_UP27  
BFD\_RELOC\_KVX\_S43\_TLS\_DTPOFF\_L010  
BFD\_RELOC\_KVX\_S43\_TLS\_DTPOFF\_UP27  
BFD\_RELOC\_KVX\_S43\_TLS\_DTPOFF\_EX6

BFD\_RELOC\_KVX\_S37\_TLS\_GD\_L010  
 BFD\_RELOC\_KVX\_S37\_TLS\_GD\_UP27  
 BFD\_RELOC\_KVX\_S43\_TLS\_GD\_L010  
 BFD\_RELOC\_KVX\_S43\_TLS\_GD\_UP27  
 BFD\_RELOC\_KVX\_S43\_TLS\_GD\_EX6  
 BFD\_RELOC\_KVX\_S37\_TLS\_LD\_L010  
 BFD\_RELOC\_KVX\_S37\_TLS\_LD\_UP27  
 BFD\_RELOC\_KVX\_S43\_TLS\_LD\_L010  
 BFD\_RELOC\_KVX\_S43\_TLS\_LD\_UP27  
 BFD\_RELOC\_KVX\_S43\_TLS\_LD\_EX6  
 BFD\_RELOC\_KVX\_64\_TPOFF  
 BFD\_RELOC\_KVX\_S37\_TLS\_IE\_L010  
 BFD\_RELOC\_KVX\_S37\_TLS\_IE\_UP27  
 BFD\_RELOC\_KVX\_S43\_TLS\_IE\_L010  
 BFD\_RELOC\_KVX\_S43\_TLS\_IE\_UP27  
 BFD\_RELOC\_KVX\_S43\_TLS\_IE\_EX6  
 BFD\_RELOC\_KVX\_S37\_TLS\_LE\_L010  
 BFD\_RELOC\_KVX\_S37\_TLS\_LE\_UP27  
 BFD\_RELOC\_KVX\_S43\_TLS\_LE\_L010  
 BFD\_RELOC\_KVX\_S43\_TLS\_LE\_UP27  
 BFD\_RELOC\_KVX\_S43\_TLS\_LE\_EX6  
 BFD\_RELOC\_KVX\_8

KVX Relocations.

BFD\_RELOC\_KVX\_RELOC\_END

KVX pseudo relocation code to mark the end of the KVX relocation enumerators that have direct mapping to ELF reloc codes. There are a few more enumerators after this one; those are mainly used by the KVX assembler for the internal fixup or to select one of the above enumerators.

BFD\_RELOC\_AARCH64\_RELOC\_START

AArch64 pseudo relocation code to mark the start of the AArch64 relocation enumerators. N.B. the order of the enumerators is important as several tables in the AArch64 bfd backend are indexed by these enumerators; make sure they are all synced.

BFD\_RELOC\_AARCH64\_NULL

Deprecated AArch64 null relocation code.

BFD\_RELOC\_AARCH64\_NONE

AArch64 null relocation code.

BFD\_RELOC\_AARCH64\_64

BFD\_RELOC\_AARCH64\_32

BFD\_RELOC\_AARCH64\_16

Basic absolute relocations of N bits. These are equivalent to BFD\_RELOC\_N and they were added to assist the indexing of the howto table.

BFD\_RELOC\_AARCH64\_64\_PCREL

BFD\_RELOC\_AARCH64\_32\_PCREL

**BFD\_RELOC\_AARCH64\_16\_PCREL**

PC-relative relocations. These are equivalent to `BFD_RELOC_N_PCREL` and they were added to assist the indexing of the howto table.

**BFD\_RELOC\_AARCH64\_MOVV\_G0**

AArch64 `MOV[NZK]` instruction with most significant bits 0 to 15 of an unsigned address/value.

**BFD\_RELOC\_AARCH64\_MOVV\_G0\_NC**

AArch64 `MOV[NZK]` instruction with less significant bits 0 to 15 of an address/value. No overflow checking.

**BFD\_RELOC\_AARCH64\_MOVV\_G1**

AArch64 `MOV[NZK]` instruction with most significant bits 16 to 31 of an unsigned address/value.

**BFD\_RELOC\_AARCH64\_MOVV\_G1\_NC**

AArch64 `MOV[NZK]` instruction with less significant bits 16 to 31 of an address/value. No overflow checking.

**BFD\_RELOC\_AARCH64\_MOVV\_G2**

AArch64 `MOV[NZK]` instruction with most significant bits 32 to 47 of an unsigned address/value.

**BFD\_RELOC\_AARCH64\_MOVV\_G2\_NC**

AArch64 `MOV[NZK]` instruction with less significant bits 32 to 47 of an address/value. No overflow checking.

**BFD\_RELOC\_AARCH64\_MOVV\_G3**

AArch64 `MOV[NZK]` instruction with most significant bits 48 to 64 of a signed or unsigned address/value.

**BFD\_RELOC\_AARCH64\_MOVV\_G0\_S**

AArch64 `MOV[NZ]` instruction with most significant bits 0 to 15 of a signed value. Changes instruction to `MOVZ` or `MOVN` depending on the value's sign.

**BFD\_RELOC\_AARCH64\_MOVV\_G1\_S**

AArch64 `MOV[NZ]` instruction with most significant bits 16 to 31 of a signed value. Changes instruction to `MOVZ` or `MOVN` depending on the value's sign.

**BFD\_RELOC\_AARCH64\_MOVV\_G2\_S**

AArch64 `MOV[NZ]` instruction with most significant bits 32 to 47 of a signed value. Changes instruction to `MOVZ` or `MOVN` depending on the value's sign.

**BFD\_RELOC\_AARCH64\_MOVV\_PREL\_G0**

AArch64 `MOV[NZ]` instruction with most significant bits 0 to 15 of a signed value. Changes instruction to `MOVZ` or `MOVN` depending on the value's sign.

**BFD\_RELOC\_AARCH64\_MOVV\_PREL\_G0\_NC**

AArch64 `MOV[NZ]` instruction with most significant bits 0 to 15 of a signed value. Changes instruction to `MOVZ` or `MOVN` depending on the value's sign.



**BFD\_RELOC\_AARCH64\_MOVW\_PREL\_G1**

AArch64 MOVK instruction with most significant bits 16 to 31 of a signed value.

**BFD\_RELOC\_AARCH64\_MOVW\_PREL\_G1\_NC**

AArch64 MOVK instruction with most significant bits 16 to 31 of a signed value.

**BFD\_RELOC\_AARCH64\_MOVW\_PREL\_G2**

AArch64 MOVK instruction with most significant bits 32 to 47 of a signed value.

**BFD\_RELOC\_AARCH64\_MOVW\_PREL\_G2\_NC**

AArch64 MOVK instruction with most significant bits 32 to 47 of a signed value.

**BFD\_RELOC\_AARCH64\_MOVW\_PREL\_G3**

AArch64 MOVK instruction with most significant bits 47 to 63 of a signed value.

**BFD\_RELOC\_AARCH64\_LD\_L019\_PCREL**

AArch64 Load Literal instruction, holding a 19 bit pc-relative word offset. The lowest two bits must be zero and are not stored in the instruction, giving a 21 bit signed byte offset.

**BFD\_RELOC\_AARCH64\_ADR\_L021\_PCREL**

AArch64 ADR instruction, holding a simple 21 bit pc-relative byte offset.

**BFD\_RELOC\_AARCH64\_ADR\_HI21\_PCREL**

AArch64 ADRP instruction, with bits 12 to 32 of a pc-relative page offset, giving a 4KB aligned page base address.

**BFD\_RELOC\_AARCH64\_ADR\_HI21\_NC\_PCREL**

AArch64 ADRP instruction, with bits 12 to 32 of a pc-relative page offset, giving a 4KB aligned page base address, but with no overflow checking.

**BFD\_RELOC\_AARCH64\_ADD\_L012**

AArch64 ADD immediate instruction, holding bits 0 to 11 of the address. Used in conjunction with BFD\_RELOC\_AARCH64\_ADR\_HI21\_PCREL.

**BFD\_RELOC\_AARCH64\_LDST8\_L012**

AArch64 8-bit load/store instruction, holding bits 0 to 11 of the address. Used in conjunction with BFD\_RELOC\_AARCH64\_ADR\_HI21\_PCREL.

**BFD\_RELOC\_AARCH64\_TSTBR14**

AArch64 14 bit pc-relative test bit and branch. The lowest two bits must be zero and are not stored in the instruction, giving a 16 bit signed byte offset.

**BFD\_RELOC\_AARCH64\_BRANCH19**

AArch64 19 bit pc-relative conditional branch and compare & branch. The lowest two bits must be zero and are not stored in the instruction, giving a 21 bit signed byte offset.

**BFD\_RELOC\_AARCH64\_JUMP26**

AArch64 26 bit pc-relative unconditional branch. The lowest two bits must be zero and are not stored in the instruction, giving a 28 bit signed byte offset.

**BFD\_RELOC\_AARCH64\_CALL26**

AArch64 26 bit pc-relative unconditional branch and link. The lowest two bits must be zero and are not stored in the instruction, giving a 28 bit signed byte offset.

**BFD\_RELOC\_AARCH64\_LDST16\_L012**

AArch64 16-bit load/store instruction, holding bits 0 to 11 of the address. Used in conjunction with BFD\_RELOC\_AARCH64\_ADR\_HI21\_PCREL.

**BFD\_RELOC\_AARCH64\_LDST32\_L012**

AArch64 32-bit load/store instruction, holding bits 0 to 11 of the address. Used in conjunction with BFD\_RELOC\_AARCH64\_ADR\_HI21\_PCREL.

**BFD\_RELOC\_AARCH64\_LDST64\_L012**

AArch64 64-bit load/store instruction, holding bits 0 to 11 of the address. Used in conjunction with BFD\_RELOC\_AARCH64\_ADR\_HI21\_PCREL.

**BFD\_RELOC\_AARCH64\_LDST128\_L012**

AArch64 128-bit load/store instruction, holding bits 0 to 11 of the address. Used in conjunction with BFD\_RELOC\_AARCH64\_ADR\_HI21\_PCREL.

**BFD\_RELOC\_AARCH64\_GOT\_LD\_PREL19**

AArch64 Load Literal instruction, holding a 19 bit PC relative word offset of the global offset table entry for a symbol. The lowest two bits must be zero and are not stored in the instruction, giving a 21 bit signed byte offset. This relocation type requires signed overflow checking.

**BFD\_RELOC\_AARCH64\_ADR\_GOT\_PAGE**

Get to the page base of the global offset table entry for a symbol as part of an ADRP instruction using a 21 bit PC relative value. Used in conjunction with BFD\_RELOC\_AARCH64\_LD64\_GOT\_LO12\_NC.

**BFD\_RELOC\_AARCH64\_LD64\_GOT\_LO12\_NC**

Unsigned 12 bit byte offset for 64 bit load/store from the page of the GOT entry for this symbol. Used in conjunction with BFD\_RELOC\_AARCH64\_ADR\_GOT\_PAGE. Valid in LP64 ABI only.

**BFD\_RELOC\_AARCH64\_LD32\_GOT\_LO12\_NC**

Unsigned 12 bit byte offset for 32 bit load/store from the page of the GOT entry for this symbol. Used in conjunction with BFD\_RELOC\_AARCH64\_ADR\_GOT\_PAGE. Valid in ILP32 ABI only.

**BFD\_RELOC\_AARCH64\_MOVW\_GOTOFF\_GO\_NC**

Unsigned 16 bit byte offset for 64 bit load/store from the GOT entry for this symbol. Valid in LP64 ABI only.

**BFD\_RELOC\_AARCH64\_MOVW\_GOTOFF\_G1**

Unsigned 16 bit byte higher offset for 64 bit load/store from the GOT entry for this symbol. Valid in LP64 ABI only.

**BFD\_RELOC\_AARCH64\_LD64\_GOTOFF\_LO15**

Unsigned 15 bit byte offset for 64 bit load/store from the page of the GOT entry for this symbol. Valid in LP64 ABI only.

**BFD\_RELOC\_AARCH64\_LD32\_GOTPAGE\_L014**

Scaled 14 bit byte offset to the page base of the global offset table.

**BFD\_RELOC\_AARCH64\_LD64\_GOTPAGE\_L015**

Scaled 15 bit byte offset to the page base of the global offset table.

**BFD\_RELOC\_AARCH64\_TLSGD\_ADR\_PAGE21**

Get to the page base of the global offset table entry for a symbols `tls_index` structure as part of an `adrp` instruction using a 21 bit PC relative value. Used in conjunction with `BFD_RELOC_AARCH64_TLSGD_ADD_LO12_NC`.

**BFD\_RELOC\_AARCH64\_TLSGD\_ADR\_PREL21**

AArch64 TLS General Dynamic.

**BFD\_RELOC\_AARCH64\_TLSGD\_ADD\_LO12\_NC**

Unsigned 12 bit byte offset to global offset table entry for a symbol's `tls_index` structure. Used in conjunction with `BFD_RELOC_AARCH64_TLSGD_ADR_PAGE21`.

**BFD\_RELOC\_AARCH64\_TLSGD\_MOVW\_GO\_NC**

AArch64 TLS General Dynamic relocation.

**BFD\_RELOC\_AARCH64\_TLSGD\_MOVW\_G1**

AArch64 TLS General Dynamic relocation.

**BFD\_RELOC\_AARCH64\_TLSIE\_ADR\_GOTTPREL\_PAGE21**

AArch64 TLS INITIAL EXEC relocation.

**BFD\_RELOC\_AARCH64\_TLSIE\_LD64\_GOTTPREL\_LO12\_NC**

AArch64 TLS INITIAL EXEC relocation.

**BFD\_RELOC\_AARCH64\_TLSIE\_LD32\_GOTTPREL\_LO12\_NC**

AArch64 TLS INITIAL EXEC relocation.

**BFD\_RELOC\_AARCH64\_TLSIE\_LD\_GOTTPREL\_PREL19**

AArch64 TLS INITIAL EXEC relocation.

**BFD\_RELOC\_AARCH64\_TLSIE\_MOVW\_GOTTPREL\_GO\_NC**

AArch64 TLS INITIAL EXEC relocation.

**BFD\_RELOC\_AARCH64\_TLSIE\_MOVW\_GOTTPREL\_G1**

AArch64 TLS INITIAL EXEC relocation.

**BFD\_RELOC\_AARCH64\_TLSLD\_ADD\_DTPREL\_HI12**

bit[23:12] of byte offset to module TLS base address.

**BFD\_RELOC\_AARCH64\_TLSLD\_ADD\_DTPREL\_LO12**

Unsigned 12 bit byte offset to module TLS base address.

**BFD\_RELOC\_AARCH64\_TLSLD\_ADD\_DTPREL\_LO12\_NC**

No overflow check version of `BFD_RELOC_AARCH64_TLSLD_ADD_DTPREL_LO12`.■

**BFD\_RELOC\_AARCH64\_TLSLD\_ADD\_L012\_NC**

Unsigned 12 bit byte offset to global offset table entry for a symbol's `tls.index` structure. Used in conjunction with `BFD_RELOC_AARCH64_TLSLD_ADR_PAGE21`.

**BFD\_RELOC\_AARCH64\_TLSLD\_ADR\_PAGE21**

GOT entry page address for AArch64 TLS Local Dynamic, used with `ADRP` instruction.

**BFD\_RELOC\_AARCH64\_TLSLD\_ADR\_PREL21**

GOT entry address for AArch64 TLS Local Dynamic, used with `ADR` instruction.

**BFD\_RELOC\_AARCH64\_TLSLD\_LDST16\_DTPREL\_L012**

bit[11:1] of byte offset to module TLS base address, encoded in `ldst` instructions.

**BFD\_RELOC\_AARCH64\_TLSLD\_LDST16\_DTPREL\_L012\_NC**

Similar to `BFD_RELOC_AARCH64_TLSLD_LDST16_DTPREL_L012`, but no overflow check.

**BFD\_RELOC\_AARCH64\_TLSLD\_LDST32\_DTPREL\_L012**

bit[11:2] of byte offset to module TLS base address, encoded in `ldst` instructions.

**BFD\_RELOC\_AARCH64\_TLSLD\_LDST32\_DTPREL\_L012\_NC**

Similar to `BFD_RELOC_AARCH64_TLSLD_LDST32_DTPREL_L012`, but no overflow check.

**BFD\_RELOC\_AARCH64\_TLSLD\_LDST64\_DTPREL\_L012**

bit[11:3] of byte offset to module TLS base address, encoded in `ldst` instructions.

**BFD\_RELOC\_AARCH64\_TLSLD\_LDST64\_DTPREL\_L012\_NC**

Similar to `BFD_RELOC_AARCH64_TLSLD_LDST64_DTPREL_L012`, but no overflow check.

**BFD\_RELOC\_AARCH64\_TLSLD\_LDST8\_DTPREL\_L012**

bit[11:0] of byte offset to module TLS base address, encoded in `ldst` instructions.

**BFD\_RELOC\_AARCH64\_TLSLD\_LDST8\_DTPREL\_L012\_NC**

Similar to `BFD_RELOC_AARCH64_TLSLD_LDST8_DTPREL_L012`, but no overflow check.

**BFD\_RELOC\_AARCH64\_TLSLD\_MOVW\_DTPREL\_G0**

bit[15:0] of byte offset to module TLS base address.

**BFD\_RELOC\_AARCH64\_TLSLD\_MOVW\_DTPREL\_G0\_NC**

No overflow check version of `BFD_RELOC_AARCH64_TLSLD_MOVW_DTPREL_G0`.■

**BFD\_RELOC\_AARCH64\_TLSLD\_MOVW\_DTPREL\_G1**

bit[31:16] of byte offset to module TLS base address.

**BFD\_RELOC\_AARCH64\_TLSLD\_MOVW\_DTPREL\_G1\_NC**

No overflow check version of `BFD_RELOC_AARCH64_TLSLD_MOVW_DTPREL_G1`.■

- BFD\_RELOC\_AARCH64\_TLSLD\_MOVW\_DTPREL\_G2**  
bit[47:32] of byte offset to module TLS base address.
- BFD\_RELOC\_AARCH64\_TLSLE\_MOVW\_TPREL\_G2**  
AArch64 TLS LOCAL EXEC relocation.
- BFD\_RELOC\_AARCH64\_TLSLE\_MOVW\_TPREL\_G1**  
AArch64 TLS LOCAL EXEC relocation.
- BFD\_RELOC\_AARCH64\_TLSLE\_MOVW\_TPREL\_G1\_NC**  
AArch64 TLS LOCAL EXEC relocation.
- BFD\_RELOC\_AARCH64\_TLSLE\_MOVW\_TPREL\_GO**  
AArch64 TLS LOCAL EXEC relocation.
- BFD\_RELOC\_AARCH64\_TLSLE\_MOVW\_TPREL\_GO\_NC**  
AArch64 TLS LOCAL EXEC relocation.
- BFD\_RELOC\_AARCH64\_TLSLE\_ADD\_TPREL\_HI12**  
AArch64 TLS LOCAL EXEC relocation.
- BFD\_RELOC\_AARCH64\_TLSLE\_ADD\_TPREL\_LO12**  
AArch64 TLS LOCAL EXEC relocation.
- BFD\_RELOC\_AARCH64\_TLSLE\_ADD\_TPREL\_LO12\_NC**  
AArch64 TLS LOCAL EXEC relocation.
- BFD\_RELOC\_AARCH64\_TLSLE\_LDST16\_TPREL\_LO12**  
bit[11:1] of byte offset to module TLS base address, encoded in ldst instructions.
- BFD\_RELOC\_AARCH64\_TLSLE\_LDST16\_TPREL\_LO12\_NC**  
Similar to BFD\_RELOC\_AARCH64\_TLSLE\_LDST16\_TPREL\_LO12, but no overflow check.
- BFD\_RELOC\_AARCH64\_TLSLE\_LDST32\_TPREL\_LO12**  
bit[11:2] of byte offset to module TLS base address, encoded in ldst instructions.
- BFD\_RELOC\_AARCH64\_TLSLE\_LDST32\_TPREL\_LO12\_NC**  
Similar to BFD\_RELOC\_AARCH64\_TLSLE\_LDST32\_TPREL\_LO12, but no overflow check.
- BFD\_RELOC\_AARCH64\_TLSLE\_LDST64\_TPREL\_LO12**  
bit[11:3] of byte offset to module TLS base address, encoded in ldst instructions.
- BFD\_RELOC\_AARCH64\_TLSLE\_LDST64\_TPREL\_LO12\_NC**  
Similar to BFD\_RELOC\_AARCH64\_TLSLE\_LDST64\_TPREL\_LO12, but no overflow check.
- BFD\_RELOC\_AARCH64\_TLSLE\_LDST8\_TPREL\_LO12**  
bit[11:0] of byte offset to module TLS base address, encoded in ldst instructions.

**BFD\_RELOC\_AARCH64\_TLSLE\_LDST8\_TPREL\_L012\_NC**

Similar to BFD\_RELOC\_AARCH64\_TLSLE\_LDST8\_TPREL\_L012, but no overflow check.

**BFD\_RELOC\_AARCH64\_TLSDESC\_LD\_PREL19**

**BFD\_RELOC\_AARCH64\_TLSDESC\_ADR\_PREL21**

**BFD\_RELOC\_AARCH64\_TLSDESC\_ADR\_PAGE21**

**BFD\_RELOC\_AARCH64\_TLSDESC\_LD64\_L012**

**BFD\_RELOC\_AARCH64\_TLSDESC\_LD32\_L012\_NC**

**BFD\_RELOC\_AARCH64\_TLSDESC\_ADD\_L012**

**BFD\_RELOC\_AARCH64\_TLSDESC\_OFF\_G1**

**BFD\_RELOC\_AARCH64\_TLSDESC\_OFF\_GO\_NC**

**BFD\_RELOC\_AARCH64\_TLSDESC\_LDR**

**BFD\_RELOC\_AARCH64\_TLSDESC\_ADD**

**BFD\_RELOC\_AARCH64\_TLSDESC\_CALL**

AArch64 TLS DESC relocations.

**BFD\_RELOC\_AARCH64\_COPY**

**BFD\_RELOC\_AARCH64\_GLOB\_DAT**

**BFD\_RELOC\_AARCH64\_JUMP\_SLOT**

**BFD\_RELOC\_AARCH64\_RELATIVE**

AArch64 DSO relocations.

**BFD\_RELOC\_AARCH64\_TLS\_DTPMOD**

**BFD\_RELOC\_AARCH64\_TLS\_DTPREL**

**BFD\_RELOC\_AARCH64\_TLS\_TPREL**

**BFD\_RELOC\_AARCH64\_TLSDESC**

AArch64 TLS relocations.

**BFD\_RELOC\_AARCH64\_IRELATIVE**

AArch64 support for STT\_GNU\_IFUNC.

**BFD\_RELOC\_AARCH64\_RELOC\_END**

AArch64 pseudo relocation code to mark the end of the AArch64 relocation enumerators that have direct mapping to ELF reloc codes. There are a few more enumerators after this one; those are mainly used by the AArch64 assembler for the internal fixup or to select one of the above enumerators.

**BFD\_RELOC\_AARCH64\_GAS\_INTERNAL\_FIXUP**

AArch64 pseudo relocation code to be used internally by the AArch64 assembler and not (currently) written to any object files.

**BFD\_RELOC\_AARCH64\_LDST\_L012**

AArch64 unspecified load/store instruction, holding bits 0 to 11 of the address. Used in conjunction with BFD\_RELOC\_AARCH64\_ADR\_HI21\_PCREL.

**BFD\_RELOC\_AARCH64\_TLSD\_L012**

AArch64 pseudo relocation code for TLS local dynamic mode. It's to be used internally by the AArch64 assembler and not (currently) written to any object files.

**BFD\_RELOC\_AARCH64\_TLSLD\_LDST\_DTPREL\_L012\_NC**

Similar to BFD\_RELOC\_AARCH64\_TLSLD\_LDST\_DTPREL\_L012, but no overflow check.

**BFD\_RELOC\_AARCH64\_TLSLE\_LDST\_TPREL\_L012**

AArch64 pseudo relocation code for TLS local exec mode. It's to be used internally by the AArch64 assembler and not (currently) written to any object files.

**BFD\_RELOC\_AARCH64\_TLSLE\_LDST\_TPREL\_L012\_NC**

Similar to BFD\_RELOC\_AARCH64\_TLSLE\_LDST\_TPREL\_L012, but no overflow check.

**BFD\_RELOC\_AARCH64\_LD\_GOT\_L012\_NC**

AArch64 pseudo relocation code to be used internally by the AArch64 assembler and not (currently) written to any object files.

**BFD\_RELOC\_AARCH64\_TLSIE\_LD\_GOTTPREL\_L012\_NC**

AArch64 pseudo relocation code to be used internally by the AArch64 assembler and not (currently) written to any object files.

**BFD\_RELOC\_AARCH64\_TLSDESC\_LD\_L012\_NC**

AArch64 pseudo relocation code to be used internally by the AArch64 assembler and not (currently) written to any object files.

**BFD\_RELOC\_AARCH64\_BRANCH9**

AArch64 9 bit pc-relative conditional branch and compare & branch. The lowest two bits must be zero and are not stored in the instruction, giving an 11 bit signed byte offset.

**BFD\_RELOC\_TILEPRO\_BROFF\_X1****BFD\_RELOC\_TILEPRO\_JOFFLONG\_X1****BFD\_RELOC\_TILEPRO\_JOFFLONG\_X1\_PLT****BFD\_RELOC\_TILEPRO\_IMM8\_X0****BFD\_RELOC\_TILEPRO\_IMM8\_Y0****BFD\_RELOC\_TILEPRO\_IMM8\_X1****BFD\_RELOC\_TILEPRO\_IMM8\_Y1****BFD\_RELOC\_TILEPRO\_DEST\_IMM8\_X1****BFD\_RELOC\_TILEPRO\_MT\_IMM15\_X1****BFD\_RELOC\_TILEPRO\_MF\_IMM15\_X1****BFD\_RELOC\_TILEPRO\_IMM16\_X0****BFD\_RELOC\_TILEPRO\_IMM16\_X1****BFD\_RELOC\_TILEPRO\_IMM16\_X0\_LO****BFD\_RELOC\_TILEPRO\_IMM16\_X1\_LO****BFD\_RELOC\_TILEPRO\_IMM16\_X0\_HI****BFD\_RELOC\_TILEPRO\_IMM16\_X1\_HI****BFD\_RELOC\_TILEPRO\_IMM16\_X0\_HA****BFD\_RELOC\_TILEPRO\_IMM16\_X1\_HA****BFD\_RELOC\_TILEPRO\_IMM16\_X0\_PCREL****BFD\_RELOC\_TILEPRO\_IMM16\_X1\_PCREL**

BFD\_RELOC\_TILEPRO\_IMM16\_X0\_LO\_PCREL  
BFD\_RELOC\_TILEPRO\_IMM16\_X1\_LO\_PCREL  
BFD\_RELOC\_TILEPRO\_IMM16\_X0\_HI\_PCREL  
BFD\_RELOC\_TILEPRO\_IMM16\_X1\_HI\_PCREL  
BFD\_RELOC\_TILEPRO\_IMM16\_X0\_HA\_PCREL  
BFD\_RELOC\_TILEPRO\_IMM16\_X1\_HA\_PCREL  
BFD\_RELOC\_TILEPRO\_IMM16\_X0\_GOT  
BFD\_RELOC\_TILEPRO\_IMM16\_X1\_GOT  
BFD\_RELOC\_TILEPRO\_IMM16\_X0\_GOT\_LO  
BFD\_RELOC\_TILEPRO\_IMM16\_X1\_GOT\_LO  
BFD\_RELOC\_TILEPRO\_IMM16\_X0\_GOT\_HI  
BFD\_RELOC\_TILEPRO\_IMM16\_X1\_GOT\_HI  
BFD\_RELOC\_TILEPRO\_IMM16\_X0\_GOT\_HA  
BFD\_RELOC\_TILEPRO\_IMM16\_X1\_GOT\_HA  
BFD\_RELOC\_TILEPRO\_MMSTART\_X0  
BFD\_RELOC\_TILEPRO\_MMEND\_X0  
BFD\_RELOC\_TILEPRO\_MMSTART\_X1  
BFD\_RELOC\_TILEPRO\_MMEND\_X1  
BFD\_RELOC\_TILEPRO\_SHAMT\_X0  
BFD\_RELOC\_TILEPRO\_SHAMT\_X1  
BFD\_RELOC\_TILEPRO\_SHAMT\_Y0  
BFD\_RELOC\_TILEPRO\_SHAMT\_Y1  
BFD\_RELOC\_TILEPRO\_TLS\_GD\_CALL  
BFD\_RELOC\_TILEPRO\_IMM8\_X0\_TLS\_GD\_ADD  
BFD\_RELOC\_TILEPRO\_IMM8\_X1\_TLS\_GD\_ADD  
BFD\_RELOC\_TILEPRO\_IMM8\_Y0\_TLS\_GD\_ADD  
BFD\_RELOC\_TILEPRO\_IMM8\_Y1\_TLS\_GD\_ADD  
BFD\_RELOC\_TILEPRO\_TLS\_IE\_LOAD  
BFD\_RELOC\_TILEPRO\_IMM16\_X0\_TLS\_GD  
BFD\_RELOC\_TILEPRO\_IMM16\_X1\_TLS\_GD  
BFD\_RELOC\_TILEPRO\_IMM16\_X0\_TLS\_GD\_LO  
BFD\_RELOC\_TILEPRO\_IMM16\_X1\_TLS\_GD\_LO  
BFD\_RELOC\_TILEPRO\_IMM16\_X0\_TLS\_GD\_HI  
BFD\_RELOC\_TILEPRO\_IMM16\_X1\_TLS\_GD\_HI  
BFD\_RELOC\_TILEPRO\_IMM16\_X0\_TLS\_GD\_HA  
BFD\_RELOC\_TILEPRO\_IMM16\_X1\_TLS\_GD\_HA  
BFD\_RELOC\_TILEPRO\_IMM16\_X0\_TLS\_IE  
BFD\_RELOC\_TILEPRO\_IMM16\_X1\_TLS\_IE  
BFD\_RELOC\_TILEPRO\_IMM16\_X0\_TLS\_IE\_LO  
BFD\_RELOC\_TILEPRO\_IMM16\_X1\_TLS\_IE\_LO  
BFD\_RELOC\_TILEPRO\_IMM16\_X0\_TLS\_IE\_HI  
BFD\_RELOC\_TILEPRO\_IMM16\_X1\_TLS\_IE\_HI  
BFD\_RELOC\_TILEPRO\_IMM16\_X0\_TLS\_IE\_HA  
BFD\_RELOC\_TILEPRO\_IMM16\_X1\_TLS\_IE\_HA  
BFD\_RELOC\_TILEPRO\_TLS\_DTPMOD32  
BFD\_RELOC\_TILEPRO\_TLS\_DTPOFF32  
BFD\_RELOC\_TILEPRO\_TLS\_TPOFF32



BFD\_RELOC\_TILEPRO\_IMM16\_X0\_TLS\_LE  
BFD\_RELOC\_TILEPRO\_IMM16\_X1\_TLS\_LE  
BFD\_RELOC\_TILEPRO\_IMM16\_X0\_TLS\_LE\_LO  
BFD\_RELOC\_TILEPRO\_IMM16\_X1\_TLS\_LE\_LO  
BFD\_RELOC\_TILEPRO\_IMM16\_X0\_TLS\_LE\_HI  
BFD\_RELOC\_TILEPRO\_IMM16\_X1\_TLS\_LE\_HI  
BFD\_RELOC\_TILEPRO\_IMM16\_X0\_TLS\_LE\_HA  
BFD\_RELOC\_TILEPRO\_IMM16\_X1\_TLS\_LE\_HA

Tilera TILEPro Relocations.

BFD\_RELOC\_TILEGX\_HW0  
BFD\_RELOC\_TILEGX\_HW1  
BFD\_RELOC\_TILEGX\_HW2  
BFD\_RELOC\_TILEGX\_HW3  
BFD\_RELOC\_TILEGX\_HW0\_LAST  
BFD\_RELOC\_TILEGX\_HW1\_LAST  
BFD\_RELOC\_TILEGX\_HW2\_LAST  
BFD\_RELOC\_TILEGX\_BROFF\_X1  
BFD\_RELOC\_TILEGX\_JUMPOFF\_X1  
BFD\_RELOC\_TILEGX\_JUMPOFF\_X1\_PLT  
BFD\_RELOC\_TILEGX\_IMM8\_X0  
BFD\_RELOC\_TILEGX\_IMM8\_Y0  
BFD\_RELOC\_TILEGX\_IMM8\_X1  
BFD\_RELOC\_TILEGX\_IMM8\_Y1  
BFD\_RELOC\_TILEGX\_DEST\_IMM8\_X1  
BFD\_RELOC\_TILEGX\_MT\_IMM14\_X1  
BFD\_RELOC\_TILEGX\_MF\_IMM14\_X1  
BFD\_RELOC\_TILEGX\_MMSTART\_X0  
BFD\_RELOC\_TILEGX\_MMEND\_X0  
BFD\_RELOC\_TILEGX\_SHAMT\_X0  
BFD\_RELOC\_TILEGX\_SHAMT\_X1  
BFD\_RELOC\_TILEGX\_SHAMT\_Y0  
BFD\_RELOC\_TILEGX\_SHAMT\_Y1  
BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW0  
BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW0  
BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW1  
BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW1  
BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW2  
BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW2  
BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW3  
BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW3  
BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW0\_LAST  
BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW0\_LAST  
BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW1\_LAST  
BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW1\_LAST  
BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW2\_LAST  
BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW2\_LAST

BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW0\_PCREL  
BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW0\_PCREL  
BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW1\_PCREL  
BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW1\_PCREL  
BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW2\_PCREL  
BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW2\_PCREL  
BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW3\_PCREL  
BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW3\_PCREL  
BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW0\_LAST\_PCREL  
BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW0\_LAST\_PCREL  
BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW1\_LAST\_PCREL  
BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW1\_LAST\_PCREL  
BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW2\_LAST\_PCREL  
BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW2\_LAST\_PCREL  
BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW0\_GOT  
BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW0\_GOT  
BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW0\_PLT\_PCREL  
BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW0\_PLT\_PCREL  
BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW1\_PLT\_PCREL  
BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW1\_PLT\_PCREL  
BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW2\_PLT\_PCREL  
BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW2\_PLT\_PCREL  
BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW0\_LAST\_GOT  
BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW0\_LAST\_GOT  
BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW1\_LAST\_GOT  
BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW1\_LAST\_GOT  
BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW3\_PLT\_PCREL  
BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW3\_PLT\_PCREL  
BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW0\_TLS\_GD  
BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW0\_TLS\_GD  
BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW0\_TLS\_LE  
BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW0\_TLS\_LE  
BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW0\_LAST\_TLS\_LE  
BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW0\_LAST\_TLS\_LE  
BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW1\_LAST\_TLS\_LE  
BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW1\_LAST\_TLS\_LE  
BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW0\_LAST\_TLS\_GD  
BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW0\_LAST\_TLS\_GD  
BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW1\_LAST\_TLS\_GD  
BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW1\_LAST\_TLS\_GD  
BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW0\_TLS\_IE  
BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW0\_TLS\_IE  
BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW0\_LAST\_PLT\_PCREL  
BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW0\_LAST\_PLT\_PCREL  
BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW1\_LAST\_PLT\_PCREL  
BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW1\_LAST\_PLT\_PCREL  
BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW2\_LAST\_PLT\_PCREL

BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW2\_LAST\_PLT\_PCREL  
 BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW0\_LAST\_TLS\_IE  
 BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW0\_LAST\_TLS\_IE  
 BFD\_RELOC\_TILEGX\_IMM16\_X0\_HW1\_LAST\_TLS\_IE  
 BFD\_RELOC\_TILEGX\_IMM16\_X1\_HW1\_LAST\_TLS\_IE  
 BFD\_RELOC\_TILEGX\_TLS\_DTPMOD64  
 BFD\_RELOC\_TILEGX\_TLS\_DTPOFF64  
 BFD\_RELOC\_TILEGX\_TLS\_TPOFF64  
 BFD\_RELOC\_TILEGX\_TLS\_DTPMOD32  
 BFD\_RELOC\_TILEGX\_TLS\_DTPOFF32  
 BFD\_RELOC\_TILEGX\_TLS\_TPOFF32  
 BFD\_RELOC\_TILEGX\_TLS\_GD\_CALL  
 BFD\_RELOC\_TILEGX\_IMM8\_X0\_TLS\_GD\_ADD  
 BFD\_RELOC\_TILEGX\_IMM8\_X1\_TLS\_GD\_ADD  
 BFD\_RELOC\_TILEGX\_IMM8\_Y0\_TLS\_GD\_ADD  
 BFD\_RELOC\_TILEGX\_IMM8\_Y1\_TLS\_GD\_ADD  
 BFD\_RELOC\_TILEGX\_TLS\_IE\_LOAD  
 BFD\_RELOC\_TILEGX\_IMM8\_X0\_TLS\_ADD  
 BFD\_RELOC\_TILEGX\_IMM8\_X1\_TLS\_ADD  
 BFD\_RELOC\_TILEGX\_IMM8\_Y0\_TLS\_ADD  
 BFD\_RELOC\_TILEGX\_IMM8\_Y1\_TLS\_ADD

Tilera TILE-Gx Relocations.

BFD\_RELOC\_BPF\_64  
 BFD\_RELOC\_BPF\_DISP32  
 BFD\_RELOC\_BPF\_DISPCALL32  
 BFD\_RELOC\_BPF\_DISP16

Linux eBPF relocations.

BFD\_RELOC\_EIPPHANY\_SIMM8

Adapteva EIPPHANY - 8 bit signed pc-relative displacement.

BFD\_RELOC\_EIPPHANY\_SIMM24

Adapteva EIPPHANY - 24 bit signed pc-relative displacement.

BFD\_RELOC\_EIPPHANY\_HIGH

Adapteva EIPPHANY - 16 most-significant bits of absolute address.

BFD\_RELOC\_EIPPHANY\_LOW

Adapteva EIPPHANY - 16 least-significant bits of absolute address.

BFD\_RELOC\_EIPPHANY\_SIMM11

Adapteva EIPPHANY - 11 bit signed number - add/sub immediate.

BFD\_RELOC\_EIPPHANY\_IMM11

Adapteva EIPPHANY - 11 bit sign-magnitude number (ld/st displacement).

BFD\_RELOC\_EIPPHANY\_IMM8

Adapteva EIPPHANY - 8 bit immediate for 16 bit mov instruction.

BFD\_RELOC\_VISIUM\_HI16  
BFD\_RELOC\_VISIUM\_LO16  
BFD\_RELOC\_VISIUM\_IM16  
BFD\_RELOC\_VISIUM\_REL16  
BFD\_RELOC\_VISIUM\_HI16\_PCREL  
BFD\_RELOC\_VISIUM\_LO16\_PCREL  
BFD\_RELOC\_VISIUM\_IM16\_PCREL

Visium Relocations.

BFD\_RELOC\_WASM32\_LEB128  
BFD\_RELOC\_WASM32\_LEB128\_GOT  
BFD\_RELOC\_WASM32\_LEB128\_GOT\_CODE  
BFD\_RELOC\_WASM32\_LEB128\_PLT  
BFD\_RELOC\_WASM32\_PLT\_INDEX  
BFD\_RELOC\_WASM32\_ABS32\_CODE  
BFD\_RELOC\_WASM32\_CODE\_POINTER  
BFD\_RELOC\_WASM32\_INDEX  
BFD\_RELOC\_WASM32\_PLT\_SIG

WebAssembly relocations.

BFD\_RELOC\_CKCORE\_NONE  
BFD\_RELOC\_CKCORE\_ADDR32  
BFD\_RELOC\_CKCORE\_PCREL\_IMM8BY4  
BFD\_RELOC\_CKCORE\_PCREL\_IMM11BY2  
BFD\_RELOC\_CKCORE\_PCREL\_IMM4BY2  
BFD\_RELOC\_CKCORE\_PCREL32  
BFD\_RELOC\_CKCORE\_PCREL\_JSR\_IMM11BY2  
BFD\_RELOC\_CKCORE\_GNU\_VTINHERIT  
BFD\_RELOC\_CKCORE\_GNU\_VTENTRY  
BFD\_RELOC\_CKCORE\_RELATIVE  
BFD\_RELOC\_CKCORE\_COPY  
BFD\_RELOC\_CKCORE\_GLOB\_DAT  
BFD\_RELOC\_CKCORE\_JUMP\_SLOT  
BFD\_RELOC\_CKCORE\_GOTOFF  
BFD\_RELOC\_CKCORE\_GOTPC  
BFD\_RELOC\_CKCORE\_GOT32  
BFD\_RELOC\_CKCORE\_PLT32  
BFD\_RELOC\_CKCORE\_ADDRGOT  
BFD\_RELOC\_CKCORE\_ADDRPLT  
BFD\_RELOC\_CKCORE\_PCREL\_IMM26BY2  
BFD\_RELOC\_CKCORE\_PCREL\_IMM16BY2  
BFD\_RELOC\_CKCORE\_PCREL\_IMM16BY4  
BFD\_RELOC\_CKCORE\_PCREL\_IMM10BY2  
BFD\_RELOC\_CKCORE\_PCREL\_IMM10BY4  
BFD\_RELOC\_CKCORE\_ADDR\_HI16  
BFD\_RELOC\_CKCORE\_ADDR\_LO16  
BFD\_RELOC\_CKCORE\_GOTPC\_HI16  
BFD\_RELOC\_CKCORE\_GOTPC\_LO16

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BFD_RELOC_CKCORE_GOTOFF_HI16
BFD_RELOC_CKCORE_GOTOFF_LO16
BFD_RELOC_CKCORE_GOT12
BFD_RELOC_CKCORE_GOT_HI16
BFD_RELOC_CKCORE_GOT_LO16
BFD_RELOC_CKCORE_PLT12
BFD_RELOC_CKCORE_PLT_HI16
BFD_RELOC_CKCORE_PLT_LO16
BFD_RELOC_CKCORE_ADDRGOT_HI16
BFD_RELOC_CKCORE_ADDRGOT_LO16
BFD_RELOC_CKCORE_ADDRPLT_HI16
BFD_RELOC_CKCORE_ADDRPLT_LO16
BFD_RELOC_CKCORE_PCREL_JSR_IMM26BY2
BFD_RELOC_CKCORE_TOFFSET_LO16
BFD_RELOC_CKCORE_DOFFSET_LO16
BFD_RELOC_CKCORE_PCREL_IMM18BY2
BFD_RELOC_CKCORE_DOFFSET_IMM18
BFD_RELOC_CKCORE_DOFFSET_IMM18BY2
BFD_RELOC_CKCORE_DOFFSET_IMM18BY4
BFD_RELOC_CKCORE_GOTOFF_IMM18
BFD_RELOC_CKCORE_GOT_IMM18BY4
BFD_RELOC_CKCORE_PLT_IMM18BY4
BFD_RELOC_CKCORE_PCREL_IMM7BY4
BFD_RELOC_CKCORE_TLS_LE32
BFD_RELOC_CKCORE_TLS_IE32
BFD_RELOC_CKCORE_TLS_GD32
BFD_RELOC_CKCORE_TLS_LDM32
BFD_RELOC_CKCORE_TLS_LD032
BFD_RELOC_CKCORE_TLS_DTPMOD32
BFD_RELOC_CKCORE_TLS_DTPOFF32
BFD_RELOC_CKCORE_TLS_TPOFF32
BFD_RELOC_CKCORE_PCREL_FLRW_IMM8BY4
BFD_RELOC_CKCORE_NOJSRI
BFD_RELOC_CKCORE_CALLGRAPH
BFD_RELOC_CKCORE_IRELATIVE
BFD_RELOC_CKCORE_PCREL_BLOOP_IMM4BY4
BFD_RELOC_CKCORE_PCREL_BLOOP_IMM12BY4

```

C-SKY relocations.

```
BFD_RELOC_S12Z_OPR
```

Freescale S12Z relocations.

```
BFD_RELOC_S12Z_15_PCREL
```

This is a 15 bit relative address. If the most significant bits are all zero then it may be truncated to 8 bits.

```
BFD_RELOC_LARCH_TLS_DTPMOD32
```

```
BFD_RELOC_LARCH_TLS_DTPREL32
```

BFD\_RELOC\_LARCH\_TLS\_DTPMOD64  
BFD\_RELOC\_LARCH\_TLS\_DTPREL64  
BFD\_RELOC\_LARCH\_TLS\_TPREL32  
BFD\_RELOC\_LARCH\_TLS\_TPREL64  
BFD\_RELOC\_LARCH\_TLS\_DESC32  
BFD\_RELOC\_LARCH\_TLS\_DESC64  
BFD\_RELOC\_LARCH\_MARK\_LA  
BFD\_RELOC\_LARCH\_MARK\_PCREL  
BFD\_RELOC\_LARCH\_SOP\_PUSH\_PCREL  
BFD\_RELOC\_LARCH\_SOP\_PUSH\_ABSOLUTE  
BFD\_RELOC\_LARCH\_SOP\_PUSH\_DUP  
BFD\_RELOC\_LARCH\_SOP\_PUSH\_GPREL  
BFD\_RELOC\_LARCH\_SOP\_PUSH\_TLS\_TPREL  
BFD\_RELOC\_LARCH\_SOP\_PUSH\_TLS\_GOT  
BFD\_RELOC\_LARCH\_SOP\_PUSH\_TLS\_GD  
BFD\_RELOC\_LARCH\_SOP\_PUSH\_PLT\_PCREL  
BFD\_RELOC\_LARCH\_SOP\_ASSERT  
BFD\_RELOC\_LARCH\_SOP\_NOT  
BFD\_RELOC\_LARCH\_SOP\_SUB  
BFD\_RELOC\_LARCH\_SOP\_SL  
BFD\_RELOC\_LARCH\_SOP\_SR  
BFD\_RELOC\_LARCH\_SOP\_ADD  
BFD\_RELOC\_LARCH\_SOP\_AND  
BFD\_RELOC\_LARCH\_SOP\_IF\_ELSE  
BFD\_RELOC\_LARCH\_SOP\_POP\_32\_S\_10\_5  
BFD\_RELOC\_LARCH\_SOP\_POP\_32\_U\_10\_12  
BFD\_RELOC\_LARCH\_SOP\_POP\_32\_S\_10\_12  
BFD\_RELOC\_LARCH\_SOP\_POP\_32\_S\_10\_16  
BFD\_RELOC\_LARCH\_SOP\_POP\_32\_S\_10\_16\_S2  
BFD\_RELOC\_LARCH\_SOP\_POP\_32\_S\_5\_20  
BFD\_RELOC\_LARCH\_SOP\_POP\_32\_S\_0\_5\_10\_16\_S2  
BFD\_RELOC\_LARCH\_SOP\_POP\_32\_S\_0\_10\_10\_16\_S2  
BFD\_RELOC\_LARCH\_SOP\_POP\_32\_U  
BFD\_RELOC\_LARCH\_ADD8  
BFD\_RELOC\_LARCH\_ADD16  
BFD\_RELOC\_LARCH\_ADD24  
BFD\_RELOC\_LARCH\_ADD32  
BFD\_RELOC\_LARCH\_ADD64  
BFD\_RELOC\_LARCH\_SUB8  
BFD\_RELOC\_LARCH\_SUB16  
BFD\_RELOC\_LARCH\_SUB24  
BFD\_RELOC\_LARCH\_SUB32  
BFD\_RELOC\_LARCH\_SUB64  
BFD\_RELOC\_LARCH\_B16  
BFD\_RELOC\_LARCH\_B21  
BFD\_RELOC\_LARCH\_B26  
BFD\_RELOC\_LARCH\_ABS\_HI20

BFD\_RELOC\_LARCH\_ABS\_L012  
BFD\_RELOC\_LARCH\_ABS64\_L020  
BFD\_RELOC\_LARCH\_ABS64\_HI12  
BFD\_RELOC\_LARCH\_PCALA\_HI20  
BFD\_RELOC\_LARCH\_PCALA\_L012  
BFD\_RELOC\_LARCH\_PCALA64\_L020  
BFD\_RELOC\_LARCH\_PCALA64\_HI12  
BFD\_RELOC\_LARCH\_GOT\_PC\_HI20  
BFD\_RELOC\_LARCH\_GOT\_PC\_L012  
BFD\_RELOC\_LARCH\_GOT64\_PC\_L020  
BFD\_RELOC\_LARCH\_GOT64\_PC\_HI12  
BFD\_RELOC\_LARCH\_GOT\_HI20  
BFD\_RELOC\_LARCH\_GOT\_L012  
BFD\_RELOC\_LARCH\_GOT64\_L020  
BFD\_RELOC\_LARCH\_GOT64\_HI12  
BFD\_RELOC\_LARCH\_TLS\_LE\_HI20  
BFD\_RELOC\_LARCH\_TLS\_LE\_L012  
BFD\_RELOC\_LARCH\_TLS\_LE64\_L020  
BFD\_RELOC\_LARCH\_TLS\_LE64\_HI12  
BFD\_RELOC\_LARCH\_TLS\_IE\_PC\_HI20  
BFD\_RELOC\_LARCH\_TLS\_IE\_PC\_L012  
BFD\_RELOC\_LARCH\_TLS\_IE64\_PC\_L020  
BFD\_RELOC\_LARCH\_TLS\_IE64\_PC\_HI12  
BFD\_RELOC\_LARCH\_TLS\_IE\_HI20  
BFD\_RELOC\_LARCH\_TLS\_IE\_L012  
BFD\_RELOC\_LARCH\_TLS\_IE64\_L020  
BFD\_RELOC\_LARCH\_TLS\_IE64\_HI12  
BFD\_RELOC\_LARCH\_TLS\_LD\_PC\_HI20  
BFD\_RELOC\_LARCH\_TLS\_LD\_HI20  
BFD\_RELOC\_LARCH\_TLS\_GD\_PC\_HI20  
BFD\_RELOC\_LARCH\_TLS\_GD\_HI20  
BFD\_RELOC\_LARCH\_32\_PCREL  
BFD\_RELOC\_LARCH\_RELAX  
BFD\_RELOC\_LARCH\_DELETE  
BFD\_RELOC\_LARCH\_ALIGN  
BFD\_RELOC\_LARCH\_PCREL20\_S2  
BFD\_RELOC\_LARCH\_CFA  
BFD\_RELOC\_LARCH\_ADD6  
BFD\_RELOC\_LARCH\_SUB6  
BFD\_RELOC\_LARCH\_ADD\_ULEB128  
BFD\_RELOC\_LARCH\_SUB\_ULEB128  
BFD\_RELOC\_LARCH\_64\_PCREL  
BFD\_RELOC\_LARCH\_CALL36  
BFD\_RELOC\_LARCH\_TLS\_DESC\_PC\_HI20  
BFD\_RELOC\_LARCH\_TLS\_DESC\_PC\_L012  
BFD\_RELOC\_LARCH\_TLS\_DESC64\_PC\_L020  
BFD\_RELOC\_LARCH\_TLS\_DESC64\_PC\_HI12

```

BFD_RELOC_LARCH_TLS_DESC_HI20
BFD_RELOC_LARCH_TLS_DESC_LO12
BFD_RELOC_LARCH_TLS_DESC64_LO20
BFD_RELOC_LARCH_TLS_DESC64_HI12
BFD_RELOC_LARCH_TLS_DESC_LD
BFD_RELOC_LARCH_TLS_DESC_CALL
BFD_RELOC_LARCH_TLS_LE_HI20_R
BFD_RELOC_LARCH_TLS_LE_ADD_R
BFD_RELOC_LARCH_TLS_LE_LO12_R
BFD_RELOC_LARCH_TLS_LD_PCREL20_S2
BFD_RELOC_LARCH_TLS_GD_PCREL20_S2
BFD_RELOC_LARCH_TLS_DESC_PCREL20_S2
BFD_RELOC_LARCH_CALL30
BFD_RELOC_LARCH_PCADD_HI20
BFD_RELOC_LARCH_PCADD_LO12
BFD_RELOC_LARCH_GOT_PCADD_HI20
BFD_RELOC_LARCH_GOT_PCADD_LO12
BFD_RELOC_LARCH_TLS_IE_PCADD_HI20
BFD_RELOC_LARCH_TLS_IE_PCADD_LO12
BFD_RELOC_LARCH_TLS_LD_PCADD_HI20
BFD_RELOC_LARCH_TLS_LD_PCADD_LO12
BFD_RELOC_LARCH_TLS_GD_PCADD_HI20
BFD_RELOC_LARCH_TLS_GD_PCADD_LO12
BFD_RELOC_LARCH_TLS_DESC_PCADD_HI20
BFD_RELOC_LARCH_TLS_DESC_PCADD_LO12
    LARCH relocations.

```

```
typedef enum bfd_reloc_code_real bfd_reloc_code_real_type;
```

### 2.10.2.2 bfd\_reloc\_type\_lookup

```

reloc_howto_type *bfd_reloc_type_lookup (bfd *abfd,           [Function]
    bfd_reloc_code_real_type code); reloc_howto_type
    *bfd_reloc_name_lookup (bfd *abfd, const char *reloc_name);
    Return a pointer to a howto structure which, when invoked, will perform the relocation code on data from the architecture noted.

```

### 2.10.2.3 bfd\_default\_reloc\_type\_lookup

```

reloc_howto_type *bfd_default_reloc_type_lookup (bfd *abfd,   [Function]
    bfd_reloc_code_real_type code);
    Provides a default relocation lookup routine for any architecture.

```



#### 2.10.2.4 `bfd_get_reloc_code_name`

`const char *bfd_get_reloc_code_name (bfd_reloc_code_real_type [Function]  
code);`

Provides a printable name for the supplied relocation code. Useful mainly for printing error messages.

#### 2.10.2.5 `bfd_generic_relax_section`

`bool bfd_generic_relax_section (bfd *abfd, asection *section, [Function]  
struct bfd_link_info *, bool *);`

Provides default handling for relaxing for back ends which don't do relaxing.

#### 2.10.2.6 `bfd_generic_gc_sections`

`bool bfd_generic_gc_sections (bfd *, struct bfd_link_info *); [Function]`

Provides default handling for relaxing for back ends which don't do section gc – i.e., does nothing.

#### 2.10.2.7 `bfd_generic_lookup_section_flags`

`bool bfd_generic_lookup_section_flags (struct bfd_link_info *, [Function]  
struct flag_info *, asection *);`

Provides default handling for section flags lookup – i.e., does nothing. Returns FALSE if the section should be omitted, otherwise TRUE.

#### 2.10.2.8 `bfd_generic_get_relocated_section_contents`

`bfd_byte *bfd_generic_get_relocated_section_contents (bfd [Function]  
*abfd, struct bfd_link_info *link_info, struct bfd_link_order *link_order,  
bfd_byte *data, bool relocatable, asymbol **symbols);`

Provides default handling of relocation effort for back ends which can't be bothered to do it efficiently.

#### 2.10.2.9 `_bfd_generic_finalize_section_relocs`

`bool _bfd_generic_finalize_section_relocs (bfd *abfd, sec_ptr [Function]  
section, arelent **relptr, unsigned int count);`

Installs a new set of internal relocations in SECTION.

#### 2.10.2.10 `_bfd_unrecognized_reloc`

`bool _bfd_unrecognized_reloc (bfd * abfd, sec_ptr section, [Function]  
unsigned int r_type);`

Reports an unrecognized reloc. Written as a function in order to reduce code duplication. Returns FALSE so that it can be called from a return statement.

### 2.10.2.11 \_bfd\_link\_reloc\_status\_error

```
void _bfd_link_reloc_status_error (bfd *abfd, struct          [Function]
    bfd_link_info *link_info, asection *input_section, arelent *reloc_entry,
    char *error_message, bfd_reloc_status_type r);
```

Mark a link relocation error according to R, with a suitable message as applicable.  
Written as a function in order to reduce code duplication.

## 2.11 Core files

### 2.11.1 Core file functions

These are functions pertaining to core files.

#### 2.11.1.1 bfd\_core\_file\_failing\_command

```
const char *bfd_core_file_failing_command (bfd *abfd);          [Function]
```

Return a read-only string explaining which program was running when it failed and produced the core file *abfd*.

#### 2.11.1.2 bfd\_core\_file\_failing\_signal

```
int bfd_core_file_failing_signal (bfd *abfd);                  [Function]
```

Returns the signal number which caused the core dump which generated the file the BFD *abfd* is attached to.

#### 2.11.1.3 bfd\_core\_file\_pid

```
int bfd_core_file_pid (bfd *abfd);                             [Function]
```

Returns the PID of the process the core dump the BFD *abfd* is attached to was generated from.

#### 2.11.1.4 core\_file\_matches\_executable\_p

```
bool core_file_matches_executable_p (bfd *core_bfd, bfd        [Function]
    *exec_bfd);
```

Return TRUE if the core file attached to *core\_bfd* was generated by a run of the executable file attached to *exec\_bfd*, FALSE otherwise.

#### 2.11.1.5 generic\_core\_file\_matches\_executable\_p

```
bool generic_core_file_matches_executable_p (bfd *core_bfd,    [Function]
    bfd *exec_bfd);
```

Return TRUE if the core file attached to *core\_bfd* was generated by a run of the executable file attached to *exec\_bfd*. The match is based on executable basenames only.

Note: When not able to determine the core file failing command or the executable name, we still return TRUE even though we're not sure that core file and executable match. This is to avoid generating a false warning in situations where we really don't know whether they match or not.

## 2.12 Targets

Each port of BFD to a different machine requires the creation of a target back end. All the back end provides to the root part of BFD is a structure containing pointers to functions which perform certain low level operations on files. BFD translates the applications's requests through a pointer into calls to the back end routines.

When a file is opened with `bfd_openr`, its format and target are unknown. BFD uses various mechanisms to determine how to interpret the file. The operations performed are:

- Create a BFD by calling the internal routine `_bfd_new_bfd`, then call `bfd_find_target` with the target string supplied to `bfd_openr` and the new BFD pointer.
- If a null target string was provided to `bfd_find_target`, look up the environment variable `GNUTARGET` and use that as the target string.
- If the target string is still `NULL`, or the target string is `default`, then use the first item in the target vector as the target type, and set `target_defaulted` in the BFD to cause `bfd_check_format` to loop through all the targets. See Section 2.12.1 [`bfd_target`], page 138. See Section 2.9 [Formats], page 49.
- Otherwise, inspect the elements in the target vector one by one, until a match on target name is found. When found, use it.
- Otherwise return the error `bfd_error_invalid_target` to `bfd_openr`.
- `bfd_openr` attempts to open the file using `bfd_open_file`, and returns the BFD.

Once the BFD has been opened and the target selected, the file format may be determined. This is done by calling `bfd_check_format` on the BFD with a suggested format. If `target_defaulted` has been set, each possible target type is tried to see if it recognizes the specified format. `bfd_check_format` returns `TRUE` when the caller guesses right.

### 2.12.1 `bfd_target`

This structure contains everything that BFD knows about a target. It includes things like its byte order, name, and which routines to call to do various operations.

Every BFD points to a target structure with its `xvec` member.

The macros below are used to dispatch to functions through the `bfd_target` vector. They are used in a number of macros further down in `bfd.h`, and are also used when calling various routines by hand inside the BFD implementation. The `arglist` argument must be parenthesized; it contains all the arguments to the called function.

They make the documentation (more) unpleasant to read, so if someone wants to fix this and not break the above, please do.

```
#define BFD_SEND(bfd, message, arglist) \
    ((*((bfd)->xvec->message)) arglist)

#ifdef DEBUG_BFD_SEND
#undef BFD_SEND
#define BFD_SEND(bfd, message, arglist) \
    (((bfd) && (bfd)->xvec && (bfd)->xvec->message) ? \
     ((*((bfd)->xvec->message)) arglist) : \
     (bfd_assert (__FILE__, __LINE__), NULL))
```

```
#endif
```

For operations which index on the BFD format:

```
#define BFD_SEND_FMT(bfd, message, arglist) \
    (((bfd)->xvec->message[(int) ((bfd)->format)]) arglist)
```

```
#ifdef DEBUG_BFD_SEND
```

```
#undef BFD_SEND_FMT
```

```
#define BFD_SEND_FMT(bfd, message, arglist) \
    (((bfd) && (bfd)->xvec && (bfd)->xvec->message) ? \
     (((bfd)->xvec->message[(int) ((bfd)->format)]) arglist) : \
     (bfd_assert (__FILE__, __LINE__), NULL))
```

```
#endif
```

```
/* Defined to TRUE if unused section symbol should be kept. */
```

```
#ifndef TARGET_KEEP_UNUSED_SECTION_SYMBOLS
```

```
#define TARGET_KEEP_UNUSED_SECTION_SYMBOLS true
```

```
#endif
```

```
/* Define to TRUE if section merging is supported by the backend. */
```

```
#ifndef TARGET_MERGE_SECTIONS
```

```
#define TARGET_MERGE_SECTIONS false
```

```
#endif
```

This is the structure which defines the type of BFD this is. The `xvec` member of the struct `bfd` itself points here. Each module that implements access to a different target under BFD, defines one of these.

FIXME, these names should be rationalised with the names of the entry points which call them. Too bad we can't have one macro to define them both!

```
typedef struct bfd_target
{
    /* Identifies the kind of target, e.g., SunOS4, Ultrix, etc. */
    const char *name;

    /* The "flavour" of a back end is a general indication about
       the contents of a file. */
    enum bfd_flavour flavour;

    /* The order of bytes within the data area of a file. */
    enum bfd_endian byteorder;

    /* The order of bytes within the header parts of a file. */
    enum bfd_endian header_byteorder;

    /* A mask of all the flags which an executable may have set -
       from the set BFD_NO_FLAGS, HAS_RELOC, ...D_PAGED. */
    flagword object_flags;
```

```

/* A mask of all the flags which a section may have set - from
   the set SEC_NO_FLAGS, SEC_ALLOC, ...SET_NEVER_LOAD. */
flagword section_flags;

/* The character normally found at the front of a symbol.
   (if any), perhaps '_'. */
char symbol_leading_char;

/* The pad character for file names within an archive header. */
char ar_pad_char;

/* The maximum number of characters in an archive header. */
unsigned char ar_max_namelen;

/* How well this target matches, used to select between various
   possible targets when more than one target matches. */
unsigned char match_priority;

/* TRUE if unused section symbols should be kept. */
bool keep_unused_section_symbols;

/* TRUE if section merging is supported by the backend. */
bool merge_sections;

/* Entries for byte swapping for data. These are different from the
   other entry points, since they don't take a BFD as the first argument.
   Certain other handlers could do the same. */
uint64_t      (*bfd_getx64) (const void *);
int64_t       (*bfd_getx_signed_64) (const void *);
void          (*bfd_putx64) (uint64_t, void *);
bfd_vma       (*bfd_getx32) (const void *);
bfd_signed_vma (*bfd_getx_signed_32) (const void *);
void          (*bfd_putx32) (bfd_vma, void *);
bfd_vma       (*bfd_getx16) (const void *);
bfd_signed_vma (*bfd_getx_signed_16) (const void *);
void          (*bfd_putx16) (bfd_vma, void *);

/* Byte swapping for the headers. */
uint64_t      (*bfd_h_getx64) (const void *);
int64_t       (*bfd_h_getx_signed_64) (const void *);
void          (*bfd_h_putx64) (uint64_t, void *);
bfd_vma       (*bfd_h_getx32) (const void *);
bfd_signed_vma (*bfd_h_getx_signed_32) (const void *);
void          (*bfd_h_putx32) (bfd_vma, void *);
bfd_vma       (*bfd_h_getx16) (const void *);
bfd_signed_vma (*bfd_h_getx_signed_16) (const void *);

```

```

void                (*bfd_h_putx16) (bfd_vma, void *);

/* Format dependent routines: these are vectors of entry points
   within the target vector structure, one for each format to check. */

/* Check the format of a file being read. Return a bfd_cleanup on
   success or zero on failure. */
bfd_cleanup (*bfd_check_format[bfd_type_end]) (bfd *);

/* Set the format of a file being written. */
bool (*bfd_set_format[bfd_type_end]) (bfd *);

/* Write cached information into a file being written, at bfd_close. */
bool (*bfd_write_contents[bfd_type_end]) (bfd *);

```

The general target vector. These vectors are initialized using the BFD\_JUMP\_TABLE macros.

```

/* Generic entry points. */
#define BFD_JUMP_TABLE_GENERIC(NAME) \
NAME##_close_and_cleanup, \
NAME##_bfd_free_cached_info, \
NAME##_new_section_hook, \
NAME##_get_section_contents

/* Called when the BFD is being closed to do any necessary cleanup. */
bool (*_close_and_cleanup) (bfd *);
/* Ask the BFD to free all cached information. */
bool (*_bfd_free_cached_info) (bfd *);
/* Called when a new section is created. */
bool (*_new_section_hook) (bfd *, sec_ptr);
/* Read the contents of a section. */
bool (*_bfd_get_section_contents) (bfd *, sec_ptr, void *, file_ptr,
                                   bfd_size_type);

/* Entry points to copy private data. */
#define BFD_JUMP_TABLE_COPY(NAME) \
NAME##_bfd_copy_private_bfd_data, \
NAME##_bfd_merge_private_bfd_data, \
NAME##_bfd_copy_private_section_data, \
NAME##_bfd_copy_private_symbol_data, \
NAME##_bfd_copy_private_header_data, \
NAME##_bfd_set_private_flags, \
NAME##_bfd_print_private_bfd_data

/* Called to copy BFD general private data from one object file
   to another. */

```

```

bool (*_bfd_copy_private_bfd_data) (bfd *, bfd *);
/* Called to merge BFD general private data from one object file
   to a common output file when linking. */
bool (*_bfd_merge_private_bfd_data) (bfd *, struct bfd_link_info *);
/* Called to copy BFD private section data from one object file
   to another. */
bool (*_bfd_copy_private_section_data) (bfd *, sec_ptr, bfd *, sec_ptr,
                                         struct bfd_link_info *);
/* Called to copy BFD private symbol data from one symbol
   to another. */
bool (*_bfd_copy_private_symbol_data) (bfd *, asymbol **,
                                         bfd *, asymbol **);
/* Called to copy BFD private header data from one object file
   to another. */
bool (*_bfd_copy_private_header_data) (bfd *, bfd *);
/* Called to set private backend flags. */
bool (*_bfd_set_private_flags) (bfd *, flagword);

/* Called to print private BFD data. */
bool (*_bfd_print_private_bfd_data) (bfd *, void *);

/* Core file entry points. */
#define BFD_JUMP_TABLE_CORE(NAME) \
  NAME##_core_file_failing_command, \
  NAME##_core_file_failing_signal, \
  NAME##_core_file_matches_executable_p, \
  NAME##_core_file_pid

char (*_core_file_failing_command) (bfd *);
int (*_core_file_failing_signal) (bfd *);
bool (*_core_file_matches_executable_p) (bfd *, bfd *);
int (*_core_file_pid) (bfd *);

/* Archive entry points. */
#define BFD_JUMP_TABLE_ARCHIVE(NAME) \
  NAME##_slurp_armap, \
  NAME##_slurp_extended_name_table, \
  NAME##_construct_extended_name_table, \
  NAME##_truncate_arname, \
  NAME##_write_armap, \
  NAME##_read_ar_hdr, \
  NAME##_write_ar_hdr, \
  NAME##_openr_next_archived_file, \
  NAME##_get_elt_at_index, \
  NAME##_generic_stat_arch_elt, \
  NAME##_update_armap_timestamp

```

```

bool (*_bfd_slurp_armap) (bfd *);
bool (*_bfd_slurp_extended_name_table) (bfd *);
bool (*_bfd_construct_extended_name_table) (bfd *, char **,
                                           bfd_size_type *,
                                           const char **);

void (*_bfd_truncate_arname) (bfd *, const char *, char *);
bool (*write_armap) (bfd *, unsigned, struct orl *, unsigned, int);
void (*_bfd_read_ar_hdr_fn) (bfd *);
bool (*_bfd_write_ar_hdr_fn) (bfd *, bfd *);
bfd *(*openr_next_archived_file) (bfd *, bfd *);
#define bfd_get_elt_at_index(b,i) \
    BFD_SEND (b, _bfd_get_elt_at_index, (b,i))
bfd *(*_bfd_get_elt_at_index) (bfd *, symindex);
int (*_bfd_stat_arch_elt) (bfd *, struct stat *);
bool (*_bfd_update_armap_timestamp) (bfd *);

/* Entry points used for symbols. */
#define BFD_JUMP_TABLE_SYMBOLS(NAME) \
    NAME##_get_symtab_upper_bound, \
    NAME##_canonicalize_symtab, \
    NAME##_make_empty_symbol, \
    NAME##_print_symbol, \
    NAME##_get_symbol_info, \
    NAME##_get_symbol_version_string, \
    NAME##_bfd_is_local_label_name, \
    NAME##_bfd_is_target_special_symbol, \
    NAME##_get_lineno, \
    NAME##_find_nearest_line, \
    NAME##_find_nearest_line_with_alt, \
    NAME##_find_line, \
    NAME##_find_inliner_info, \
    NAME##_bfd_make_debug_symbol, \
    NAME##_read_minisymbols, \
    NAME##_minisymbol_to_symbol

long (*_bfd_get_symtab_upper_bound) (bfd *);
long (*_bfd_canonicalize_symtab) (bfd *, struct bfd_symbol **);
struct bfd_symbol *
    (*_bfd_make_empty_symbol) (bfd *);
void (*_bfd_print_symbol) (bfd *, void *, struct bfd_symbol *,
                           bfd_print_symbol_type);
#define bfd_print_symbol(b,p,s,e) \
    BFD_SEND (b, _bfd_print_symbol, (b,p,s,e))
void (*_bfd_get_symbol_info) (bfd *, struct bfd_symbol *, symbol_info *);
#define bfd_get_symbol_info(b,p,e) \
    BFD_SEND (b, _bfd_get_symbol_info, (b,p,e))
const char *

```



```

        (*_bfd_get_symbol_version_string) (bfd *, struct bfd_symbol *,
                                           bool, bool *);
#define bfd_get_symbol_version_string(b,s,p,h) \
        BFD_SEND (b, _bfd_get_symbol_version_string, (b,s,p,h))
bool (*_bfd_is_local_label_name) (bfd *, const char *);
bool (*_bfd_is_target_special_symbol) (bfd *, asymbol *);
arelent *
        (*_get_lineno) (bfd *, struct bfd_symbol *);
bool (*_bfd_find_nearest_line) (bfd *, struct bfd_symbol **,
                                struct bfd_section *, bfd_vma,
                                const char **, const char **,
                                unsigned int *, unsigned int *);
bool (*_bfd_find_nearest_line_with_alt) (bfd *, const char *,
                                          struct bfd_symbol **,
                                          struct bfd_section *, bfd_vma,
                                          const char **, const char **,
                                          unsigned int *, unsigned int *);
bool (*_bfd_find_line) (bfd *, struct bfd_symbol **,
                        struct bfd_symbol *, const char **,
                        unsigned int *);
bool (*_bfd_find_inliner_info)
        (bfd *, const char **, const char **, unsigned int *);
/* Back-door to allow format-aware applications to create debug symbols
   while using BFD for everything else.  Currently used by the assembler
   when creating COFF files.  */
asymbol *
        (*_bfd_make_debug_symbol) (bfd *);
#define bfd_read_minisymbols(b, d, m, s) \
        BFD_SEND (b, _read_minisymbols, (b, d, m, s))
long (*_read_minisymbols) (bfd *, bool, void **, unsigned int *);
#define bfd_minisymbol_to_symbol(b, d, m, f) \
        BFD_SEND (b, _minisymbol_to_symbol, (b, d, m, f))
asymbol *
        (*_minisymbol_to_symbol) (bfd *, bool, const void *, asymbol *);

/* Routines for relocs.  */
#define BFD_JUMP_TABLE_RELOCS(NAME) \
        NAME##_get_reloc_upper_bound, \
        NAME##_canonicalize_reloc, \
        NAME##_finalize_section_relocs, \
        NAME##_bfd_reloc_type_lookup, \
        NAME##_bfd_reloc_name_lookup

long (*_get_reloc_upper_bound) (bfd *, sec_ptr);
long (*_bfd_canonicalize_reloc) (bfd *, sec_ptr, arelent **,
                                struct bfd_symbol **);
bool (*_bfd_finalize_section_relocs) (bfd *, sec_ptr, arelent **,

```

[illegible]

```

bool (*_bfd_relax_section) (bfd *, struct bfd_section *,
                           struct bfd_link_info *, bool *);

/* Create a hash table for the linker.  Different backends store
   different information in this table.  */
struct bfd_link_hash_table *
  (*_bfd_link_hash_table_create) (bfd *);

/* Add symbols from this object file into the hash table.  */
bool (*_bfd_link_add_symbols) (bfd *, struct bfd_link_info *);

/* Indicate that we are only retrieving symbol values from this section.  */
void (*_bfd_link_just_syms) (asection *, struct bfd_link_info *);

/* Copy the symbol type and other attributes for a linker script
   assignment of one symbol to another.  */
#define bfd_copy_link_hash_symbol_type(b, t, f) \
  BFD_SEND (b, _bfd_copy_link_hash_symbol_type, (b, t, f))
void (*_bfd_copy_link_hash_symbol_type) (bfd *,
                                         struct bfd_link_hash_entry *,
                                         struct bfd_link_hash_entry *);

/* Do a link based on the link_order structures attached to each
   section of the BFD.  */
bool (*_bfd_final_link) (bfd *, struct bfd_link_info *);

/* Should this section be split up into smaller pieces during linking.  */
bool (*_bfd_link_split_section) (bfd *, struct bfd_section *);

/* Check the relocations in the bfd for validity.  */
bool (*_bfd_link_check_relocs) (bfd *, struct bfd_link_info *);

/* Remove sections that are not referenced from the output.  */
bool (*_bfd_gc_sections) (bfd *, struct bfd_link_info *);

/* Sets the bitmask of allowed and disallowed section flags.  */
bool (*_bfd_lookup_section_flags) (struct bfd_link_info *,
                                   struct flag_info *, asection *);

/* Is this section a member of a group?  */
bool (*_bfd_is_group_section) (bfd *, const struct bfd_section *);

/* The group name, if section is a member of a group.  */
const char *(*_bfd_group_name) (bfd *, const struct bfd_section *);

/* Discard members of a group.  */
bool (*_bfd_discard_group) (bfd *, struct bfd_section *);

```

```

/* Check if SEC has been already linked during a reloceatable or
   final link. */
bool (*_section_already_linked) (bfd *, asection *,
                                  struct bfd_link_info *);

/* Define a common symbol. */
bool (*_bfd_define_common_symbol) (bfd *, struct bfd_link_info *,
                                    struct bfd_link_hash_entry *);

/* Hide a symbol. */
void (*_bfd_link_hide_symbol) (bfd *, struct bfd_link_info *,
                                struct bfd_link_hash_entry *);

/* Define a __start, __stop, .startof. or .sizeof. symbol. */
struct bfd_link_hash_entry *
  (*_bfd_define_start_stop) (struct bfd_link_info *, const char *,
                             asection *);

/* Routines to handle dynamic symbols and relocs. */
#define BFD_JUMP_TABLE_DYNAMIC(NAME) \
  NAME##_get_dynamic_symtab_upper_bound, \
  NAME##_canonicalize_dynamic_symtab, \
  NAME##_get_synthetic_symtab, \
  NAME##_get_dynamic_reloc_upper_bound, \
  NAME##_canonicalize_dynamic_reloc

/* Get the amount of memory required to hold the dynamic symbols. */
long (*_bfd_get_dynamic_symtab_upper_bound) (bfd *);
/* Read in the dynamic symbols. */
long (*_bfd_canonicalize_dynamic_symtab) (bfd *, struct bfd_symbol **);
/* Create synthetized symbols. */
long (*_bfd_get_synthetic_symtab) (bfd *, long, struct bfd_symbol **,
                                    long, struct bfd_symbol **,
                                    struct bfd_symbol **);
/* Get the amount of memory required to hold the dynamic relocs. */
long (*_bfd_get_dynamic_reloc_upper_bound) (bfd *);
/* Read in the dynamic relocs. */
long (*_bfd_canonicalize_dynamic_reloc) (bfd *, arelent **,
                                         struct bfd_symbol **);

```

A pointer to an alternative `bfd_target` in case the current one is not satisfactory. This can happen when the target cpu supports both big and little endian code, and target chosen by the linker has the wrong endianness. The function `open_output()` in `ld/ldlang.c` uses this field to find an alternative output format that is suitable.

```

/* Opposite endian version of this target. */

```

```
    const struct bfd_target *alternative_target;

    /* Data for use by back-end routines, which isn't
       generic enough to belong in this structure. */
    const void *backend_data;
} bfd_target;

static inline const char *
bfd_get_target (const bfd *abfd)
{
    return abfd->xvec->name;
}

static inline enum bfd_flavour
bfd_get_flavour (const bfd *abfd)
{
    return abfd->xvec->flavour;
}

static inline flagword
bfd_applicable_file_flags (const bfd *abfd)
{
    return abfd->xvec->object_flags;
}

static inline bool
bfd_family_coff (const bfd *abfd)
{
    return (bfd_get_flavour (abfd) == bfd_target_coff_flavour
            || bfd_get_flavour (abfd) == bfd_target_xcoff_flavour);
}

static inline bool
bfd_big_endian (const bfd *abfd)
{
    return abfd->xvec->byteorder == BFD_ENDIAN_BIG;
}

static inline bool
bfd_little_endian (const bfd *abfd)
{
    return abfd->xvec->byteorder == BFD_ENDIAN_LITTLE;
}

static inline bool
bfd_header_big_endian (const bfd *abfd)
{

```

```
    return abfd->xvec->header_byteorder == BFD_ENDIAN_BIG;
}

static inline bool
bfd_header_little_endian (const bfd *abfd)
{
    return abfd->xvec->header_byteorder == BFD_ENDIAN_LITTLE;
}

static inline flagword
bfd_applicable_section_flags (const bfd *abfd)
{
    return abfd->xvec->section_flags;
}

static inline char
bfd_get_symbol_leading_char (const bfd *abfd)
{
    return abfd->xvec->symbol_leading_char;
}

static inline enum bfd_flavour
bfd_asybol_flavour (const asymbol *sy)
{
    if ((sy->flags & BSF_SYNTHETIC) != 0)
        return bfd_target_unknown_flavour;
    return sy->the_bfd->xvec->flavour;
}

static inline bool
bfd_keep_unused_section_symbols (const bfd *abfd)
{
    return abfd->xvec->keep_unused_section_symbols;
}

static inline bool
bfd_target_supports_archives (const bfd *abfd)
{
    return (abfd->xvec->_bfd_check_format[bfd_archive]
            != abfd->xvec->_bfd_check_format[bfd_unknown]);
}
```

### 2.12.1.1 bfd\_set\_default\_target

**bool** **bfd\_set\_default\_target** (*const char \*name*); [Function]  
 Set the default target vector to use when recognizing a BFD. This takes the name of the target, which may be a BFD target name or a configuration triplet.

### 2.12.1.2 bfd\_find\_target

**const bfd\_target \****bfd\_find\_target* (*const char \*target\_name, bfd \*abfd*); [Function]  
 Return a pointer to the transfer vector for the object target named *target\_name*. If *target\_name* is NULL, choose the one in the environment variable GNUTARGET; if that is null or not defined, then choose the first entry in the target list. Passing in the string "default" or setting the environment variable to "default" will cause the first entry in the target list to be returned, and "target\_defaulted" will be set in the BFD if *abfd* isn't NULL. This causes **bfd\_check\_format** to loop over all the targets to find the one that matches the file being read.

### 2.12.1.3 bfd\_get\_target\_info

**const bfd\_target \****bfd\_get\_target\_info* (*const char \*target\_name, bfd \*abfd, bool \*is\_bigendian, int \*underscoring, const char \*\*def\_target\_arch*); [Function]  
 Return a pointer to the transfer vector for the object target named *target\_name*. If *target\_name* is NULL, choose the one in the environment variable GNUTARGET; if that is null or not defined, then choose the first entry in the target list. Passing in the string "default" or setting the environment variable to "default" will cause the first entry in the target list to be returned, and "target\_defaulted" will be set in the BFD if *abfd* isn't NULL. This causes **bfd\_check\_format** to loop over all the targets to find the one that matches the file being read. If *is\_bigendian* is not NULL, then set this value to target's endian mode. True for big-endian, FALSE for little-endian or for invalid target. If *underscoring* is not NULL, then set this value to target's underscoring mode. Zero for none-underscoring, -1 for invalid target, else the value of target vector's symbol underscoring. If *def\_target\_arch* is not NULL, then set it to the architecture string specified by the target\_name.

### 2.12.1.4 bfd\_target\_list

**const char \*\****bfd\_target\_list* (*void*); [Function]  
 Return a freshly malloced NULL-terminated vector of the names of all the valid BFD targets. Do not modify the names.

### 2.12.1.5 bfd\_iterate\_over\_targets

**const bfd\_target \****bfd\_iterate\_over\_targets* (*int (\*func) (const bfd\_target \*, void \*), void \*data*); [Function]  
 Call *func* for each target in the list of BFD target vectors, passing *data* to *func*. Stop iterating if *func* returns a non-zero result, and return that target vector. Return NULL if *func* always returns zero.

### 2.12.1.6 bfd\_flavour\_name

`const char *bfd_flavour_name (enum bfd_flavour flavour);` [Function]  
 Return the string form of *flavour*.

## 2.13 Architectures

BFD keeps one atom in a BFD describing the architecture of the data attached to the BFD: a pointer to a `bfd_arch_info_type`.

Pointers to structures can be requested independently of a BFD so that an architecture's information can be interrogated without access to an open BFD.

The architecture information is provided by each architecture package. The set of default architectures is selected by the macro `SELECT_ARCHITECTURES`. This is normally set up in the `config/target.mt` file of your choice. If the name is not defined, then all the architectures supported are included.

When BFD starts up, all the architectures are called with an initialize method. It is up to the architecture back end to insert as many items into the list of architectures as it wants to; generally this would be one for each machine and one for the default case (an item with a machine field of 0).

BFD's idea of an architecture is implemented in `archures.c`.

### 2.13.1 bfd\_architecture

This enum gives the object file's CPU architecture, in a global sense—i.e., what processor family does it belong to? Another field indicates which processor within the family is in use. The machine gives a number which distinguishes different versions of the architecture, containing, for example, 68020 for Motorola 68020.

```
enum bfd_architecture
{
    bfd_arch_unknown,    /* File arch not known.  */
    bfd_arch_obscure,    /* Arch known, not one of these.  */
    bfd_arch_m68k,       /* Motorola 68xxx.  */
#define bfd_mach_m68000      1
#define bfd_mach_m68008      2
#define bfd_mach_m68010      3
#define bfd_mach_m68020      4
#define bfd_mach_m68030      5
#define bfd_mach_m68040      6
#define bfd_mach_m68060      7
#define bfd_mach_cpu32       8
#define bfd_mach_fido        9
#define bfd_mach_mcf_isa_a_nodiv 10
#define bfd_mach_mcf_isa_a    11
#define bfd_mach_mcf_isa_a_mac 12
#define bfd_mach_mcf_isa_a_emac 13
#define bfd_mach_mcf_isa_a_plus 14
#define bfd_mach_mcf_isa_a_plus_mac 15
}
```



```

#define bfd_mach_mcf_isa_aplus_emac    16
#define bfd_mach_mcf_isa_b_nousp      17
#define bfd_mach_mcf_isa_b_nousp_mac  18
#define bfd_mach_mcf_isa_b_nousp_emac 19
#define bfd_mach_mcf_isa_b            20
#define bfd_mach_mcf_isa_b_mac        21
#define bfd_mach_mcf_isa_b_emac       22
#define bfd_mach_mcf_isa_b_float      23
#define bfd_mach_mcf_isa_b_float_mac  24
#define bfd_mach_mcf_isa_b_float_emac 25
#define bfd_mach_mcf_isa_c            26
#define bfd_mach_mcf_isa_c_mac        27
#define bfd_mach_mcf_isa_c_emac       28
#define bfd_mach_mcf_isa_c_nodiv      29
#define bfd_mach_mcf_isa_c_nodiv_mac  30
#define bfd_mach_mcf_isa_c_nodiv_emac 31
    bfd_arch_vax,          /* DEC Vax. */

    bfd_arch_or1k,        /* OpenRISC 1000. */
#define bfd_mach_or1k      1
#define bfd_mach_or1knd    2

    bfd_arch_sparc,       /* SPARC. */
#define bfd_mach_sparc      1
/* The difference between v8plus and v9 is that v9 is a true 64 bit env. */
#define bfd_mach_sparc_sparclet    2
#define bfd_mach_sparc_sparclite   3
#define bfd_mach_sparc_v8plus      4
#define bfd_mach_sparc_v8plusa     5 /* with ultrasparc add'ns. */
#define bfd_mach_sparc_sparclite_le 6
#define bfd_mach_sparc_v9          7
#define bfd_mach_sparc_v9a         8 /* with ultrasparc add'ns. */
#define bfd_mach_sparc_v8plusb     9 /* with cheetah add'ns. */
#define bfd_mach_sparc_v9b        10 /* with cheetah add'ns. */
#define bfd_mach_sparc_v8plusc     11 /* with UA2005 and T1 add'ns. */
#define bfd_mach_sparc_v9c        12 /* with UA2005 and T1 add'ns. */
#define bfd_mach_sparc_v8plusd     13 /* with UA2007 and T3 add'ns. */
#define bfd_mach_sparc_v9d        14 /* with UA2007 and T3 add'ns. */
#define bfd_mach_sparc_v8pluse     15 /* with OSA2001 and T4 add'ns (no IMA). */
#define bfd_mach_sparc_v9e        16 /* with OSA2001 and T4 add'ns (no IMA). */
#define bfd_mach_sparc_v8plusv     17 /* with OSA2011 and T4 and IMA and FJMAU add'ns. */
#define bfd_mach_sparc_v9v        18 /* with OSA2011 and T4 and IMA and FJMAU add'ns. */
#define bfd_mach_sparc_v8plusm     19 /* with OSA2015 and M7 add'ns. */
#define bfd_mach_sparc_v9m        20 /* with OSA2015 and M7 add'ns. */
#define bfd_mach_sparc_v8plusm8    21 /* with OSA2017 and M8 add'ns. */
#define bfd_mach_sparc_v9m8       22 /* with OSA2017 and M8 add'ns. */
/* Nonzero if MACH has the v9 instruction set. */

```

```

#define bfd_mach_sparc_v9_p(mach) \
    ((mach) >= bfd_mach_sparc_v8plus && (mach) <= bfd_mach_sparc_v9m8 \
     && (mach) != bfd_mach_sparc_sparclite_le)
/* Nonzero if MACH is a 64 bit sparc architecture. */
#define bfd_mach_sparc_64bit_p(mach) \
    ((mach) >= bfd_mach_sparc_v9 \
     && (mach) != bfd_mach_sparc_v8plusb \
     && (mach) != bfd_mach_sparc_v8plusc \
     && (mach) != bfd_mach_sparc_v8plusd \
     && (mach) != bfd_mach_sparc_v8pluse \
     && (mach) != bfd_mach_sparc_v8plusv \
     && (mach) != bfd_mach_sparc_v8plusm \
     && (mach) != bfd_mach_sparc_v8plusm8)
    bfd_arch_spu,          /* PowerPC SPU. */
#define bfd_mach_spu          256
    bfd_arch_mips,        /* MIPS Rxxxx. */
#define bfd_mach_mips3000          3000
#define bfd_mach_mips3900          3900
#define bfd_mach_mips4000          4000
#define bfd_mach_mips4010          4010
#define bfd_mach_mips4100          4100
#define bfd_mach_mips4111          4111
#define bfd_mach_mips4120          4120
#define bfd_mach_mips4300          4300
#define bfd_mach_mips4400          4400
#define bfd_mach_mips4600          4600
#define bfd_mach_mips4650          4650
#define bfd_mach_mips5000          5000
#define bfd_mach_mips5400          5400
#define bfd_mach_mips5500          5500
#define bfd_mach_mips5900          5900
#define bfd_mach_mips6000          6000
#define bfd_mach_mips7000          7000
#define bfd_mach_mips8000          8000
#define bfd_mach_mips9000          9000
#define bfd_mach_mips10000         10000
#define bfd_mach_mips12000         12000
#define bfd_mach_mips14000         14000
#define bfd_mach_mips16000         16000
#define bfd_mach_mips16           16
#define bfd_mach_mips5             5
#define bfd_mach_mips_allegrex     10111431 /* octal 'AL', 31. */
#define bfd_mach_mips_loongson_2e  3001
#define bfd_mach_mips_loongson_2f  3002
#define bfd_mach_mips_gs464        3003
#define bfd_mach_mips_gs464e       3004
#define bfd_mach_mips_gs264e       3005

```

```

#define bfd_mach_mips_sb1          12310201 /* octal 'SB', 01. */
#define bfd_mach_mips_octeon       6501
#define bfd_mach_mips_octeonp      6601
#define bfd_mach_mips_octeon2      6502
#define bfd_mach_mips_octeon3      6503
#define bfd_mach_mips_xlr          887682 /* decimal 'XLR'. */
#define bfd_mach_mips_interaptiv_mr2 736550 /* decimal 'IA2'. */
#define bfd_mach_mipsisa32         32
#define bfd_mach_mipsisa32r2       33
#define bfd_mach_mipsisa32r3       34
#define bfd_mach_mipsisa32r5       36
#define bfd_mach_mipsisa32r6       37
#define bfd_mach_mipsisa64         64
#define bfd_mach_mipsisa64r2       65
#define bfd_mach_mipsisa64r3       66
#define bfd_mach_mipsisa64r5       68
#define bfd_mach_mipsisa64r6       69
#define bfd_mach_mips_micromips     96
    bfd_arch_i386,          /* Intel 386. */
#define bfd_mach_i386_intel_syntax (1 << 0)
#define bfd_mach_i386_i8086        (1 << 1)
#define bfd_mach_i386_i386         (1 << 2)
#define bfd_mach_x86_64            (1 << 3)
#define bfd_mach_x64_32            (1 << 4)
#define bfd_mach_i386_i386_intel_syntax (bfd_mach_i386_i386 | bfd_mach_i386_intel_synt
#define bfd_mach_x86_64_intel_syntax (bfd_mach_x86_64 | bfd_mach_i386_intel_syntax)
#define bfd_mach_x64_32_intel_syntax (bfd_mach_x64_32 | bfd_mach_i386_intel_syntax)
    bfd_arch_iamcu,        /* Intel MCU. */
#define bfd_mach_iamcu         (1 << 8)
#define bfd_mach_i386_iamcu    (bfd_mach_i386_i386 | bfd_mach_iamcu)
#define bfd_mach_i386_iamcu_intel_syntax (bfd_mach_i386_iamcu | bfd_mach_i386_intel_sy
    bfd_arch_romp,        /* IBM ROMP PC/RT. */
    bfd_arch_convex,      /* Convex. */
    bfd_arch_m98k,        /* Motorola 98xxx. */
    bfd_arch_pyramid,     /* Pyramid Technology. */
    bfd_arch_h8300,       /* Renesas H8/300 (formerly Hitachi H8/300). */
#define bfd_mach_h8300         1
#define bfd_mach_h8300h        2
#define bfd_mach_h8300s        3
#define bfd_mach_h8300hn       4
#define bfd_mach_h8300sn       5
#define bfd_mach_h8300sx       6
#define bfd_mach_h8300sxn      7
    bfd_arch_pdp11,       /* DEC PDP-11. */
    bfd_arch_powerpc,     /* PowerPC. */
#define bfd_mach_ppc          32
#define bfd_mach_ppc64        64

```

```

#define bfd_mach_ppc_403      403
#define bfd_mach_ppc_403gc   4030
#define bfd_mach_ppc_405     405
#define bfd_mach_ppc_505     505
#define bfd_mach_ppc_601     601
#define bfd_mach_ppc_602     602
#define bfd_mach_ppc_603     603
#define bfd_mach_ppc_ec603e  6031
#define bfd_mach_ppc_604     604
#define bfd_mach_ppc_620     620
#define bfd_mach_ppc_630     630
#define bfd_mach_ppc_750     750
#define bfd_mach_ppc_860     860
#define bfd_mach_ppc_a35     35
#define bfd_mach_ppc_rs64ii  642
#define bfd_mach_ppc_rs64iii 643
#define bfd_mach_ppc_7400    7400
#define bfd_mach_ppc_e500    500
#define bfd_mach_ppc_e500mc  5001
#define bfd_mach_ppc_e500mc64 5005
#define bfd_mach_ppc_e5500   5006
#define bfd_mach_ppc_e6500   5007
#define bfd_mach_ppc_titan   83
#define bfd_mach_ppc_vle     84
    bfd_arch_rs6000, /* IBM RS/6000. */
#define bfd_mach_rs6k        6000
#define bfd_mach_rs6k_rs1    6001
#define bfd_mach_rs6k_rsc    6003
#define bfd_mach_rs6k_rs2    6002
    bfd_arch_hppa, /* HP PA RISC. */
#define bfd_mach_hppa10      10
#define bfd_mach_hppa11      11
#define bfd_mach_hppa20      20
#define bfd_mach_hppa20w     25
    bfd_arch_d10v, /* Mitsubishi D10V. */
#define bfd_mach_d10v        1
#define bfd_mach_d10v_ts2    2
#define bfd_mach_d10v_ts3    3
    bfd_arch_d30v, /* Mitsubishi D30V. */
    bfd_arch_dlx, /* DLX. */
    bfd_arch_m68hc11, /* Motorola 68HC11. */
    bfd_arch_m68hc12, /* Motorola 68HC12. */
#define bfd_mach_m6812_default 0
#define bfd_mach_m6812       1
#define bfd_mach_m6812s      2
    bfd_arch_m9s12x, /* Freescale S12X. */
    bfd_arch_m9s12xg, /* Freescale XGATE. */

```

```

    bfd_arch_s12z,    /* Freescale S12Z.  */
#define bfd_mach_s12z_default 0
    bfd_arch_z8k,     /* Zilog Z8000.  */
#define bfd_mach_z8001      1
#define bfd_mach_z8002      2
    bfd_arch_sh,      /* Renesas / SuperH SH (formerly Hitachi SH).  */
#define bfd_mach_sh          1
#define bfd_mach_sh2         0x20
#define bfd_mach_sh_dsp      0x2d
#define bfd_mach_sh2a        0x2a
#define bfd_mach_sh2a_nofpu   0x2b
#define bfd_mach_sh2a_nofpu_or_sh4_nommu_nofpu 0x2a1
#define bfd_mach_sh2a_nofpu_or_sh3_nommu       0x2a2
#define bfd_mach_sh2a_or_sh4                   0x2a3
#define bfd_mach_sh2a_or_sh3e                  0x2a4
#define bfd_mach_sh2e                         0x2e
#define bfd_mach_sh3                         0x30
#define bfd_mach_sh3_nommu                   0x31
#define bfd_mach_sh3_dsp                     0x3d
#define bfd_mach_sh3e                       0x3e
#define bfd_mach_sh4                       0x40
#define bfd_mach_sh4_nofpu                  0x41
#define bfd_mach_sh4_nommu_nofpu            0x42
#define bfd_mach_sh4a                      0x4a
#define bfd_mach_sh4a_nofpu                 0x4b
#define bfd_mach_sh4al_dsp                  0x4d
    bfd_arch_alpha,   /* Dec Alpha.  */
#define bfd_mach_alpha_ev4      0x10
#define bfd_mach_alpha_ev5      0x20
#define bfd_mach_alpha_ev6      0x30
    bfd_arch_arm,     /* Advanced Risc Machines ARM.  */
#define bfd_mach_arm_unknown    0
#define bfd_mach_arm_2          1
#define bfd_mach_arm_2a        2
#define bfd_mach_arm_3          3
#define bfd_mach_arm_3M        4
#define bfd_mach_arm_4          5
#define bfd_mach_arm_4T        6
#define bfd_mach_arm_5          7
#define bfd_mach_arm_5T        8
#define bfd_mach_arm_5TE       9
#define bfd_mach_arm_XScale    10
#define bfd_mach_arm_ep9312    11
#define bfd_mach_arm_iWMMXt    12
#define bfd_mach_arm_iWMMXt2   13
#define bfd_mach_arm_5TEJ      14
#define bfd_mach_arm_6         15

```

```

#define bfd_mach_arm_6KZ      16
#define bfd_mach_arm_6T2      17
#define bfd_mach_arm_6K       18
#define bfd_mach_arm_7        19
#define bfd_mach_arm_6M       20
#define bfd_mach_arm_6SM      21
#define bfd_mach_arm_7EM      22
#define bfd_mach_arm_8        23
#define bfd_mach_arm_8R       24
#define bfd_mach_arm_8M_BASE  25
#define bfd_mach_arm_8M_MAIN  26
#define bfd_mach_arm_8_1M_MAIN 27
#define bfd_mach_arm_9        28
    bfd_arch_nds32,          /* Andes NDS32. */
#define bfd_mach_n1           1
#define bfd_mach_n1h         2
#define bfd_mach_n1h_v2      3
#define bfd_mach_n1h_v3      4
#define bfd_mach_n1h_v3m     5
    bfd_arch_ns32k,          /* National Semiconductors ns32000. */
    bfd_arch_tic30,          /* Texas Instruments TMS320C30. */
    bfd_arch_tic4x,          /* Texas Instruments TMS320C3X/4X. */
#define bfd_mach_tic3x        30
#define bfd_mach_tic4x        40
    bfd_arch_tic54x,          /* Texas Instruments TMS320C54X. */
    bfd_arch_tic6x,          /* Texas Instruments TMS320C6X. */
    bfd_arch_v850,           /* NEC V850. */
    bfd_arch_v850_rh850,     /* NEC V850 (using RH850 ABI). */
#define bfd_mach_v850         1
#define bfd_mach_v850e        'E'
#define bfd_mach_v850e1       '1'
#define bfd_mach_v850e2       0x4532
#define bfd_mach_v850e2v3     0x45325633
#define bfd_mach_v850e3v5     0x45335635 /* ('E'|'3'|'V'|'5'). */
    bfd_arch_arc,            /* ARC Cores. */
#define bfd_mach_arc_a4        0
#define bfd_mach_arc_a5        1
#define bfd_mach_arc_arc600    2
#define bfd_mach_arc_arc601    4
#define bfd_mach_arc_arc700    3
#define bfd_mach_arc_arcv2     5
    bfd_arch_m32c,           /* Renesas M16C/M32C. */
#define bfd_mach_m16c          0x75
#define bfd_mach_m32c          0x78
    bfd_arch_m32r,           /* Renesas M32R (formerly Mitsubishi M32R/D). */
#define bfd_mach_m32r          1 /* For backwards compatibility. */
#define bfd_mach_m32rx         'x'

```

```

#define bfd_mach_m32r2      '2'
    bfd_arch_mn10200,      /* Matsushita MN10200.  */
    bfd_arch_mn10300,      /* Matsushita MN10300.  */
#define bfd_mach_mn10300    300
#define bfd_mach_am33       330
#define bfd_mach_am33_2     332
    bfd_arch_fr30,
#define bfd_mach_fr30       0x46523330
    bfd_arch_frv,
#define bfd_mach_frv        1
#define bfd_mach_frvsimple   2
#define bfd_mach_fr300      300
#define bfd_mach_fr400      400
#define bfd_mach_fr450      450
#define bfd_mach_frvtomcat  499      /* fr500 prototype.  */
#define bfd_mach_fr500      500
#define bfd_mach_fr550      550
    bfd_arch_moxie,        /* The moxie processor.  */
#define bfd_mach_moxie      1
    bfd_arch_ft32,         /* The ft32 processor.  */
#define bfd_mach_ft32       1
#define bfd_mach_ft32b      2
    bfd_arch_mcore,
    bfd_arch_mep,
#define bfd_mach_mep        1
#define bfd_mach_mep_h1     0x6831
#define bfd_mach_mep_c5     0x6335
    bfd_arch_metag,
#define bfd_mach_metag      1
    bfd_arch_ia64,         /* HP/Intel ia64.  */
#define bfd_mach_ia64_elf64  64
#define bfd_mach_ia64_elf32  32
    bfd_arch_ip2k,         /* Ubicom IP2K microcontrollers.  */
#define bfd_mach_ip2022     1
#define bfd_mach_ip2022ext  2
    bfd_arch_iq2000,       /* Vitesse IQ2000.  */
#define bfd_mach_iq2000     1
#define bfd_mach_iq10       2
    bfd_arch_bpf,          /* Linux eBPF.  */
#define bfd_mach_bpf        1
#define bfd_mach_xbpf       2
    bfd_arch_epiphany,     /* Adapteva EPIPHANY.  */
#define bfd_mach_epiphany16  1
#define bfd_mach_epiphany32  2
    bfd_arch_mt,
#define bfd_mach_ms1        1
#define bfd_mach_mrisc2     2

```

```

#define bfd_mach_ms2          3
    bfd_arch_pj,
    bfd_arch_avr,             /* Atmel AVR microcontrollers. */
#define bfd_mach_avr1         1
#define bfd_mach_avr2         2
#define bfd_mach_avr25        25
#define bfd_mach_avr3         3
#define bfd_mach_avr31        31
#define bfd_mach_avr35        35
#define bfd_mach_avr4         4
#define bfd_mach_avr5         5
#define bfd_mach_avr51        51
#define bfd_mach_avr6         6
#define bfd_mach_avrtiny      100
#define bfd_mach_avrxmega1    101
#define bfd_mach_avrxmega2    102
#define bfd_mach_avrxmega3    103
#define bfd_mach_avrxmega4    104
#define bfd_mach_avrxmega5    105
#define bfd_mach_avrxmega6    106
#define bfd_mach_avrxmega7    107
    bfd_arch_bfin,           /* ADI Blackfin. */
#define bfd_mach_bfin         1
    bfd_arch_cr16,           /* National Semiconductor CompactRISC (ie CR16). */
#define bfd_mach_cr16         1
    bfd_arch_crx,            /* National Semiconductor CRX. */
#define bfd_mach_crx          1
    bfd_arch_cris,           /* Axis CRIS. */
#define bfd_mach_cris_v0_v10  255
#define bfd_mach_cris_v32     32
#define bfd_mach_cris_v10_v32 1032
    bfd_arch_riscv,
#define bfd_mach_riscv32       132
#define bfd_mach_riscv64       164
    bfd_arch_rl78,
#define bfd_mach_rl78          0x75
    bfd_arch_rx,             /* Renesas RX. */
#define bfd_mach_rx            0x75
#define bfd_mach_rx_v2         0x76
#define bfd_mach_rx_v3         0x77
    bfd_arch_s390,           /* IBM s390. */
#define bfd_mach_s390_31       31
#define bfd_mach_s390_64       64
    bfd_arch_score,          /* Sunplus score. */
#define bfd_mach_score3        3
#define bfd_mach_score7        7
    bfd_arch_mmix,           /* Donald Knuth's educational processor. */

```



```

    bfd_arch_xstormy16,
#define bfd_mach_xstormy16      1
    bfd_arch_msp430,      /* Texas Instruments MSP430 architecture. */
#define bfd_mach_msp11        11
#define bfd_mach_msp110       110
#define bfd_mach_msp12        12
#define bfd_mach_msp13        13
#define bfd_mach_msp14        14
#define bfd_mach_msp15        15
#define bfd_mach_msp16        16
#define bfd_mach_msp20        20
#define bfd_mach_msp21        21
#define bfd_mach_msp22        22
#define bfd_mach_msp23        23
#define bfd_mach_msp24        24
#define bfd_mach_msp26        26
#define bfd_mach_msp31        31
#define bfd_mach_msp32        32
#define bfd_mach_msp33        33
#define bfd_mach_msp41        41
#define bfd_mach_msp42        42
#define bfd_mach_msp43        43
#define bfd_mach_msp44        44
#define bfd_mach_msp430x      45
#define bfd_mach_msp46        46
#define bfd_mach_msp47        47
#define bfd_mach_msp54        54
    bfd_arch_xgate,      /* Freescale XGATE. */
#define bfd_mach_xgate        1
    bfd_arch_xtensa,      /* Tensilica's Xtensa cores. */
#define bfd_mach_xtensa      1
    bfd_arch_z80,
/* Zilog Z80 without undocumented opcodes. */
#define bfd_mach_z80strict    1
/* Zilog Z180: successor with additional instructions, but without
   halves of ix and iy. */
#define bfd_mach_z180        2
/* Zilog Z80 with ixl, ixh, iyl, and iyh. */
#define bfd_mach_z80        3
/* Zilog eZ80 (successor of Z80 & Z180) in Z80 (16-bit address) mode. */
#define bfd_mach_ez80_z80    4
/* Zilog eZ80 (successor of Z80 & Z180) in ADL (24-bit address) mode. */
#define bfd_mach_ez80_adl    5
/* Z80N */
#define bfd_mach_z80n        6
/* Zilog Z80 with all undocumented instructions. */
#define bfd_mach_z80full     7

```

```

/* GameBoy Z80 (reduced instruction set). */
#define bfd_mach_gbz80      8
/* ASCII R800: successor with multiplication. */
#define bfd_mach_r800      11
    bfd_arch_lm32,        /* Lattice Mico32. */
#define bfd_mach_lm32      1
    bfd_arch_microblaze, /* Xilinx MicroBlaze. */
    bfd_arch_kvxx,       /* Kalray VLIW core of the MPPA processor family */
#define bfd_mach_kv3_unknown 0
#define bfd_mach_kv3_1      1
#define bfd_mach_kv3_1_64   2
#define bfd_mach_kv3_1_usr  3
#define bfd_mach_kv3_2      4
#define bfd_mach_kv3_2_64   5
#define bfd_mach_kv3_2_usr  6
#define bfd_mach_kv4_1      7
#define bfd_mach_kv4_1_64   8
#define bfd_mach_kv4_1_usr  9
    bfd_arch_tilepro,    /* Tilera TILEPro. */
    bfd_arch_tilegx,     /* Tilera TILE-Gx. */
#define bfd_mach_tilepro    1
#define bfd_mach_tilegx    1
#define bfd_mach_tilegx32   2
    bfd_arch_aarch64,    /* AArch64. */
#define bfd_mach_aarch64 0
#define bfd_mach_aarch64_8R 1
#define bfd_mach_aarch64_ilp32 32
#define bfd_mach_aarch64_llp64 64
    bfd_arch_visium,     /* Visium. */
#define bfd_mach_visium    1
    bfd_arch_wasm32,     /* WebAssembly. */
#define bfd_mach_wasm32    1
    bfd_arch_pru,        /* PRU. */
#define bfd_mach_pru       0
    bfd_arch_nfp,        /* Netronome Flow Processor */
#define bfd_mach_nfp3200   0x3200
#define bfd_mach_nfp6000   0x6000
    bfd_arch_csky,       /* C-SKY. */
#define bfd_mach_ck_unknown 0
#define bfd_mach_ck510     1
#define bfd_mach_ck610     2
#define bfd_mach_ck801     3
#define bfd_mach_ck802     4
#define bfd_mach_ck803     5
#define bfd_mach_ck807     6
#define bfd_mach_ck810     7
#define bfd_mach_ck860     8

```

```

    bfd_arch_loongarch,      /* LoongArch */
#define bfd_mach_loongarch32 1
#define bfd_mach_loongarch64 2
    bfd_arch_amdgc,         /* AMDGCN */
#define bfd_mach_amdgc_unkown 0x000
#define bfd_mach_amdgc_gfx9_generic 0x051
#define bfd_mach_amdgc_gfx900 0x02c
#define bfd_mach_amdgc_gfx904 0x02e
#define bfd_mach_amdgc_gfx906 0x02f
#define bfd_mach_amdgc_gfx908 0x030
#define bfd_mach_amdgc_gfx90a 0x03f
#define bfd_mach_amdgc_gfx9_4_generic 0x05f
#define bfd_mach_amdgc_gfx942 0x04c
#define bfd_mach_amdgc_gfx950 0x04f
#define bfd_mach_amdgc_gfx10_1_generic 0x052
#define bfd_mach_amdgc_gfx1010 0x033
#define bfd_mach_amdgc_gfx1011 0x034
#define bfd_mach_amdgc_gfx1012 0x035
#define bfd_mach_amdgc_gfx10_3_generic 0x053
#define bfd_mach_amdgc_gfx1030 0x036
#define bfd_mach_amdgc_gfx1031 0x037
#define bfd_mach_amdgc_gfx1032 0x038
#define bfd_mach_amdgc_gfx11_generic 0x054
#define bfd_mach_amdgc_gfx1100 0x041
#define bfd_mach_amdgc_gfx1101 0x046
#define bfd_mach_amdgc_gfx1102 0x047
#define bfd_mach_amdgc_gfx1150 0x043
#define bfd_mach_amdgc_gfx1151 0x04a
#define bfd_mach_amdgc_gfx1152 0x055
#define bfd_mach_amdgc_gfx1153 0x058
#define bfd_mach_amdgc_gfx12_generic 0x059
#define bfd_mach_amdgc_gfx1200 0x048
#define bfd_mach_amdgc_gfx1201 0x04e
    bfd_arch_last
};

```

### 2.13.2 bfd\_arch\_info

This structure contains information on architectures for use within BFD.

```

typedef struct bfd_arch_info
{
    int bits_per_word;
    int bits_per_address;
    int bits_per_byte;
    enum bfd_architecture arch;
    unsigned long mach;

```

```

const char *arch_name;
const char *printable_name;
unsigned int section_align_power;
/* TRUE if this is the default machine for the architecture.
   The default arch should be the first entry for an arch so that
   all the entries for that arch can be accessed via next. */
bool the_default;
const struct bfd_arch_info * (*compatible) (const struct bfd_arch_info *,
                                             const struct bfd_arch_info *);

bool (*scan) (const struct bfd_arch_info *, const char *);

/* Allocate via bfd_malloc and return a fill buffer of size COUNT.  If
   IS_BIGENDIAN is TRUE, the order of bytes is big endian.  If CODE is
   TRUE, the buffer contains code. */
void *(*fill) (bfd_size_type count, bool is_bigendian, bool code);

const struct bfd_arch_info *next;

/* On some architectures the offset for a relocation can point into
   the middle of an instruction.  This field specifies the maximum
   offset such a relocation can have (in octets).  This affects the
   behaviour of the disassembler, since a value greater than zero
   means that it may need to disassemble an instruction twice, once
   to get its length and then a second time to display it.  If the
   value is negative then this has to be done for every single
   instruction, regardless of the offset of the reloc. */
signed int max_reloc_offset_into_insn;
}
bfd_arch_info_type;

```

### 2.13.2.1 bfd\_printable\_name

`const char *bfd_printable_name (bfd *abfd);` [Function]  
 Return a printable string representing the architecture and machine from the pointer to the architecture info structure.

### 2.13.2.2 bfd\_scan\_arch

`const bfd_arch_info_type *bfd_scan_arch (const char *string);` [Function]  
 Figure out if BFD supports any cpu which could be described with the name *string*.  
 Return a pointer to an `arch_info` structure if a machine is found, otherwise NULL.

### 2.13.2.3 bfd\_arch\_list

`const char **bfd_arch_list (void);` [Function]  
 Return a freshly malloced NULL-terminated vector of the names of all the valid BFD architectures. Do not modify the names.

### 2.13.2.4 bfd\_arch\_get\_compatible

`const bfd_arch_info_type *bfd_arch_get_compatible (const bfd [Function]  
*abfd, const bfd *bbfd, bool accept_unknowns);`

Determine whether two BFDs' architectures and machine types are compatible. Calculates the lowest common denominator between the two architectures and machine types implied by the BFDs and returns a pointer to an `arch_info` structure describing the compatible machine.

### 2.13.2.5 bfd\_default\_arch\_struct

The `bfd_default_arch_struct` is an item of `bfd_arch_info_type` which has been initialized to a fairly generic state. A BFD starts life by pointing to this structure, until the correct back end has determined the real architecture of the file.

```
extern const bfd_arch_info_type bfd_default_arch_struct;
```

### 2.13.2.6 bfd\_set\_arch\_info

`void bfd_set_arch_info (bfd *abfd, const bfd_arch_info_type [Function]  
*arg);`

Set the architecture info of *abfd* to *arg*.

### 2.13.2.7 bfd\_default\_set\_arch\_mach

`bool bfd_default_set_arch_mach (bfd *abfd, enum [Function]  
bfd_architecture arch, unsigned long mach);`

Set the architecture and machine type in BFD *abfd* to *arch* and *mach*. Find the correct pointer to a structure and insert it into the `arch_info` pointer.

### 2.13.2.8 bfd\_get\_arch

`enum bfd_architecture bfd_get_arch (const bfd *abfd); [Function]`

Return the enumerated type which describes the BFD *abfd*'s architecture.

### 2.13.2.9 bfd\_get\_mach

`unsigned long bfd_get_mach (const bfd *abfd); [Function]`

Return the long type which describes the BFD *abfd*'s machine.

### 2.13.2.10 bfd\_arch\_bits\_per\_byte

`unsigned int bfd_arch_bits_per_byte (const bfd *abfd); [Function]`

Return the number of bits in one of the BFD *abfd*'s architecture's bytes.

### 2.13.2.11 bfd\_arch\_bits\_per\_address

`unsigned int bfd_arch_bits_per_address (const bfd *abfd); [Function]`

Return the number of bits in one of the BFD *abfd*'s architecture's addresses.

**2.13.2.12 bfd\_default\_compatible**

```
const bfd_arch_info_type *bfd_default_compatible (const      [Function]
          bfd_arch_info_type *a, const bfd_arch_info_type *b);
```

The default function for testing for compatibility.

**2.13.2.13 bfd\_default\_scan**

```
bool bfd_default_scan (const struct bfd_arch_info *info, const      [Function]
          char *string);
```

The default function for working out whether this is an architecture hit and a machine hit.

**2.13.2.14 bfd\_get\_arch\_info**

```
const bfd_arch_info_type *bfd_get_arch_info (bfd *abfd);      [Function]
```

Return the architecture info struct in *abfd*.

**2.13.2.15 bfd\_lookup\_arch**

```
const bfd_arch_info_type *bfd_lookup_arch (enum                [Function]
          bfd_architecture arch, unsigned long machine);
```

Look for the architecture info structure which matches the arguments *arch* and *machine*. A machine of 0 matches the machine/architecture structure which marks itself as the default.

**2.13.2.16 bfd\_printable\_arch\_mach**

```
const char *bfd_printable_arch_mach (enum bfd_architecture arch,      [Function]
          unsigned long machine);
```

Return a printable string representing the architecture and machine type.

This routine is depreciated.

**2.13.2.17 bfd\_octets\_per\_byte**

```
unsigned int bfd_octets_per_byte (const bfd *abfd, const asection      [Function]
          *sec);
```

Return the number of octets (8-bit quantities) per target byte (minimum addressable unit). In most cases, this will be one, but some DSP targets have 16, 32, or even 48 bits per byte.

**2.13.2.18 bfd\_arch\_mach\_octets\_per\_byte**

```
unsigned int bfd_arch_mach_octets_per_byte (enum                [Function]
          bfd_architecture arch, unsigned long machine);
```

See `bfd_octets_per_byte`.

This routine is provided for those cases where a `bfd *` is not available

### 2.13.2.19 bfd\_arch\_default\_fill

`void *bfd_arch_default_fill (bfd_size_type count, bool is_bigendian, bool code);` [Function]

Allocate via `bfd_malloc` and return a fill buffer of size `COUNT`. If `IS_BIGENDIAN` is `TRUE`, the order of bytes is big endian. If `CODE` is `TRUE`, the buffer contains code.

## 2.14 Opening and closing BFDs

### 2.14.1 Functions for opening and closing

#### 2.14.1.1 \_bfd\_new\_bfd

`bfd *_bfd_new_bfd (void);` [Function]

Return a new BFD. All BFD's are allocated through this routine.

#### 2.14.1.2 \_bfd\_new\_bfd\_contained\_in

`bfd *_bfd_new_bfd_contained_in (bfd *);` [Function]

Allocate a new BFD as a member of archive OBFD.

#### 2.14.1.3 \_bfd\_free\_cached\_info

`bool _bfd_free_cached_info (bfd *);` [Function]

Free objalloc memory.

#### 2.14.1.4 bfd\_fopen

`bfd *bfd_fopen (const char *filename, const char *target, const char *mode, int fd);` [Function]

Open the file *filename* with the target *target*. Return a pointer to the created BFD. If *fd* is not -1, then `fdopen` is used to open the file; otherwise, `fopen` is used. *mode* is passed directly to `fopen` or `fdopen`.

Calls `bfd_find_target`, so *target* is interpreted as by that function.

The new BFD is marked as cacheable iff *fd* is -1.

If `NULL` is returned then an error has occurred. Possible errors are `bfd_error_no_memory`, `bfd_error_invalid_target` or `system_call` error.

On error, *fd* is always closed.

A copy of the *filename* argument is stored in the newly created BFD. It can be accessed via the `bfd_get_filename()` macro.

#### 2.14.1.5 bfd\_openr

`bfd *bfd_openr (const char *filename, const char *target);` [Function]

Open the file *filename* (using `fopen`) with the target *target*. Return a pointer to the created BFD.

Calls `bfd_find_target`, so *target* is interpreted as by that function.

If NULL is returned then an error has occurred. Possible errors are `bfd_error_no_memory`, `bfd_error_invalid_target` or `system_call` error.

A copy of the *filename* argument is stored in the newly created BFD. It can be accessed via the `bfd_get_filename()` macro.

### 2.14.1.6 `bfd_fdopenr`

**`bfd *bfd_fdopenr (const char *filename, const char *target, int fd);`** [Function]  
`bfd_fdopenr` is to `bfd_fopenr` much like `fdopen` is to `fopen`. It opens a BFD on a file already described by the *fd* supplied.

When the file is later `bfd_close`d, the file descriptor will be closed. If the caller desires that this file descriptor be cached by BFD (opened as needed, closed as needed to free descriptors for other opens), with the supplied *fd* used as an initial file descriptor (but subject to closure at any time), call `bfd_set_cacheable(bfd, 1)` on the returned BFD. The default is to assume no caching; the file descriptor will remain open until `bfd_close`, and will not be affected by BFD operations on other files.

Possible errors are `bfd_error_no_memory`, `bfd_error_invalid_target` and `bfd_error_system_call`.

On error, *fd* is closed.

A copy of the *filename* argument is stored in the newly created BFD. It can be accessed via the `bfd_get_filename()` macro.

### 2.14.1.7 `bfd_fdopenw`

**`bfd *bfd_fdopenw (const char *filename, const char *target, int fd);`** [Function]  
`bfd_fdopenw` is exactly like `bfd_fdopenr` with the exception that the resulting BFD is suitable for output.

### 2.14.1.8 `bfd_openstreamr`

**`bfd *bfd_openstreamr (const char * filename, const char * target, void * stream);`** [Function]

Open a BFD for read access on an existing stdio stream. When the BFD is passed to `bfd_close`, the stream will be closed.

A copy of the *filename* argument is stored in the newly created BFD. It can be accessed via the `bfd_get_filename()` macro.

### 2.14.1.9 `bfd_openr_fake_archive`

**`bfd *bfd_openr_fake_archive (bfd *fbfd);`** [Function]

Open a list of BFDs starting from *fbfd* as an artificial archive. Subsequent members of the archive are indicated by each BFD's `proxy_handle.abfd` member, with the final one holding a NULL pointer there. The newly opened archive will necessarily also be a thin archive as there will be member files only to refer to and no containing archive file.



### 2.14.1.10 bfd\_openr\_iovec

```
bfd *bfd_openr_iovec (const char *filename, const char *target,          [Function]
                     void (*open_func) (struct bfd *nbfd, void *open_closure, void
                     *open_closure, file_ptr (*pread_func) (struct bfd *nbfd, void *stream,
                     void *buf, file_ptr nbytes, file_ptr offset), int (*close_func) (struct bfd
                     *nbfd, void *stream), int (*stat_func) (struct bfd *abfd, void *stream,
                     struct stat *sb));
```

Create and return a BFD backed by a read-only *stream*. The *stream* is created using *open\_func*, accessed using *pread\_func* and destroyed using *close\_func*.

Calls *bfd\_find\_target*, so *target* is interpreted as by that function.

Calls *open\_func* (which can call *bfd\_zalloc* and *bfd\_get\_filename*) to obtain the read-only stream backing the BFD. *open\_func* either succeeds returning the non-NULL *stream*, or fails returning NULL (setting *bfd\_error*).

Calls *pread\_func* to request *nbytes* of data from *stream* starting at *offset* (e.g., via a call to *bfd\_read*). *pread\_func* either succeeds returning the number of bytes read (which can be less than *nbytes* when end-of-file), or fails returning -1 (setting *bfd\_error*).

Calls *close\_func* when the BFD is later closed using *bfd\_close*. *close\_func* either succeeds returning 0, or fails returning -1 (setting *bfd\_error*).

Calls *stat\_func* to fill in a stat structure for *bfd\_stat*, *bfd\_get\_size*, and *bfd\_get\_mtime* calls. *stat\_func* returns 0 on success, or returns -1 on failure (setting *bfd\_error*).

If *bfd\_openr\_iovec* returns NULL then an error has occurred. Possible errors are *bfd\_error\_no\_memory*, *bfd\_error\_invalid\_target* and *bfd\_error\_system\_call*.

A copy of the *filename* argument is stored in the newly created BFD. It can be accessed via the *bfd\_get\_filename*() macro.

### 2.14.1.11 bfd\_openw

```
bfd *bfd_openw (const char *filename, const char *target);          [Function]
```

Create a BFD, associated with file *filename*, using the file format *target*, and return a pointer to it.

Possible errors are *bfd\_error\_system\_call*, *bfd\_error\_no\_memory*, *bfd\_error\_invalid\_target*.

A copy of the *filename* argument is stored in the newly created BFD. It can be accessed via the *bfd\_get\_filename*() macro.

### 2.14.1.12 bfd\_elf\_bfd\_from\_remote\_memory

```
bfd *bfd_elf_bfd_from_remote_memory (bfd *templ, bfd_vma          [Function]
                                     ehdr_vma, bfd_size_type size, bfd_vma *loadbasep, int
                                     (*target_read_memory) (bfd_vma vma, bfd_byte *myaddr, bfd_size_type
                                     len));
```

Create a new BFD as if by *bfd\_openr*. Rather than opening a file, reconstruct an ELF file by reading the segments out of remote memory based on the ELF file header at *EHDR\_VMA* and the ELF program headers it points to. If non-zero, *SIZE* is the

known extent of the object. If not null, \*LOADBASEP is filled in with the difference between the VMAs from which the segments were read, and the VMAs the file headers (and hence BFD's idea of each section's VMA) put them at.

The function TARGET\_READ\_MEMORY is called to copy LEN bytes from the remote memory at target address VMA into the local buffer at MYADDR; it should return zero on success or an errno code on failure. TEMPL must be a BFD for an ELF target with the word size and byte order found in the remote memory.

#### 2.14.1.13 bfd\_close

**bool bfd\_close (bfd \*abfd);** [Function]

Close a BFD. If the BFD was open for writing, then pending operations are completed and the file written out and closed. If the created file is executable, then `chmod` is called to mark it as such.

All memory attached to the BFD is released.

The file descriptor associated with the BFD is closed (even if it was passed in to BFD by `bfd_fdopenr`).

TRUE is returned if all is ok, otherwise FALSE.

#### 2.14.1.14 bfd\_close\_all\_done

**bool bfd\_close\_all\_done (bfd \*);** [Function]

Close a BFD. Differs from `bfd_close` since it does not complete any pending operations. This routine would be used if the application had just used BFD for swapping and didn't want to use any of the writing code.

If the created file is executable, then `chmod` is called to mark it as such.

All memory attached to the BFD is released.

TRUE is returned if all is ok, otherwise FALSE.

#### 2.14.1.15 bfd\_create

**bfd \*bfd\_create (const char \*filename, bfd \*templ);** [Function]

Create a new BFD in the manner of `bfd_openw`, but without opening a file. The new BFD takes the target from the target used by `templ`. The format is always set to `bfd_object`.

A copy of the `filename` argument is stored in the newly created BFD. It can be accessed via the `bfd_get_filename()` macro.

#### 2.14.1.16 bfd\_make\_writable

**bool bfd\_make\_writable (bfd \*abfd);** [Function]

Takes a BFD as created by `bfd_create` and converts it into one like as returned by `bfd_openw`. It does this by converting the BFD to BFD\_IN\_MEMORY. It's assumed that you will call `bfd_make_readable` on this bfd later.

TRUE is returned if all is ok, otherwise FALSE.

**2.14.1.17 bfd\_make\_readable**

**bool bfd\_make\_readable** (*bfd \*abfd*); [Function]

Takes a BFD as created by `bfd_create` and `bfd_make_writable` and converts it into one like as returned by `bfd_openr`. It does this by writing the contents out to the memory buffer, then reversing the direction.

TRUE is returned if all is ok, otherwise FALSE.

**2.14.1.18 bfd\_calc\_gnu\_debuglink\_crc32**

**uint32\_t bfd\_calc\_gnu\_debuglink\_crc32** (*uint32\_t crc, const bfd\_byte \*buf, bfd\_size\_type len*); [Function]

Computes a CRC value as used in the `.gnu_debuglink` section. Advances the previously computed `crc` value by computing and adding in the `crc32` for `len` bytes of `buf`.

Return the updated CRC32 value.

**2.14.1.19 bfd\_get\_debug\_link\_info**

**char \*bfd\_get\_debug\_link\_info** (*bfd \*abfd, uint32\_t \*crc32\_out*); [Function]

Extracts the filename and CRC32 value for any separate debug information file associated with *abfd*.

Returns the filename of the associated debug information file, or NULL if there is no such file. If the filename was found then the contents of `crc32_out` are updated to hold the corresponding CRC32 value for the file.

The returned filename is allocated with `malloc`; freeing it is the responsibility of the caller.

**2.14.1.20 bfd\_get\_alt\_debug\_link\_info**

**char \*bfd\_get\_alt\_debug\_link\_info** (*bfd \*abfd, bfd\_size\_type \*buildid\_len, bfd\_byte \*\*buildid\_out*); [Function]

Fetch the filename and BuildID value for any alternate debuginfo associated with *abfd*. Return NULL if no such info found, otherwise return filename and update `buildid_len` and `buildid_out`. The returned filename and `build_id` are allocated with `malloc`; freeing them is the responsibility of the caller.

**2.14.1.21 bfd\_follow\_gnu\_debuglink**

**char \*bfd\_follow\_gnu\_debuglink** (*bfd \*abfd, const char \*dir*); [Function]

Takes a BFD and searches it for a `.gnu_debuglink` section. If this section is found, it examines the section for the name and checksum of a `'debug'` file containing auxiliary debugging information. It then searches the filesystem for this `.debug` file in some standard locations, including the directory tree rooted at *dir*, and if found returns the full filename.

If *dir* is NULL, the search will take place starting at the current directory.

Returns NULL on any errors or failure to locate the `.debug` file, otherwise a pointer to a heap-allocated string containing the filename. The caller is responsible for freeing this string.

### 2.14.1.22 `bfd_follow_gnu_debugaltlink`

**char \*bfd\_follow\_gnu\_debugaltlink** (*bfd \*abfd, const char \*dir*); [Function]

Takes a BFD and searches it for a `.gnu.debugaltlink` section. If this section is found, it examines the section for the name of a file containing auxiliary debugging information. It then searches the filesystem for this file in a set of standard locations, including the directory tree rooted at *dir*, and if found returns the full filename.

If *dir* is NULL, the search will take place starting at the current directory.

Returns NULL on any errors or failure to locate the debug file, otherwise a pointer to a heap-allocated string containing the filename. The caller is responsible for freeing this string.

### 2.14.1.23 `bfd_create_gnu_debuglink_section`

**struct bfd\_section \*bfd\_create\_gnu\_debuglink\_section** (*bfd \*abfd, const char \*filename*); [Function]

Takes a BFD and adds a `.gnu.debuglink` section to it. The section is sized to be big enough to contain a link to the specified *filename*.

A pointer to the new section is returned if all is ok. Otherwise NULL is returned and `bfd_error` is set.

### 2.14.1.24 `bfd_fill_in_gnu_debuglink_section`

**bool bfd\_fill\_in\_gnu\_debuglink\_section** (*bfd \*abfd, struct bfd\_section \*sect, const char \*filename*); [Function]

Takes a BFD and containing a `.gnu.debuglink` section *SECT* and fills in the contents of the section to contain a link to the specified *filename*. The filename should be absolute or relative to the current directory.

TRUE is returned if all is ok. Otherwise FALSE is returned and `bfd_error` is set.

### 2.14.1.25 `bfd_follow_build_id_debuglink`

**char \*bfd\_follow\_build\_id\_debuglink** (*bfd \*abfd, const char \*dir*); [Function]

Takes *abfd* and searches it for a `.note.gnu.build-id` section. If this section is found, it extracts the value of the `NT_GNU_BUILD_ID` note, which should be a hexadecimal value `NNNN+NN` (for 32+ hex digits). It then searches the filesystem for a file named `.build-id/NN/NN+NN.debug` in a set of standard locations, including the directory tree rooted at *dir*. The filename of the first matching file to be found is returned. A matching file should contain a `.note.gnu.build-id` section with the same `NNNN+NN` note as *abfd*, although this check is currently not implemented.

If *dir* is NULL, the search will take place starting at the current directory.

Returns NULL on any errors or failure to locate the debug file, otherwise a pointer to a heap-allocated string containing the filename. The caller is responsible for freeing this string.

### 2.14.1.26 `bfd_set_filename`

`const char *bfd_set_filename (bfd *abfd, const char *filename);` [Function]  
 Set the filename of *abfd*, copying the `FILENAME` parameter to `bfd_alloc`'d memory owned by *abfd*. Returns a pointer the newly allocated name, or `NULL` if the allocation failed.

### 2.14.1.27 `bfd_extract_object_only_section`

`const char *bfd_extract_object_only_section (bfd *abfd);` [Function]  
 Takes a *ABFD* and extract the `.gnu_object-only` section into a temporary file.

## 2.15 Implementation details

### 2.15.1 Internal functions

These routines are used within BFD. They are not intended for export, but are documented here for completeness.

#### 2.15.1.1 `bfd_malloc`

`void *bfd_malloc (bfd_size_type *size*);` [Function]  
 Returns a pointer to an allocated block of memory that is at least `SIZE` bytes long. If `SIZE` is 0 then it will be treated as if it were 1. If `SIZE` is too big then `NULL` will be returned. Returns `NULL` upon error and sets `bfd_error`.

#### 2.15.1.2 `bfd_realloc`

`void *bfd_realloc (void **mem*, bfd_size_type *size*);` [Function]  
 Returns a pointer to an allocated block of memory that is at least `SIZE` bytes long. If `SIZE` is 0 then it will be treated as if it were 1. If `SIZE` is too big then `NULL` will be returned. If `MEM` is not `NULL` then it must point to an allocated block of memory. If this block is large enough then `MEM` may be used as the return value for this function, but this is not guaranteed.  
 If `MEM` is not returned then the first `N` bytes in the returned block will be identical to the first `N` bytes in region pointed to by `MEM`, where `N` is the lessor of `SIZE` and the length of the region of memory currently addressed by `MEM`.  
 Returns `NULL` upon error and sets `bfd_error`.

#### 2.15.1.3 `bfd_realloc_or_free`

`void *bfd_realloc_or_free (void **mem*, bfd_size_type *size*);` [Function]  
 Returns a pointer to an allocated block of memory that is at least `SIZE` bytes long. If `SIZE` is 0 then no memory will be allocated, `MEM` will be freed, and `NULL` will be returned. This will not cause `bfd_error` to be set.  
 If `SIZE` is too big then `NULL` will be returned and `bfd_error` will be set. If `MEM` is not `NULL` then it must point to an allocated block of memory. If this block is large enough then `MEM` may be used as the return value for this function, but this is not guaranteed.

If MEM is not returned then the first N bytes in the returned block will be identical to the first N bytes in region pointed to by MEM, where N is the lessor of SIZE and the length of the region of memory currently addressed by MEM.

#### 2.15.1.4 bfd\_zmalloc

**void \*bfd\_zmalloc** (*bfd\_size\_type \*size*); [Function]

Returns a pointer to an allocated block of memory that is at least SIZE bytes long. If SIZE is 0 then it will be treated as if it were 1. If SIZE is too big then NULL will be returned. Returns NULL upon error and sets bfd\_error.

If NULL is not returned then the allocated block of memory will have been cleared.

#### 2.15.1.5 bfd\_alloc

**void \*bfd\_alloc** (*bfd \*abfd, bfd\_size\_type wanted*); [Function]

Allocate a block of *wanted* bytes of memory attached to *abfd* and return a pointer to it.

#### 2.15.1.6 bfd\_zalloc

**void \*bfd\_zalloc** (*bfd \*abfd, bfd\_size\_type wanted*); [Function]

Allocate a block of *wanted* bytes of zeroed memory attached to *abfd* and return a pointer to it.

#### 2.15.1.7 bfd\_release

**void bfd\_release** (*bfd \*, void \**); [Function]

Free a block allocated for a BFD. Note: Also frees all more recently allocated blocks!

#### 2.15.1.8 bfd\_write\_bigendian\_4byte\_int

**bool bfd\_write\_bigendian\_4byte\_int** (*bfd \*, unsigned int*); [Function]

Write a 4 byte integer *i* to the output BFD *abfd*, in big endian order regardless of what else is going on. This is useful in archives.

#### 2.15.1.9 bfd\_put\_size

#### 2.15.1.10 bfd\_get\_size

These macros are used for reading and writing raw data in sections; each access (except for bytes) is vectored through the target format of the BFD and mangled accordingly. The mangling performs any necessary endian translations and removes alignment restrictions. Note that types accepted and returned by these macros are identical so they can be swapped around in macros—for example, *libaout.h* defines GET\_WORD to either *bfd\_get\_32* or *bfd\_get\_64*.

In the put routines, *val* must be a *bfd\_vma*. If we are on a system without prototypes, the caller is responsible for making sure that is true, with a cast if necessary. We don't cast them in the macro definitions because that would prevent *lint* or *gcc -Wall* from detecting

sins such as passing a pointer. To detect calling these with less than a `bfd_vma`, use `gcc -Wconversion` on a host with 64 bit `bfd_vma`'s.

```

/* Byte swapping macros for user section data.  */

#define bfd_put_8(abfd, val, ptr) \
  ((void) (*((bfd_byte *) (ptr)) = (val) & 0xff))
#define bfd_put_signed_8 \
  bfd_put_8
#define bfd_get_8(abfd, ptr) \
  ((bfd_vma) *(const bfd_byte *) (ptr) & 0xff)
#define bfd_get_signed_8(abfd, ptr) \
  (((bfd_signed_vma) *(const bfd_byte *) (ptr) & 0xff) ^ 0x80) - 0x80)

#define bfd_put_16(abfd, val, ptr) \
  BFD_SEND (abfd, bfd_putx16, ((val), (ptr)))
#define bfd_put_signed_16 \
  bfd_put_16
#define bfd_get_16(abfd, ptr) \
  BFD_SEND (abfd, bfd_getx16, (ptr))
#define bfd_get_signed_16(abfd, ptr) \
  BFD_SEND (abfd, bfd_getx_signed_16, (ptr))

#define bfd_put_24(abfd, val, ptr) \
  do \
    if (bfd_big_endian (abfd)) \
      bfd_putb24 ((val), (ptr)); \
    else \
      bfd_putl24 ((val), (ptr)); \
  while (0)

bfd_vma bfd_getb24 (const void *p);
bfd_vma bfd_getl24 (const void *p);

#define bfd_get_24(abfd, ptr) \
  (bfd_big_endian (abfd) ? bfd_getb24 (ptr) : bfd_getl24 (ptr))

#define bfd_put_32(abfd, val, ptr) \
  BFD_SEND (abfd, bfd_putx32, ((val), (ptr)))
#define bfd_put_signed_32 \
  bfd_put_32
#define bfd_get_32(abfd, ptr) \
  BFD_SEND (abfd, bfd_getx32, (ptr))
#define bfd_get_signed_32(abfd, ptr) \
  BFD_SEND (abfd, bfd_getx_signed_32, (ptr))

```

```

#define bfd_put_64(abfd, val, ptr) \
    BFD_SEND (abfd, bfd_putx64, ((val), (ptr)))
#define bfd_put_signed_64 \
    bfd_put_64
#define bfd_get_64(abfd, ptr) \
    BFD_SEND (abfd, bfd_getx64, (ptr))
#define bfd_get_signed_64(abfd, ptr) \
    BFD_SEND (abfd, bfd_getx_signed_64, (ptr))

#define bfd_get(bits, abfd, ptr) \
    ((bits) == 8 ? bfd_get_8 (abfd, ptr) \
     : (bits) == 16 ? bfd_get_16 (abfd, ptr) \
     : (bits) == 32 ? bfd_get_32 (abfd, ptr) \
     : (bits) == 64 ? bfd_get_64 (abfd, ptr) \
     : (abort (), (bfd_vma) - 1))

#define bfd_put(bits, abfd, val, ptr) \
    ((bits) == 8 ? bfd_put_8 (abfd, val, ptr) \
     : (bits) == 16 ? bfd_put_16 (abfd, val, ptr) \
     : (bits) == 32 ? bfd_put_32 (abfd, val, ptr) \
     : (bits) == 64 ? bfd_put_64 (abfd, val, ptr) \
     : (abort (), (void) 0))

```

### 2.15.1.11 bfd\_h\_put\_size

These macros have the same function as their `bfd_get_x` brethren, except that they are used for removing information for the header records of object files. Believe it or not, some object files keep their header records in big endian order and their data in little endian order.

```

/* Byte swapping macros for file header data. */

#define bfd_h_put_8(abfd, val, ptr) \
    bfd_put_8 (abfd, val, ptr)
#define bfd_h_put_signed_8(abfd, val, ptr) \
    bfd_put_8 (abfd, val, ptr)
#define bfd_h_get_8(abfd, ptr) \
    bfd_get_8 (abfd, ptr)
#define bfd_h_get_signed_8(abfd, ptr) \
    bfd_get_signed_8 (abfd, ptr)

#define bfd_h_put_16(abfd, val, ptr) \
    BFD_SEND (abfd, bfd_h_putx16, (val, ptr))
#define bfd_h_put_signed_16 \
    bfd_h_put_16
#define bfd_h_get_16(abfd, ptr) \

```



```

    BFD_SEND (abfd, bfd_h_getx16, (ptr))
#define bfd_h_get_signed_16(abfd, ptr) \
    BFD_SEND (abfd, bfd_h_getx_signed_16, (ptr))

#define bfd_h_put_32(abfd, val, ptr) \
    BFD_SEND (abfd, bfd_h_putx32, (val, ptr))
#define bfd_h_put_signed_32 \
    bfd_h_put_32
#define bfd_h_get_32(abfd, ptr) \
    BFD_SEND (abfd, bfd_h_getx32, (ptr))
#define bfd_h_get_signed_32(abfd, ptr) \
    BFD_SEND (abfd, bfd_h_getx_signed_32, (ptr))

#define bfd_h_put_64(abfd, val, ptr) \
    BFD_SEND (abfd, bfd_h_putx64, (val, ptr))
#define bfd_h_put_signed_64 \
    bfd_h_put_64
#define bfd_h_get_64(abfd, ptr) \
    BFD_SEND (abfd, bfd_h_getx64, (ptr))
#define bfd_h_get_signed_64(abfd, ptr) \
    BFD_SEND (abfd, bfd_h_getx_signed_64, (ptr))

/* Aliases for the above, which should eventually go away. */

#define H_PUT_64    bfd_h_put_64
#define H_PUT_32    bfd_h_put_32
#define H_PUT_16    bfd_h_put_16
#define H_PUT_8     bfd_h_put_8
#define H_PUT_S64   bfd_h_put_signed_64
#define H_PUT_S32   bfd_h_put_signed_32
#define H_PUT_S16   bfd_h_put_signed_16
#define H_PUT_S8    bfd_h_put_signed_8
#define H_GET_64    bfd_h_get_64
#define H_GET_32    bfd_h_get_32
#define H_GET_16    bfd_h_get_16
#define H_GET_8     bfd_h_get_8
#define H_GET_S64   bfd_h_get_signed_64
#define H_GET_S32   bfd_h_get_signed_32
#define H_GET_S16   bfd_h_get_signed_16
#define H_GET_S8    bfd_h_get_signed_8

```

### 2.15.1.12 Byte swapping routines.

```
uint64_t bfd_getb64 (const void *); uint64_t bfd_getl64 (const void * [Function]
*); int64_t bfd_getb_signed_64 (const void *); int64_t bfd_getl_signed_64
(const void *); bfd_vma bfd_getb32 (const void *); bfd_vma bfd_getl32
(const void *); bfd_signed_vma bfd_getb_signed_32 (const void *);
bfd_signed_vma bfd_getl_signed_32 (const void *); bfd_vma bfd_getb16
(const void *); bfd_vma bfd_getl16 (const void *); bfd_signed_vma
bfd_getb_signed_16 (const void *); bfd_signed_vma bfd_getl_signed_16
(const void *); void bfd_putb64 (uint64_t, void *); void bfd_putl64
(uint64_t, void *); void bfd_putb32 (bfd_vma, void *); void bfd_putl32
(bfd_vma, void *); void bfd_putb24 (bfd_vma, void *); void bfd_putl24
(bfd_vma, void *); void bfd_putb16 (bfd_vma, void *); void bfd_putl16
(bfd_vma, void *); uint64_t bfd_get_bits (const void *, int, bool); void
bfd_put_bits (uint64_t, void *, int, bool);
```

Read and write integers in a particular endian order. `getb` and `putb` functions handle big-endian, `getl` and `putl` handle little-endian. `bfd_get_bits` and `bfd_put_bits` specify big-endian by passing `TRUE` in the last parameter, little-endian by passing `FALSE`.

### 2.15.1.13 `bfd_log2`

```
unsigned int bfd_log2 (bfd_vma x); [Function]
Return the log base 2 of the value supplied, rounded up. E.g., an x of 1025 returns
11. A x of 0 returns 0.
```

## 2.16 File caching

The file caching mechanism is embedded within BFD and allows the application to open as many BFDs as it wants without regard to the underlying operating system's file descriptor limit (often as low as 20 open files). The module in `cache.c` maintains a least recently used list of `bfd_cache_max_open` files, and exports the name `bfd_cache_lookup`, which runs around and makes sure that the required BFD is open. If not, then it chooses a file to close, closes it and opens the one wanted, returning its file handle.

### 2.16.1 Caching functions

#### 2.16.1.1 `bfd_cache_init`

```
bool bfd_cache_init (bfd *abfd); [Function]
Add a newly opened BFD to the cache.
```

#### 2.16.1.2 `bfd_cache_close`

```
bool bfd_cache_close (bfd *abfd); [Function]
Remove the BFD abfd from the cache. If the attached file is open, then close it too.
FALSE is returned if closing the file fails, TRUE is returned if all is well.
```

### 2.16.1.3 bfd\_cache\_close\_all

**bool bfd\_cache\_close\_all (void);** [Function]  
 Remove all BFDs from the cache. If the attached file is open, then close it too. Note - despite its name this function will close a BFD even if it is not marked as being cacheable, ie even if `bfd_get_cacheable()` returns false.  
 FALSE is returned if closing one of the file fails, TRUE is returned if all is well.

### 2.16.1.4 bfd\_cache\_set\_uncloseable

**bool bfd\_cache\_set\_uncloseable (bfd \*abfd, bool value, bool \*old);** [Function]  
 Internal function to mark ABFD as either closeable or not. This is used by `bfd_check_format_matches` to avoid races where `bfd_cache_close_all` is called in another thread. VALUE is true to mark the BFD as temporarily uncloseable by the cache; false to mark it as closeable once again. OLD, if non-NULL, is set to the previous value of the flag. Returns false on error, true on success.

### 2.16.1.5 bfd\_cache\_size

**unsigned bfd\_cache\_size (void);** [Function]  
 Return the number of open files in the cache.

### 2.16.1.6 bfd\_open\_file

**FILE\* bfd\_open\_file (bfd \*abfd);** [Function]  
 Call the OS to open a file for *abfd*. Return the FILE \* (possibly NULL) that results from this operation. Set up the BFD so that future accesses know the file is open. If the FILE \* returned is NULL, then it won't have been put in the cache, so it won't have to be removed from it.

## 2.17 Linker Functions

The linker uses three special entry points in the BFD target vector. It is not necessary to write special routines for these entry points when creating a new BFD back end, since generic versions are provided. However, writing them can speed up linking and make it use significantly less runtime memory.

The first routine creates a hash table used by the other routines. The second routine adds the symbols from an object file to the hash table. The third routine takes all the object files and links them together to create the output file. These routines are designed so that the linker proper does not need to know anything about the symbols in the object files that it is linking. The linker merely arranges the sections as directed by the linker script and lets BFD handle the details of symbols and relocs.

The second routine and third routines are passed a pointer to a `struct bfd_link_info` structure (defined in `bfdlink.h`) which holds information relevant to the link, including the linker hash table (which was created by the first routine) and a set of callback functions to the linker proper.

The generic linker routines are in `linker.c`, and use the header file `genlink.h`. As of this writing, the only back ends which have implemented versions of these routines are a.out (in `aoutx.h`) and ECOFF (in `ecoff.c`). The a.out routines are used as examples throughout this section.

### 2.17.1 Creating a linker hash table

The linker routines must create a hash table, which must be derived from `struct bfd_link_hash_table` described in `bfdlink.c`. See Section 2.18 [Hash Tables], page 185, for information on how to create a derived hash table. This entry point is called using the target vector of the linker output file.

The `_bfd_link_hash_table_create` entry point must allocate and initialize an instance of the desired hash table. If the back end does not require any additional information to be stored with the entries in the hash table, the entry point may simply create a `struct bfd_link_hash_table`. Most likely, however, some additional information will be needed.

For example, with each entry in the hash table the a.out linker keeps the index the symbol has in the final output file (this index number is used so that when doing a relocatable link the symbol index used in the output file can be quickly filled in when copying over a reloc). The a.out linker code defines the required structures and functions for a hash table derived from `struct bfd_link_hash_table`. The a.out linker hash table is created by the function `NAME(aout,link_hash_table_create)`; it simply allocates space for the hash table, initializes it, and returns a pointer to it.

When writing the linker routines for a new back end, you will generally not know exactly which fields will be required until you have finished. You should simply create a new hash table which defines no additional fields, and then simply add fields as they become necessary.

### 2.17.2 Adding symbols to the hash table

The linker proper will call the `_bfd_link_add_symbols` entry point for each object file or archive which is to be linked (typically these are the files named on the command line, but some may also come from the linker script). The entry point is responsible for examining the file. For an object file, BFD must add any relevant symbol information to the hash table. For an archive, BFD must determine which elements of the archive should be used and adding them to the link.

The a.out version of this entry point is `NAME(aout,link_add_symbols)`.

#### 2.17.2.1 Differing file formats

Normally all the files involved in a link will be of the same format, but it is also possible to link together different format object files, and the back end must support that. The `_bfd_link_add_symbols` entry point is called via the target vector of the file to be added. This has an important consequence: the function may not assume that the hash table is the type created by the corresponding `_bfd_link_hash_table_create` vector. All the `_bfd_link_add_symbols` function can assume about the hash table is that it is derived from `struct bfd_link_hash_table`.

Sometimes the `_bfd_link_add_symbols` function must store some information in the hash table entry to be used by the `_bfd_final_link` function. In such a case the output bfd xvec must be checked to make sure that the hash table was created by an object file of the same format.

The `_bfd_final_link` routine must be prepared to handle a hash entry without any extra information added by the `_bfd_link_add_symbols` function. A hash entry without extra information will also occur when the linker script directs the linker to create a symbol. Note that, regardless of how a hash table entry is added, all the fields will be initialized to some sort of null value by the hash table entry initialization function.

See `ecoff_link_add_externals` for an example of how to check the output bfd before saving information (in this case, the ECOFF external symbol debugging information) in a hash table entry.

### 2.17.2.2 Adding symbols from an object file

When the `_bfd_link_add_symbols` routine is passed an object file, it must add all externally visible symbols in that object file to the hash table. The actual work of adding the symbol to the hash table is normally handled by the function `_bfd_generic_link_add_one_symbol`. The `_bfd_link_add_symbols` routine is responsible for reading all the symbols from the object file and passing the correct information to `_bfd_generic_link_add_one_symbol`.

The `_bfd_link_add_symbols` routine should not use `bfd_canonicalize_syntab` to read the symbols. The point of providing this routine is to avoid the overhead of converting the symbols into generic `asymbol` structures.

`_bfd_generic_link_add_one_symbol` handles the details of combining common symbols, warning about multiple definitions, and so forth. It takes arguments which describe the symbol to add, notably symbol flags, a section, and an offset. The symbol flags include such things as `BSF_WEAK` or `BSF_INDIRECT`. The section is a section in the object file, or something like `bfd_und_section_ptr` for an undefined symbol or `bfd_com_section_ptr` for a common symbol.

If the `_bfd_final_link` routine is also going to need to read the symbol information, the `_bfd_link_add_symbols` routine should save it somewhere attached to the object file BFD. However, the information should only be saved if the `keep_memory` field of the `info` argument is `TRUE`, so that the `-no-keep-memory` linker switch is effective.

The `a.out` function which adds symbols from an object file is `aout_link_add_object_symbols`, and most of the interesting work is in `aout_link_add_symbols`. The latter saves pointers to the hash tables entries created by `_bfd_generic_link_add_one_symbol` indexed by symbol number, so that the `_bfd_final_link` routine does not have to call the hash table lookup routine to locate the entry.

### 2.17.2.3 Adding symbols from an archive

When the `_bfd_link_add_symbols` routine is passed an archive, it must look through the symbols defined by the archive and decide which elements of the archive should be included in the link. For each such element it must call the `add_archive_element` linker callback, and it must add the symbols from the object file to the linker hash table. (The callback may in fact indicate that a replacement BFD should be used, in which case the symbols from that BFD should be added to the linker hash table instead.)

In most cases the work of looking through the symbols in the archive should be done by the `_bfd_generic_link_add_archive_symbols` function. `_bfd_generic_link_add_archive_symbols` is passed a function to call to make the final decision about adding an

archive element to the link and to do the actual work of adding the symbols to the linker hash table. If the element is to be included, the `add_archive_element` linker callback routine must be called with the element as an argument, and the element's symbols must be added to the linker hash table just as though the element had itself been passed to the `_bfd_link_add_symbols` function.

When the `a.out _bfd_link_add_symbols` function receives an archive, it calls `_bfd_generic_link_add_archive_symbols` passing `aout_link_check_archive_element` as the function argument. `aout_link_check_archive_element` calls `aout_link_check_ar_symbols`. If the latter decides to add the element (an element is only added if it provides a real, non-common, definition for a previously undefined or common symbol) it calls the `add_archive_element` callback and then `aout_link_check_archive_element` calls `aout_link_add_symbols` to actually add the symbols to the linker hash table - possibly those of a substitute BFD, if the `add_archive_element` callback avails itself of that option.

The ECOFF back end is unusual in that it does not normally call `_bfd_generic_link_add_archive_symbols`, because ECOFF archives already contain a hash table of symbols. The ECOFF back end searches the archive itself to avoid the overhead of creating a new hash table.

### 2.17.3 Performing the final link

When all the input files have been processed, the linker calls the `_bfd_final_link` entry point of the output BFD. This routine is responsible for producing the final output file, which has several aspects. It must relocate the contents of the input sections and copy the data into the output sections. It must build an output symbol table including any local symbols from the input files and the global symbols from the hash table. When producing relocatable output, it must modify the input relocs and write them into the output file. There may also be object format dependent work to be done.

The linker will also call the `write_object_contents` entry point when the BFD is closed. The two entry points must work together in order to produce the correct output file.

The details of how this works are inevitably dependent upon the specific object file format. The `a.out _bfd_final_link` routine is `NAME(aout,final_link)`.

#### 2.17.3.1 Information provided by the linker

Before the linker calls the `_bfd_final_link` entry point, it sets up some data structures for the function to use.

The `input_bfds` field of the `bfd_link_info` structure will point to a list of all the input files included in the link. These files are linked through the `link.next` field of the `bfd` structure.

Each section in the output file will have a list of `link_order` structures attached to the `map_head.link_order` field (the `link_order` structure is defined in `bfdlink.h`). These structures describe how to create the contents of the output section in terms of the contents of various input sections, fill constants, and, eventually, other types of information. They also describe relocs that must be created by the BFD backend, but do not correspond to any input file; this is used to support `-Ur`, which builds constructors while generating a relocatable object file.

### 2.17.3.2 Relocating the section contents

The `_bfd_final_link` function should look through the `link_order` structures attached to each section of the output file. Each `link_order` structure should either be handled specially, or it should be passed to the function `_bfd_default_link_order` which will do the right thing (`_bfd_default_link_order` is defined in `linker.c`).

For efficiency, a `link_order` of type `bfd_indirect_link_order` whose associated section belongs to a BFD of the same format as the output BFD must be handled specially. This type of `link_order` describes part of an output section in terms of a section belonging to one of the input files. The `_bfd_final_link` function should read the contents of the section and any associated relocs, apply the relocs to the section contents, and write out the modified section contents. If performing a relocatable link, the relocs themselves must also be modified and written out.

The functions `_bfd_relocate_contents` and `_bfd_final_link_relocate` provide some general support for performing the actual relocations, notably overflow checking. Their arguments include information about the symbol the relocation is against and a `reloc_howto_type` argument which describes the relocation to perform. These functions are defined in `reloc.c`.

The `a.out` function which handles reading, relocating, and writing section contents is `aout_link_input_section`. The actual relocation is done in `aout_link_input_section_std` and `aout_link_input_section_ext`.

### 2.17.3.3 Writing the symbol table

The `_bfd_final_link` function must gather all the symbols in the input files and write them out. It must also write out all the symbols in the global hash table. This must be controlled by the `strip` and `discard` fields of the `bfd_link_info` structure.

The local symbols of the input files will not have been entered into the linker hash table. The `_bfd_final_link` routine must consider each input file and include the symbols in the output file. It may be convenient to do this when looking through the `link_order` structures, or it may be done by stepping through the `input_bfds` list.

The `_bfd_final_link` routine must also traverse the global hash table to gather all the externally visible symbols. It is possible that most of the externally visible symbols may be written out when considering the symbols of each input file, but it is still necessary to traverse the hash table since the linker script may have defined some symbols that are not in any of the input files.

The `strip` field of the `bfd_link_info` structure controls which symbols are written out. The possible values are listed in `bfdlink.h`. If the value is `strip_some`, then the `keep_hash` field of the `bfd_link_info` structure is a hash table of symbols to keep; each symbol should be looked up in this hash table, and only symbols which are present should be included in the output file.

If the `strip` field of the `bfd_link_info` structure permits local symbols to be written out, the `discard` field is used to further controls which local symbols are included in the output file. If the value is `discard_1`, then all local symbols which begin with a certain prefix are discarded; this is controlled by the `bfd_is_local_label_name` entry point.

The `a.out` backend handles symbols by calling `aout_link_write_symbols` on each input BFD and then traversing the global hash table with the function `aout_link_write_other_`

symbol. It builds a string table while writing out the symbols, which is written to the output file at the end of `NAME(aout,final_link)`.

#### 2.17.3.4 `_bfd_generic_link_add_one_symbol`

```
bool _bfd_generic_link_add_one_symbol (struct bfd_link_info [Function]
                                     *info, bfd *abfd, const char *name, flagword flags, asection *section,
                                     bfd_vma value, const char *string, bool copy, bool collect, struct
                                     bfd_link_hash_entry **hashp);
```

Add a symbol to the global hash table. ABFD is the BFD the symbol comes from. NAME is the name of the symbol. FLAGS is the BSF\_\* bits associated with the symbol. SECTION is the section in which the symbol is defined; this may be `bfd_und_section_ptr` or `bfd_com_section_ptr`. VALUE is the value of the symbol, relative to the section. STRING is used for either an indirect symbol, in which case it is the name of the symbol to indirect to, or a warning symbol, in which case it is the warning string. COPY is TRUE if NAME or STRING must be copied into locally allocated memory if they need to be saved. COLLECT is TRUE if we should automatically collect gcc constructor or destructor names as collect2 does. HASHP, if not NULL, is a place to store the created hash table entry; if \*HASHP is not NULL, the caller has already looked up the hash table entry, and stored it in \*HASHP.

#### 2.17.3.5 `bfd_link_align_section`

```
bool bfd_link_align_section (asection *, unsigned int); [Function]
```

Increase section alignment if the current section alignment is less than the requested value. Adjust output section alignment too, so that linker layout adjusts for alignment on the current `lang_size_sections` pass. This is important for `lang_size_relo_segment`. If the output section alignment isn't adjusted, the linker will place the output section at an address depending on its current alignment. When sizing the output section, input sections attached transfer any increase in alignment to the output section, which will affect layout for the next sizing pass. Which is all well and good except that `lang_size_relo_segment` for the current sizing pass uses that possibly increased alignment with a layout that doesn't suit.

#### 2.17.3.6 `bfd_link_split_section`

```
bool bfd_link_split_section (bfd *abfd, asection *sec); [Function]
```

Return nonzero if sec should be split during a relocatable or final link.

```
#define bfd_link_split_section(abfd, sec) \
    BFD_SEND (abfd, _bfd_link_split_section, (abfd, sec))
```

#### 2.17.3.7 `bfd_section_already_linked`

```
bool bfd_section_already_linked (bfd *abfd, asection *sec, struct [Function]
                                bfd_link_info *info);
```

Check if data has been already linked during a relocatable or final link. Return TRUE if it has.

```
#define bfd_section_already_linked(abfd, sec, info) \
```



```
BFD_SEND (abfd, _section_already_linked, (abfd, sec, info))
```

### 2.17.3.8 bfd\_generic\_define\_common\_symbol

```
bool bfd_generic_define_common_symbol (bfd *output_bfd, struct [Function]
    bfd_link_info *info, struct bfd_link_hash_entry *h);
```

Convert common symbol *h* into a defined symbol. Return TRUE on success and FALSE on failure.

```
#define bfd_define_common_symbol(output_bfd, info, h) \
    BFD_SEND (output_bfd, _bfd_define_common_symbol, (output_bfd, info, h))■
```

### 2.17.3.9 \_bfd\_generic\_link\_hide\_symbol

```
void _bfd_generic_link_hide_symbol (bfd *output_bfd, struct [Function]
    bfd_link_info *info, struct bfd_link_hash_entry *h);
```

Hide symbol *h*. This is an internal function. It should not be called from outside the BFD library.

```
#define bfd_link_hide_symbol(output_bfd, info, h) \
    BFD_SEND (output_bfd, _bfd_link_hide_symbol, (output_bfd, info, h))■
```

### 2.17.3.10 bfd\_generic\_define\_start\_stop

```
struct bfd_link_hash_entry *bfd_generic_define_start_stop (struct [Function]
    bfd_link_info *info, const char *symbol, asection *sec);
```

Define a `__start`, `__stop`, `.startof`, or `.sizeof` symbol. Return the symbol or NULL if no such undefined symbol exists.

```
#define bfd_define_start_stop(output_bfd, info, symbol, sec) \
    BFD_SEND (output_bfd, _bfd_define_start_stop, (info, symbol, sec))■
```

### 2.17.3.11 bfd\_find\_version\_for\_sym

```
struct bfd_elf_version_tree *bfd_find_version_for_sym (struct [Function]
    bfd_elf_version_tree *verdefs, const char *sym_name, bool *hide);
```

Search an elf version script tree for symbol versioning info and export / don't-export status for a given symbol. Return non-NULL on success and NULL on failure; also sets the output 'hide' boolean parameter.

### 2.17.3.12 bfd\_hide\_sym\_by\_version

```
bool bfd_hide_sym_by_version (struct bfd_elf_version_tree [Function]
    *verdefs, const char *sym_name);
```

Search an elf version script tree for symbol versioning info for a given symbol. Return TRUE if the symbol is hidden.

### 2.17.3.13 `bfd_link_check_relocs`

`bool bfd_link_check_relocs (bfd *abfd, struct bfd_link_info *info);` [Function]

Checks the relocs in ABFD for validity. Does not execute the relocs. Return TRUE if everything is OK, FALSE otherwise. This is the external entry point to this code.

### 2.17.3.14 `_bfd_generic_link_check_relocs`

`bool _bfd_generic_link_check_relocs (bfd *abfd, struct bfd_link_info *info);` [Function]

Stub function for targets that do not implement reloc checking. Return TRUE. This is an internal function. It should not be called from outside the BFD library.

### 2.17.3.15 `bfd_merge_private_bfd_data`

Merge private BFD information from the BFD *ibfd* to the the output file BFD when linking. Return TRUE on success, FALSE on error. Possible error returns are:

- `bfd_error_no_memory` - Not enough memory exists to create private data for *obfd*.

```
#define bfd_merge_private_bfd_data(ibfd, info) \
    BFD_SEND ((info)->output_bfd, _bfd_merge_private_bfd_data, \
              (ibfd, info))
```

### 2.17.3.16 `_bfd_generic_verify_endian_match`

`bool _bfd_generic_verify_endian_match (bfd *ibfd, struct bfd_link_info *info);` [Function]

Can be used from / for `bfd_merge_private_bfd_data` to check that endianness matches between input and output file. Returns TRUE for a match, otherwise returns FALSE and emits an error.

## 2.18 Hash Tables

BFD provides a simple set of hash table functions. Routines are provided to initialize a hash table, to free a hash table, to look up a string in a hash table and optionally create an entry for it, and to traverse a hash table. There is currently no routine to delete a string from a hash table.

The basic hash table does not permit any data to be stored with a string. However, a hash table is designed to present a base class from which other types of hash tables may be derived. These derived types may store additional information with the string. Hash tables were implemented in this way, rather than simply providing a data pointer in a hash table entry, because they were designed for use by the linker back ends. The linker may create thousands of hash table entries, and the overhead of allocating private data and storing and following pointers becomes noticeable.

The basic hash table code is in `hash.c`.

### 2.18.1 Creating and freeing a hash table

To create a hash table, create an instance of a `struct bfd_hash_table` (defined in `bfd.h`) and call `bfd_hash_table_init` (if you know approximately how many entries you will need, the function `bfd_hash_table_init_n`, which takes a *size* argument, may be used). `bfd_hash_table_init` returns `FALSE` if some sort of error occurs.

The function `bfd_hash_table_init` take as an argument a function to use to create new entries. For a basic hash table, use the function `bfd_hash_newfunc`. See Section 2.18.4 [Deriving a New Hash Table Type], page 186, for why you would want to use a different value for this argument.

`bfd_hash_table_init` will create an `objalloc` which will be used to allocate new entries. You may allocate memory on this `objalloc` using `bfd_hash_allocate`.

Use `bfd_hash_table_free` to free up all the memory that has been allocated for a hash table. This will not free up the `struct bfd_hash_table` itself, which you must provide.

Use `bfd_hash_set_default_size` to set the default size of hash table to use.

### 2.18.2 Looking up or entering a string

The function `bfd_hash_lookup` is used both to look up a string in the hash table and to create a new entry.

If the *create* argument is `FALSE`, `bfd_hash_lookup` will look up a string. If the string is found, it will return a pointer to a `struct bfd_hash_entry`. If the string is not found in the table `bfd_hash_lookup` will return `NULL`. You should not modify any of the fields in the returned `struct bfd_hash_entry`.

If the *create* argument is `TRUE`, the string will be entered into the hash table if it is not already there. Either way a pointer to a `struct bfd_hash_entry` will be returned, either to the existing structure or to a newly created one. In this case, a `NULL` return means that an error occurred.

If the *create* argument is `TRUE`, and a new entry is created, the *copy* argument is used to decide whether to copy the string onto the hash table `objalloc` or not. If *copy* is passed as `FALSE`, you must be careful not to deallocate or modify the string as long as the hash table exists.

### 2.18.3 Traversing a hash table

The function `bfd_hash_traverse` may be used to traverse a hash table, calling a function on each element. The traversal is done in a random order.

`bfd_hash_traverse` takes as arguments a function and a generic `void *` pointer. The function is called with a hash table entry (a `struct bfd_hash_entry *`) and the generic pointer passed to `bfd_hash_traverse`. The function must return a `boolean` value, which indicates whether to continue traversing the hash table. If the function returns `FALSE`, `bfd_hash_traverse` will stop the traversal and return immediately.

### 2.18.4 Deriving a new hash table type

Many uses of hash tables want to store additional information which each entry in the hash table. Some also find it convenient to store additional information with the hash table itself. This may be done using a derived hash table.

Since C is not an object oriented language, creating a derived hash table requires sticking together some boilerplate routines with a few differences specific to the type of hash table you want to create.

An example of a derived hash table is the linker hash table. The structures for this are defined in `bfdlink.h`. The functions are in `linker.c`.

You may also derive a hash table from an already derived hash table. For example, the a.out linker backend code uses a hash table derived from the linker hash table.

#### 2.18.4.1 Define the derived structures

You must define a structure for an entry in the hash table, and a structure for the hash table itself.

The first field in the structure for an entry in the hash table must be of the type used for an entry in the hash table you are deriving from. If you are deriving from a basic hash table this is `struct bfd_hash_entry`, which is defined in `bfd.h`. The first field in the structure for the hash table itself must be of the type of the hash table you are deriving from itself. If you are deriving from a basic hash table, this is `struct bfd_hash_table`.

For example, the linker hash table defines `struct bfd_link_hash_entry` (in `bfdlink.h`). The first field, `root`, is of type `struct bfd_hash_entry`. Similarly, the first field in `struct bfd_link_hash_table`, `table`, is of type `struct bfd_hash_table`.

#### 2.18.4.2 Write the derived creation routine

You must write a routine which will create and initialize an entry in the hash table. This routine is passed as the function argument to `bfd_hash_table_init`.

In order to permit other hash tables to be derived from the hash table you are creating, this routine must be written in a standard way.

The first argument to the creation routine is a pointer to a hash table entry. This may be `NULL`, in which case the routine should allocate the right amount of space. Otherwise the space has already been allocated by a hash table type derived from this one.

After allocating space, the creation routine must call the creation routine of the hash table type it is derived from, passing in a pointer to the space it just allocated. This will initialize any fields used by the base hash table.

Finally the creation routine must initialize any local fields for the new hash table type.

Here is a boilerplate example of a creation routine. *function\_name* is the name of the routine. *entry\_type* is the type of an entry in the hash table you are creating. *base\_newfunc* is the name of the creation routine of the hash table type your hash table is derived from.

```
struct bfd_hash_entry *
function_name (struct bfd_hash_entry *entry,
               struct bfd_hash_table *table,
               const char *string)
{
    struct entry_type *ret = (entry_type *) entry;

    /* Allocate the structure if it has not already been allocated by a
       derived class. */
}
```

```

    if (ret == NULL)
    {
        ret = bfd_hash_allocate (table, sizeof (* ret));
        if (ret == NULL)
            return NULL;
    }

    /* Call the allocation method of the base class. */
    ret = ((entry_type *)
          base_newfunc ((struct bfd_hash_entry *) ret, table, string));

    /* Initialize the local fields here. */

    return (struct bfd_hash_entry *) ret;
}

```

The creation routine for the linker hash table, which is in `linker.c`, looks just like this example. *function\_name* is `_bfd_link_hash_newfunc`. *entry\_type* is `struct bfd_link_hash_entry`. *base\_newfunc* is `bfd_hash_newfunc`, the creation routine for a basic hash table.

`_bfd_link_hash_newfunc` also initializes the local fields in a linker hash table entry: `type`, `written` and `next`.

#### 2.18.4.3 Write other derived routines

You will want to write other routines for your new hash table, as well.

You will want an initialization routine which calls the initialization routine of the hash table you are deriving from and initializes any other local fields. For the linker hash table, this is `_bfd_link_hash_table_init` in `linker.c`.

You will want a lookup routine which calls the lookup routine of the hash table you are deriving from and casts the result. The linker hash table uses `bfd_link_hash_lookup` in `linker.c` (this actually takes an additional argument which it uses to decide how to return the looked up value).

You may want a traversal routine. This should just call the traversal routine of the hash table you are deriving from with appropriate casts. The linker hash table uses `bfd_link_hash_traverse` in `linker.c`.

These routines may simply be defined as macros. For example, the a.out backend linker hash table, which is derived from the linker hash table, uses macros for the lookup and traversal routines. These are `aout_link_hash_lookup` and `aout_link_hash_traverse` in `aoutx.h`.

#### 2.18.4.4 `bfd_hash_table_init_n`

```

bool bfd_hash_table_init_n (struct bfd_hash_table *, struct
    bfd_hash_entry *(* *newfunc*) (struct bfd_hash_entry *, struct
    bfd_hash_table *, const char *), unsigned int *entsize*, unsigned int
    *size*);

```

[Function]

Create a new hash table, given a number of entries.

**2.18.4.5 bfd\_hash\_table\_init**

```
bool bfd_hash_table_init (struct bfd_hash_table *, struct      [Function]
                          bfd_hash_entry *(* newfunc) (struct bfd_hash_entry *, struct
                          bfd_hash_table *, const char *), unsigned int *entsize*);
```

Create a new hash table with the default number of entries.

**2.18.4.6 bfd\_hash\_table\_free**

```
void bfd_hash_table_free (struct bfd_hash_table *);           [Function]
```

Free a hash table.

**2.18.4.7 bfd\_hash\_lookup**

```
struct bfd_hash_entry *bfd_hash_lookup (struct bfd_hash_table *, [Function]
                                         const char *, bool *create*, bool *copy*);
```

Look up a string in a hash table.

**2.18.4.8 bfd\_hash\_insert**

```
struct bfd_hash_entry *bfd_hash_insert (struct bfd_hash_table *, [Function]
                                         const char *, unsigned long *hash*);
```

Insert an entry in a hash table.

**2.18.4.9 bfd\_hash\_rename**

```
void bfd_hash_rename (struct bfd_hash_table *, const char *, struct [Function]
                      bfd_hash_entry *);
```

Rename an entry in a hash table.

**2.18.4.10 bfd\_hash\_replace**

```
void bfd_hash_replace (struct bfd_hash_table *, struct          [Function]
                       bfd_hash_entry * *old*, struct bfd_hash_entry * *new*);
```

Replace an entry in a hash table.

**2.18.4.11 bfd\_hash\_allocate**

```
void *bfd_hash_allocate (struct bfd_hash_table *, unsigned int   [Function]
                         *size*);
```

Allocate space in a hash table.

**2.18.4.12 bfd\_hash\_newfunc**

```
struct bfd_hash_entry *bfd_hash_newfunc (struct bfd_hash_entry   [Function]
                                         *, struct bfd_hash_table *, const char *);
```

Base method for creating a new hash table entry.

**2.18.4.13 bfd\_hash\_traverse**

`void bfd_hash_traverse (struct bfd_hash_table *, bool (*) (struct  
                   bfd_hash_entry *, void *), void *);` [Function]

Traverse a hash table.

**2.18.4.14 bfd\_hash\_set\_default\_size**

`unsigned int bfd_hash_set_default_size (unsigned int);` [Function]

Set hash table default size.

**2.18.4.15 \_bfd\_stringtab\_init**

`struct bfd_strtab_hash *_bfd_stringtab_init (void);` [Function]

Create a new strtab.

**2.18.4.16 \_bfd\_xcoff\_stringtab\_init**

`struct bfd_strtab_hash *_bfd_xcoff_stringtab_init (bool  
                   *isxcoff64*);` [Function]

Create a new strtab in which the strings are output in the format used in the XCOFF .debug section: a two byte length precedes each string.

**2.18.4.17 \_bfd\_stringtab\_free**

`void _bfd_stringtab_free (struct bfd_strtab_hash *);` [Function]

Free a strtab.

**2.18.4.18 \_bfd\_stringtab\_add**

`bfd_size_type _bfd_stringtab_add (struct bfd_strtab_hash *,  
                   const char *, bool *hash*, bool *copy*);` [Function]

Get the index of a string in a strtab, adding it if it is not already present. If HASH is FALSE, we don't really use the hash table, and we don't eliminate duplicate strings. If COPY is true then store a copy of STR if creating a new entry.

**2.18.4.19 \_bfd\_stringtab\_size**

`bfd_size_type _bfd_stringtab_size (struct bfd_strtab_hash *);` [Function]

Get the number of bytes in a strtab.

**2.18.4.20 \_bfd\_stringtab\_emit**

`bool _bfd_stringtab_emit (bfd *, struct bfd_strtab_hash *);` [Function]

Write out a strtab. ABFD must already be at the right location in the file.

## 3 BFD back ends

### 3.1 What to Put Where

All of BFD lives in one directory.

### 3.2 a.out backends

BFD supports a number of different flavours of a.out format, though the major differences are only the sizes of the structures on disk, and the shape of the relocation information.

The support is split into a basic support file `aoutx.h` and other files which derive functions from the base. One derivation file is `aoutf1.h` (for a.out flavour 1), and adds to the basic a.out functions support for sun3, sun4, and 386 a.out files, to create a target jump vector for a specific target.

This information is further split out into more specific files for each machine, including `sunos.c` for sun3 and sun4, and `demo64.c` for a demonstration of a 64 bit a.out format.

The base file `aoutx.h` defines general mechanisms for reading and writing records to and from disk and various other methods which BFD requires. It is included by `aout32.c` and `aout64.c` to form the names `aout_32_swap_exec_header_in`, `aout_64_swap_exec_header_in`, etc.

As an example, this is what goes on to make the back end for a sun4, from `aout32.c`:

```
#define ARCH_SIZE 32
#include "aoutx.h"
```

Which exports names:

```
...
aout_32_canonicalize_reloc
aout_32_find_nearest_line
aout_32_get_lineno
aout_32_get_reloc_upper_bound
...
```

from `sunos.c`:

```
#define TARGET_NAME "a.out-sunos-big"
#define VECNAME      sparc_aout_sunos_be_vec
#include "aoutf1.h"
```

requires all the names from `aout32.c`, and produces the jump vector

```
sparc_aout_sunos_be_vec
```

The file `host-aout.c` is a special case. It is for a large set of hosts that use “more or less standard” a.out files, and for which cross-debugging is not interesting. It uses the standard 32-bit a.out support routines, but determines the file offsets and addresses of the text, data, and BSS sections, the machine architecture and machine type, and the entry point address, in a host-dependent manner. Once these values have been determined, generic code is used to handle the object file.

When porting it to run on a new system, you must supply:

```
HOST_PAGE_SIZE
```



```

HOST_SEGMENT_SIZE
HOST_MACHINE_ARCH      (optional)
HOST_MACHINE_MACHINE    (optional)
HOST_TEXT_START_ADDR
HOST_STACK_END_ADDR

```

in the file `../include/sys/h-XXX.h` (for your host). These values, plus the structures and macros defined in `a.out.h` on your host system, will produce a BFD target that will access ordinary `a.out` files on your host. To configure a new machine to use `host-aout.c`, specify:

```

TDEFAULTS = -DDEFAULT_VECTOR=host_aout_big_vec
TDEFILES=  host-aout.o trad-core.o

```

in the `config/XXX.mt` file, and modify `configure.ac` to use the `XXX.mt` file (by setting `"bfd_target=XXX"`) when your configuration is selected.

### 3.2.1 Relocations

The file `aoutx.h` provides for both the *standard* and *extended* forms of `a.out` relocation records.

The standard records contain only an address, a symbol index, and a type field. The extended records also have a full integer for an addend.

### 3.2.2 Internal entry points

`aoutx.h` exports several routines for accessing the contents of an `a.out` file, which are gathered and exported in turn by various format specific files (eg `sunos.c`).

#### 3.2.2.1 aout\_size\_swap\_exec\_header\_in

```

void aout_size_swap_exec_header_in, (bfd *abfd, struct      [Function]
    external_exec *bytes, struct internal_exec *execp);
    Swap the information in an executable header raw_bytes taken from a raw byte stream
    memory image into the internal exec header structure execp.

```

#### 3.2.2.2 aout\_size\_swap\_exec\_header\_out

```

bool aout_size_swap_exec_header_out (bfd *abfd, struct      [Function]
    internal_exec *execp, struct external_exec *raw_bytes);
    Swap the information in an internal exec header structure execp into the buffer
    raw_bytes ready for writing to disk.

```

#### 3.2.2.3 aout\_size\_some\_aout\_object\_p

```

bfd_cleanup aout_size_some_aout_object_p (bfd *abfd, struct  [Function]
    internal_exec *execp, bfd_cleanup (*callback_to_real_object_p) (bfd *));
    Some a.out variant thinks that the file open in abfd checking is an a.out file. Do
    some more checking, and set up for access if it really is. Call back to the calling
    environment's "finish up" function just before returning, to handle any last-minute
    setup.

```

### 3.2.2.4 aout\_size\_mkobject

`bool aout_size_mkobject, (bfd *abfd);` [Function]  
 Initialize BFD *abfd* for use with a.out files.

### 3.2.2.5 aout\_size\_machine\_type

`enum machine_type aout_size_machine_type (enum` [Function]  
     `bfd_architecture arch, unsigned long machine, bool *unknown);`  
 Keep track of machine architecture and machine type for a.out's. Return the `machine_type` for a particular architecture and machine, or `M_UNKNOWN` if that exact architecture and machine can't be represented in a.out format.  
 If the architecture is understood, machine type 0 (default) is always understood.

### 3.2.2.6 aout\_size\_set\_arch\_mach

`bool aout_size_set_arch_mach, (bfd *, enum bfd_architecture` [Function]  
     `arch, unsigned long machine);`  
 Set the architecture and the machine of the BFD *abfd* to the values *arch* and *machine*. Verify that *abfd*'s format can support the architecture required.

### 3.2.2.7 aout\_size\_new\_section\_hook

`bool aout_size_new_section_hook, (bfd *abfd, asection` [Function]  
     `*newsect);`  
 Called by the BFD in response to a `bfd_make_section` request.

## 3.3 coff backends

BFD supports a number of different flavours of coff format. The major differences between formats are the sizes and alignments of fields in structures on disk, and the occasional extra field.

Coff in all its varieties is implemented with a few common files and a number of implementation specific files. For example, the i386 coff format is implemented in the file `coff-i386.c`. This file `#includes` `coff/i386.h` which defines the external structure of the coff format for the i386, and `coff/internal.h` which defines the internal structure. `coff-i386.c` also defines the relocations used by the i386 coff format See Section 2.10 [Relocations], page 50.

### 3.3.1 Porting to a new version of coff

The recommended method is to select from the existing implementations the version of coff which is most like the one you want to use. For example, we'll say that i386 coff is the one you select, and that your coff flavour is called foo. Copy `i386coff.c` to `foocoff.c`, copy `../include/coff/i386.h` to `../include/coff/foo.h`, and add the lines to `targets.c` and `Makefile.in` so that your new back end is used. Alter the shapes of the structures in `../include/coff/foo.h` so that they match what you need. You will probably also have to add `#ifdefs` to the code in `coff/internal.h` and `coffcode.h` if your version of coff is too wild.

You can verify that your new BFD backend works quite simply by building `objdump` from the `binutils` directory, and making sure that its version of what's going on and your host system's idea (assuming it has the pretty standard coff dump utility, usually called `att-dump` or just `dump`) are the same. Then clean up your code, and send what you've done to Cygnus. Then your stuff will be in the next release, and you won't have to keep integrating it.

### 3.3.2 How the coff backend works

#### 3.3.2.1 File layout

The Coff backend is split into generic routines that are applicable to any Coff target and routines that are specific to a particular target. The target-specific routines are further split into ones which are basically the same for all Coff targets except that they use the external symbol format or use different values for certain constants.

The generic routines are in `coffgen.c`. These routines work for any Coff target. They use some hooks into the target specific code; the hooks are in a `bfd_coff_backend_data` structure, one of which exists for each target.

The essentially similar target-specific routines are in `coffcode.h`. This header file includes executable C code. The various Coff targets first include the appropriate Coff header file, make any special defines that are needed, and then include `coffcode.h`.

Some of the Coff targets then also have additional routines in the target source file itself.

#### 3.3.2.2 Coff long section names

In the standard Coff object format, section names are limited to the eight bytes available in the `s_name` field of the `SCNHDR` section header structure. The format requires the field to be NUL-padded, but not necessarily NUL-terminated, so the longest section names permitted are a full eight characters.

The Microsoft PE variants of the Coff object file format add an extension to support the use of long section names. This extension is defined in section 4 of the Microsoft PE/COFF specification (rev 8.1). If a section name is too long to fit into the section header's `s_name` field, it is instead placed into the string table, and the `s_name` field is filled with a slash ("/") followed by the ASCII decimal representation of the offset of the full name relative to the string table base.

Note that this implies that the extension can only be used in object files, as executables do not contain a string table. The standard specifies that long section names from objects emitted into executable images are to be truncated.

However, as a GNU extension, BFD can generate executable images that contain a string table and long section names. This would appear to be technically valid, as the standard only says that Coff debugging information is deprecated, not forbidden, and in practice it works, although some tools that parse PE files expecting the MS standard format may become confused; `PEview` is one known example.

The functionality is supported in BFD by code implemented under the control of the macro `COFF_LONG_SECTION_NAMES`. If not defined, the format does not support long section names in any way. If defined, it is used to initialise a flag, `_bfd_coff_long_section_names`, and a hook function pointer, `_bfd_coff_set_long_section_names`, in the Coff backend data structure. The flag controls the generation of long section names in output BFDs at runtime;

if it is false, as it will be by default when generating an executable image, long section names are truncated; if true, the long section names extension is employed. The hook points to a function that allows the value of a copy of the flag in coff object tdata to be altered at runtime, on formats that support long section names at all; on other formats it points to a stub that returns an error indication.

With input BFDs, the flag is set according to whether any long section names are detected while reading the section headers. For a completely new BFD, the flag is set to the default for the target format. This information can be used by a client of the BFD library when deciding what output format to generate, and means that a BFD that is opened for read and subsequently converted to a writeable BFD and modified in-place will retain whatever format it had on input.

If `COFF_LONG_SECTION_NAMES` is simply defined (blank), or is defined to the value "1", then long section names are enabled by default; if it is defined to the value zero, they are disabled by default (but still accepted in input BFDs). The header `coffcode.h` defines a macro, `COFF_DEFAULT_LONG_SECTION_NAMES`, which is used in the backends to initialise the backend data structure fields appropriately; see the comments for further detail.

### 3.3.2.3 Bit twiddling

Each flavour of coff supported in BFD has its own header file describing the external layout of the structures. There is also an internal description of the coff layout, in `coff/internal.h`. A major function of the coff backend is swapping the bytes and twiddling the bits to translate the external form of the structures into the normal internal form. This is all performed in the `bfd_swap_thing_direction` routines. Some elements are different sizes between different versions of coff; it is the duty of the coff version specific include file to override the definitions of various packing routines in `coffcode.h`. E.g., the size of line number entry in coff is sometimes 16 bits, and sometimes 32 bits. `#define`ing `PUT_LNSZ_LNNO` and `GET_LNSZ_LNNO` will select the correct one. No doubt, some day someone will find a version of coff which has a varying field size not catered to at the moment. To port BFD, that person will have to add more `#defines`. Three of the bit twiddling routines are exported to `gdb`; `coff_swap_aux_in`, `coff_swap_sym_in` and `coff_swap_lineno_in`. `GDB` reads the symbol table on its own, but uses BFD to fix things up. More of the bit twiddlers are exported for `gas`; `coff_swap_aux_out`, `coff_swap_sym_out`, `coff_swap_lineno_out`, `coff_swap_reloc_out`, `coff_swap_filehdr_out`, `coff_swap_aouthdr_out`, `coff_swap_scnhdr_out`. `Gas` currently keeps track of all the symbol table and reloc drudgery itself, thereby saving the internal BFD overhead, but uses BFD to swap things on the way out, making cross ports much safer. Doing so also allows BFD (and thus the linker) to use the same header files as `gas`, which makes one avenue to disaster disappear.

### 3.3.2.4 Symbol reading

The simple canonical form for symbols used by BFD is not rich enough to keep all the information available in a coff symbol table. The back end gets around this problem by keeping the original symbol table around, "behind the scenes".

When a symbol table is requested (through a call to `bfd_canonicalize_syntab`), a request gets through to `coff_get_normalized_syntab`. This reads the symbol table from the coff file and swaps all the structures inside into the internal form. It also fixes up all the pointers in the table (represented in the file by offsets from the first symbol in the table) into physical

pointers to elements in the new internal table. This involves some work since the meanings of fields change depending upon context: a field that is a pointer to another structure in the symbol table at one moment may be the size in bytes of a structure at the next. Another pass is made over the table. All symbols which mark file names (`C_FILE` symbols) are modified so that the internal string points to the value in the auxent (the real filename) rather than the normal text associated with the symbol ("`.file`").

At this time the symbol names are moved around. Coff stores all symbols less than nine characters long physically within the symbol table; longer strings are kept at the end of the file in the string table. This pass moves all strings into memory and replaces them with pointers to the strings.

The symbol table is massaged once again, this time to create the canonical table used by the BFD application. Each symbol is inspected in turn, and a decision made (using the `sclass` field) about the various flags to set in the `asymbol`. See Section 2.7 [Symbols], page 40. The generated canonical table shares strings with the hidden internal symbol table.

Any linenumbers are read from the coff file too, and attached to the symbols which own the functions the linenumbers belong to.

### 3.3.2.5 Symbol writing

Writing a symbol to a coff file which didn't come from a coff file will lose any debugging information. The `asymbol` structure remembers the BFD from which the symbol was taken, and on output the back end makes sure that the same destination target as source target is present.

When the symbols have come from a coff file then all the debugging information is preserved. Symbol tables are provided for writing to the back end in a vector of pointers to pointers. This allows applications like the linker to accumulate and output large symbol tables without having to do too much byte copying.

This function runs through the provided symbol table and patches each symbol marked as a file place holder (`C_FILE`) to point to the next file place holder in the list. It also marks each `offset` field in the list with the offset from the first symbol of the current symbol.

Another function of this procedure is to turn the canonical value form of BFD into the form used by coff. Internally, BFD expects symbol values to be offsets from a section base; so a symbol physically at 0x120, but in a section starting at 0x100, would have the value 0x20. Coff expects symbols to contain their final value, so symbols have their values changed at this point to reflect their sum with their owning section. This transformation uses the `output_section` field of the `asymbol`'s `asection`. See Section 2.6 [Sections], page 25.

- `coff_mangle_symbols`

This routine runs through the provided symbol table and uses the offsets generated by the previous pass and the pointers generated when the symbol table was read in to create the structured hierarchy required by coff. It changes each pointer to a symbol into the index into the symbol table of the `asymbol`.

- `coff_write_symbols`

This routine runs through the symbol table and patches up the symbols from their internal form into the coff way, calls the bit twiddlers, and writes out the table to the file.

### 3.3.2.6 coff\_symbol\_type

The hidden information for an `asymbol` is described in a `combined_entry_type`:

```
typedef struct coff_ptr_struct
{
    /* Remembers the offset from the first symbol in the file for
       this symbol. Generated by coff_renumber_symbols. */
    unsigned int offset;

    /* Selects between the elements of the union below. */
    unsigned int is_sym : 1;

    /* Selects between the elements of the x_sym.x_tagndx union. If set,
       p is valid and the field will be renumbered. */
    unsigned int fix_tag : 1;

    /* Selects between the elements of the x_sym.x_fcny.x_fcn.x_endndx
       union. If set, p is valid and the field will be renumbered. */
    unsigned int fix_end : 1;

    /* Selects between the elements of the x_csect.x_scnlen union. If set,
       p is valid and the field will be renumbered. */
    unsigned int fix_scnlen : 1;

    /* If set, u.syment.n_value contains a pointer to a symbol. The final
       value will be the offset field. Used for XCOFF C_BSTAT symbols. */
    unsigned int fix_value : 1;

    /* If set, u.syment.n_value is an index into the line number entries.
       Used for XCOFF C_BINCL/C_EINCL symbols. */
    unsigned int fix_line : 1;

    /* The container for the symbol structure as read and translated
       from the file. */
    union
    {
        union internal_auxent auxent;
        struct internal_syment syment;
    } u;

    /* An extra pointer which can be used by format based on COFF (like XCOFF)
       to provide extra information to their backend. */
    void *extrap;
} combined_entry_type;

/* Each canonical asymbol really looks like this: */
```

```

typedef struct coff_symbol_struct
{
    /* The actual symbol which the rest of BFD works with */
    asymbol symbol;

    /* A pointer to the hidden information for this symbol */
    combined_entry_type *native;

    /* A pointer to the lineno information for this symbol */
    struct lineno_cache_entry *lineno;

    /* Have the line numbers been relocated yet ? */
    bool done_lineno;
} coff_symbol_type;

```

### 3.3.2.7 bfd\_coff\_backend\_data

```

typedef struct
{
    void (*_bfd_coff_swap_aux_in)
        (bfd *, void *, int, int, int, int, void *);

    void (*_bfd_coff_swap_sym_in)
        (bfd *, void *, void *);

    void (*_bfd_coff_swap_lineno_in)
        (bfd *, void *, void *);

    unsigned int (*_bfd_coff_swap_aux_out)
        (bfd *, void *, int, int, int, int, void *);

    unsigned int (*_bfd_coff_swap_sym_out)
        (bfd *, void *, void *);

    unsigned int (*_bfd_coff_swap_lineno_out)
        (bfd *, void *, void *);

    unsigned int (*_bfd_coff_swap_reloc_out)
        (bfd *, void *, void *);

    unsigned int (*_bfd_coff_swap_filehdr_out)
        (bfd *, void *, void *);

    unsigned int (*_bfd_coff_swap_aouthdr_out)
        (bfd *, void *, void *);
}

```

```

unsigned int (*_bfd_coff_swap_scnhdr_out)
    (bfd *, void *, void *, const asection *);

unsigned int _bfd_filhsz;
unsigned int _bfd_aoutsz;
unsigned int _bfd_scnhsz;
unsigned int _bfd_symesz;
unsigned int _bfd_auxesz;
unsigned int _bfd_relsz;
unsigned int _bfd_linesz;
unsigned int _bfd_filnmlen;
bool _bfd_coff_long_filenames;

bool _bfd_coff_long_section_names;
bool (*_bfd_coff_set_long_section_names)
    (bfd *, int);

unsigned int _bfd_coff_default_section_alignment_power;
bool _bfd_coff_force_symnames_in_strings;
unsigned int _bfd_coff_debug_string_prefix_length;
unsigned int _bfd_coff_max_nscns;

void (*_bfd_coff_swap_filehdr_in)
    (bfd *, void *, void *);

void (*_bfd_coff_swap_aouthdr_in)
    (bfd *, void *, void *);

void (*_bfd_coff_swap_scnhdr_in)
    (bfd *, void *, void *);

void (*_bfd_coff_swap_reloc_in)
    (bfd *abfd, void *, void *);

bool (*_bfd_coff_bad_format_hook)
    (bfd *, void *);

bool (*_bfd_coff_set_arch_mach_hook)
    (bfd *, void *);

void * (*_bfd_coff_mkobject_hook)
    (bfd *, void *, void *);

bool (*_bfd_styp_to_sec_flags_hook)
    (bfd *, void *, const char *, asection *, flagword *);

void (*_bfd_set_alignment_hook)

```



```

    (bfd *, asection *, void *);

bool (*_bfd_coff_slurp_symbol_table)
    (bfd *);

bool (*_bfd_coff_symname_in_debug)
    (bfd *, struct internal_syment *);

bool (*_bfd_coff_pointerize_aux_hook)
    (bfd *, combined_entry_type *, combined_entry_type *,
     unsigned int, combined_entry_type *);

bool (*_bfd_coff_print_aux)
    (bfd *, FILE *, combined_entry_type *, combined_entry_type *,
     combined_entry_type *, unsigned int);

bool (*_bfd_coff_reloc16_extra_cases)
    (bfd *, struct bfd_link_info *, struct bfd_link_order *, arelent *,
     bfd_byte *, size_t *, size_t *);

int (*_bfd_coff_reloc16_estimate)
    (bfd *, asection *, arelent *, unsigned int,
     struct bfd_link_info *);

enum coff_symbol_classification (*_bfd_coff_classify_symbol)
    (bfd *, struct internal_syment *);

bool (*_bfd_coff_compute_section_file_positions)
    (bfd *);

bool (*_bfd_coff_start_final_link)
    (bfd *, struct bfd_link_info *);

bool (*_bfd_coff_relocate_section)
    (bfd *, struct bfd_link_info *, bfd *, asection *, bfd_byte *,
     struct internal_reloc *, struct internal_syment *, asection **);

reloc_howto_type (*_bfd_coff_rtype_to_howto)
    (bfd *, asection *, struct internal_reloc *,
     struct coff_link_hash_entry *, struct internal_syment *, bfd_vma *);

bool (*_bfd_coff_adjust_symndx)
    (bfd *, struct bfd_link_info *, bfd *, asection *,
     struct internal_reloc *, bool *);

bool (*_bfd_coff_link_output_has_begun)
    (bfd *, struct coff_final_link_info *);

```

```

    bool (*_bfd_coff_final_link_postscript)
        (bfd *, struct coff_final_link_info *);

    bool (*_bfd_coff_print_pdata)
        (bfd *, void *);

} bfd_coff_backend_data;

```

### 3.3.2.8 Writing relocations

To write relocations, the back end steps through the canonical relocation table and create an **internal\_reloc**. The symbol index to use is removed from the **offset** field in the symbol table supplied. The address comes directly from the sum of the section base address and the relocation offset; the type is dug directly from the **howto** field. Then the **internal\_reloc** is swapped into the shape of an **external\_reloc** and written out to disk.

### 3.3.2.9 Reading linenumbers

Creating the linenumbers table is done by reading in the entire coff linenumbers table, and creating another table for internal use.

A coff linenumbers table is structured so that each function is marked as having a line number of 0. Each line within the function is an offset from the first line in the function. The base of the line number information for the table is stored in the symbol associated with the function.

Note: The PE format uses line number 0 for a flag indicating a new source file.

The information is copied from the external to the internal table, and each symbol which marks a function is marked by pointing its...

How does this work ?

### 3.3.2.10 Reading relocations

Coff relocations are easily transformed into the internal BFD form (**arelent**).

Reading a coff relocation table is done in the following stages:

- Read the entire coff relocation table into memory.
- Process each relocation in turn; first swap it from the external to the internal form.
- Turn the symbol referenced in the relocation's symbol index into a pointer into the canonical symbol table. This table is the same as the one returned by a call to **bfd\_canonicalize\_sytab**. The back end will call that routine and save the result if a canonicalization hasn't been done.
- The reloc index is turned into a pointer to a howto structure, in a back end specific way. For instance, the 386 uses the **r\_type** to directly produce an index into a howto table vector.
- Note that **arelent.addend** for COFF is often not what most people understand as a relocation addend, but rather an adjustment to the relocation addend stored in section contents of relocatable object files. The value found in section contents may also be

confusing, depending on both symbol value and addend somewhat similar to the field value for a final-linked object. See `CALC_ADDEND`.

### 3.4 ELF backends

BFD support for ELF formats is being worked on. Currently, the best supported back ends are for sparc and i386 (running svr4 or Solaris 2).

Documentation of the internals of the support code still needs to be written. The code is changing quickly enough that we haven't bothered yet.

### 3.5 mmo backend

The mmo object format is used exclusively together with Professor Donald E. Knuth's educational 64-bit processor MMIX. The simulator `mmix` which is available at <http://mmix.cs.hm.edu/src/index.html> understands this format. That package also includes a combined assembler and linker called `mmixal`. The mmo format has no advantages feature-wise compared to e.g. ELF. It is a simple non-relocatable object format with no support for archives or debugging information, except for symbol value information and line numbers (which is not yet implemented in BFD). See <http://mmix.cs.hm.edu/> for more information about MMIX. The ELF format is used for intermediate object files in the BFD implementation.

#### 3.5.1 File layout

The mmo file contents is not partitioned into named sections as with e.g. ELF. Memory areas is formed by specifying the location of the data that follows. Only the memory area '0x0000...00' to '0x01ff...ff' is executable, so it is used for code (and constants) and the area '0x2000...00' to '0x20ff...ff' is used for writable data. See Section 3.5.3 [mmo section mapping], page 206.

There is provision for specifying "special data" of 65536 different types. We use type 80 (decimal), arbitrarily chosen the same as the ELF `e_machine` number for MMIX, filling it with section information normally found in ELF objects. See Section 3.5.3 [mmo section mapping], page 206.

Contents is entered as 32-bit words, xored over previous contents, always zero-initialized. A word that starts with the byte '0x98' forms a command called a 'lopcode', where the next byte distinguished between the thirteen lopcodes. The two remaining bytes, called the 'Y' and 'Z' fields, or the 'YZ' field (a 16-bit big-endian number), are used for various purposes different for each lopcode. As documented in <http://mmix.cs.hm.edu/doc/mmixal.pdf>, the lopcodes are:

##### lop\_quote

0x98000001. The next word is contents, regardless of whether it starts with 0x98 or not.

##### lop\_loc

0x9801YYZZ, where 'Z' is 1 or 2. This is a location directive, setting the location for the next data to the next 32-bit word (for  $Z = 1$ ) or 64-bit word (for  $Z = 2$ ), plus  $Y * 2^{56}$ . Normally 'Y' is 0 for the text segment and 2 for the data segment. Beware that the low bits of non- tetrabyte-aligned values are silently discarded when being automatically incremented and when storing contents (in contrast

to e.g. its use as current location when followed by `lop_fixo` et al before the next possibly-quoted tetrabyte contents).

- `lop_skip` 0x9802YYZZ. Increase the current location by ‘YZ’ bytes.
- `lop_fixo` 0x9803YYZZ, where ‘Z’ is 1 or 2. Store the current location as 64 bits into the location pointed to by the next 32-bit ( $Z = 1$ ) or 64-bit ( $Z = 2$ ) word, plus  $Y * 2^6$ .
- `lop_fixr` 0x9804YYZZ. ‘YZ’ is stored into the current location plus  $2 - 4 * YZ$ .
- `lop_fixrx` 0x980500ZZ. ‘Z’ is 16 or 24. A value ‘L’ derived from the following 32-bit word are used in a manner similar to ‘YZ’ in `lop_fixr`: it is xor’ed into the current location minus  $4 * L$ . The first byte of the word is 0 or 1. If it is 1, then  $L = (\text{lowest24bitsofword}) - 2^Z$ , if 0, then  $L = (\text{lowest24bitsofword})$ .
- `lop_file` 0x9806YYZZ. ‘Y’ is the file number, ‘Z’ is count of 32-bit words. Set the file number to ‘Y’ and the line counter to 0. The next  $Z * 4$  bytes contain the file name, padded with zeros if the count is not a multiple of four. The same ‘Y’ may occur multiple times, but ‘Z’ must be 0 for all but the first occurrence.
- `lop_line` 0x9807YYZZ. ‘YZ’ is the line number. Together with `lop_file`, it forms the source location for the next 32-bit word. Note that for each non-lopcode 32-bit word, line numbers are assumed incremented by one.
- `lop_spec` 0x9808YYZZ. ‘YZ’ is the type number. Data until the next lopcode other than `lop_quote` forms special data of type ‘YZ’. See Section 3.5.3 [mmo section mapping], page 206.
- Other types than 80, (or type 80 with a content that does not parse) is stored in sections named `.MMIX.spec_data.n` where  $n$  is the ‘YZ’-type. The flags for such a sections say not to allocate or load the data. The vma is 0. Contents of multiple occurrences of special data  $n$  is concatenated to the data of the previous `lop_spec ns`. The location in data or code at which the `lop_spec` occurred is lost.
- `lop_pre` 0x980901ZZ. The first lopcode in a file. The ‘Z’ field forms the length of header information in 32-bit words, where the first word tells the time in seconds since ‘00:00:00 GMT Jan 1 1970’.
- `lop_post` 0x980a00ZZ.  $Z > 32$ . This lopcode follows after all content-generating lpcodes in a program. The ‘Z’ field denotes the value of ‘rG’ at the beginning of the program. The following  $256 - Z$  big-endian 64-bit words are loaded into global registers ‘\$G’ ... ‘\$255’.
- `lop_stab` 0x980b0000. The next-to-last lopcode in a program. Must follow immediately after the `lop_post` lopcode and its data. After this lopcode follows all symbols in a compressed format (see Section 3.5.2 [Symbol-table], page 204).
- `lop_end` 0x980cYYZZ. The last lopcode in a program. It must follow the `lop_stab` lopcode and its data. The ‘YZ’ field contains the number of 32-bit words of symbol table information after the preceding `lop_stab` lopcode.

Note that the lopcode "fixups"; `lop_fixr`, `lop_fixrx` and `lop_fixo` are not generated by BFD, but are handled. They are generated by `mmixal`.

This trivial one-label, one-instruction file:

```
:Main TRAP 1,2,3
```

can be represented this way in mmo:

```
0x98090101 - lop_pre, one 32-bit word with timestamp.
<timestamp>
0x98010002 - lop_loc, text segment, using a 64-bit address.
    Note that mmixal does not emit this for the file above.
0x00000000 - Address, high 32 bits.
0x00000000 - Address, low 32 bits.
0x98060002 - lop_file, 2 32-bit words for file-name.
0x74657374 - "test"
0x2e730000 - ".s\0\0"
0x98070001 - lop_line, line 1.
0x00010203 - TRAP 1,2,3
0x980a00ff - lop_post, setting $255 to 0.
0x00000000
0x00000000
0x980b0000 - lop_stab for ":Main" = 0, serial 1.
0x203a4040 See Section 3.5.2 [Symbol-table], page 204.
0x10404020
0x4d206120
0x69016e00
0x81000000
0x980c0005 - lop_end; symbol table contained five 32-bit words.
```

### 3.5.2 Symbol table format

From mmixal.w (or really, the generated mmixal.tex) in the MMIXware package which also contains the mmix simulator: “Symbols are stored and retrieved by means of a ‘ternary search trie’, following ideas of Bentley and Sedgewick. (See ACM–SIAM Symp. on Discrete Algorithms ‘8’ (1997), 360–369; R.Sedgewick, ‘Algorithms in C’ (Reading, Mass. Addison–Wesley, 1998), ‘15.4’.) Each trie node stores a character, and there are branches to subtrees for the cases where a given character is less than, equal to, or greater than the character in the trie. There also is a pointer to a symbol table entry if a symbol ends at the current node.”

So it’s a tree encoded as a stream of bytes. The stream of bytes acts on a single virtual global symbol, adding and removing characters and signalling complete symbol points. Here, we read the stream and create symbols at the completion points.

First, there’s a control byte *m*. If any of the listed bits in *m* is nonzero, we execute what stands at the right, in the listed order:

```
(MM03_LEFT)
0x40 - Traverse left trie.
    (Read a new command byte and recurse.)

(MM03_SYMBITS)
0x2f - Read the next byte as a character and store it in the
```

current character position; increment character position.  
Test the bits of m:

(MM03\_WCHAR)

0x80 - The character is 16-bit (so read another byte,  
merge into current character.

(MM03\_TYPEBITS)

0xf - We have a complete symbol; parse the type, value  
and serial number and do what should be done  
with a symbol. The type and length information  
is in  $j = (m \& 0xf)$ .

(MM03\_REGQUAL\_BITS)

$j == 0xf$ : A register variable. The following  
byte tells which register.  
 $j \leq 8$ : An absolute symbol. Read  $j$  bytes as the  
big-endian number the symbol equals.  
A  $j = 2$  with two zero bytes denotes an  
unknown symbol.  
 $j > 8$ : As with  $j \leq 8$ , but add  $(0x20 \ll 56)$   
to the value in the following  $j - 8$   
bytes.

Then comes the serial number, as a variant of  
uleb128, but better named ubeb128:  
Read bytes and shift the previous value left 7  
(multiply by 128). Add in the new byte, repeat  
until a byte has bit 7 set. The serial number  
is the computed value minus 128.

(MM03\_MIDDLE)

0x20 - Traverse middle trie. (Read a new command byte  
and recurse.) Decrement character position.

(MM03\_RIGHT)

0x10 - Traverse right trie. (Read a new command byte and  
recurse.)

Let's look again at the `lop_stab` for the trivial file (see Section 3.5.1 [File layout], page 202).

```
0x980b0000 - lop_stab for ":Main" = 0, serial 1.
0x203a4040
0x10404020
0x4d206120
0x69016e00
0x81000000
```

This forms the trivial trie (note that the path between “:” and “M” is redundant):

```

203a      ":"
40        /
40        /
10        \
40        /
40        /
204d      "M"
2061      "a"
2069      "i"
016e      "n" is the last character in a full symbol, and
          with a value represented in one byte.
00        The value is 0.
81        The serial number is 1.

```

### 3.5.3 mmo section mapping

The implementation in BFD uses special data type 80 (decimal) to encapsulate and describe named sections, containing e.g. debug information. If needed, any datum in the encapsulation will be quoted using `lop_quote`. First comes a 32-bit word holding the number of 32-bit words containing the zero-terminated zero-padded segment name. After the name there's a 32-bit word holding flags describing the section type. Then comes a 64-bit big-endian word with the section length (in bytes), then another with the section start address. Depending on the type of section, the contents might follow, zero-padded to 32-bit boundary. For a loadable section (such as data or code), the contents might follow at some later point, not necessarily immediately, as a `lop_loc` with the same start address as in the section description, followed by the contents. This in effect forms a descriptor that must be emitted before the actual contents. Sections described this way must not overlap.

For areas that don't have such descriptors, synthetic sections are formed by BFD. Consecutive contents in the two memory areas '0x0000...00' to '0x01ff...ff' and '0x2000...00' to '0x20ff...ff' are entered in sections named `.text` and `.data` respectively. If an area is not otherwise described, but would together with a neighboring lower area be less than '0x40000000' bytes long, it is joined with the lower area and the gap is zero-filled. For other cases, a new section is formed, named `.MMIX.sec.n`. Here, *n* is a number, a running count through the mmo file, starting at 0.

A loadable section specified as:

```

.section secname,"ax"
TETRA 1,2,3,4,-1,-2009
BYTE 80

```

and linked to address '0x4', is represented by the sequence:

```

0x98080050 - lop_spec 80
0x00000002 - two 32-bit words for the section name
0x7365636e - "secn"
0x616d6500 - "ame\0"
0x00000033 - flags CODE, READONLY, LOAD, ALLOC
0x00000000 - high 32 bits of section length
0x0000001c - section length is 28 bytes; 6 * 4 + 1 + alignment to 32 bits
0x00000000 - high 32 bits of section address

```

```

0x00000004 - section address is 4
0x98010002 - 64 bits with address of following data
0x00000000 - high 32 bits of address
0x00000004 - low 32 bits: data starts at address 4
0x00000001 - 1
0x00000002 - 2
0x00000003 - 3
0x00000004 - 4
0xffffffff - -1
0xfffff827 - -2009
0x50000000 - 80 as a byte, padded with zeros.

```

Note that the `lop_spec` wrapping does not include the section contents. Compare this to a non-loaded section specified as:

```

.section thirdsec
TETRA 200001,100002
BYTE 38,40

```

This, when linked to address ‘0x2000000000000001c’, is represented by:

```

0x98080050 - lop_spec 80
0x00000002 - two 32-bit words for the section name
0x7365636e - "thir"
0x616d6500 - "dsec"
0x00000010 - flag READONLY
0x00000000 - high 32 bits of section length
0x0000000c - section length is 12 bytes; 2 * 4 + 2 + alignment to 32 bits
0x20000000 - high 32 bits of address
0x0000001c - low 32 bits of address 0x2000000000000001c
0x00030d41 - 200001
0x000186a2 - 100002
0x26280000 - 38, 40 as bytes, padded with zeros

```

For the latter example, the section contents must not be loaded in memory, and is therefore specified as part of the special data. The address is usually unimportant but might provide information for e.g. the DWARF 2 debugging format.

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| BFD_RELOC_32                | 57  | BFD_RELOC_390_TLS_GD32     | 100 |
| BFD_RELOC_32_BASEREL        | 58  | BFD_RELOC_390_TLS_GD64     | 100 |
| BFD_RELOC_32_GOT_PCREL      | 57  | BFD_RELOC_390_TLS_GDCALL   | 100 |
| BFD_RELOC_32_GOTOFF         | 57  | BFD_RELOC_390_TLS_GOTIE12  | 100 |
| BFD_RELOC_32_PCREL          | 57  | BFD_RELOC_390_TLS_GOTIE20  | 101 |
| BFD_RELOC_32_PCREL_S2       | 59  | BFD_RELOC_390_TLS_GOTIE32  | 100 |
| BFD_RELOC_32_PLT_PCREL      | 58  | BFD_RELOC_390_TLS_GOTIE64  | 100 |
| BFD_RELOC_32_PLTOFF         | 58  | BFD_RELOC_390_TLS_IE32     | 100 |
| BFD_RELOC_32_SECREL         | 57  | BFD_RELOC_390_TLS_IE64     | 101 |
| BFD_RELOC_386_GOT32         | 68  | BFD_RELOC_390_TLS_IEENT    | 101 |
| BFD_RELOC_386_GOT32X        | 68  | BFD_RELOC_390_TLS_LDCALL   | 100 |
| BFD_RELOC_386_GOTOFF        | 68  | BFD_RELOC_390_TLS_LDM32    | 100 |
| BFD_RELOC_386_GOTPC         | 68  | BFD_RELOC_390_TLS_LDM64    | 100 |
| BFD_RELOC_386_PLT32         | 68  | BFD_RELOC_390_TLS_LD032    | 101 |
| BFD_RELOC_386_TLS_DESC      | 68  | BFD_RELOC_390_TLS_LD064    | 101 |
| BFD_RELOC_386_TLS_DESC_CALL | 68  | BFD_RELOC_390_TLS_LE32     | 101 |
| BFD_RELOC_386_TLS_DTPMOD32  | 68  | BFD_RELOC_390_TLS_LE64     | 101 |
| BFD_RELOC_386_TLS_DTPOFF32  | 68  | BFD_RELOC_390_TLS_LOAD     | 100 |
| BFD_RELOC_386_TLS_GD        | 68  | BFD_RELOC_390_TLS_TPOFF    | 101 |
| BFD_RELOC_386_TLS_GOTDESC   | 68  | BFD_RELOC_64               | 57  |
| BFD_RELOC_386_TLS_GOTIE     | 68  | BFD_RELOC_64_PCREL         | 57  |
| BFD_RELOC_386_TLS_IE        | 68  | BFD_RELOC_64_PLT_PCREL     | 58  |
| BFD_RELOC_386_TLS_IE_32     | 68  | BFD_RELOC_64_PLTOFF        | 58  |
| BFD_RELOC_386_TLS_LDM       | 68  | BFD_RELOC_68K_TLS_GD16     | 58  |
| BFD_RELOC_386_TLS_LDO_32    | 68  | BFD_RELOC_68K_TLS_GD32     | 58  |
| BFD_RELOC_386_TLS_LE        | 68  | BFD_RELOC_68K_TLS_GD8      | 58  |
| BFD_RELOC_386_TLS_LE_32     | 68  | BFD_RELOC_68K_TLS_IE16     | 58  |
| BFD_RELOC_386_TLS_TPOFF     | 68  | BFD_RELOC_68K_TLS_IE32     | 58  |
| BFD_RELOC_386_TLS_TPOFF32   | 68  | BFD_RELOC_68K_TLS_IE8      | 58  |
| BFD_RELOC_390_12            | 99  | BFD_RELOC_68K_TLS_LDM16    | 58  |
| BFD_RELOC_390_20            | 101 | BFD_RELOC_68K_TLS_LDM32    | 58  |
| BFD_RELOC_390_GOT12         | 99  | BFD_RELOC_68K_TLS_LDM8     | 58  |
| BFD_RELOC_390_GOT16         | 99  | BFD_RELOC_68K_TLS_LD016    | 58  |
| BFD_RELOC_390_GOT20         | 101 | BFD_RELOC_68K_TLS_LD032    | 58  |
| BFD_RELOC_390_GOT64         | 100 | BFD_RELOC_68K_TLS_LD08     | 58  |
| BFD_RELOC_390_GOTENT        | 100 | BFD_RELOC_68K_TLS_LE16     | 58  |
| BFD_RELOC_390_GOTOFF64      | 100 | BFD_RELOC_68K_TLS_LE32     | 58  |
| BFD_RELOC_390_GOTPC         | 99  | BFD_RELOC_68K_TLS_LE8      | 58  |
| BFD_RELOC_390_GOTPCDBL      | 100 | BFD_RELOC_8                | 57  |
| BFD_RELOC_390_GOTPLT12      | 100 | BFD_RELOC_8_BASEREL        | 59  |
| BFD_RELOC_390_GOTPLT16      | 100 | BFD_RELOC_8_FFnn           | 59  |
| BFD_RELOC_390_GOTPLT20      | 101 | BFD_RELOC_8_GOT_PCREL      | 57  |
| BFD_RELOC_390_GOTPLT32      | 100 | BFD_RELOC_8_GOTOFF         | 58  |
| BFD_RELOC_390_GOTPLT64      | 100 | BFD_RELOC_8_PCREL          | 57  |
| BFD_RELOC_390_GOTPLTENT     | 100 | BFD_RELOC_8_PLT_PCREL      | 58  |
| BFD_RELOC_390_PC12DBL       | 99  | BFD_RELOC_8_PLTOFF         | 58  |
| BFD_RELOC_390_PC16DBL       | 99  | BFD_RELOC_AARCH64_16       | 118 |
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| BFD_RELOC_AARCH64_64                 | 118 | BFD_RELOC_AARCH64_TLSDESC_ADD          | 125 |
| BFD_RELOC_AARCH64_64_PCREL           | 118 | BFD_RELOC_AARCH64_TLSDESC_ADD_L012     | 125 |
| BFD_RELOC_AARCH64_ADD_L012           | 120 | BFD_RELOC_AARCH64_TLSDESC_ADR_PAGE21   | 125 |
| BFD_RELOC_AARCH64_ADR_GOT_PAGE       | 121 | BFD_RELOC_AARCH64_TLSDESC_ADR_PREL21   | 125 |
| BFD_RELOC_AARCH64_ADR_HI21_NC_PCREL  | 120 | BFD_RELOC_AARCH64_TLSDESC_CALL         | 125 |
| BFD_RELOC_AARCH64_ADR_HI21_PCREL     | 120 | BFD_RELOC_AARCH64_TLSDESC_LD_L012_NC   | 126 |
| BFD_RELOC_AARCH64_ADR_L021_PCREL     | 120 | BFD_RELOC_AARCH64_TLSDESC_LD_PREL19    | 125 |
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| BFD_RELOC_AARCH64_BRANCH9            | 126 | BFD_RELOC_AARCH64_TLSDESC_LD64_L012    | 125 |
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| BFD_RELOC_AARCH64_JUMP_SLOT          | 125 | BFD_RELOC_AARCH64_TLSD_MOVW_GO_NC      | 122 |
| BFD_RELOC_AARCH64_JUMP26             | 120 | BFD_RELOC_AARCH64_TLSD_MOVW_G1         | 122 |
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| BFD_RELOC_AARCH64_LD32_GOTPAGE_L014  | 122 | GOTTPREL_L012_NC                       | 126 |
| BFD_RELOC_AARCH64_LD64_GOT_L012_NC   | 121 | BFD_RELOC_AARCH64_TLSIE_LD_            |     |
| BFD_RELOC_AARCH64_LD64_GOTOFF_L015   | 121 | GOTTPREL_PREL19                        | 122 |
| BFD_RELOC_AARCH64_LD64_GOTPAGE_L015  | 122 | BFD_RELOC_AARCH64_TLSIE_LD32_          |     |
| BFD_RELOC_AARCH64_LDST_L012          | 125 | GOTTPREL_L012_NC                       | 122 |
| BFD_RELOC_AARCH64_LDST128_L012       | 121 | BFD_RELOC_AARCH64_TLSIE_LD64_          |     |
| BFD_RELOC_AARCH64_LDST16_L012        | 121 | GOTTPREL_L012_NC                       | 122 |
| BFD_RELOC_AARCH64_LDST32_L012        | 121 | BFD_RELOC_AARCH64_TLSIE_MOVW_          |     |
| BFD_RELOC_AARCH64_LDST64_L012        | 121 | GOTTPREL_GO_NC                         | 122 |
| BFD_RELOC_AARCH64_LDST8_L012         | 120 | BFD_RELOC_AARCH64_TLSIE_               |     |
| BFD_RELOC_AARCH64_MOVW_GO            | 119 | MOVW_GOTTPREL_G1                       | 122 |
| BFD_RELOC_AARCH64_MOVW_GO_NC         | 119 | BFD_RELOC_AARCH64_TSLD_                |     |
| BFD_RELOC_AARCH64_MOVW_GO_S          | 119 | ADD_DTPREL_HI12                        | 122 |
| BFD_RELOC_AARCH64_MOVW_G1            | 119 | BFD_RELOC_AARCH64_TSLD_                |     |
| BFD_RELOC_AARCH64_MOVW_G1_NC         | 119 | ADD_DTPREL_L012                        | 122 |
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| BFD_RELOC_AARCH64_MOVW_G2_NC         | 119 | BFD_RELOC_AARCH64_TSLD_ADD_L012_NC     | 123 |
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| BFD_RELOC_AARCH64_MOVW_GOTOFF_GO_NC  | 121 | BFD_RELOC_AARCH64_TSLD_                |     |
| BFD_RELOC_AARCH64_MOVW_GOTOFF_G1     | 121 | LDST_DTPREL_L012                       | 125 |
| BFD_RELOC_AARCH64_MOVW_PREL_GO       | 119 | BFD_RELOC_AARCH64_TSLD_LDST_           |     |
| BFD_RELOC_AARCH64_MOVW_PREL_GO_NC    | 119 | DTPREL_L012_NC                         | 126 |
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| BFD_RELOC_AARCH64_MOVW_PREL_G1_NC    | 120 | LDST16_DTPREL_L012                     | 123 |
| BFD_RELOC_AARCH64_MOVW_PREL_G2       | 120 | BFD_RELOC_AARCH64_TSLD_LDST16_         |     |
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| BFD_RELOC_AARCH64_NONE               | 118 | LDST32_DTPREL_L012                     | 123 |
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| BFD_RELOC_AARCH64_RELATIVE           | 125 | DTPREL_L012_NC                         | 123 |
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| BFD_RELOC_AARCH64_RELOC_START        | 118 | LDST64_DTPREL_L012                     | 123 |
| BFD_RELOC_AARCH64_TLS_DTPMOD         | 125 | BFD_RELOC_AARCH64_TSLD_LDST64_         |     |
| BFD_RELOC_AARCH64_TLS_DTPREL         | 125 | DTPREL_L012_NC                         | 123 |

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| LDST8_DTPREL_L012                      | 123 | BFD_RELOC_ALPHA_GOTDTPREL16  | 63 |
| BFD_RELOC_AARCH64_TLSLD_LDST8_         |     | BFD_RELOC_ALPHA_GOTTTPREL16  | 63 |
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| BFD_RELOC_AARCH64_TLSLD_MOVW_DTPREL_GO | 123 | BFD_RELOC_ALPHA_GDISP_HI16   | 61 |
| BFD_RELOC_AARCH64_TLSLD_               |     | BFD_RELOC_ALPHA_GDISP_L016   | 61 |
| MOVW_DTPREL_GO_NC                      | 123 | BFD_RELOC_ALPHA_GPREL_HI16   | 62 |
| BFD_RELOC_AARCH64_TLSLD_MOVW_DTPREL_G1 | 123 | BFD_RELOC_ALPHA_GPREL_L016   | 62 |
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| BFD_RELOC_AARCH64_TLSLE_ADD_TPREL_HI12 | 124 | BFD_RELOC_ALPHA_LITERAL      | 62 |
| BFD_RELOC_AARCH64_TLSLE_ADD_TPREL_L012 | 124 | BFD_RELOC_ALPHA_LITUSE       | 62 |
| BFD_RELOC_AARCH64_TLSLE_ADD_           |     | BFD_RELOC_ALPHA_NOP          | 62 |
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| TPREL_L012_NC                          | 126 | BFD_RELOC_ALPHA_TPREL16      | 63 |
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| BFD_RELOC_AARCH64_TLSLE_               |     | BFD_RELOC_ARC_JLI_SECTOFF    | 80 |
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| BFD_RELOC_AARCH64_TLSLE_               |     | BFD_RELOC_ARC_N24            | 79 |
| LDST64_TPREL_L012_NC                   | 124 | BFD_RELOC_ARC_N32            | 79 |
| BFD_RELOC_AARCH64_TLSLE_               |     | BFD_RELOC_ARC_N32_ME         | 79 |
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| BFD_RELOC_AARCH64_TLSLE_LDST8_         |     | BFD_RELOC_ARC_NPS_CMEM16     | 80 |
| TPREL_L012_NC                          | 125 | BFD_RELOC_ARC_S13_PCREL      | 79 |
| BFD_RELOC_AARCH64_TLSLE_MOVW_TPREL_GO  | 124 | BFD_RELOC_ARC_S21H_PCREL     | 79 |
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| BFD_RELOC_AARCH64_TLSLE_               |     | BFD_RELOC_ARC_S25H_PCREL     | 79 |
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The body of this manual is set in  
cmr10,  
with headings in **cmbx10**  
and examples in **cmtt10**.  
*cmti10* and  
*cmsl10*  
are used for emphasis.